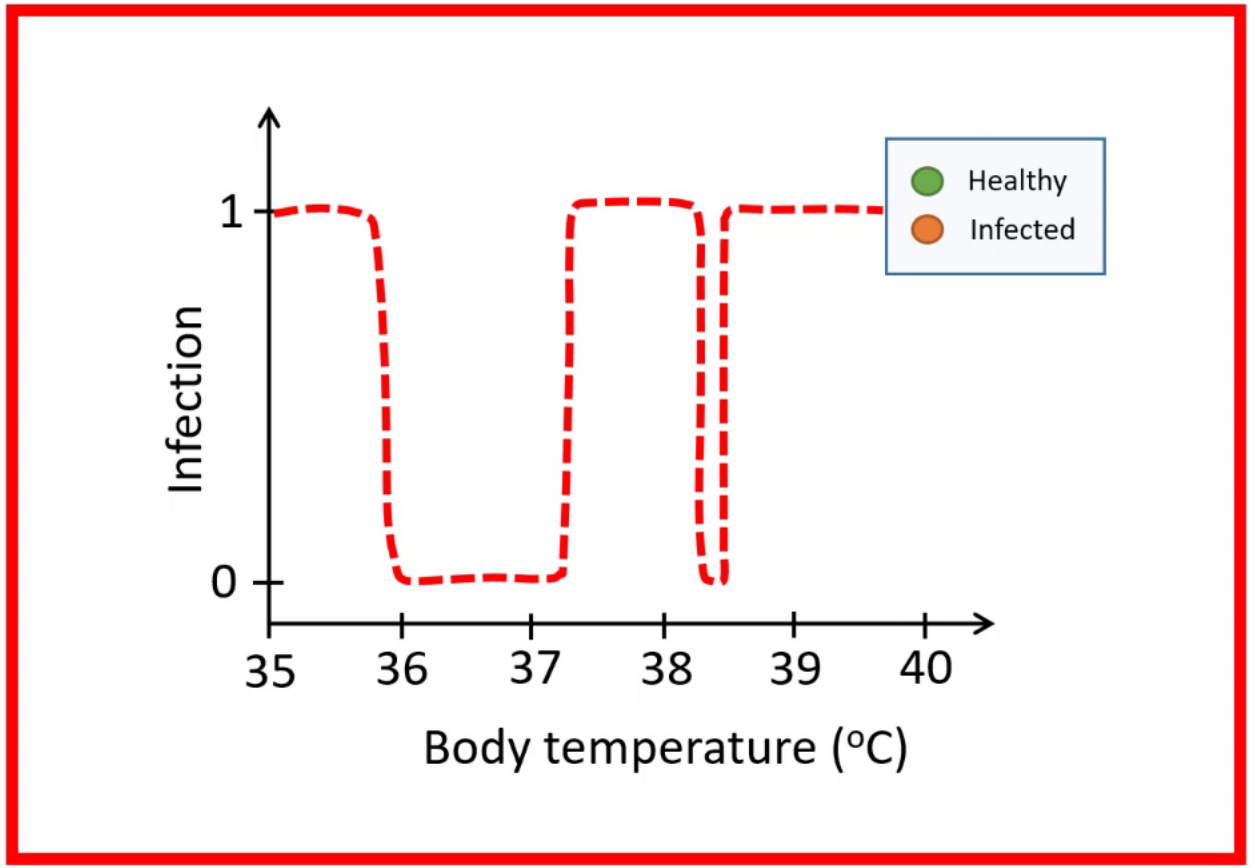
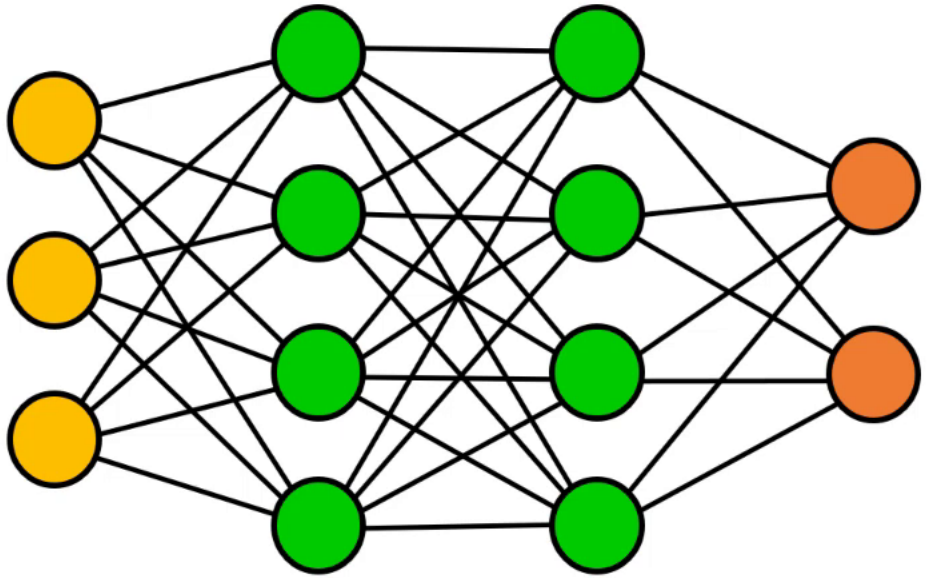
Overfitting in ANNs

Dr.Hafeez Anwar

Deep learning



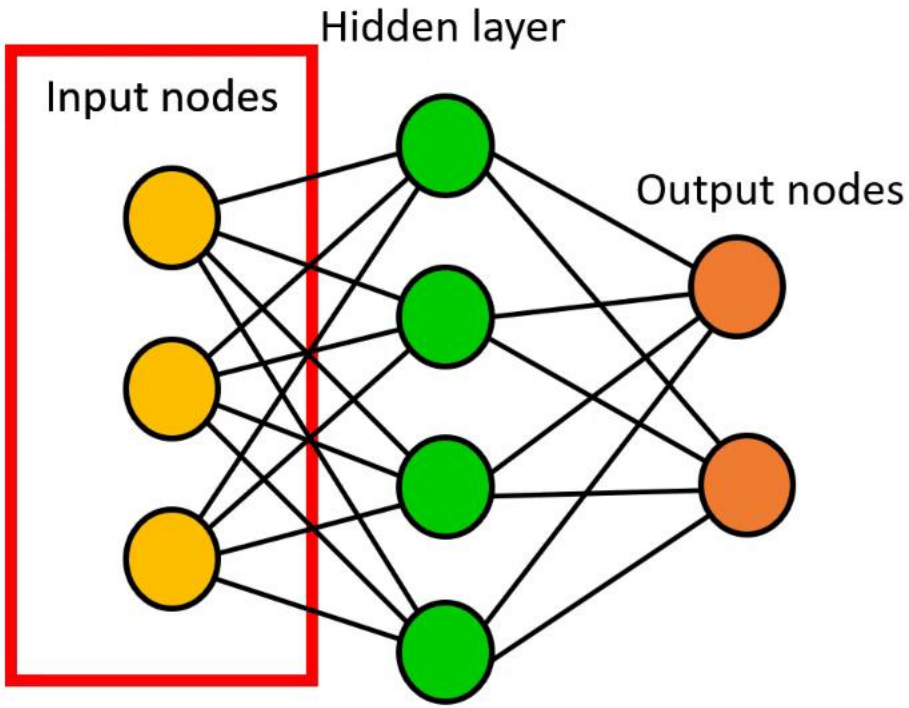


Deep learning

- Deep neural networks
- Recurrent neural networks
- Convolutional neural networks

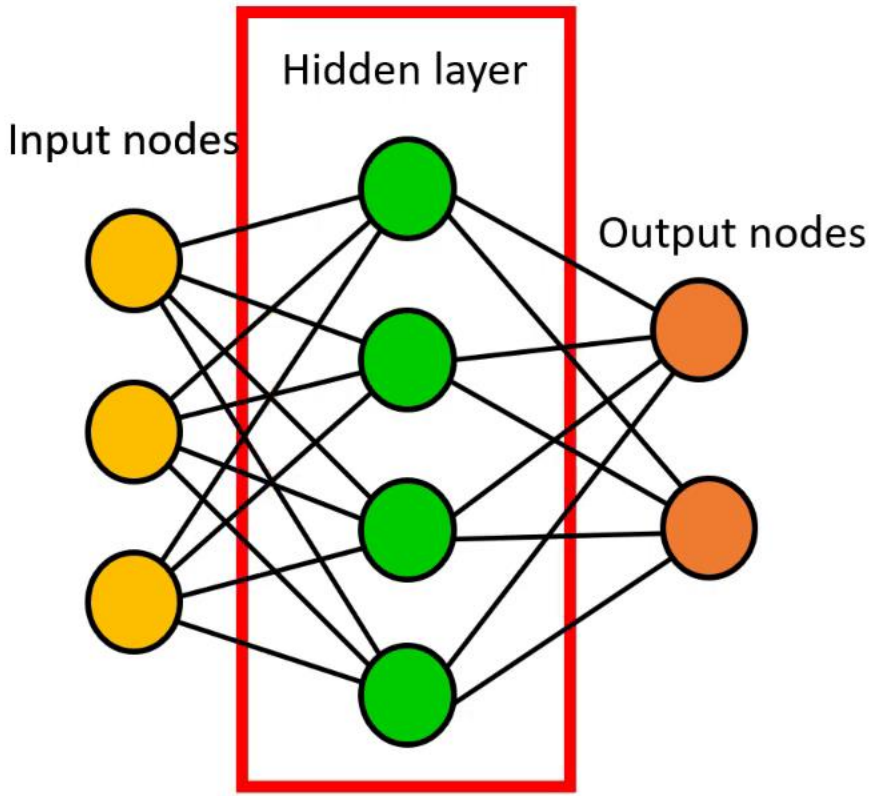
Deep learning refers to neural networks with several hidden layers that are used in for example deep neural networks, recurrent neural networks and convolutional neural networks.

An artificial neural network



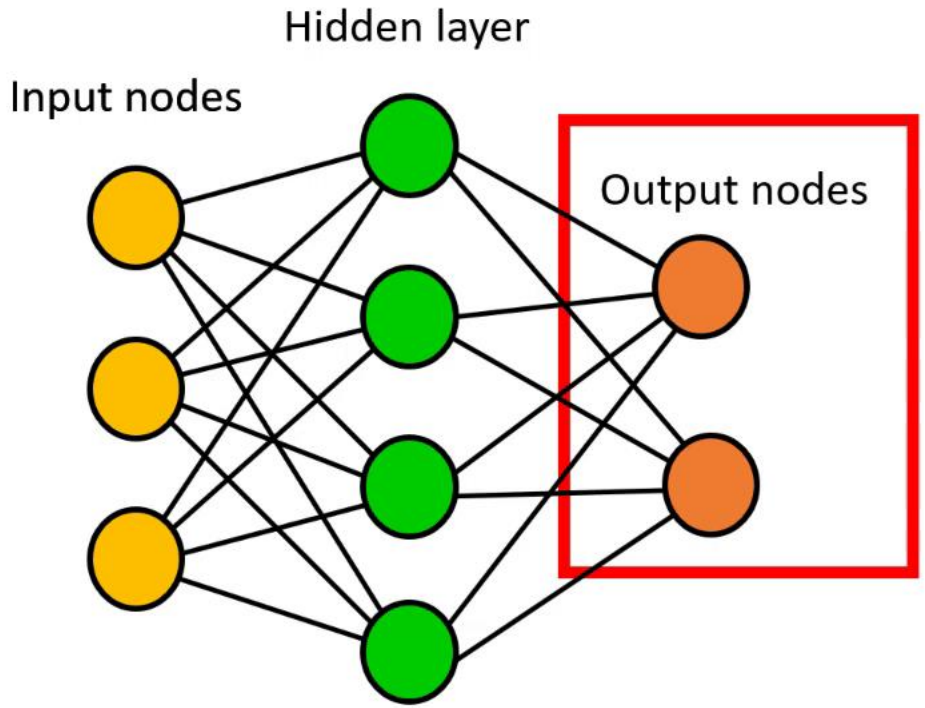
An artificial neural network generally consists of input nodes,

An artificial neural network



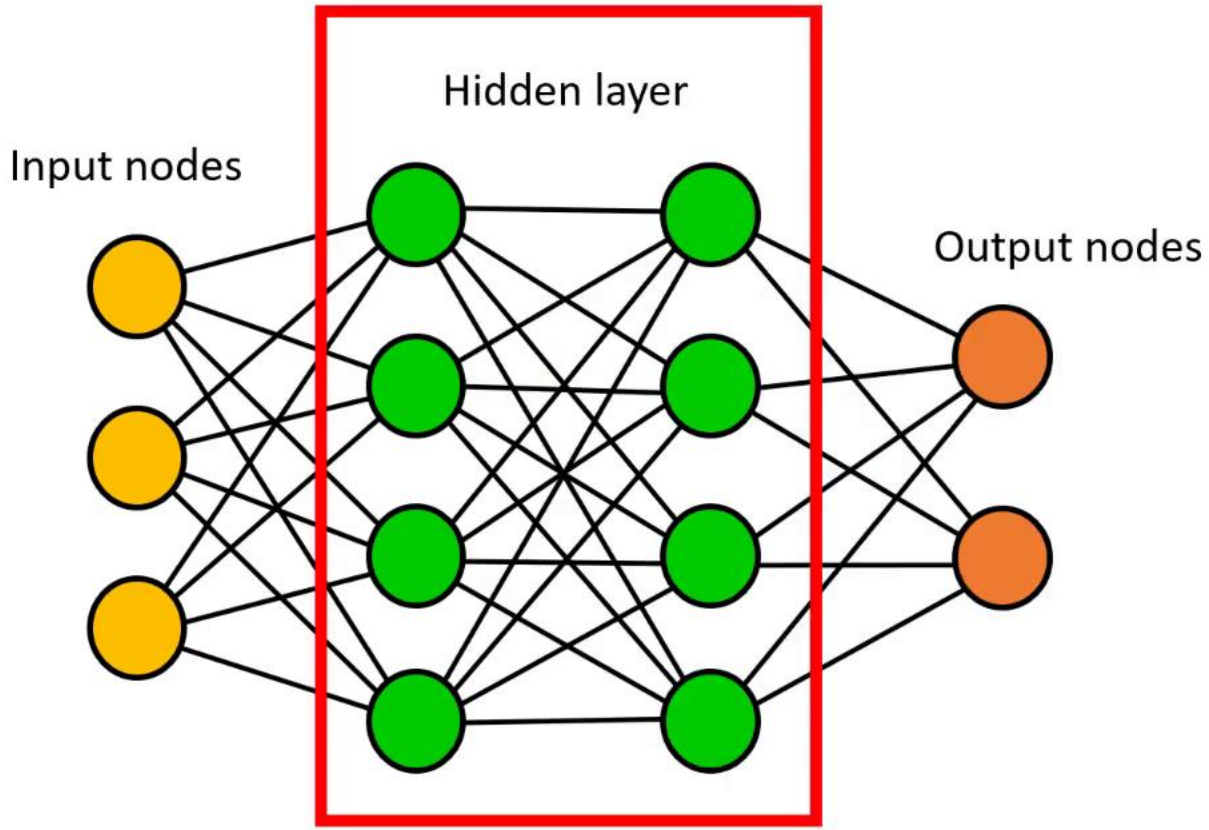
a hidden layer,

An artificial neural network



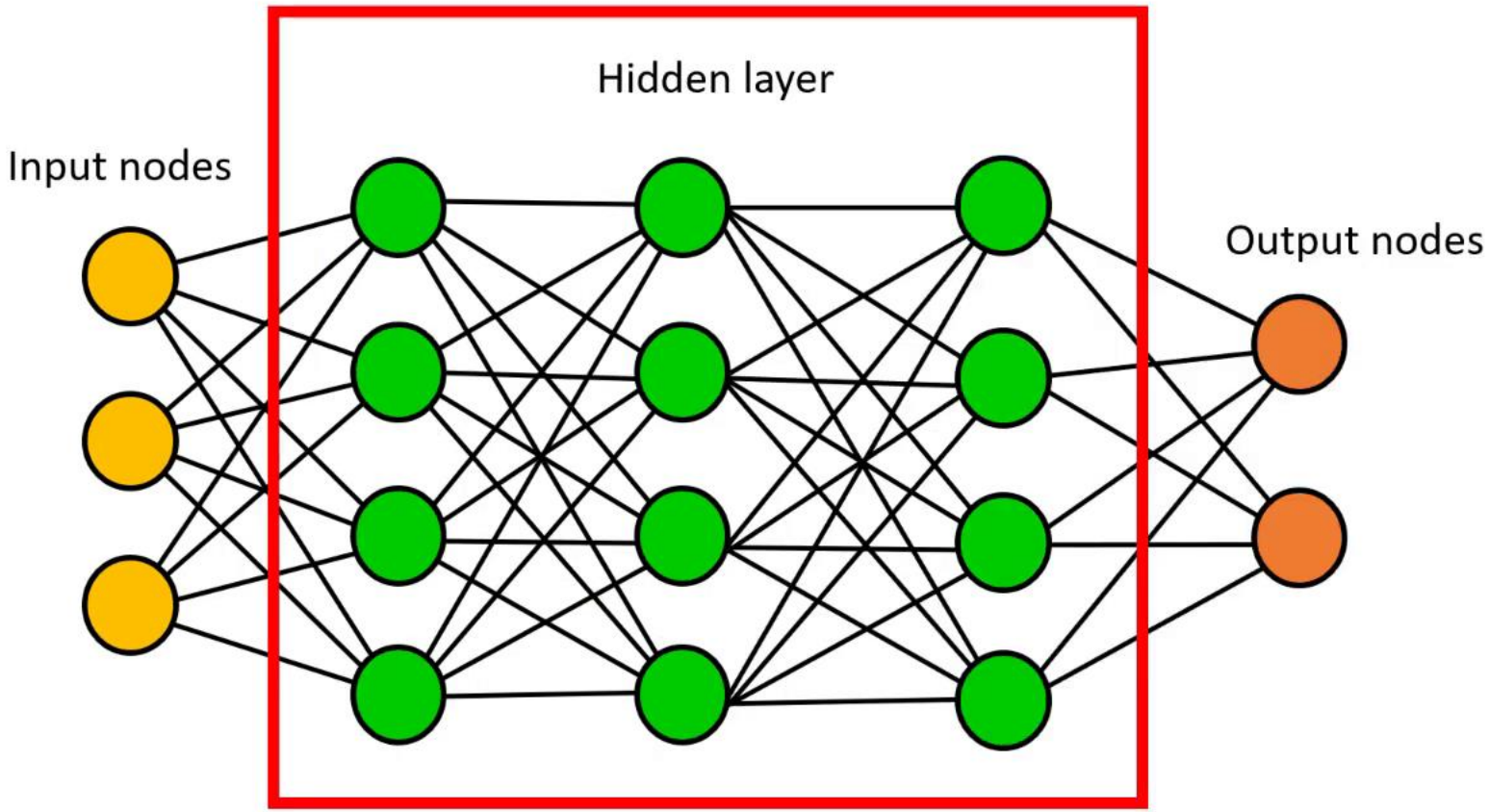
and output nodes.

A deep neural network



When the network consists of more than one hidden layer, it can be referred to as a deep neural network. In this case, the network consists of two hidden layers.

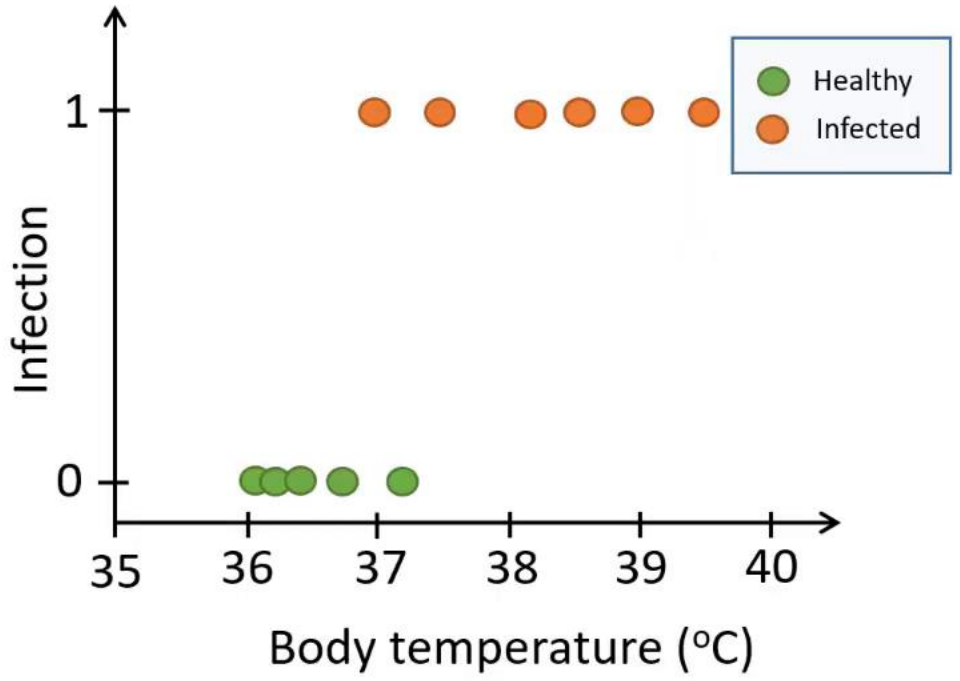
A deep neural network

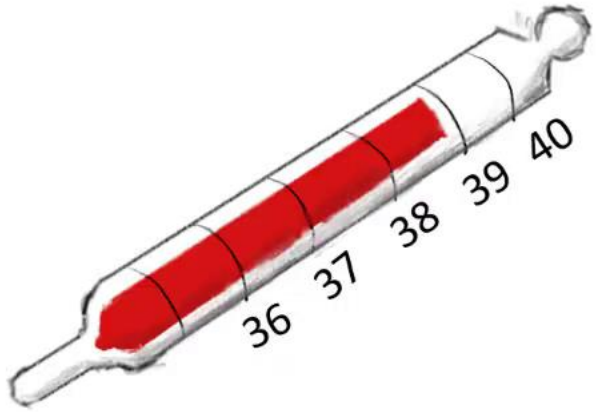


By adding more hidden layers, we will get a deeper network. A deeper network can learn more complicated patterns. In this video, I will try to explain why that is the case and how that is related to overfitting.

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

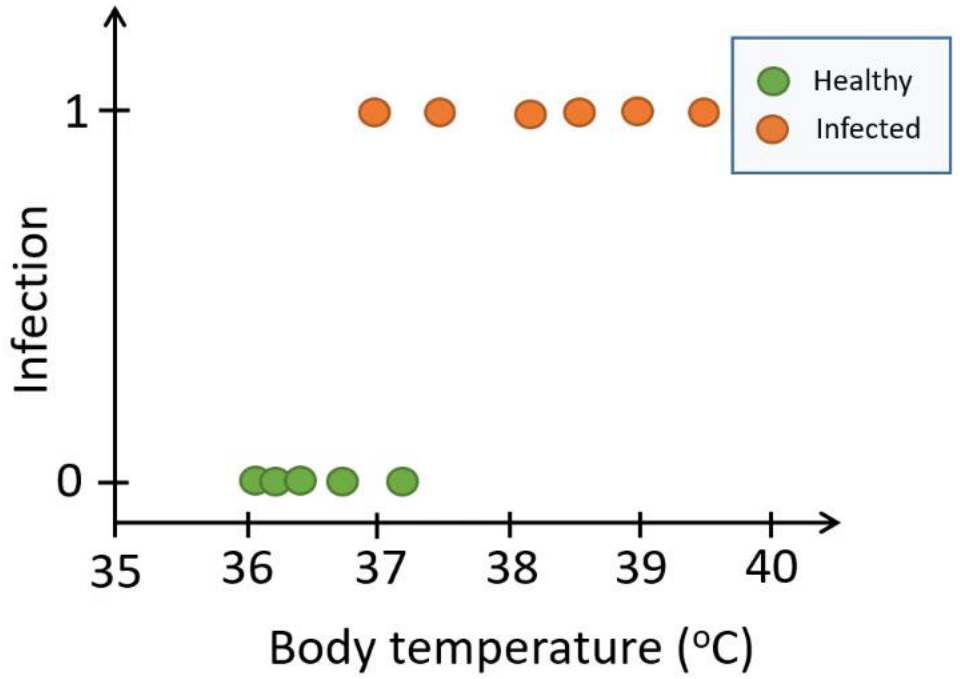


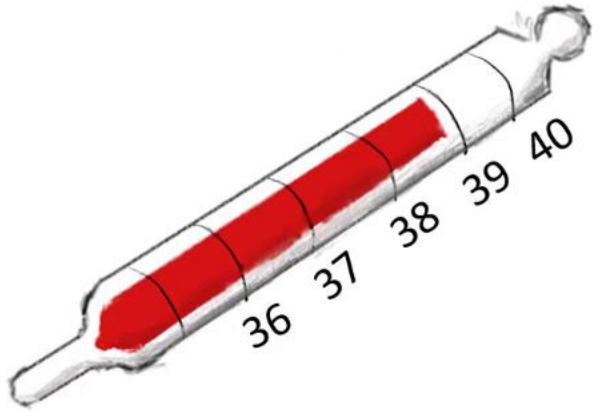


Suppose that we have the following data set where these six individuals are known to be infected with some pathogen,

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

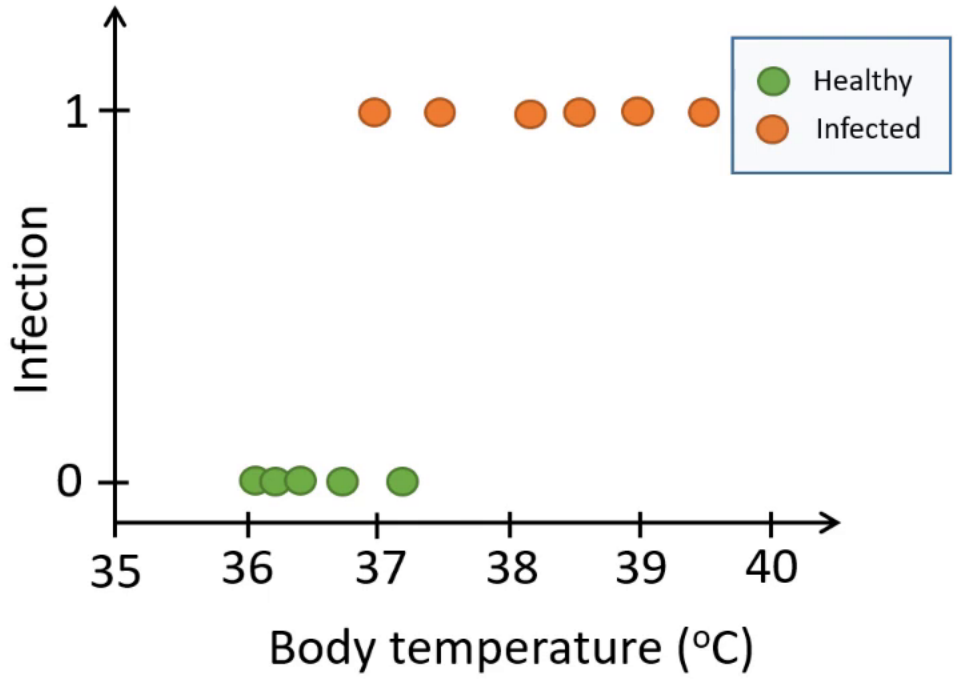


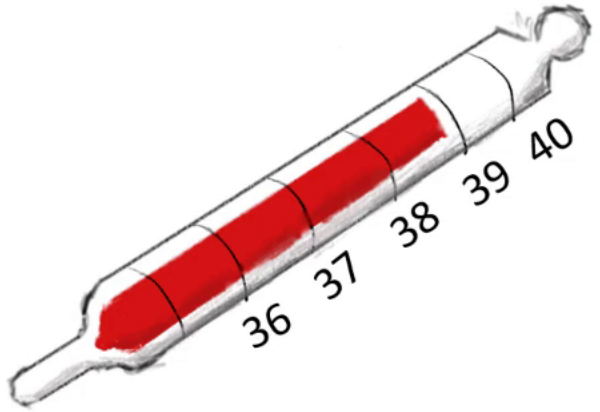


whereas these five individuals are known to be healthy.

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

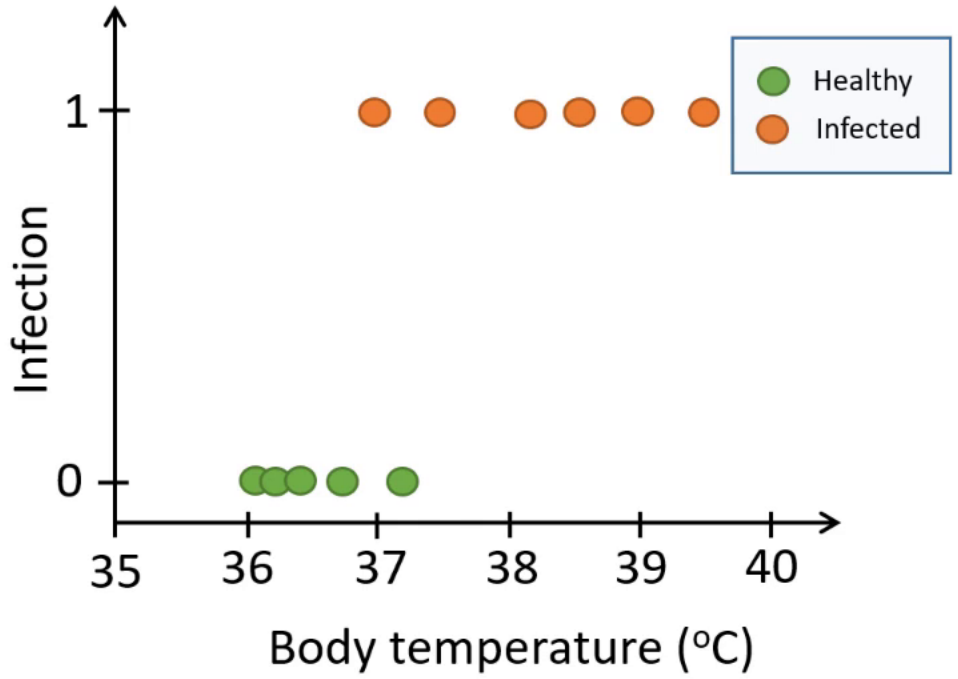


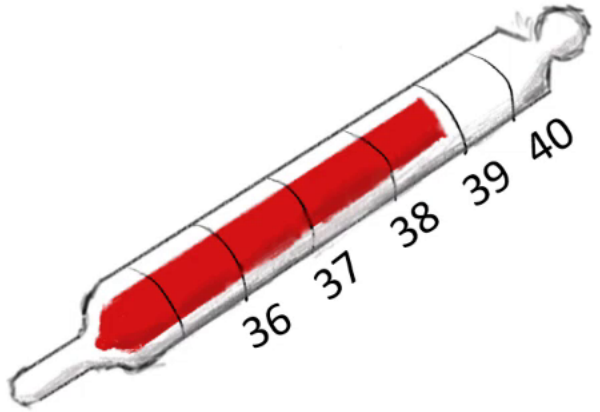


One has measured the body temperature in Celsius of these 11 individuals.

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



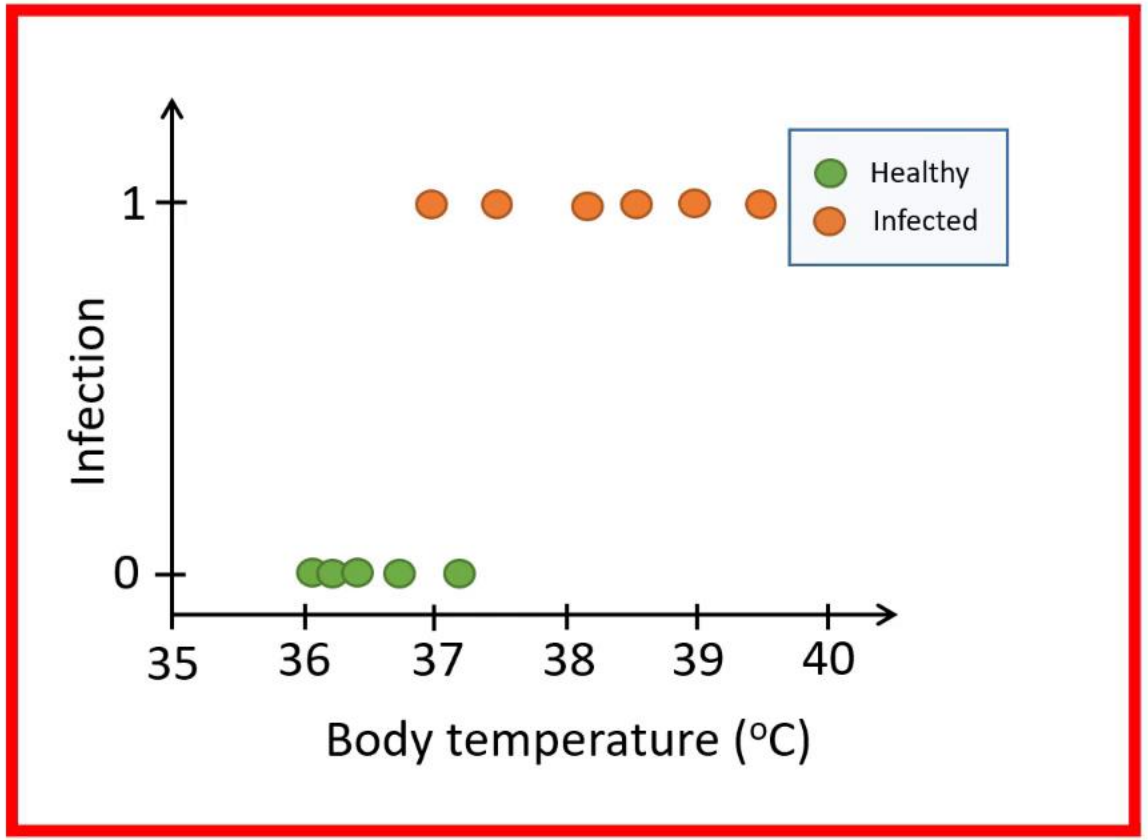


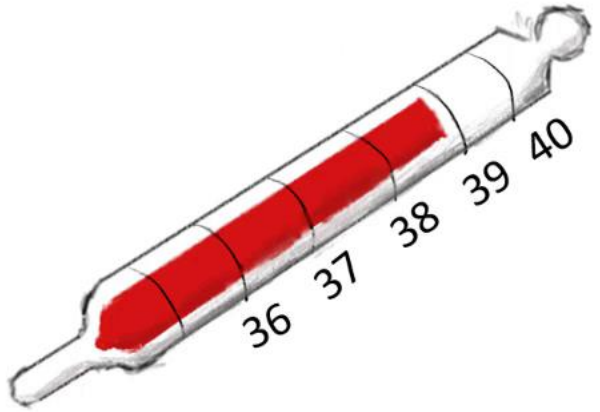
$\mathrm{BT} > 38°C\ (100.4°F)$

Fever us usually defined as a body temperature above 38 degrees Celsius.

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

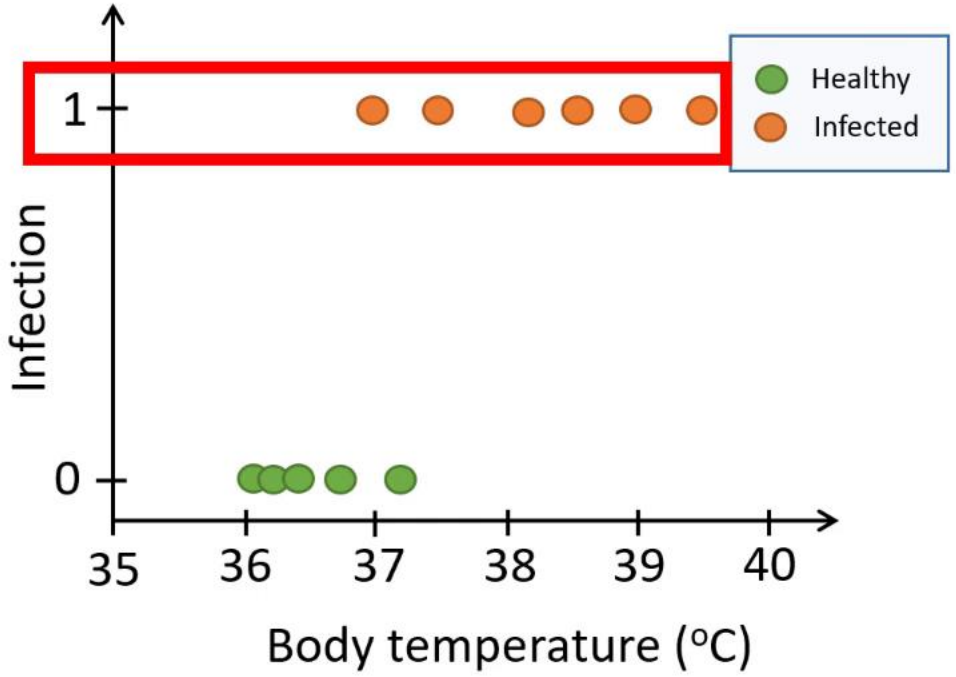


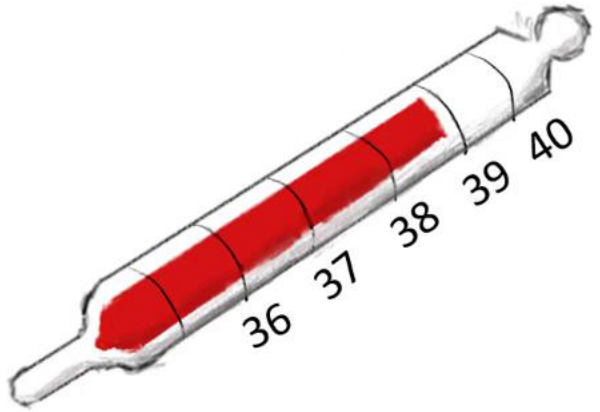


The data can be plotted like this,

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

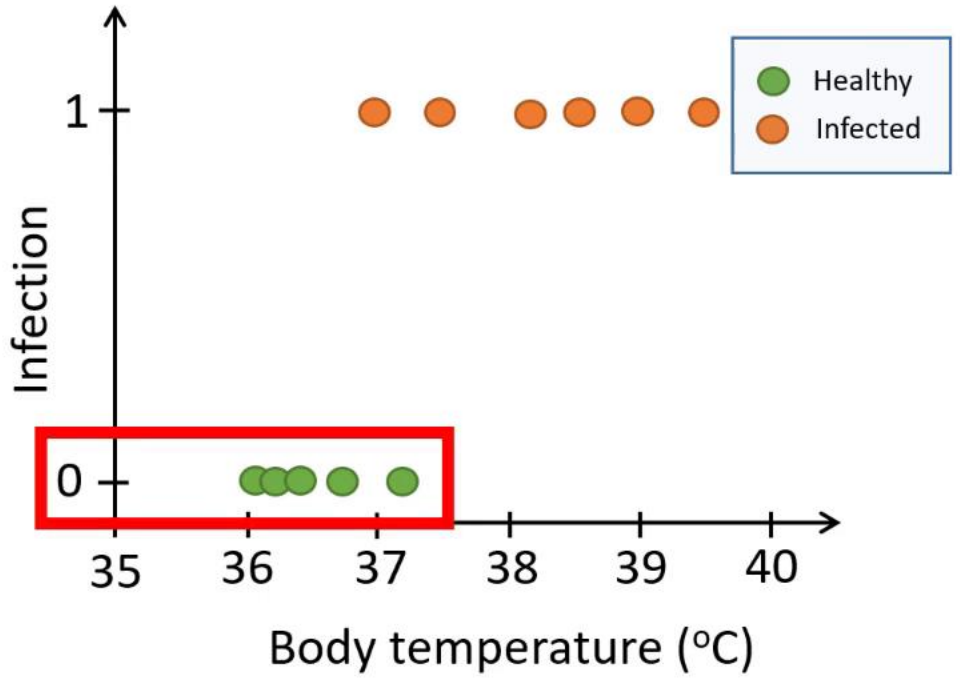


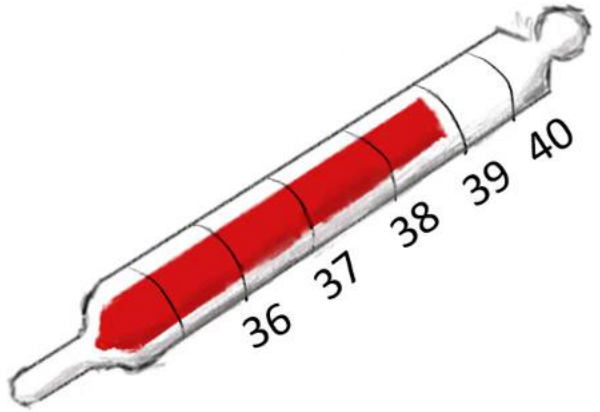


where the infected individuals are coded as ones,

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

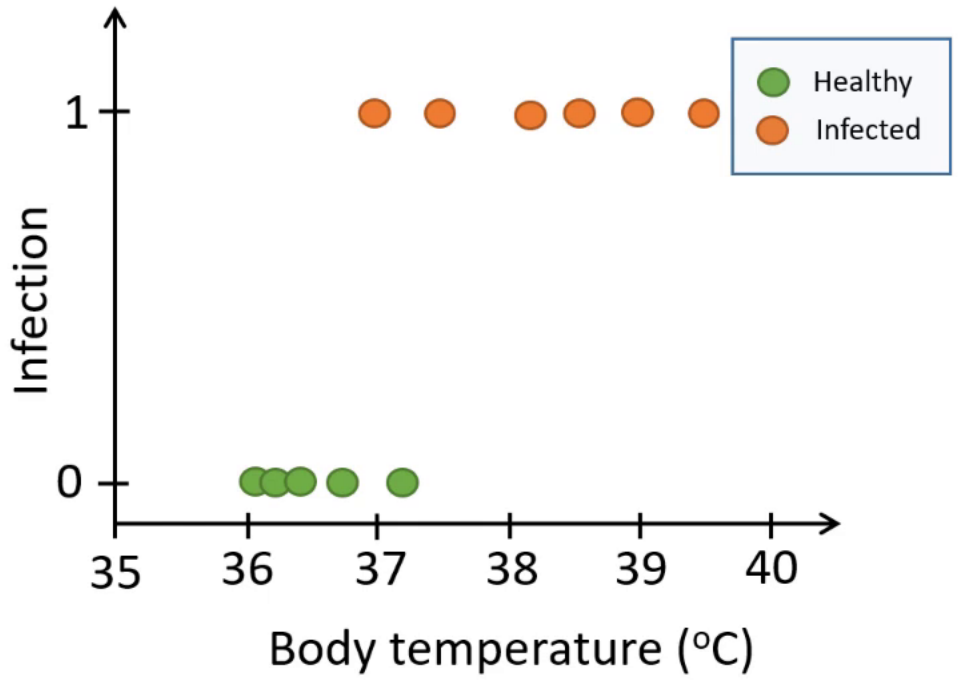


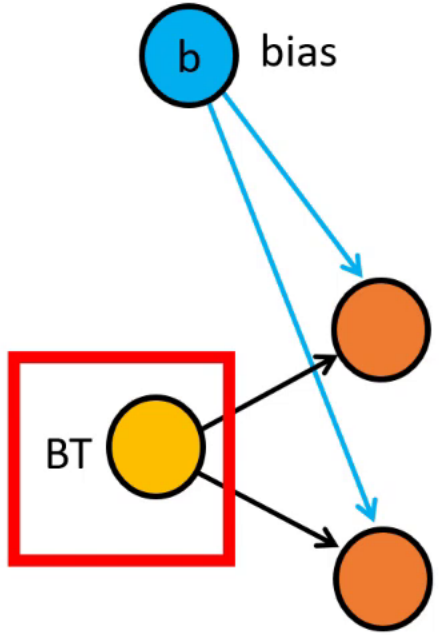


and the healthy individuals as zeros.

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

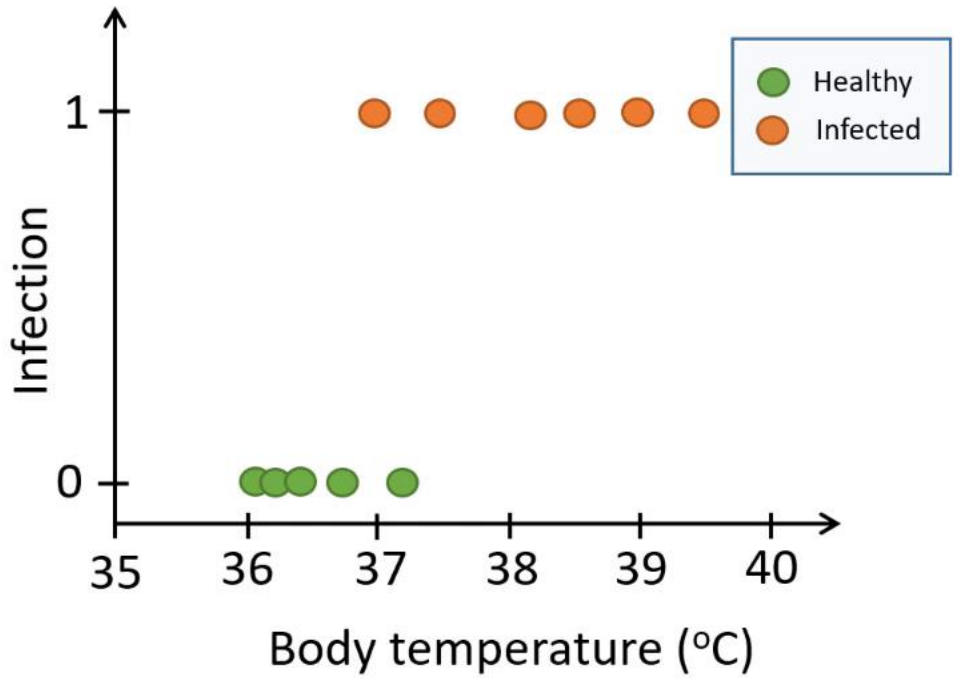


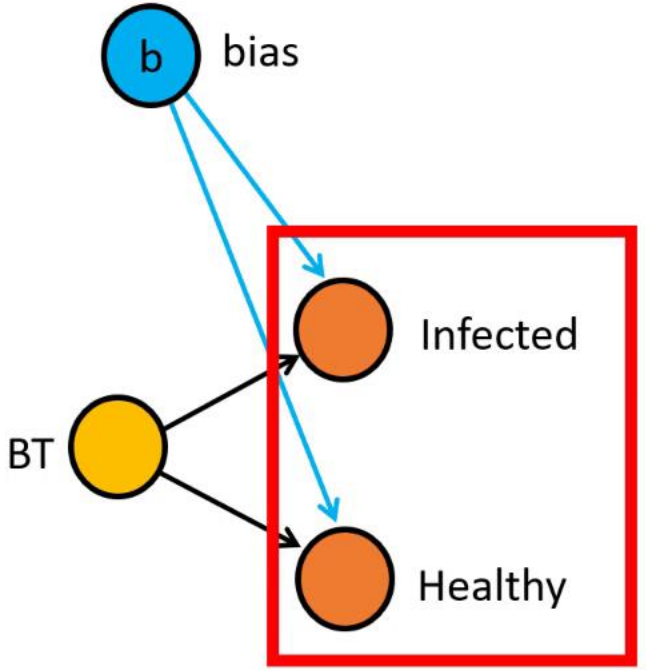


We will now use the simplest possible neural network with just one input node,

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

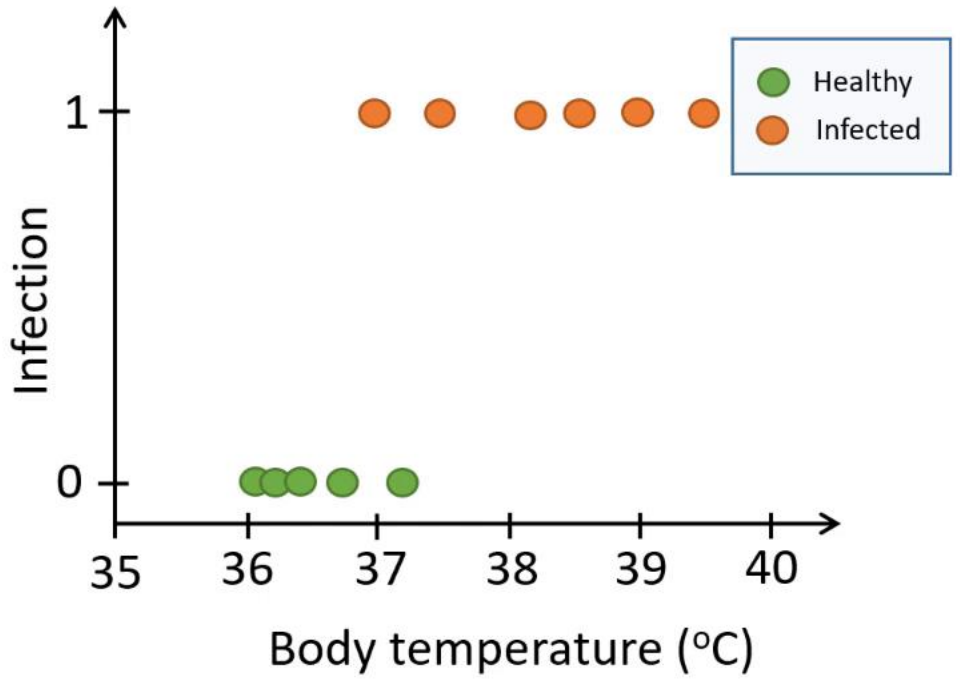


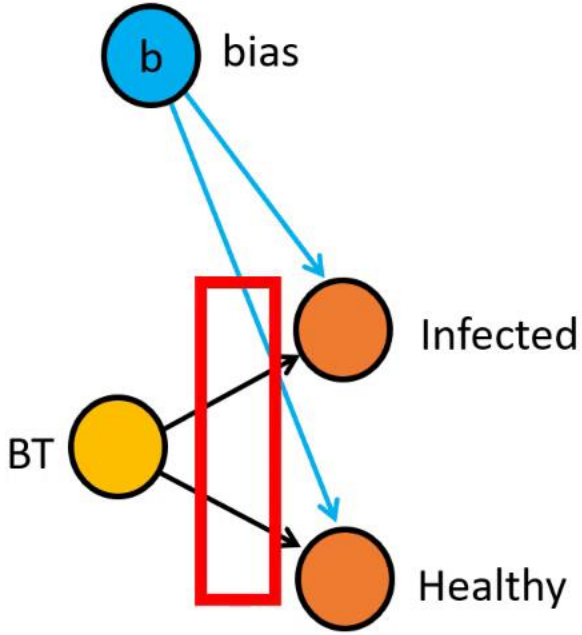


and two output nodes.

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

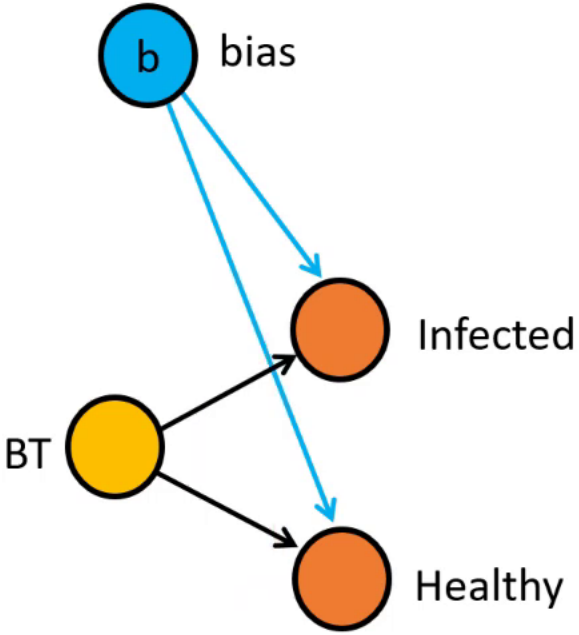




Note that this network does not include any hidden layer.

A simple example

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

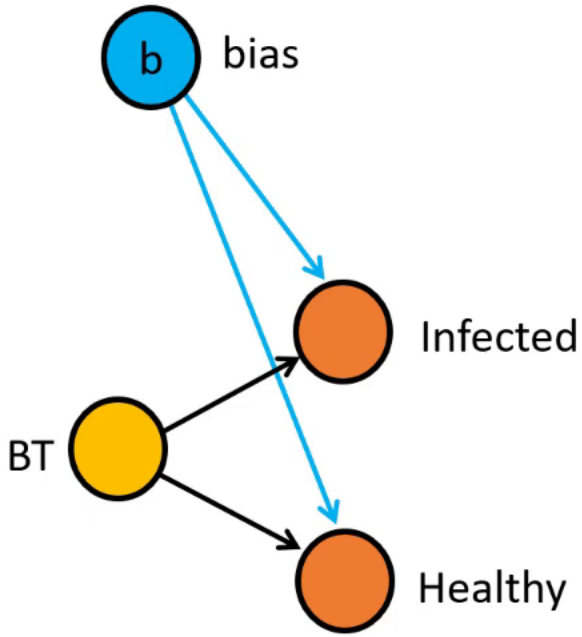


To train a neural network, one usually normalizes the data,

Normalize the data

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

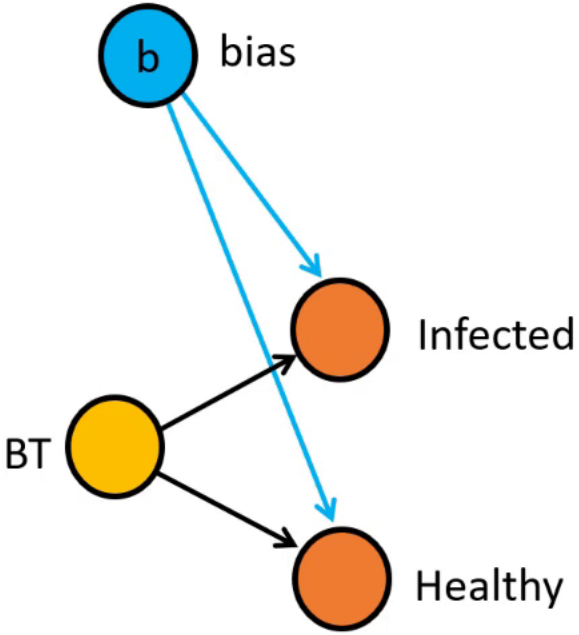


by for example using the min-max normalization.

Normalize the data

Status	BT	nBT
Infected	37.0	0.265
Infected	37.4	0.382
Infected	38.2	0.618
Infected	38.5	0.706
Infected	39.0	0.853
Infected	39.5	1
Healthy	36.1	0
Healthy	36.2	0.029
Healthy	36.4	0.088
Healthy	36.7	0.176
Healthy	37.2	0.324

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

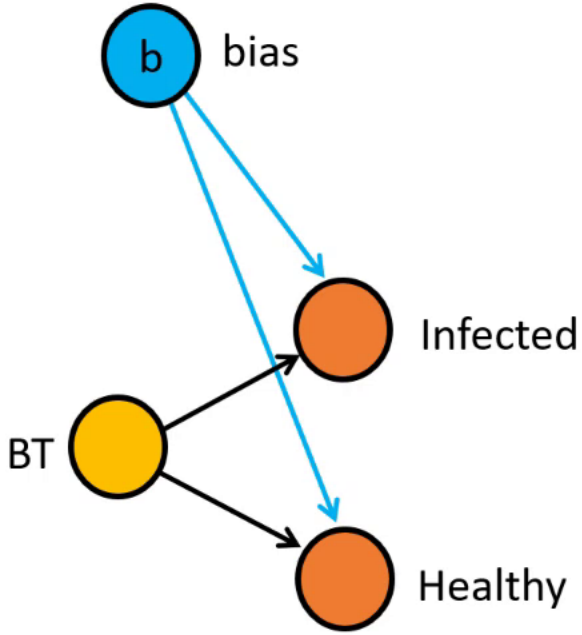


This is the normalized data that will be used as input values in the input node.

Normalize the data

Status	BT	nBT
Infected	37.0	0.265
Infected	37.4	0.382
Infected	38.2	0.618
Infected	38.5	0.706
Infected	39.0	0.853
Infected	39.5	1
Healthy	36.1	0
Healthy	36.2	0.029
Healthy	36.4	0.088
Healthy	36.7	0.176
Healthy	37.2	0.324

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

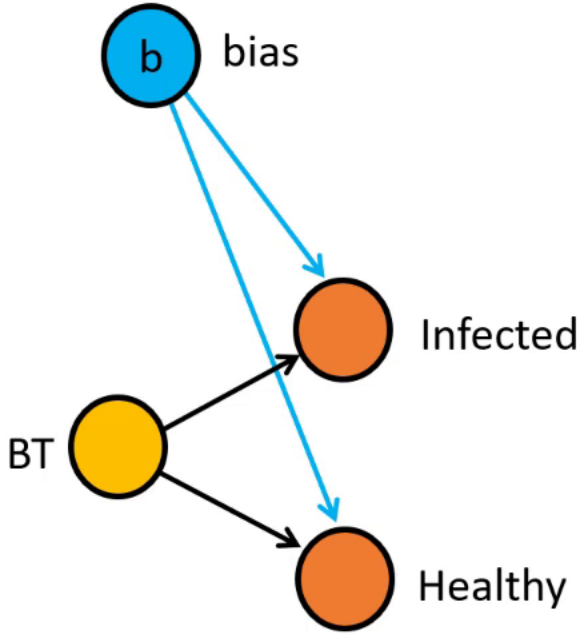


For example, this normalized value has been computed,

Normalize the data

Status	BT	nBT
Infected	37.0	0.265
Infected	37.4	0.382
Infected	38.2	0.618
Infected	38.5	0.706
Infected	39.0	0.853
Infected	39.5	1
Healthy	36.1	0
Healthy	36.2	0.029
Healthy	36.4	0.088
Healthy	36.7	0.176
Healthy	37.2	0.324

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

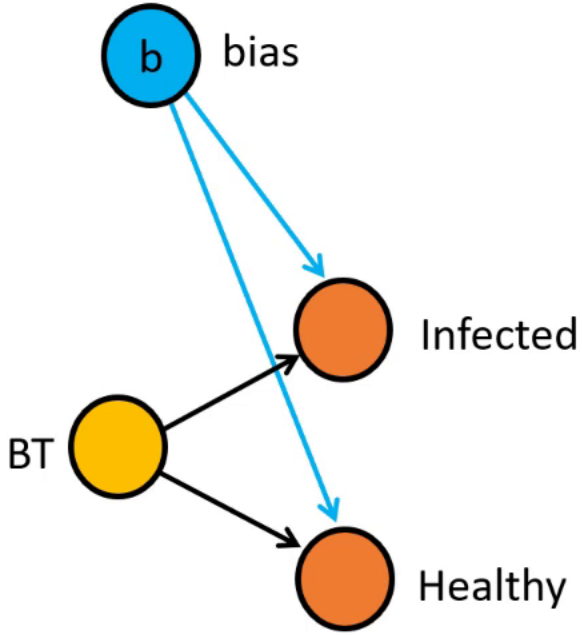


the minimum value,

Normalize the data

Status	BT	nBT
Infected	37.0	0.265
Infected	37.4	0.382
Infected	38.2	0.618
Infected	38.5	0.706
Infected	39.0	0.853
Infected	39.5	1
Healthy	36.1	0
Healthy	36.2	0.029
Healthy	36.4	0.088
Healthy	36.7	0.176
Healthy	37.2	0.324

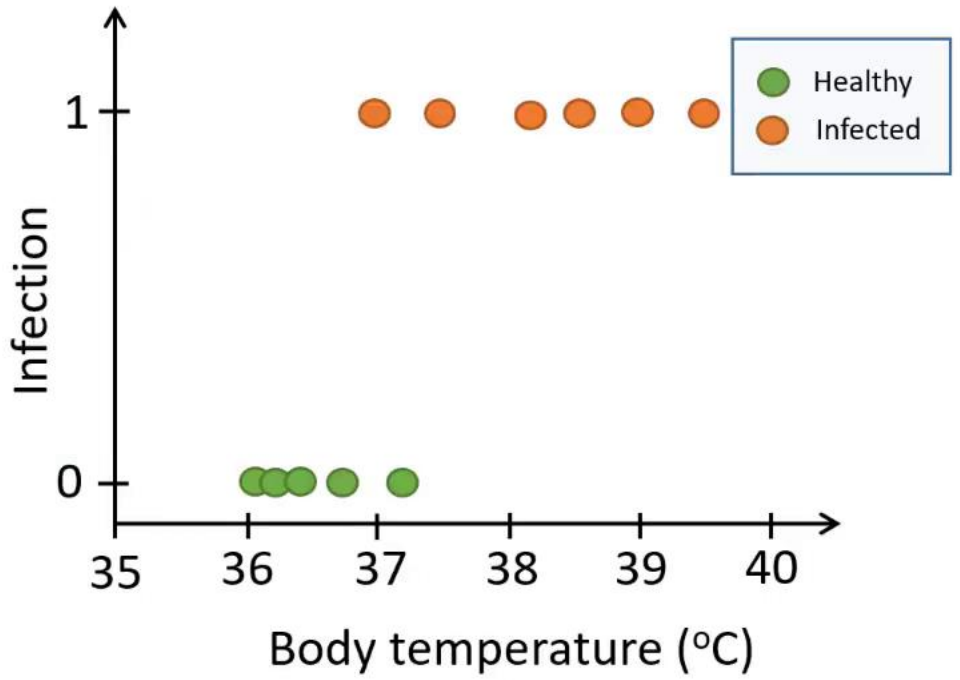
$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

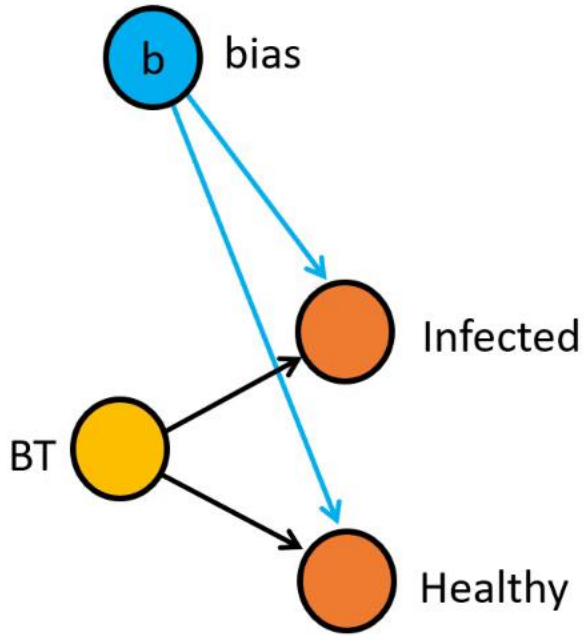


and the maximum value in the data set.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



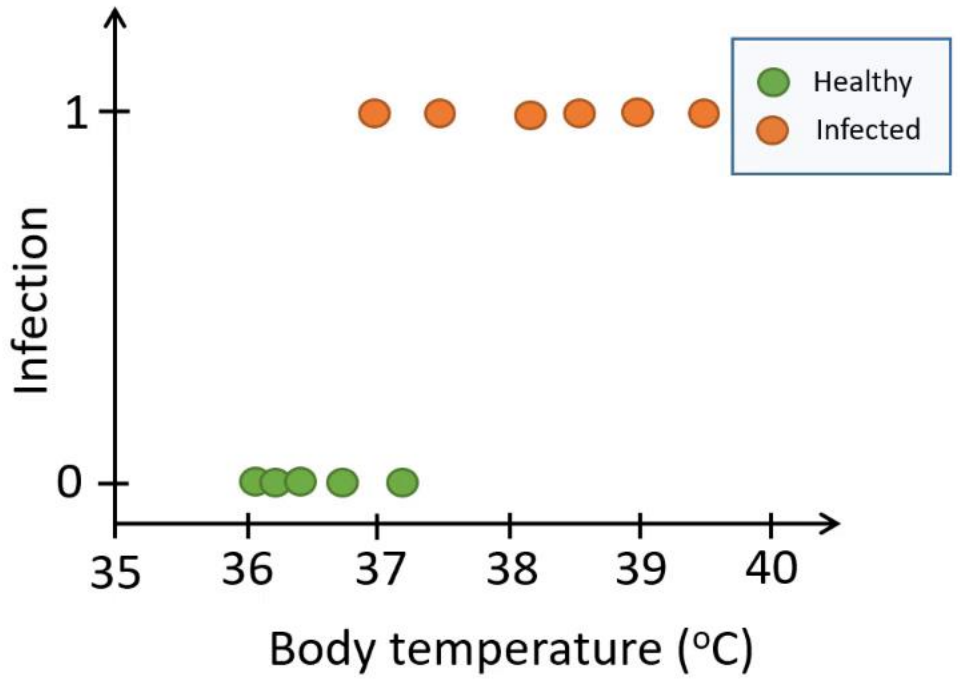


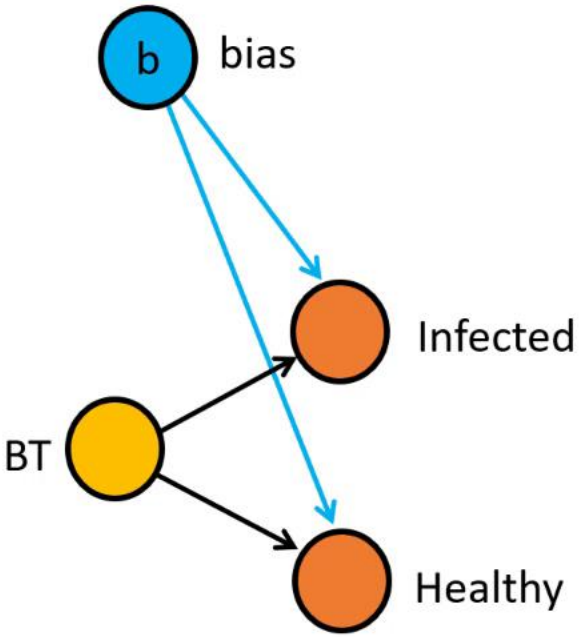
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

Training a neural network means that it finds the optimal values of its weights,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



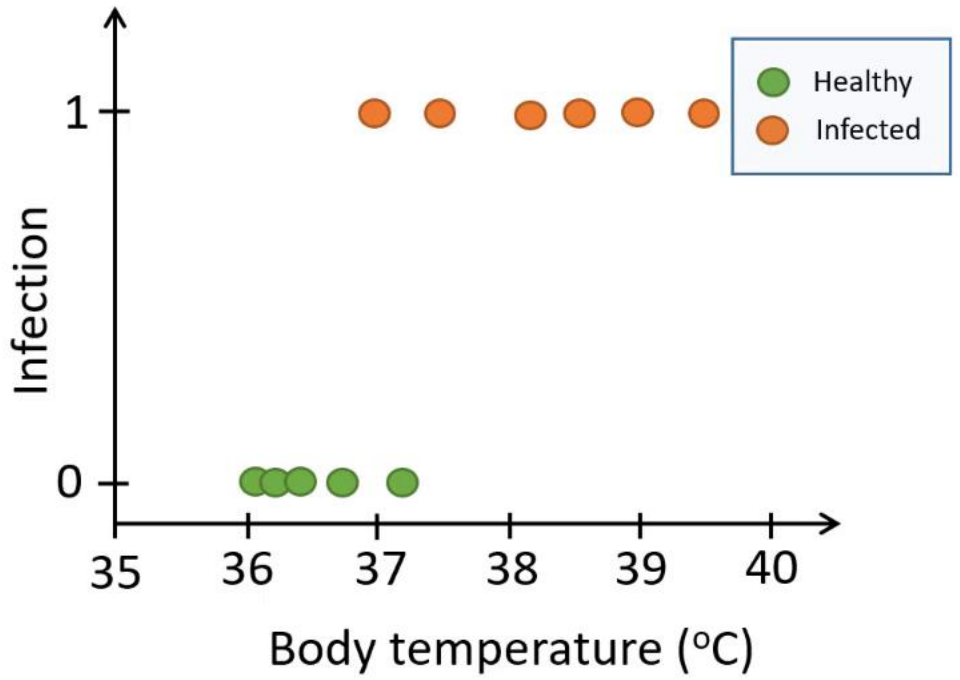


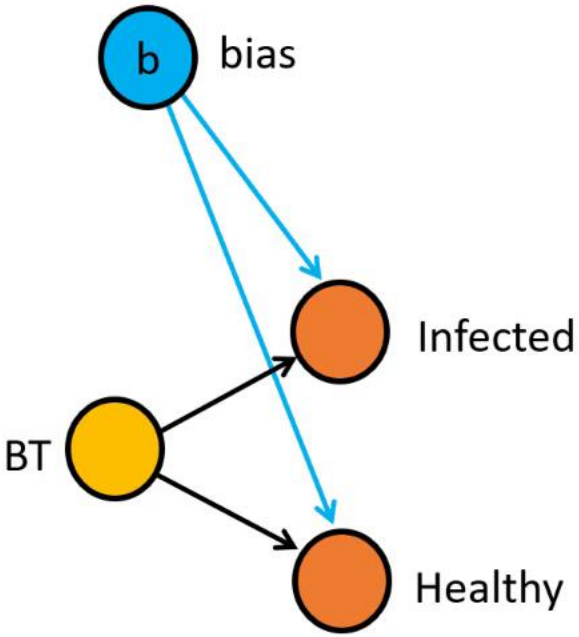
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

that are multiplied with the input nodes.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



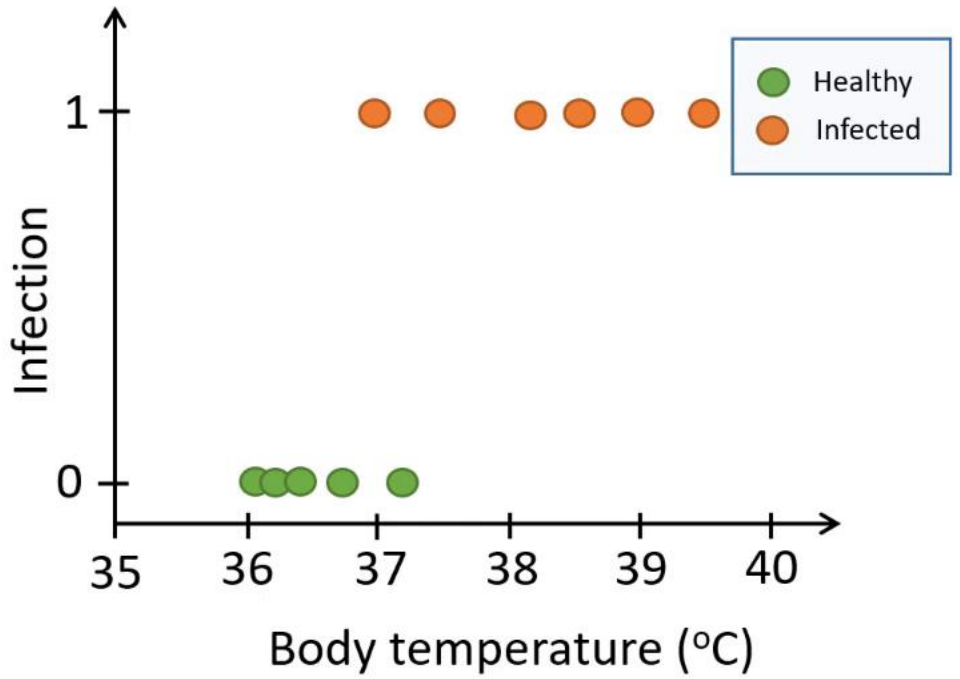


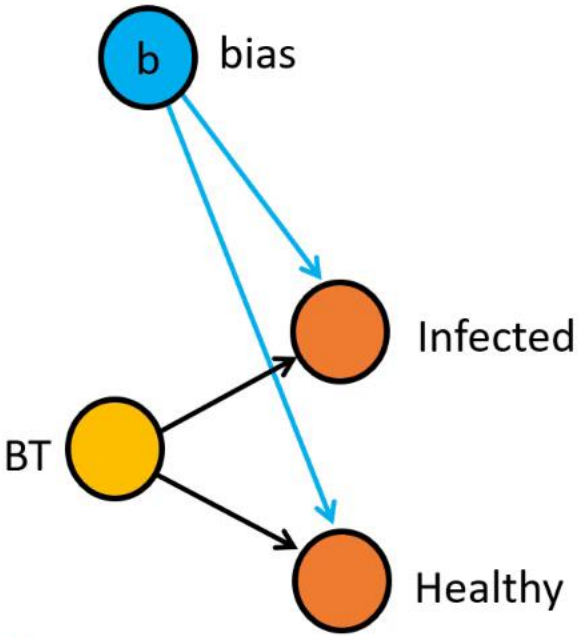
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

The network also finds the optimal value for the so-called bias weights.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



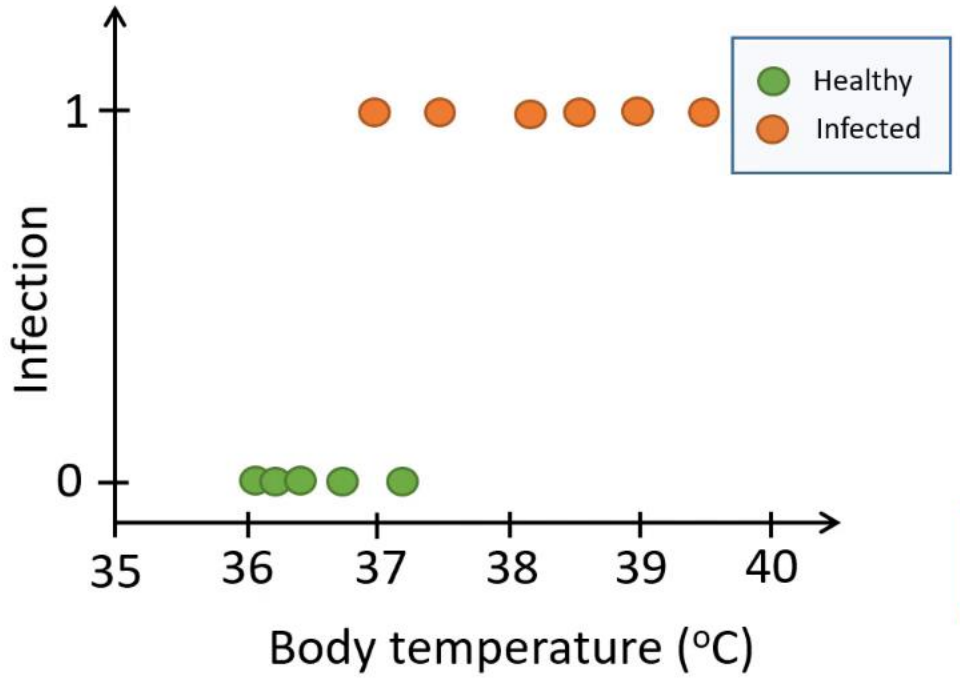


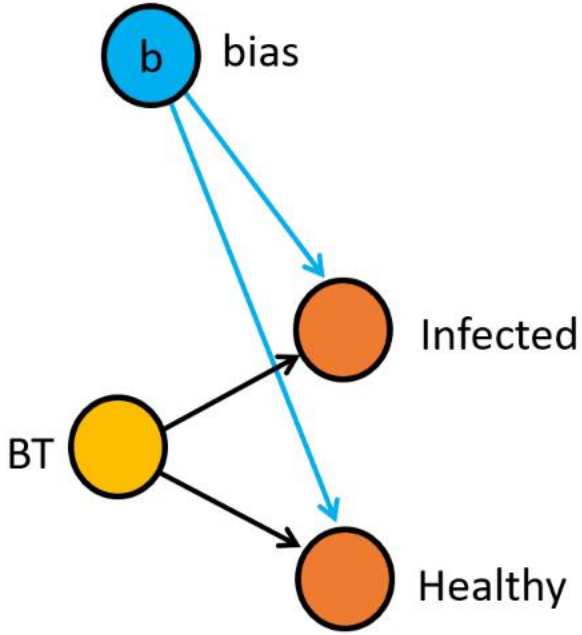
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

Since we only have one input node in this example, we can remove this summation notation,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



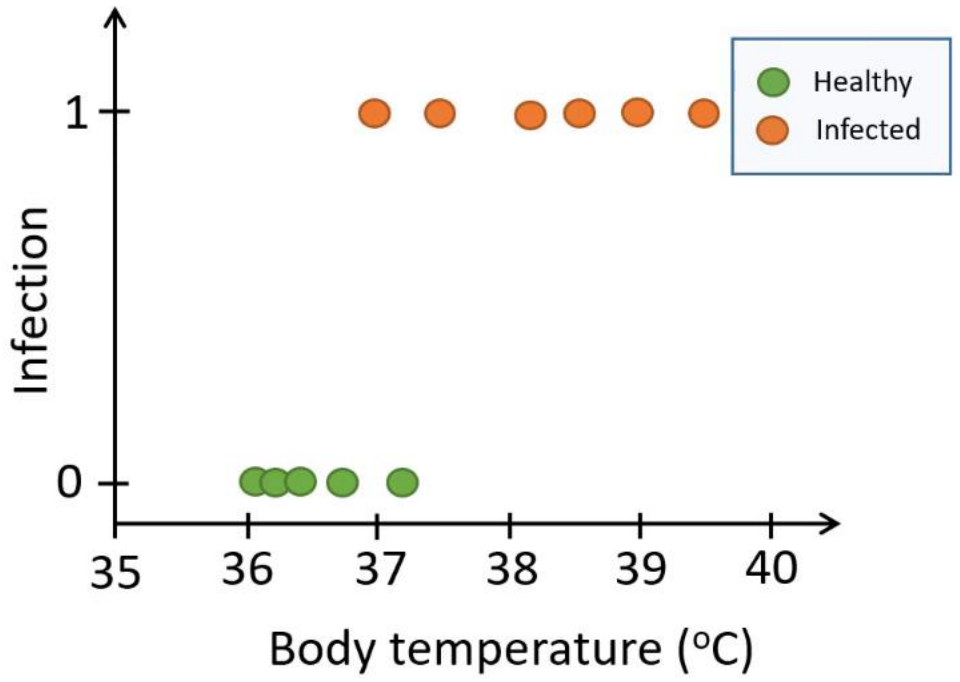


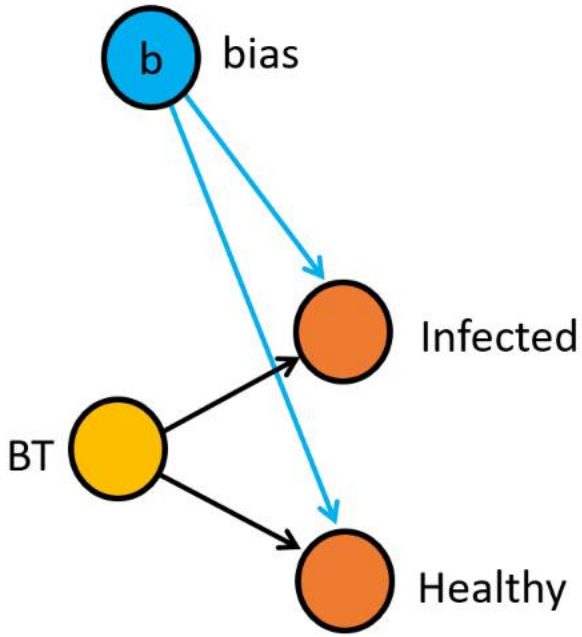
$$z = w_0 + w_1 \cdot x_{norm}$$

and write the equation like this.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2





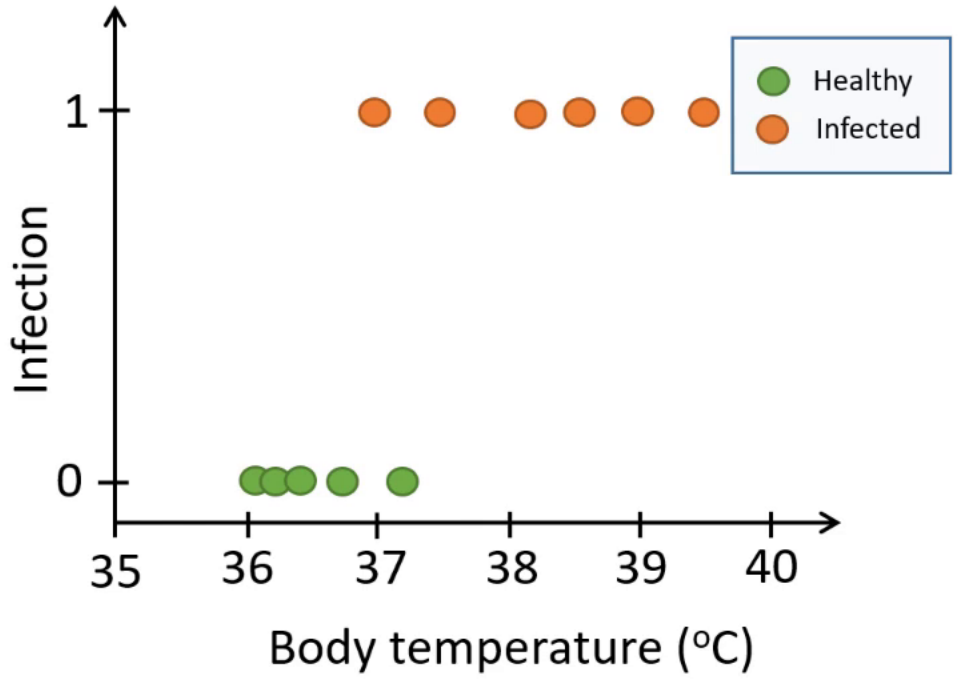
$$z = w_0 + w_1 \cdot x_{norm}$$

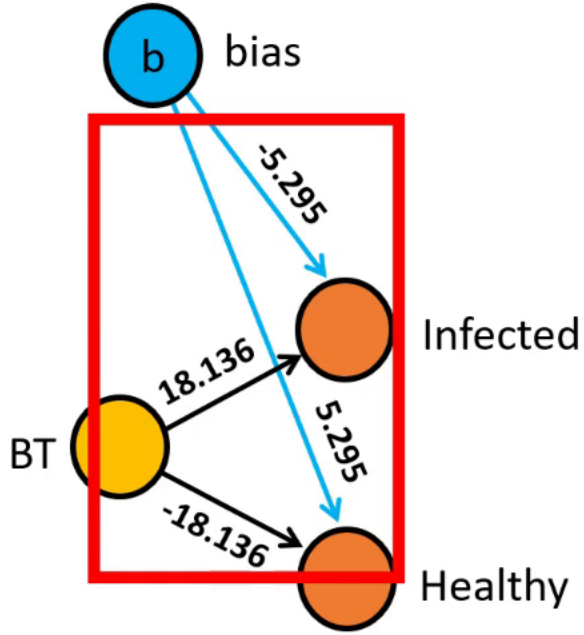
$$f(z) = \frac{1}{1 + e^{-z}}$$

A network is also based on some sort of activation function. In this case, we use the sigmoid activation function that looks like this.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2





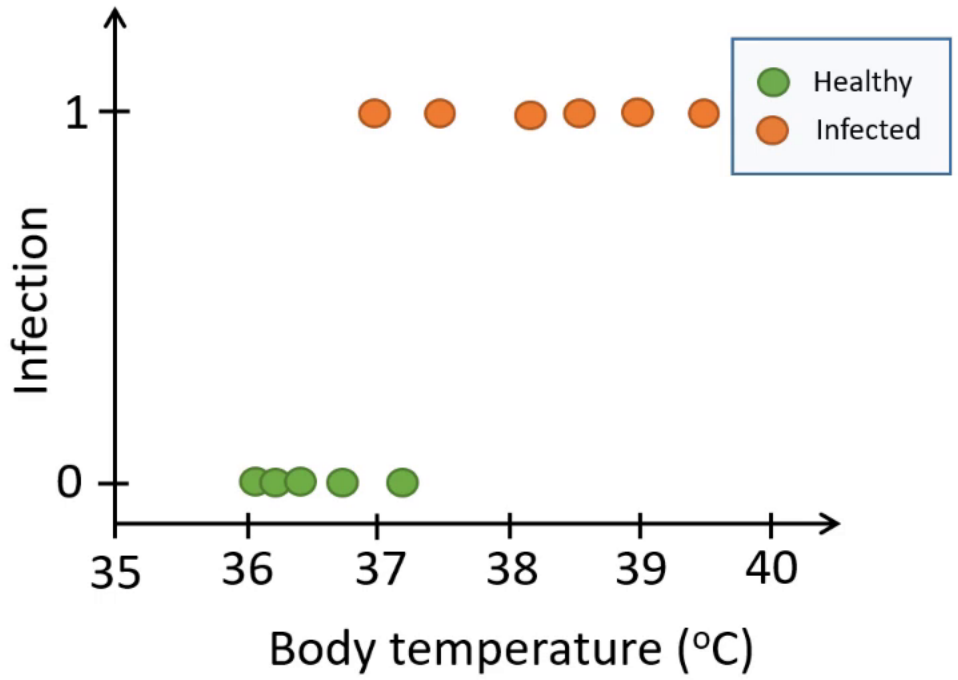
$$z = w_0 + w_1 \cdot x_{norm}$$

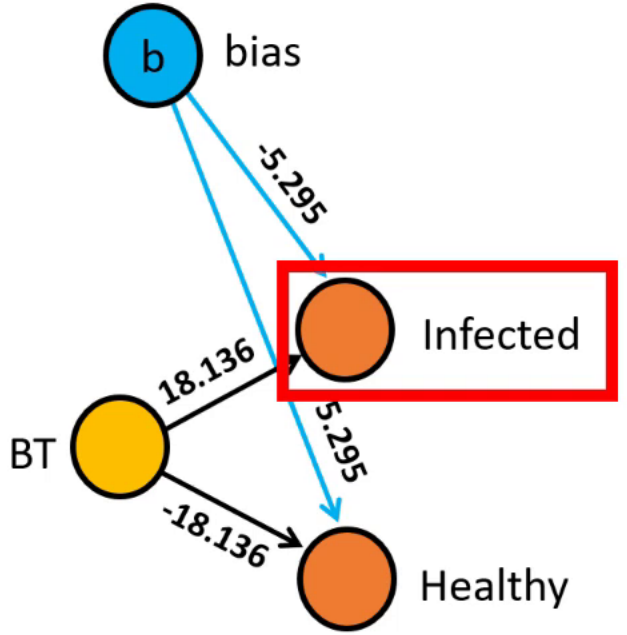
$$f(z) = \frac{1}{1 + e^{-z}}$$

If we train this network, we might obtain the following weights.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2





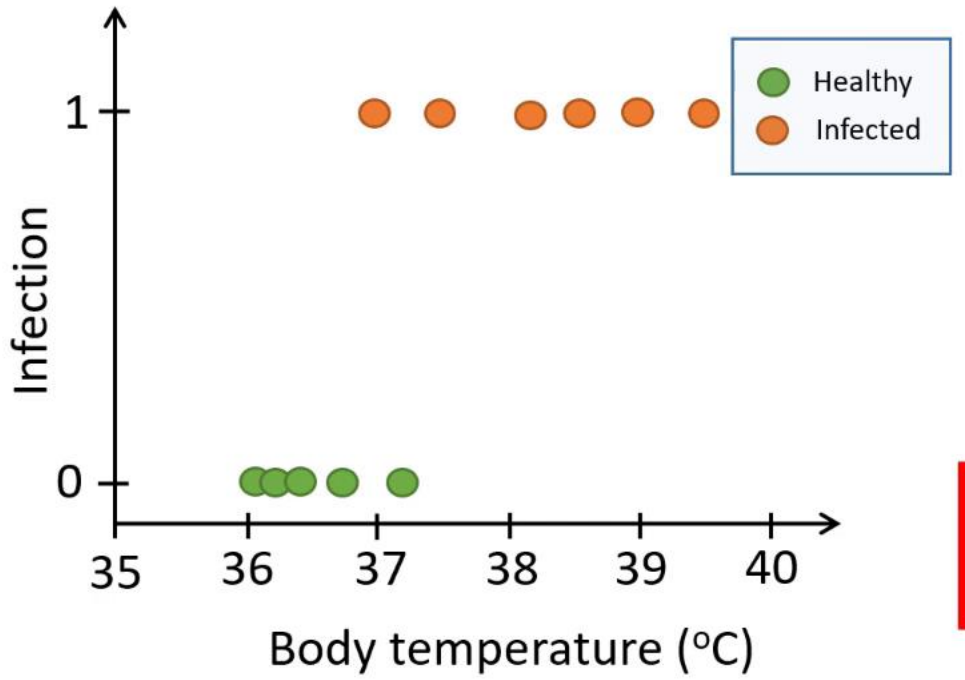
$$z = w_0 + w_1 \cdot x_{norm}$$

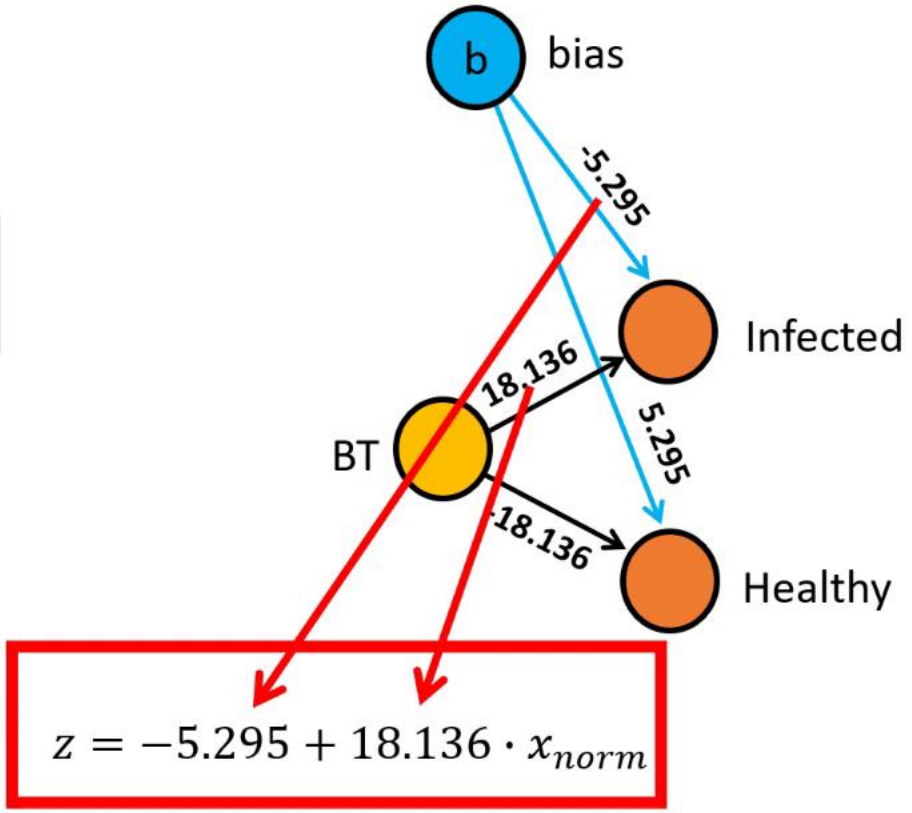
$$f(z) = \frac{1}{1 + e^{-z}}$$

If we like to predict the probability that a person is infected,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



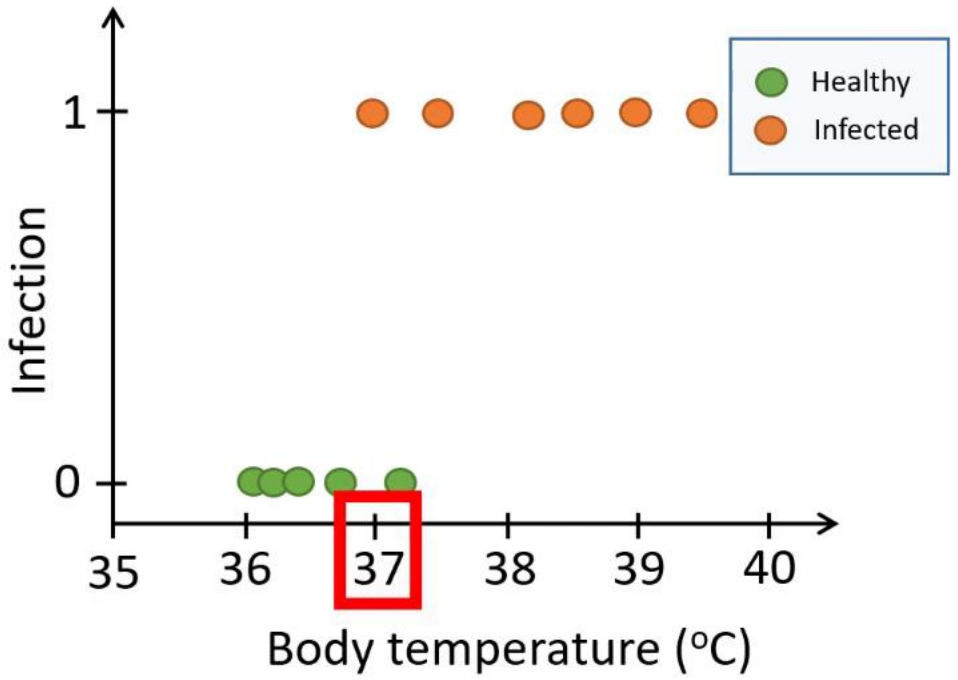


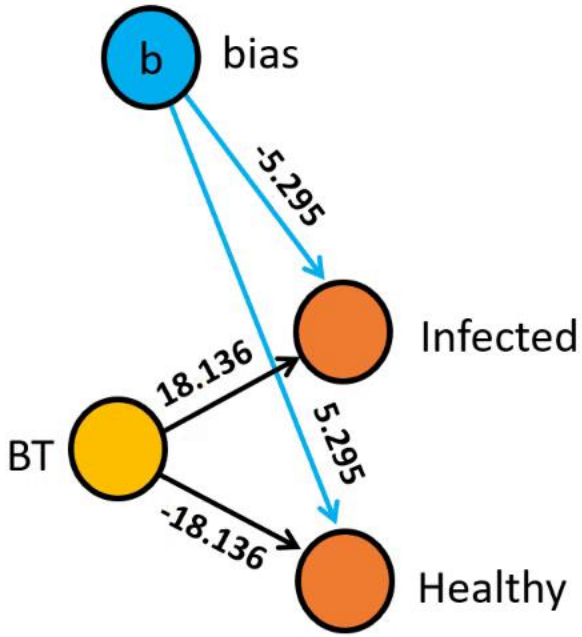
$$f(z) = \frac{1}{1 + e^{-z}}$$

we plug in the weights in this function.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2





$$z = -5.295 + 18.136 \cdot x_{norm}$$

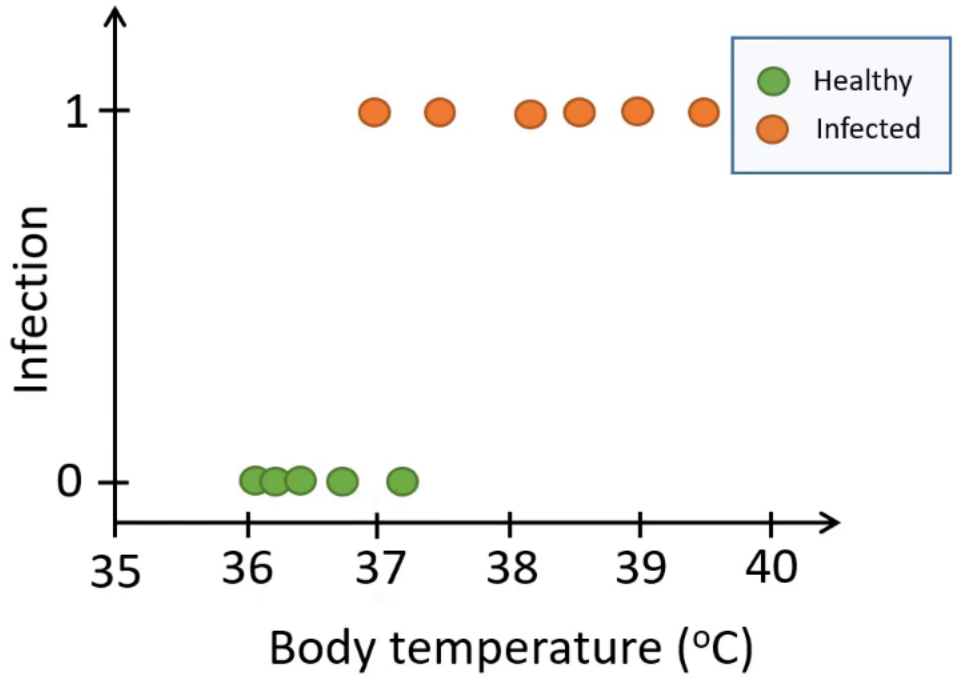
$$f(z) = \frac{1}{1 + e^{-z}}$$

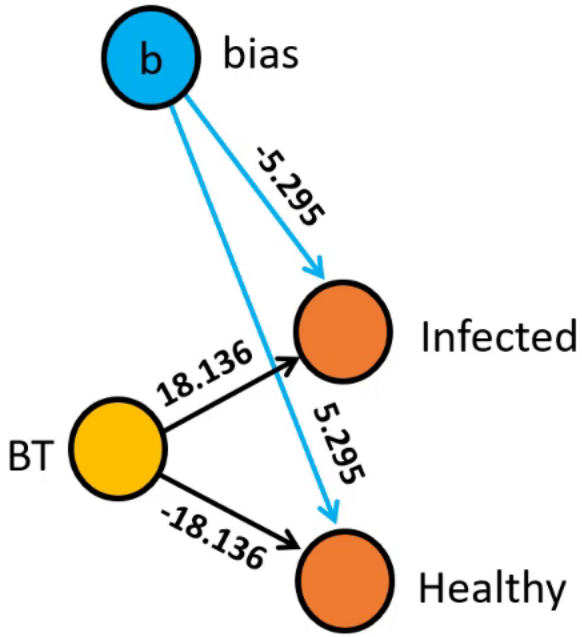
Suppose that we like to compute the probability that a person with a body temperature of 37 is infected.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{37 - 36.1}{39.5 - 36.1} = 0.265$$





$$z = -5.295 + 18.136 \cdot x_{norm}$$

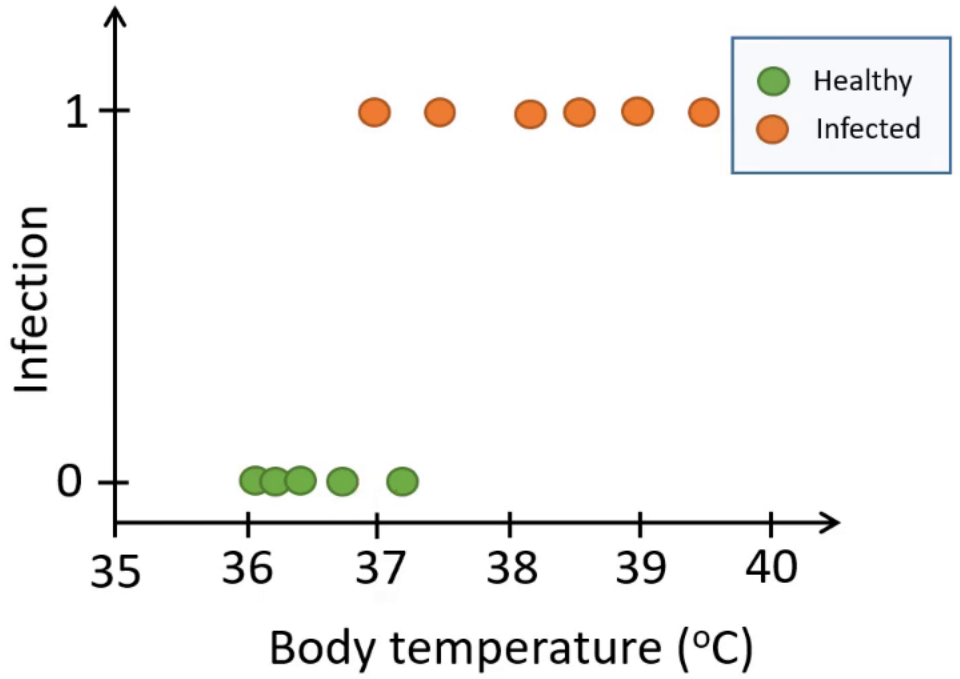
$$f(z) = \frac{1}{1 + e^{-z}}$$

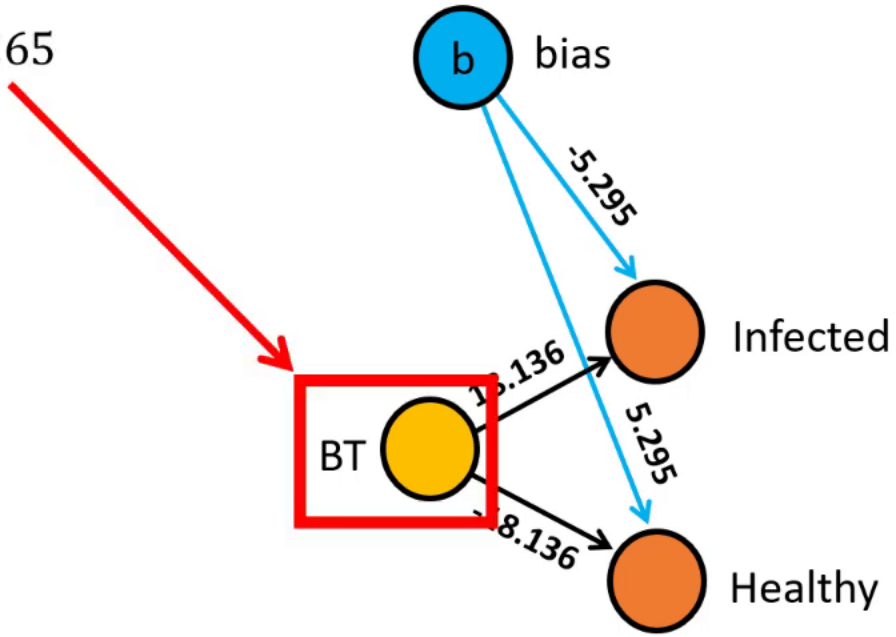
We therefore first normalize this value,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{37 - 36.1}{39.5 - 36.1} = 0.265$$





$$z = -5.295 + 18.136 \cdot x_{norm}$$

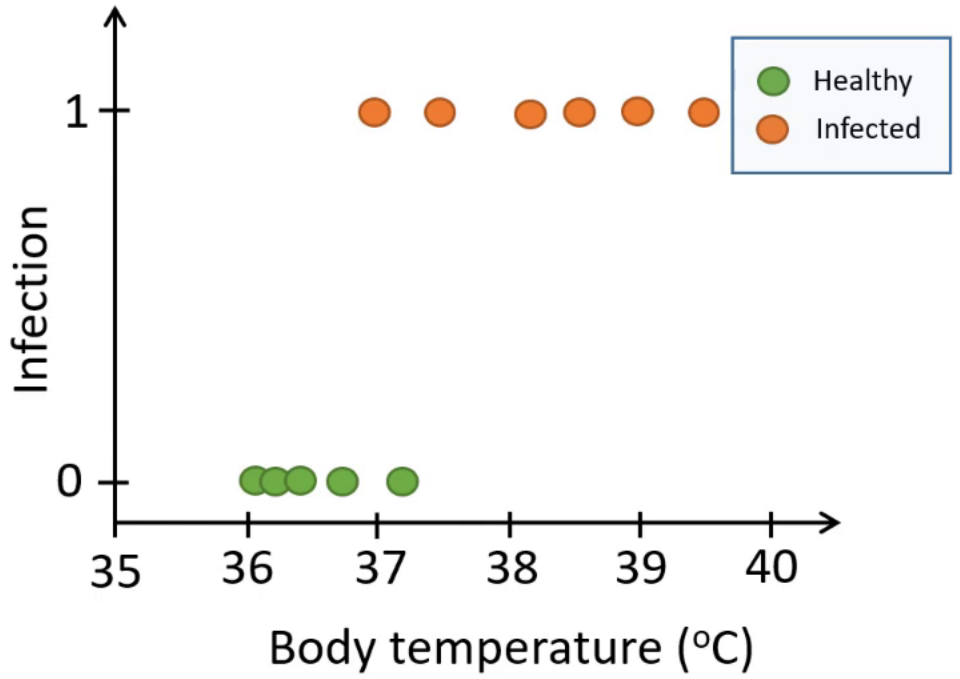
$$f(z) = \frac{1}{1 + e^{-z}}$$

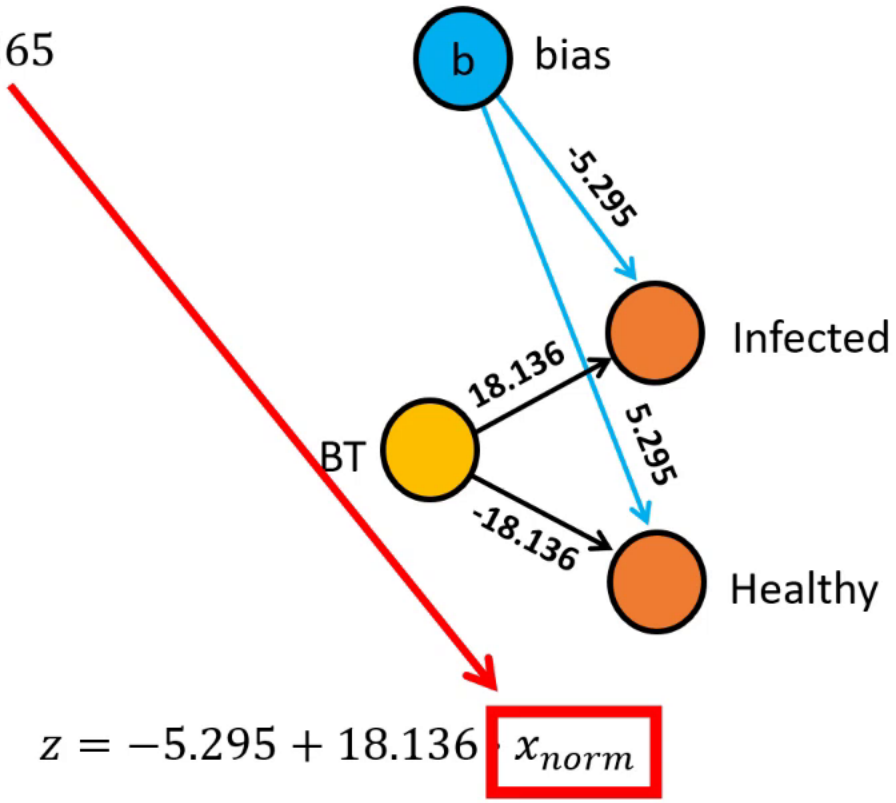
and plug in this value into the input node,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{37 - 36.1}{39.5 - 36.1} = 0.265$$





$$z = -5.295 + 18.136 \cdot x_{norm}$$

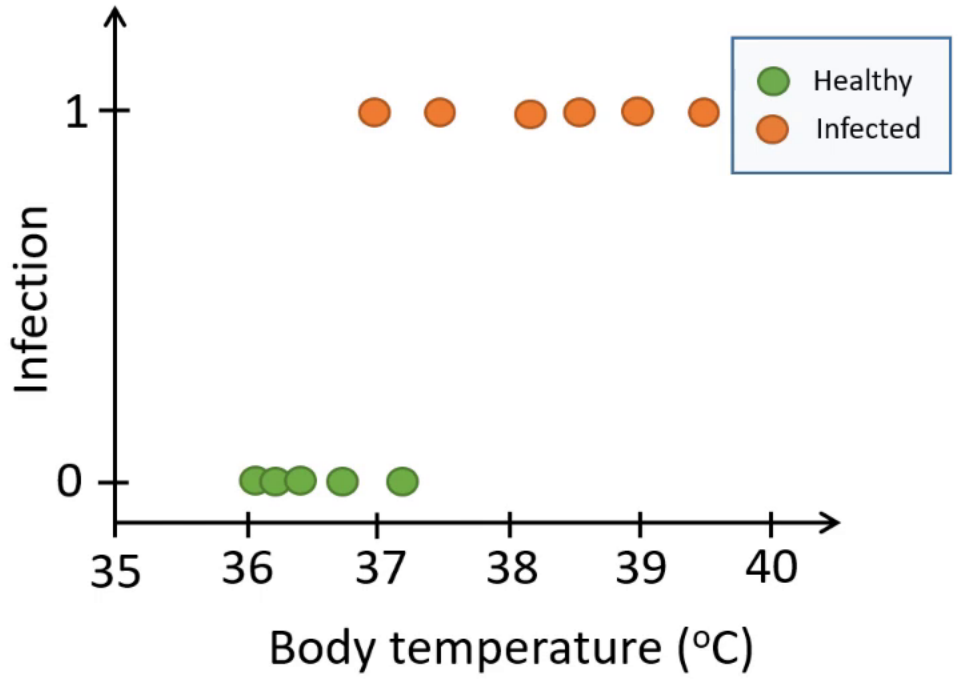
$$f(z) = \frac{1}{1 + e^{-z}}$$

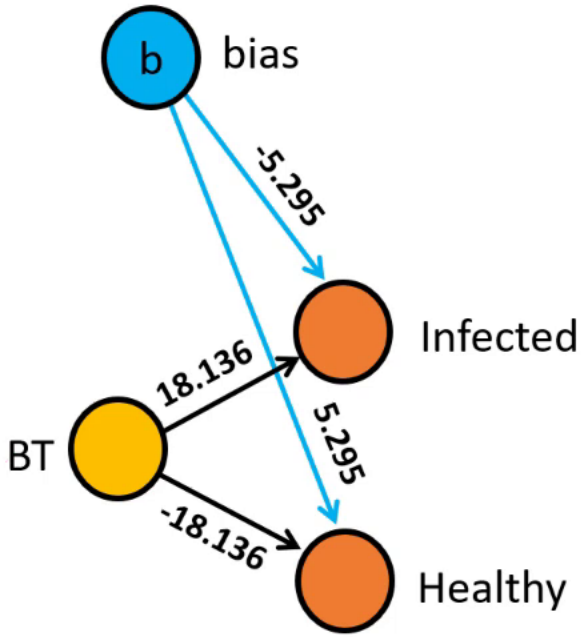
which means that the value is then used in this equation.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{37 - 36.1}{39.5 - 36.1} = 0.265$$





$$z = -5.295 + 18.136 \cdot 0.265 = -0.489$$

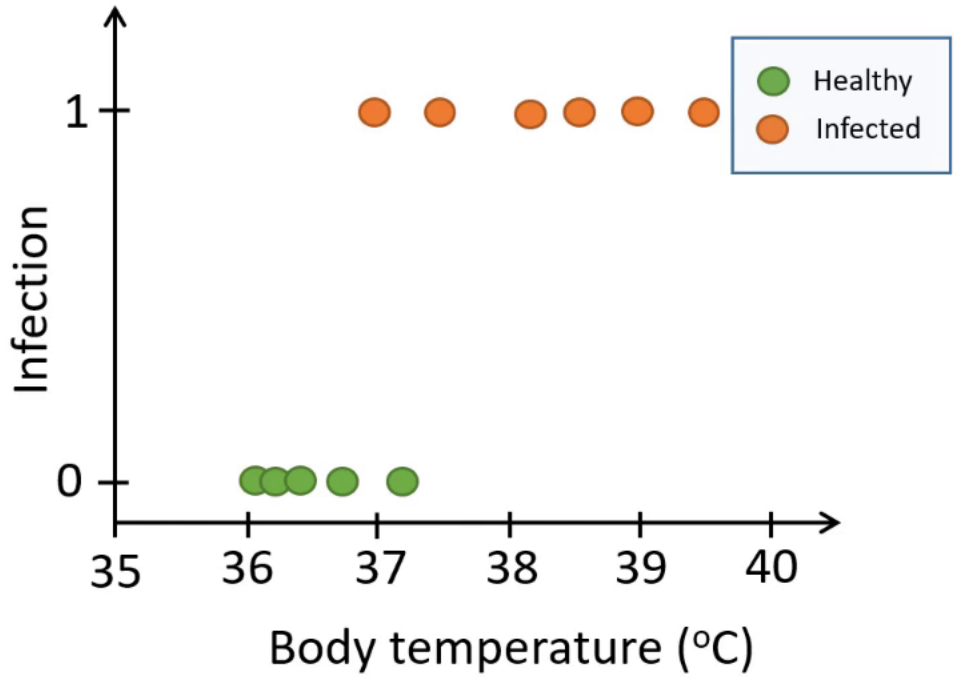
$$f(z) = \frac{1}{1 + e^{-z}}$$

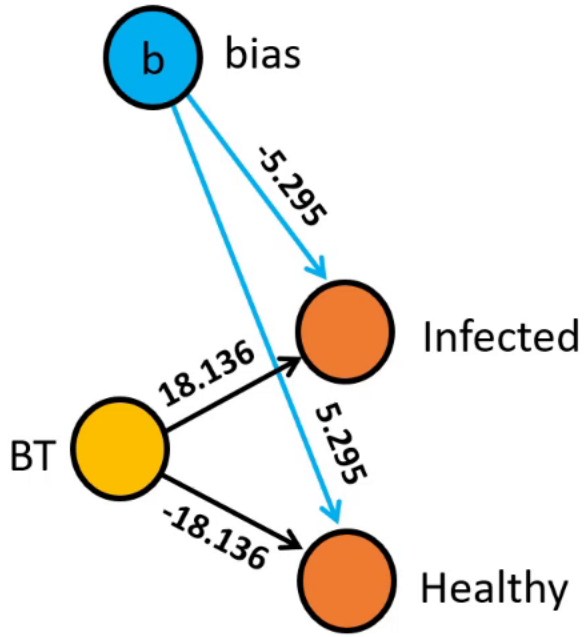
If we do the math,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{37 - 36.1}{39.5 - 36.1} = 0.265$$





$$z = -5.295 + 18.136 \cdot 0.265 = -0.489$$

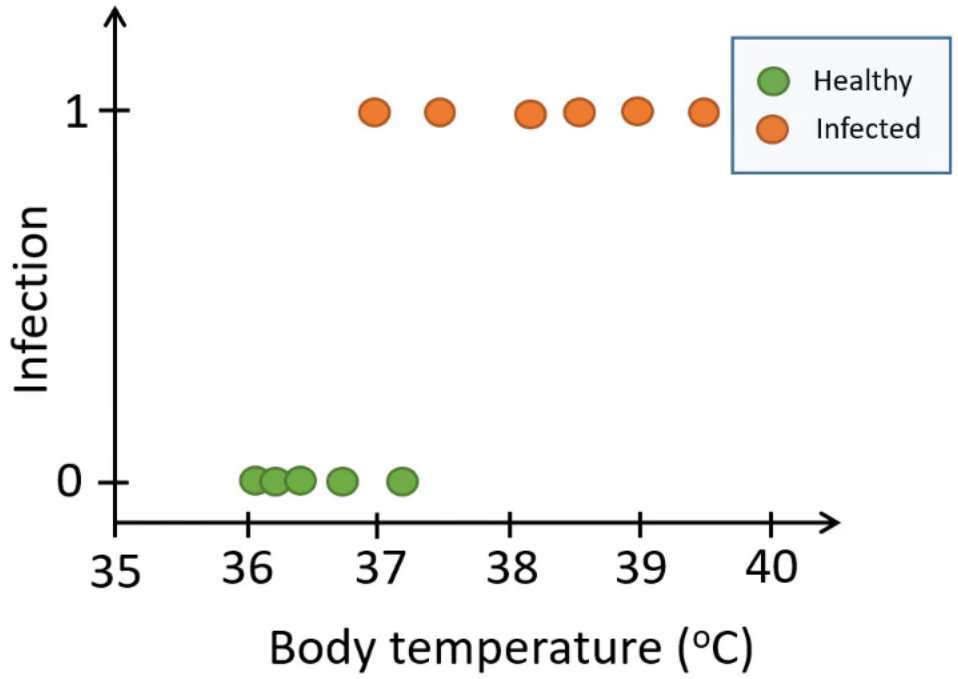
$$f(z) = \frac{1}{1 + e^{-(-0.489)}}$$

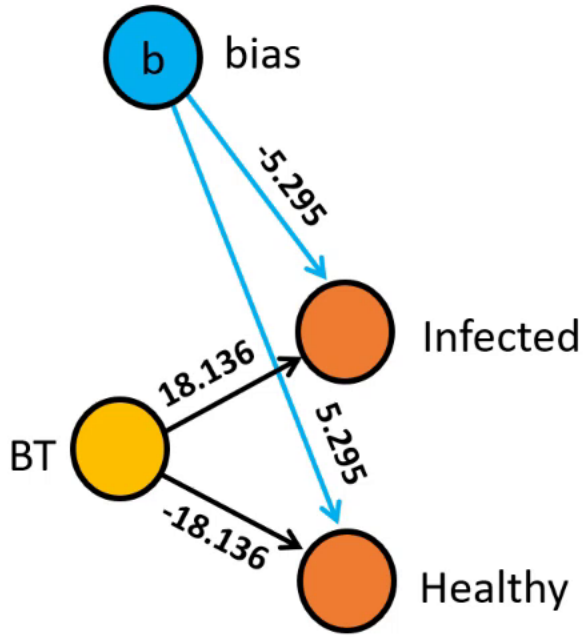
and plug in the value of z in the activation function,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{37 - 36.1}{39.5 - 36.1} = 0.265$$





$$z = -5.295 + 18.136 \cdot 0.265 = -0.489$$

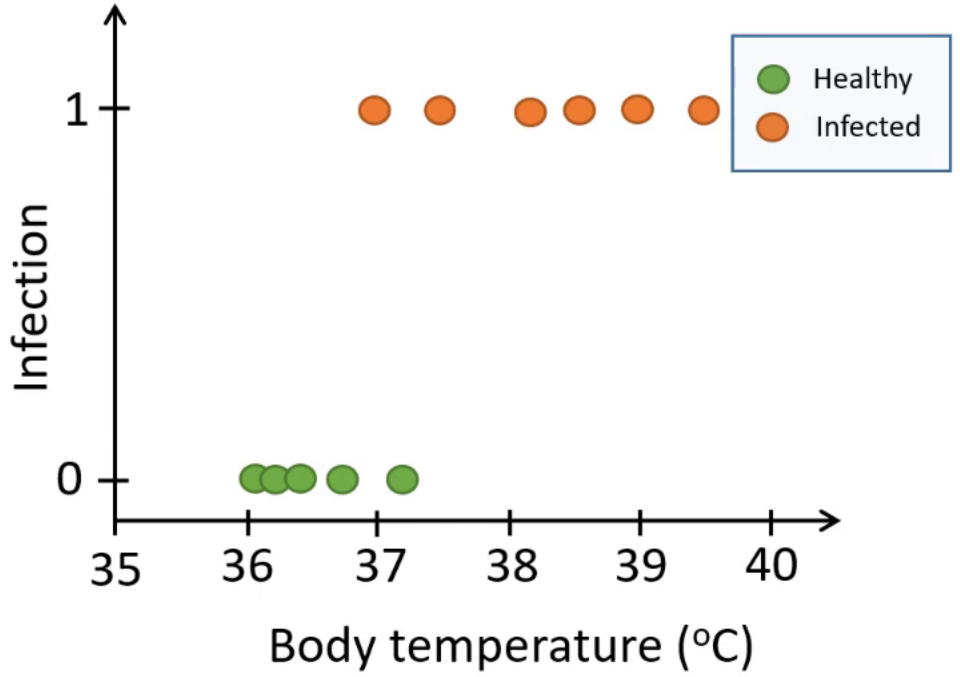
$$f(z) = \frac{1}{1 + e^{-(-0.489)}} = 0.38$$

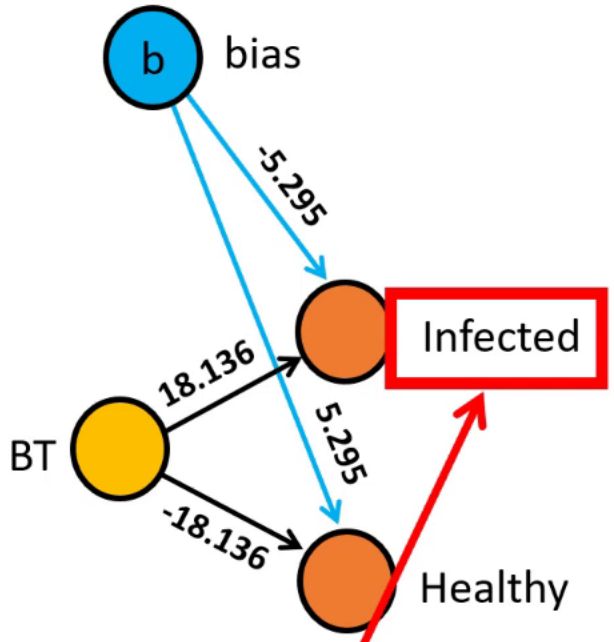
we will get a value of about 0.38.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{37 - 36.1}{39.5 - 36.1} = 0.265$$





$$z = -5.295 + 18.136 \cdot 0.265 = -0.489$$

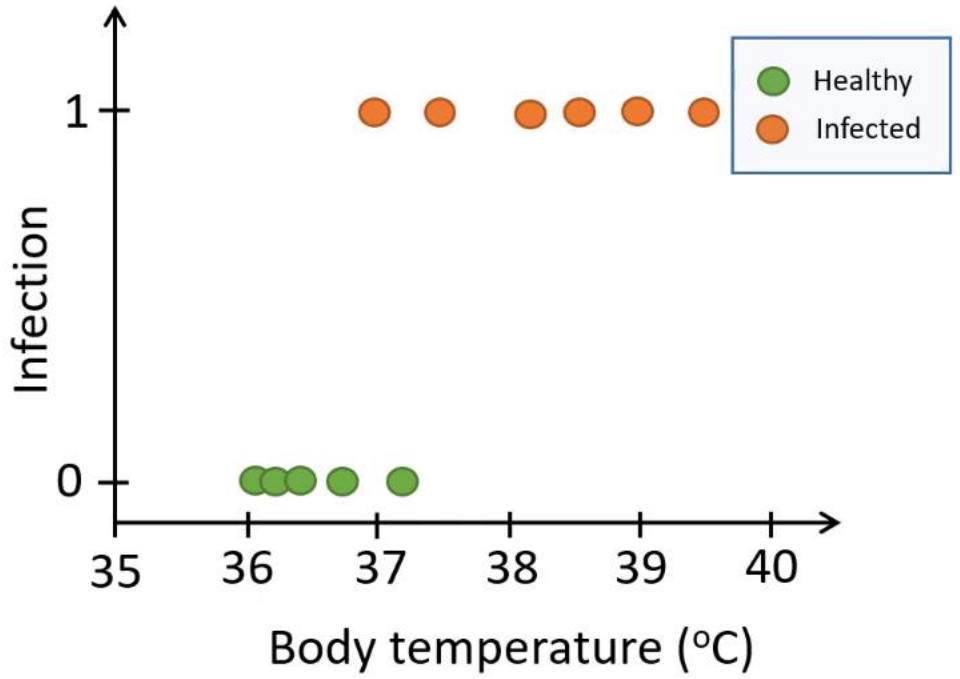
$$f(z) = \frac{1}{1 + e^{-(-0.489)}} = 0.38$$

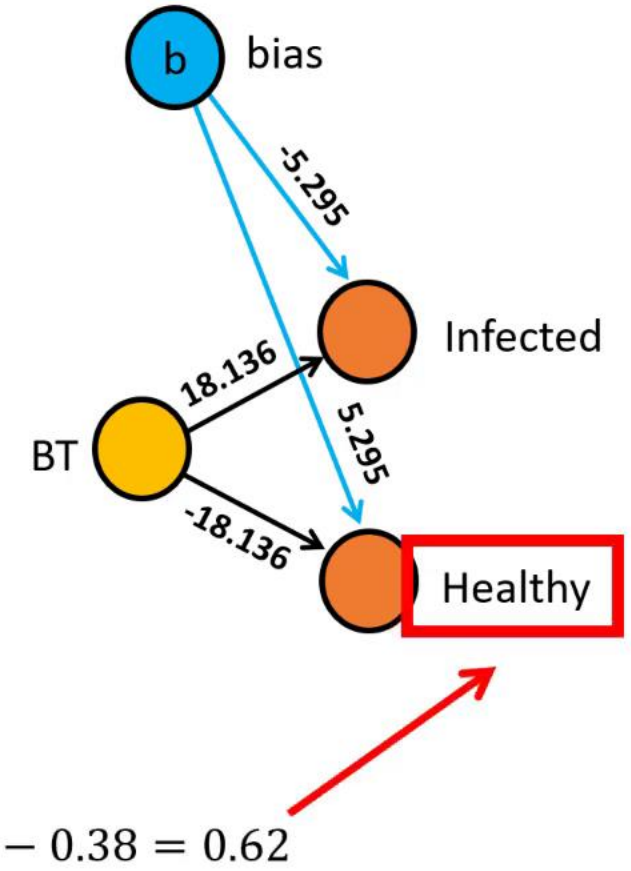
This means that the probability that a person with a body temperature of 37 is infected is about 38%.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{37 - 36.1}{39.5 - 36.1} = 0.265$$



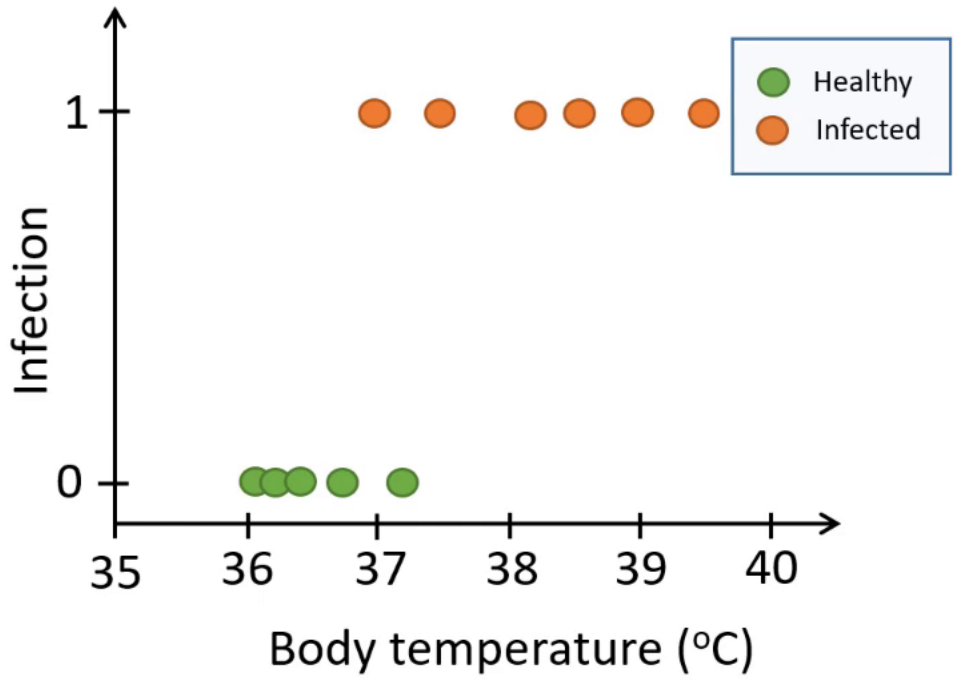


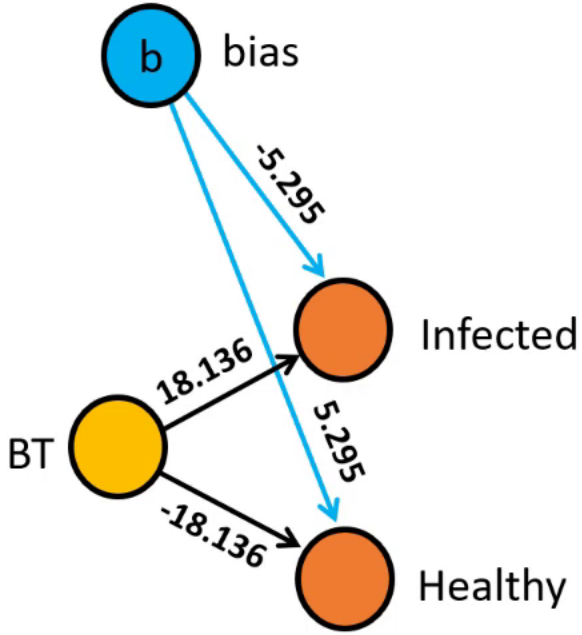
$$1 - 0.38 = 0.62$$

The probability that the person is healthy is simply 1 minus 0.38.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2





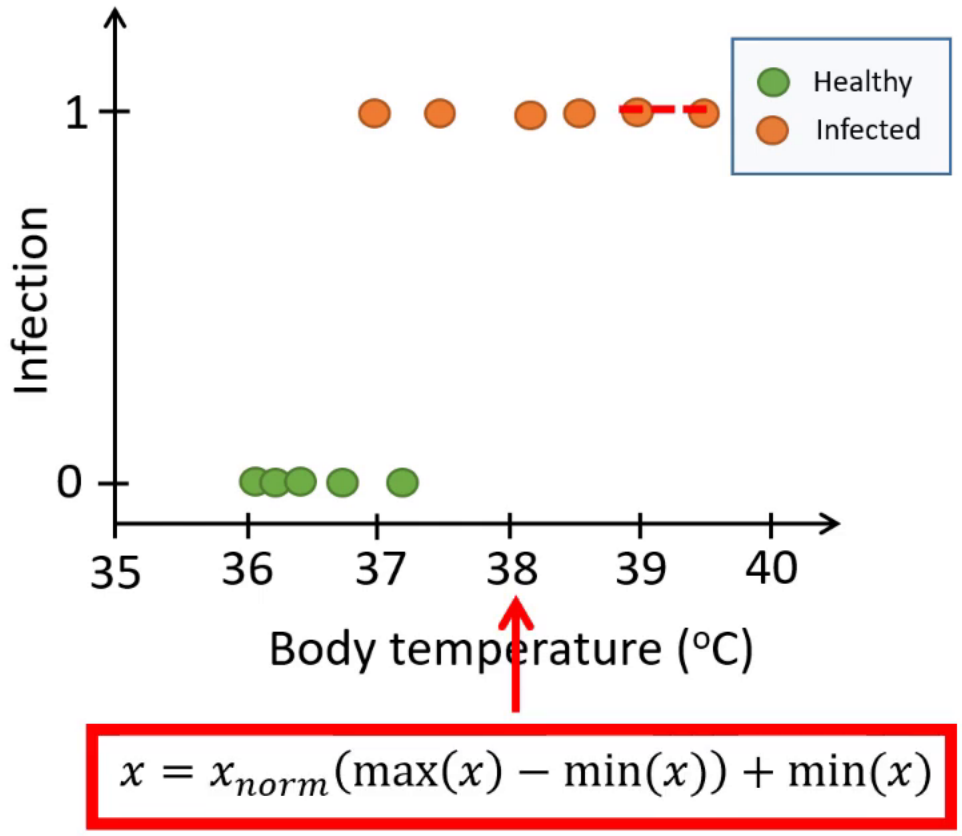
$$z = -5.295 + 18.136 \cdot x_{norm}$$

$$f(z) = \frac{1}{1 + e^{-z}}$$

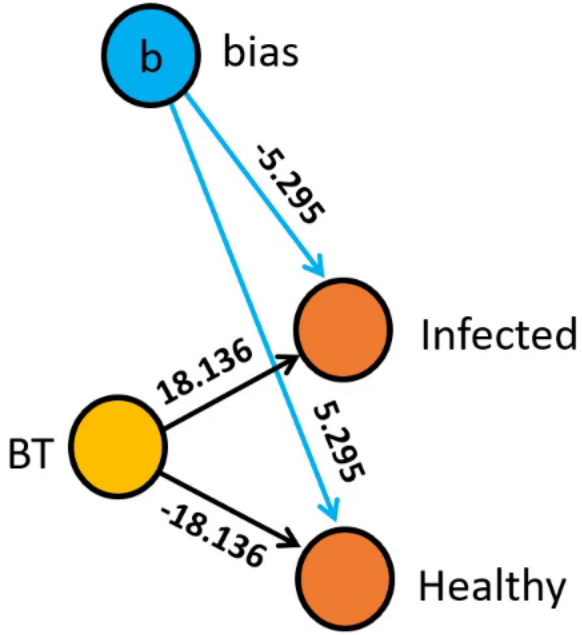
If we compute the same thing for a range of different values between zero and one,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$



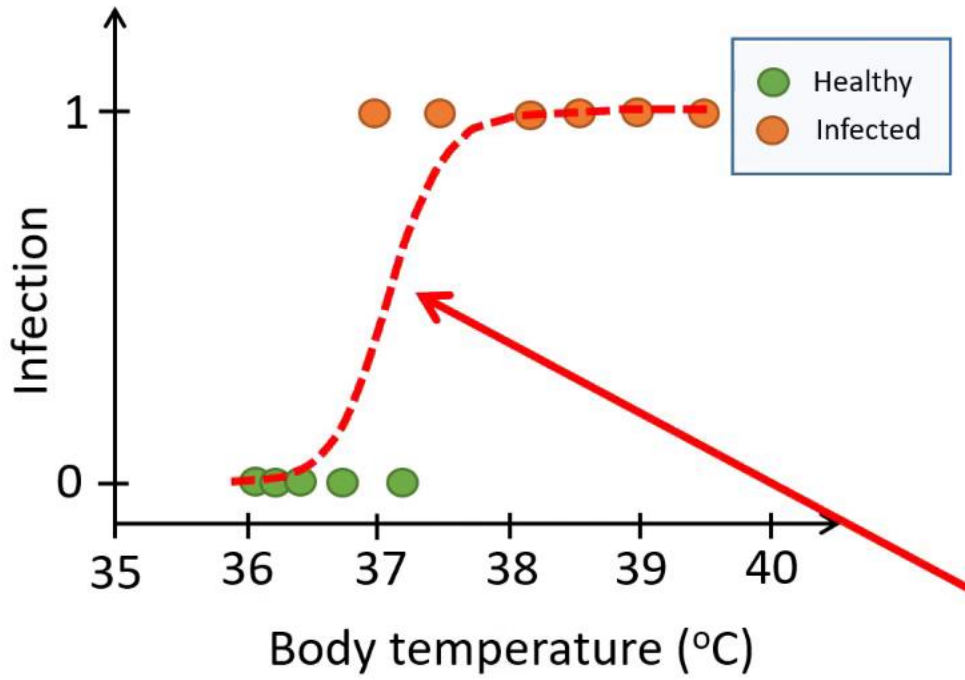
$$z = -5.295 + 18.136 \cdot x_{norm}$$

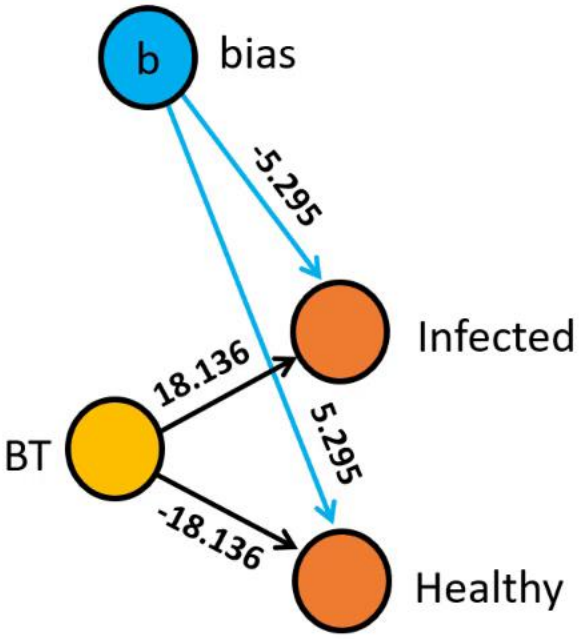
$$f(z) = \frac{1}{1 + e^{-z}}$$

and convert these normalized values between zero and one back to the original scale,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2





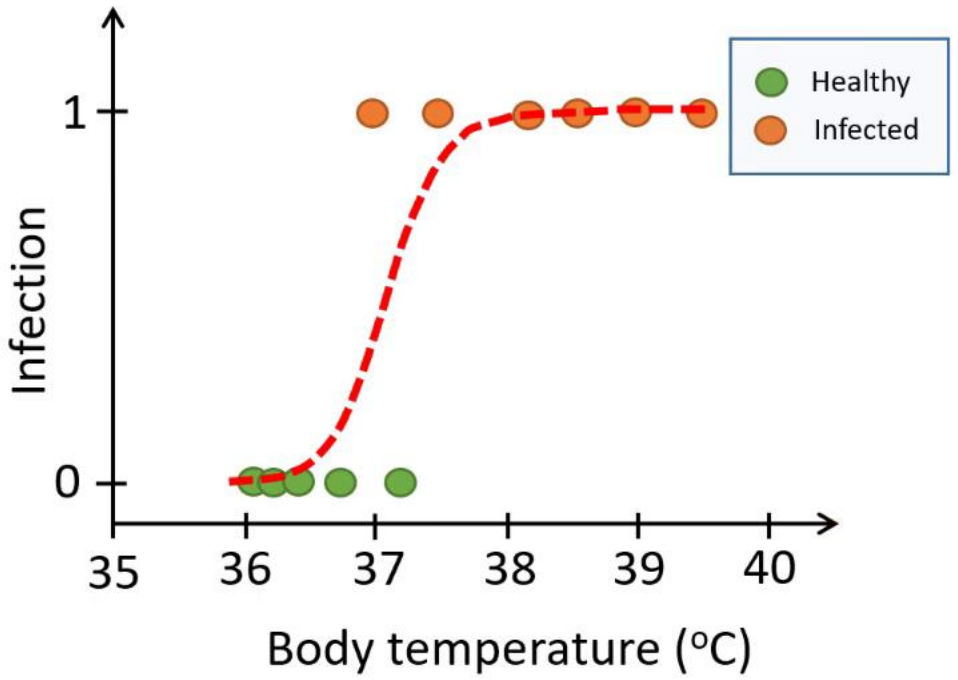
$$z = -5.295 + 18.136 \cdot x_{norm}$$

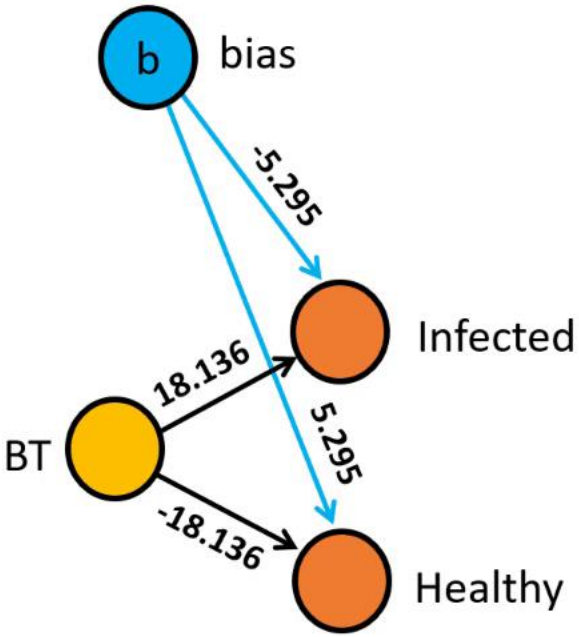
$$f(z) = \frac{1}{1 + e^{-z}}$$

we will be able to draw the following curve that shows the probability that a person is infected given a certain body temperature.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2





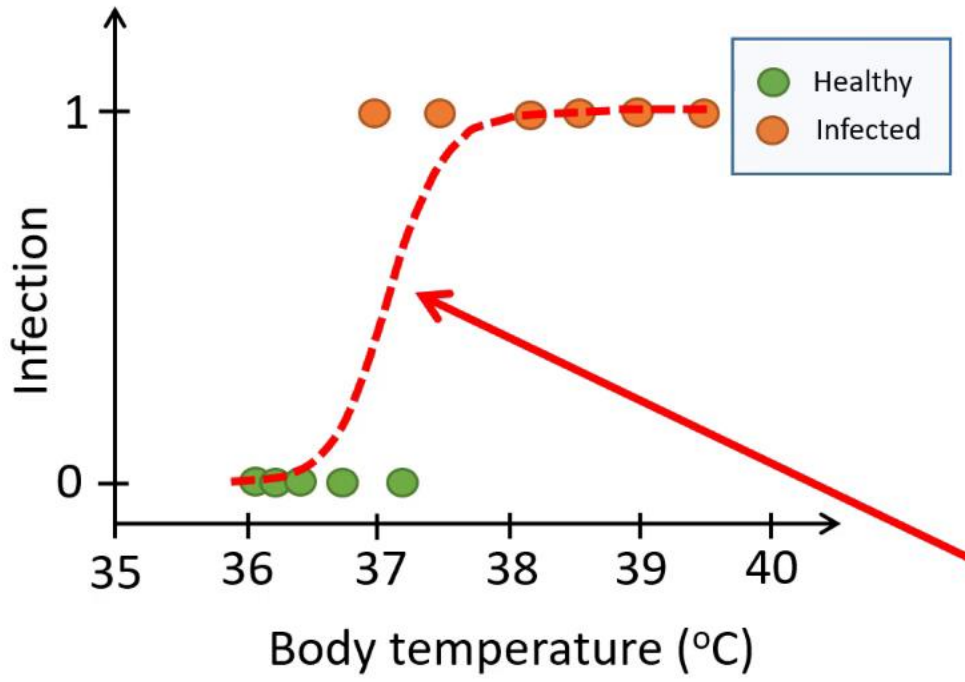
$$z = -5.295 + 18.136 \cdot x_{norm}$$

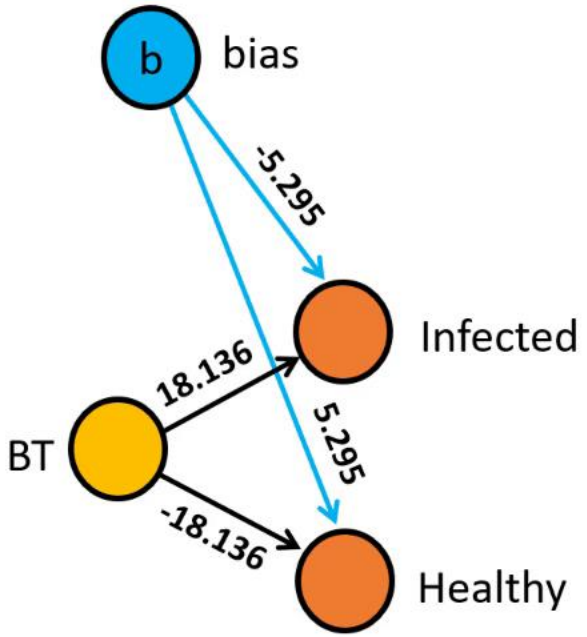
$$f(z) = \frac{1}{1 + e^{-z}}$$

Let's plug this expression here,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



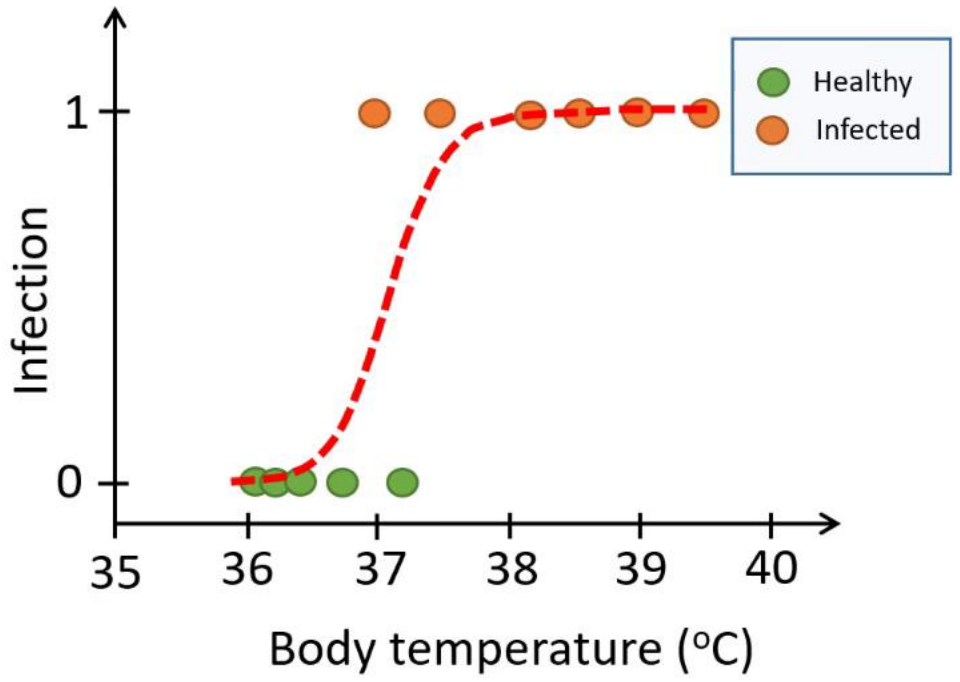


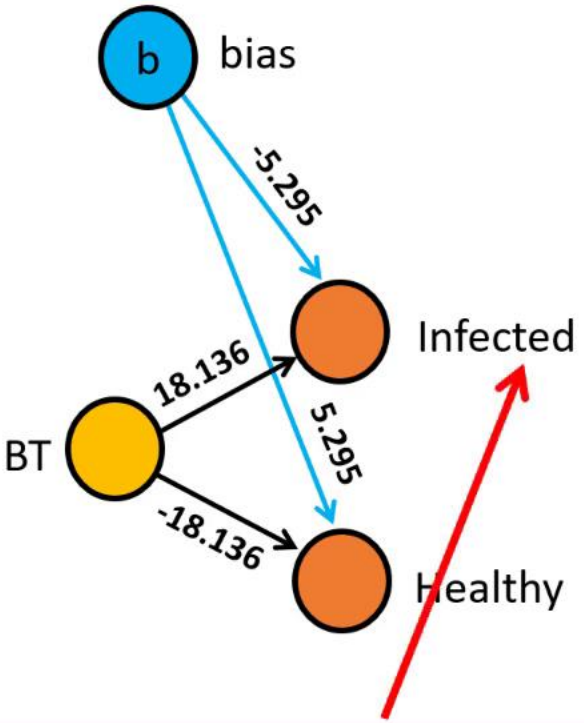
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295+18.136 \cdot x_{norm})}}$$

so that we have the function of the curve.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



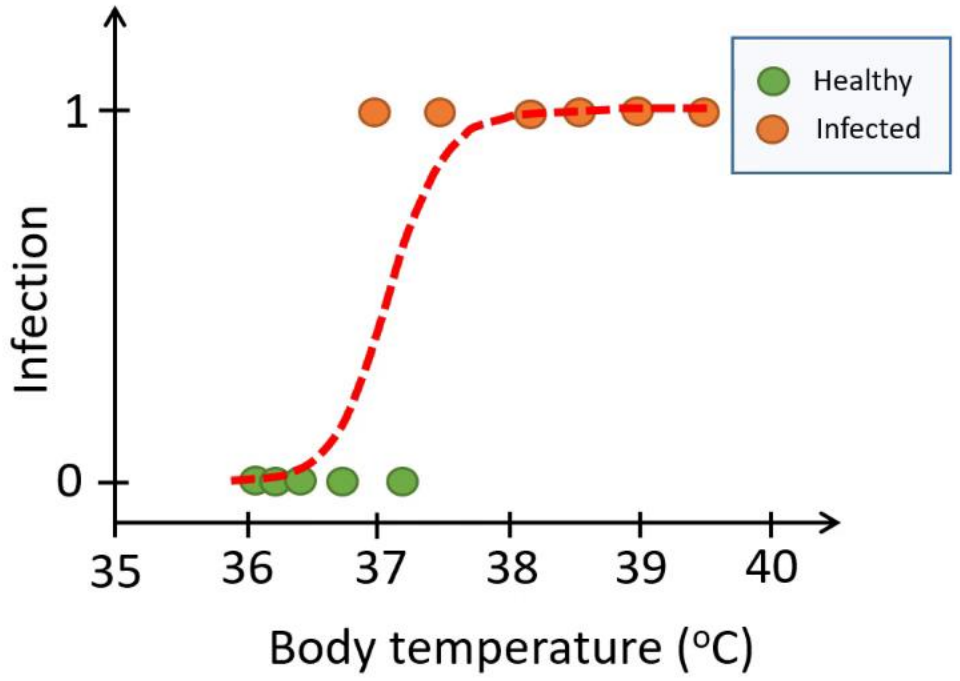


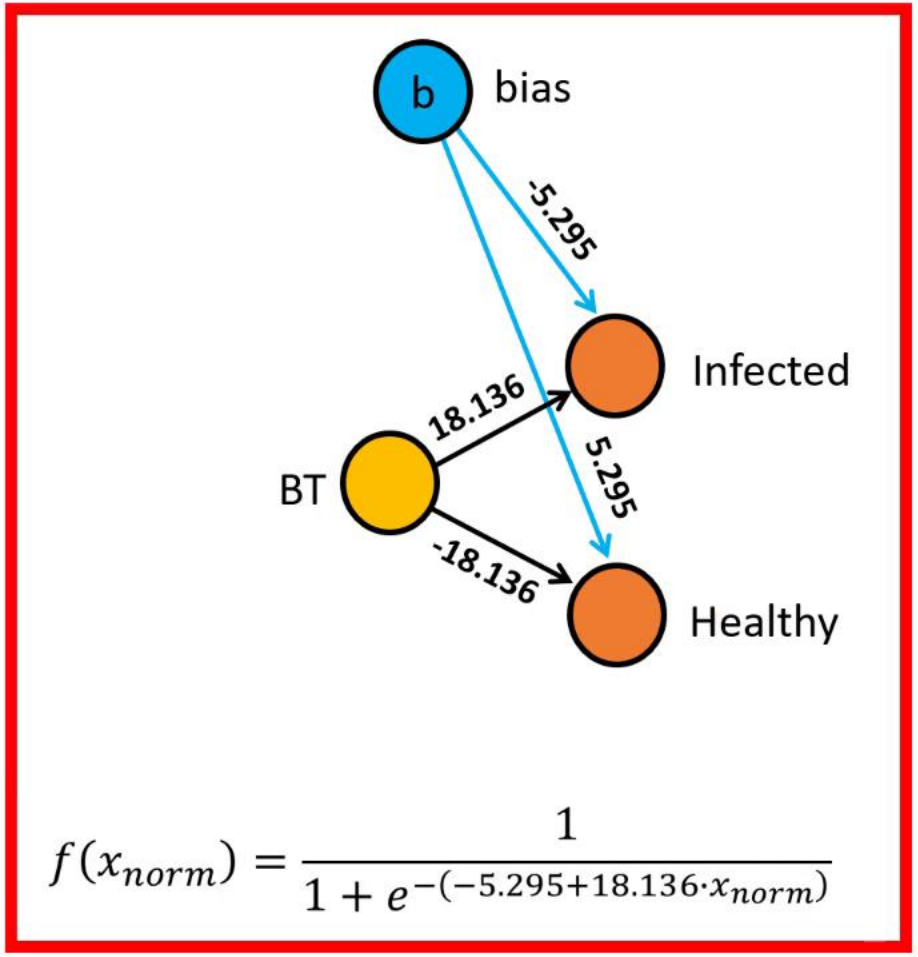
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 + 18.136 \cdot x_{norm})}}$$

This mathematical function is actually the one that is used by the network when the value of this output node is calculated, which is the predicted probability of being infected.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



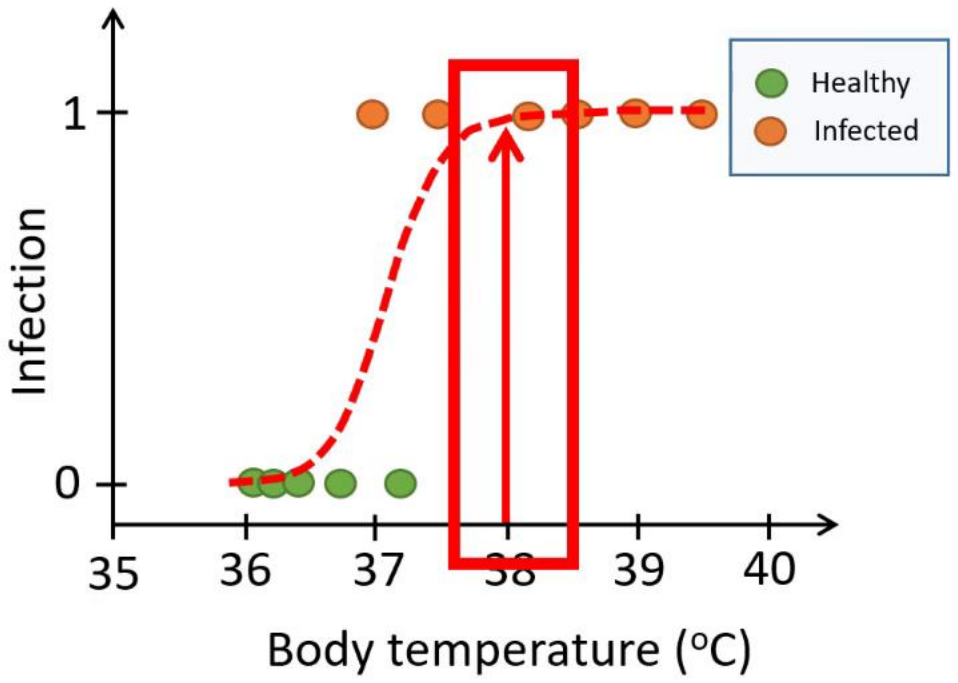


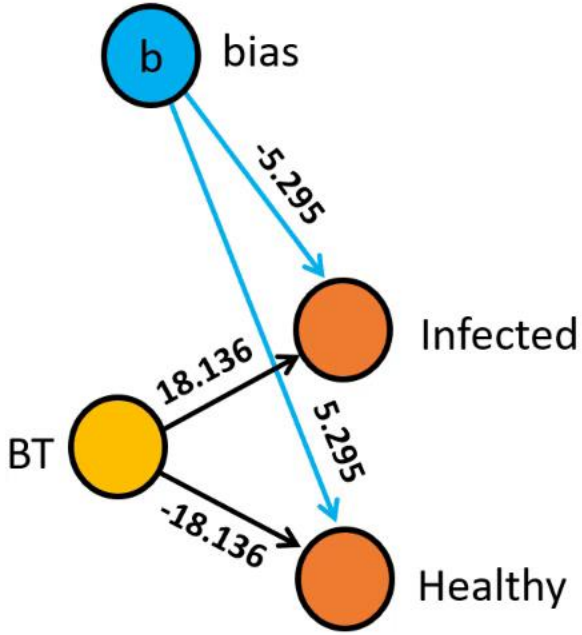
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295+18.136 \cdot x_{norm})}}$$

Note that this simple network is basically the same as logistic regression.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



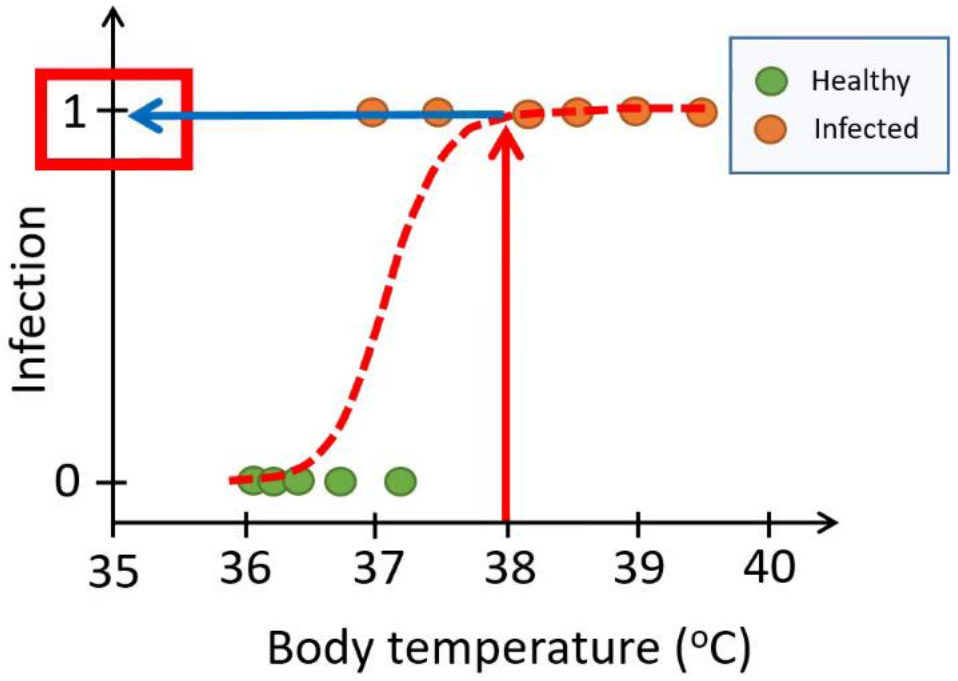


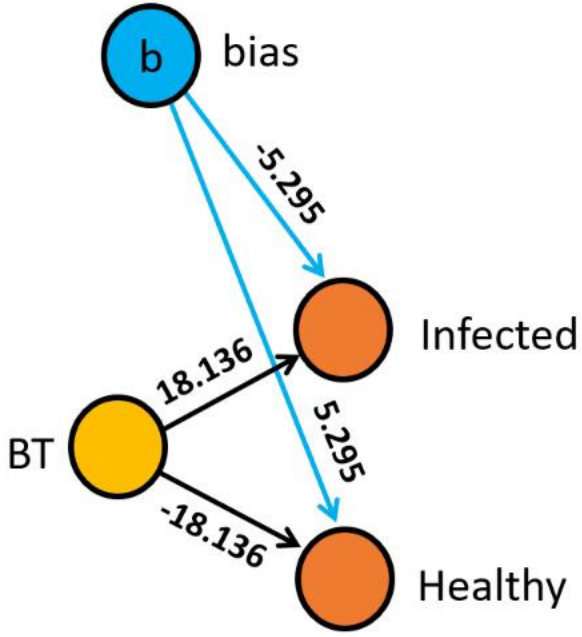
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295+18.136 \cdot x_{norm})}}$$

To predict if a person with, for example, a body temperature of 38 is infected, we can draw a vertical line from 38,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2





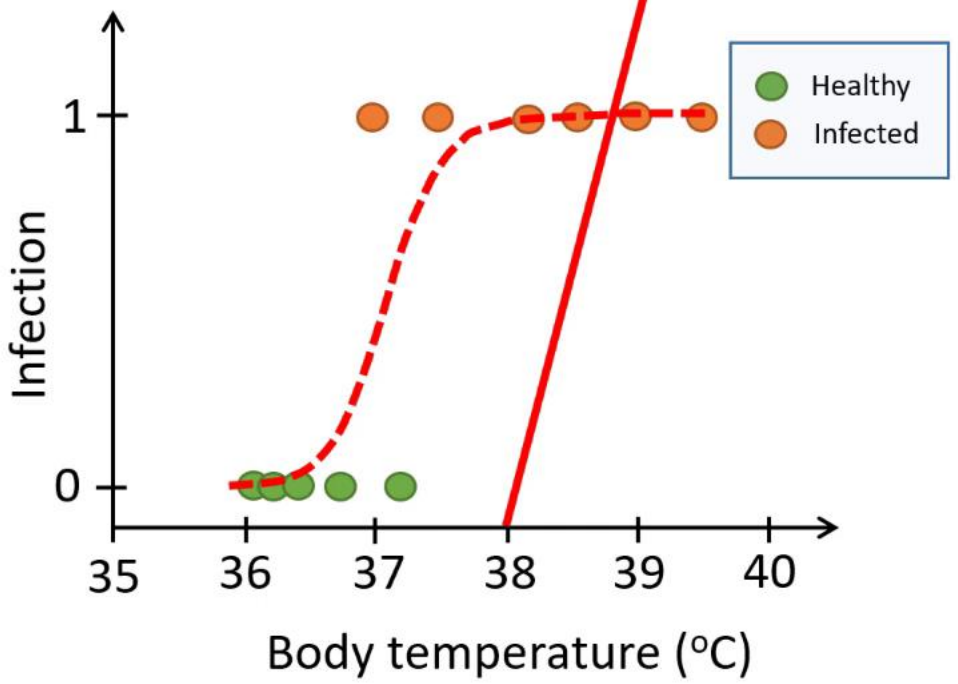
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295+18.136 \cdot x_{norm})}}$$

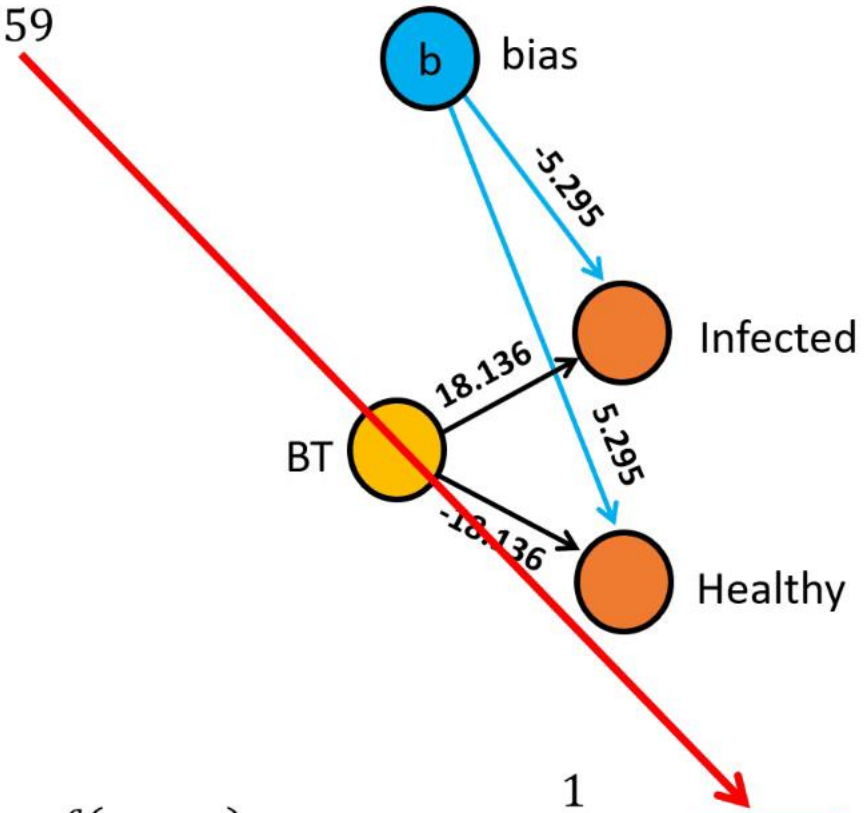
and then a horizontal line from the curve to the y-axis. We see that the corresponding value of the y-axis is about one.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{38 - 36.1}{39.5 - 36.1} = 0.559$$





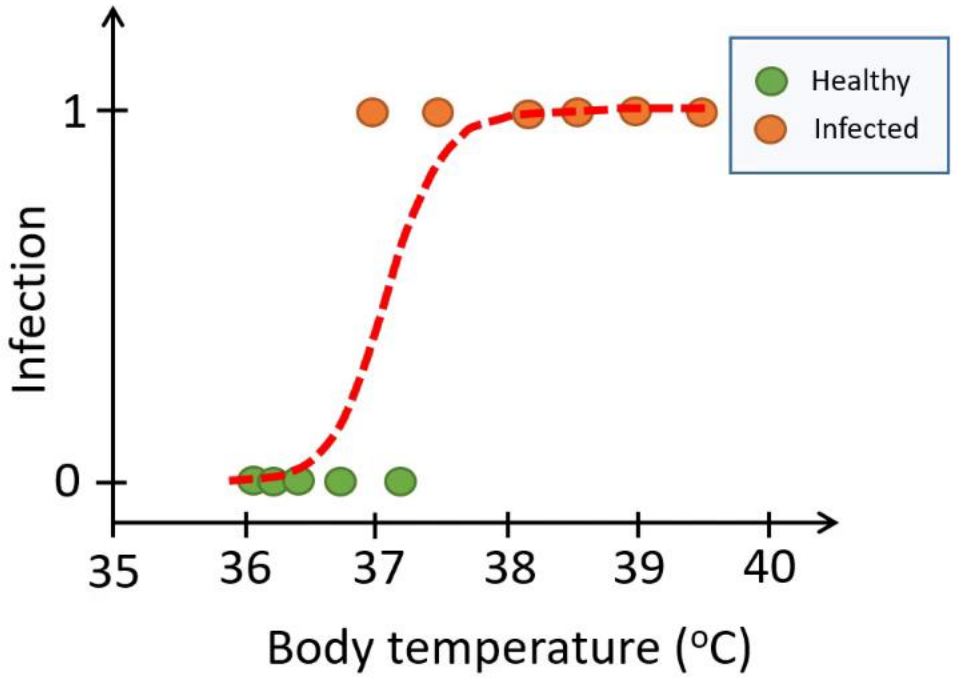
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 + 18.136 \cdot x_{norm})}}$$

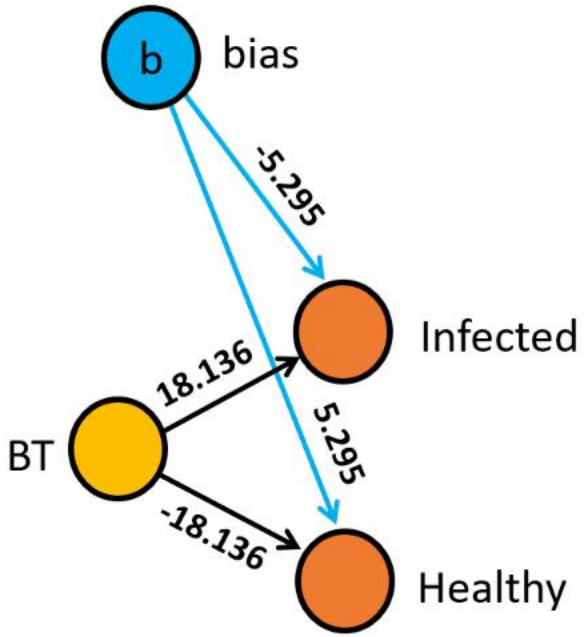
We will get a more accurate value if we plug in the normalized value here,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{38 - 36.1}{39.5 - 36.1} = 0.559$$





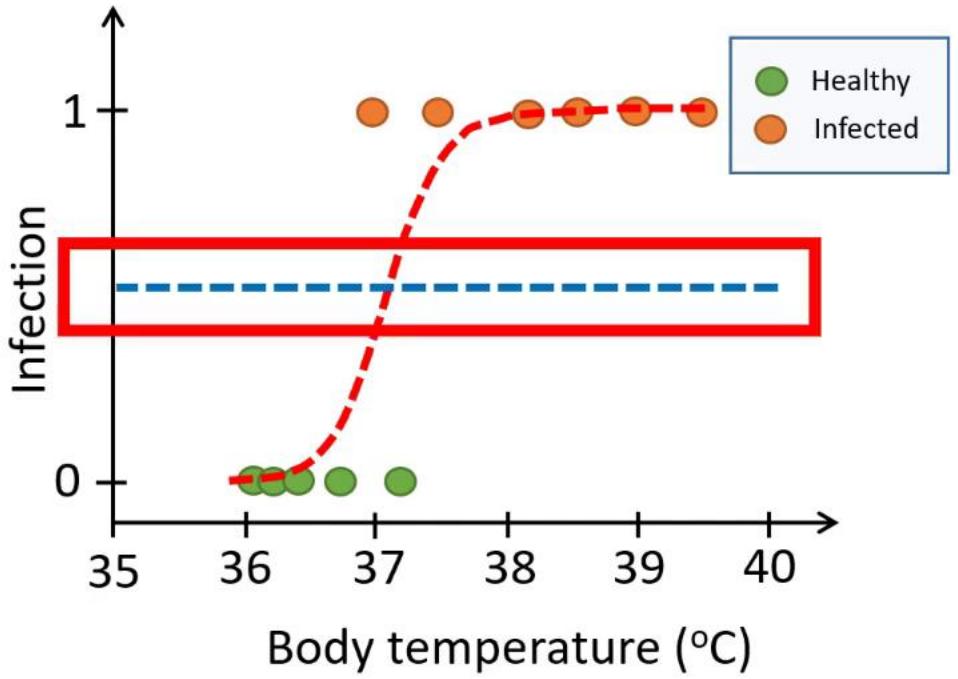
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 + 18.136 \cdot 0.559)}} = 0.99$$

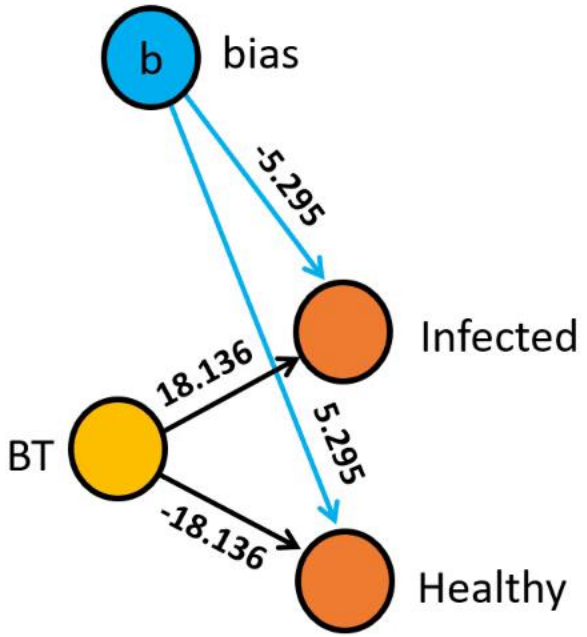
and do the math.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{38 - 36.1}{39.5 - 36.1} = 0.559$$





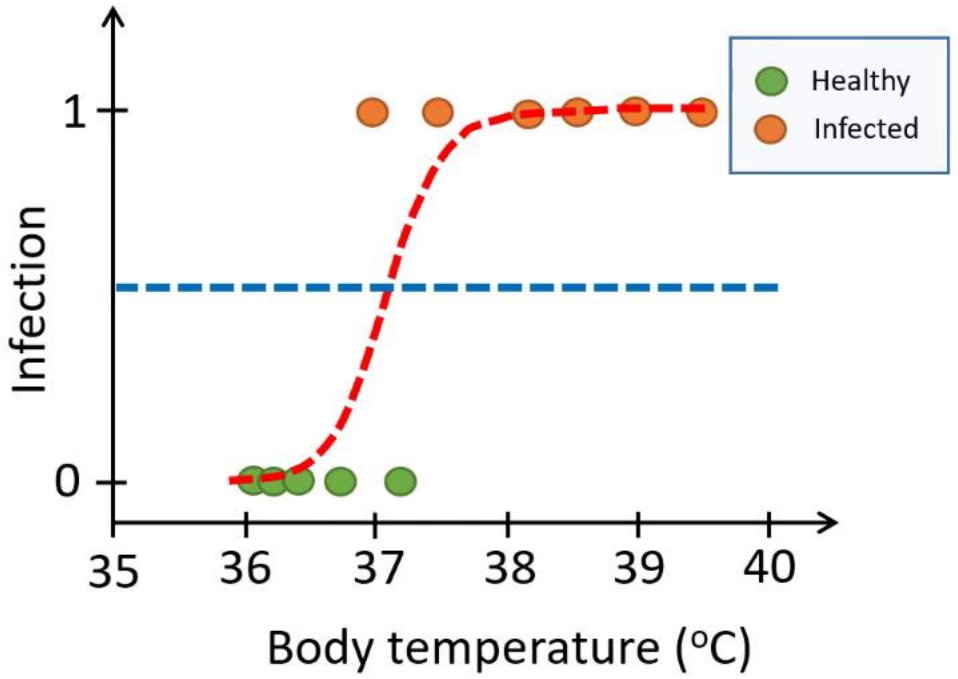
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 + 18.136 \cdot 0.559)}} = 0.99$$

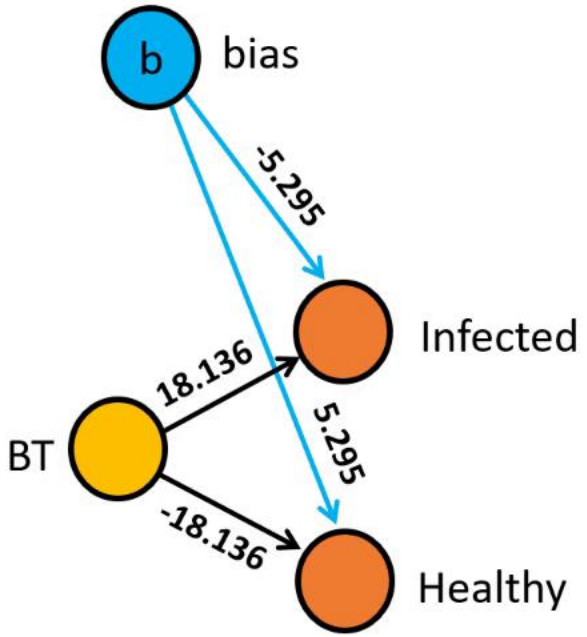
A common threshold to make a categorical prediction based on a probability is 0.5.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{38 - 36.1}{39.5 - 36.1} = 0.559$$



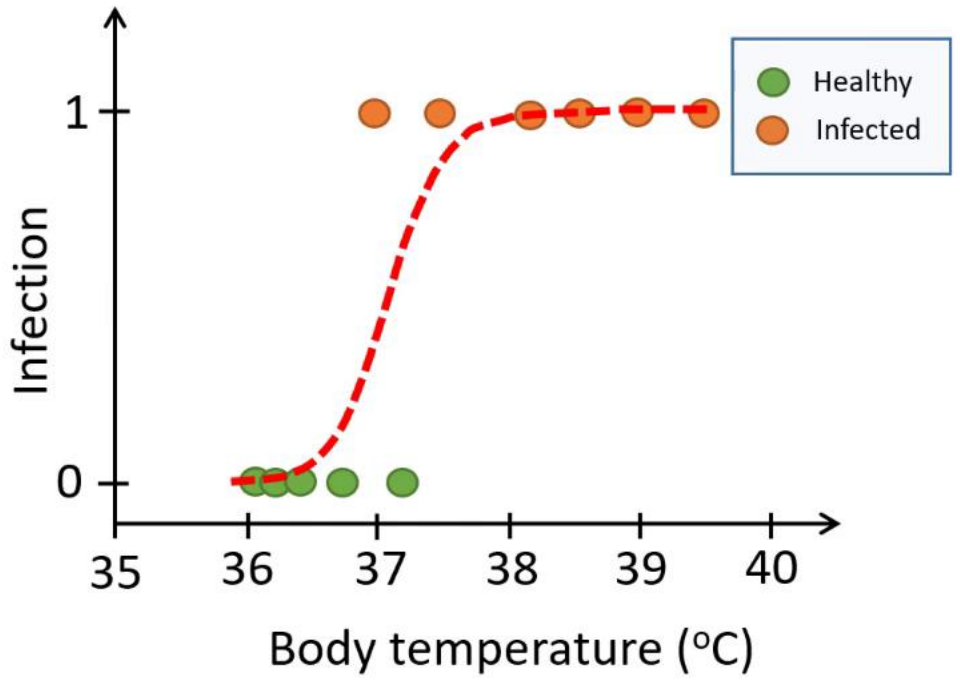


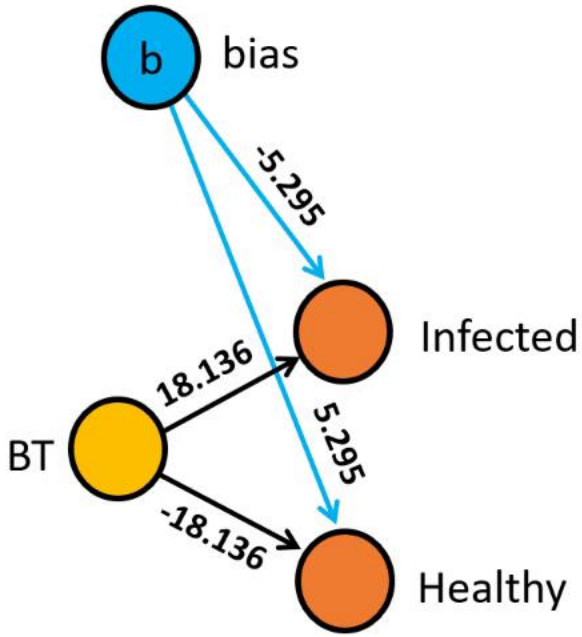
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 + 18.136 \cdot 0.559)}} = 0.99$$

Since this probability is greater than 0.5, we predict that a person with a body temperature of 38 is infected.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



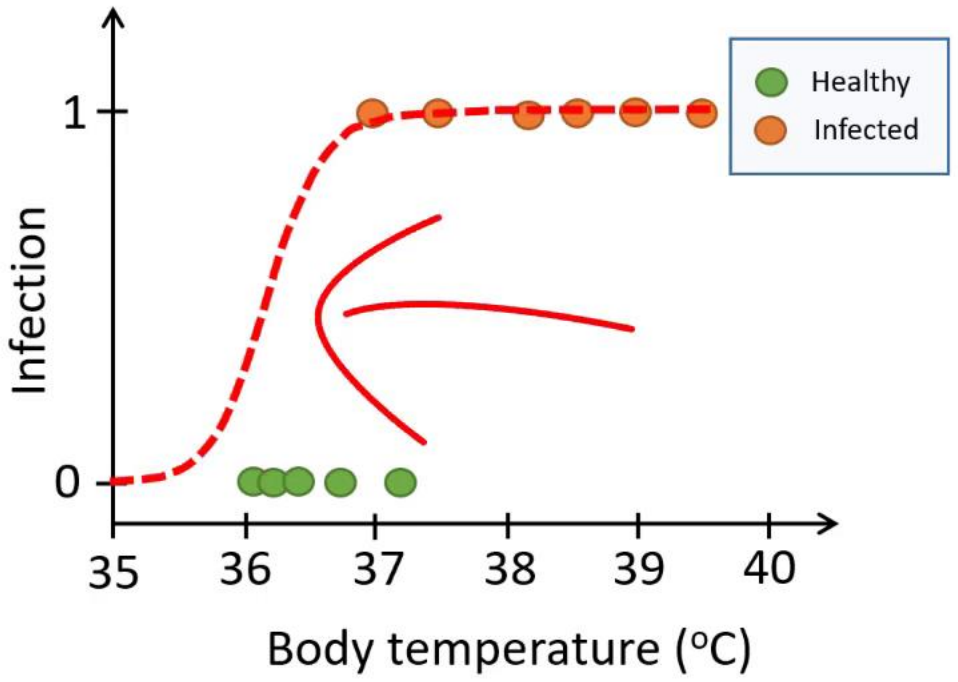


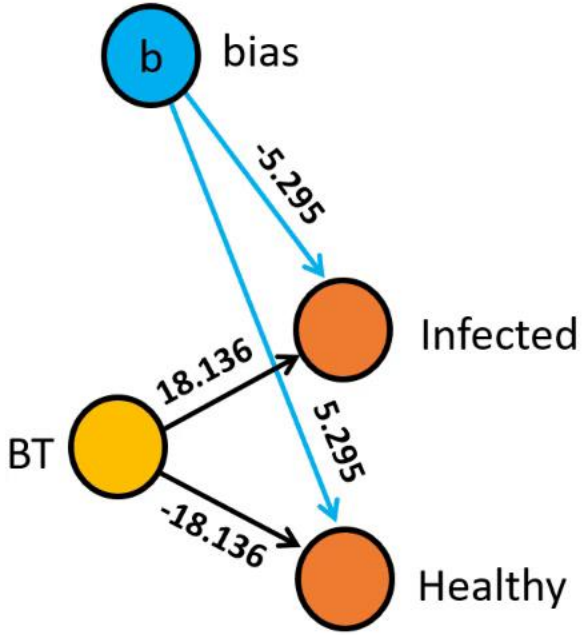
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 + 18.136 \cdot x_{norm})}}$$

By changing the bias weight,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



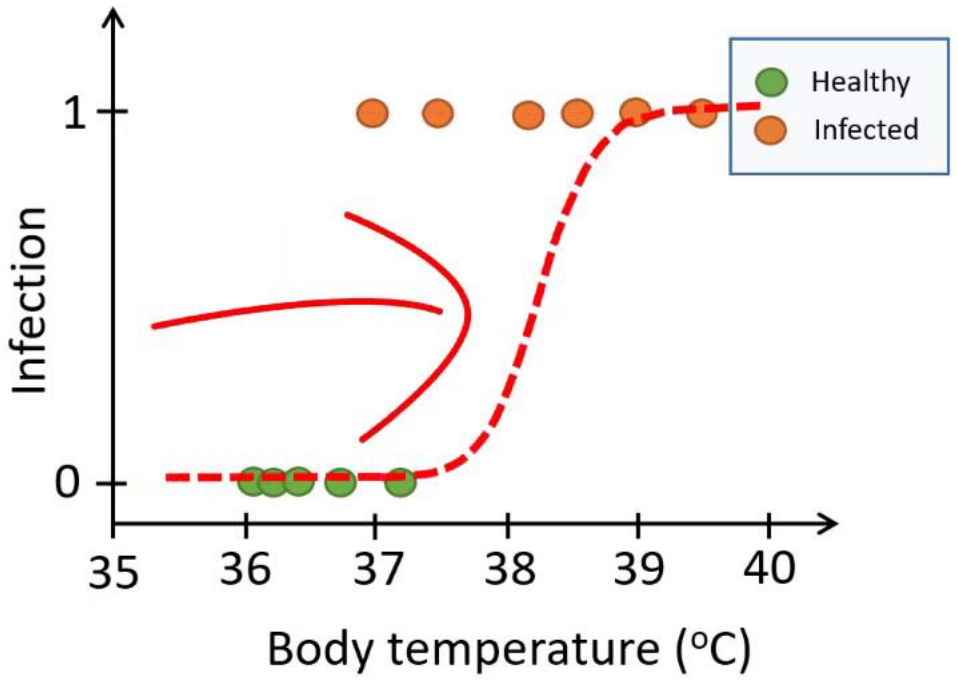


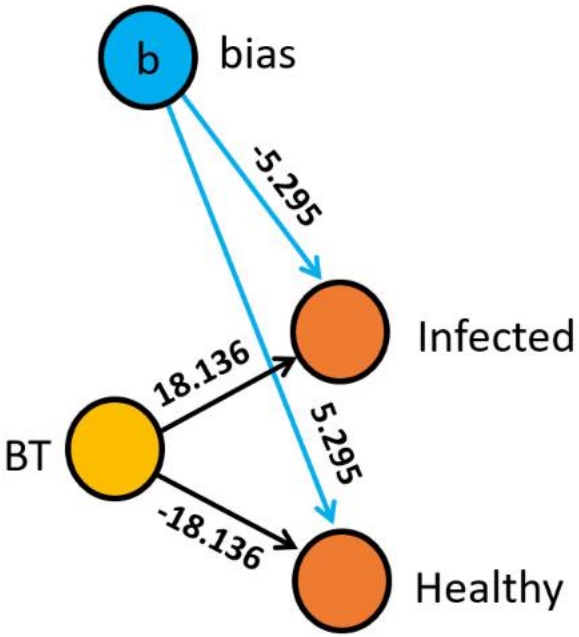
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295+18.136 \cdot x_{norm})}}$$

we can move the curve to the left,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



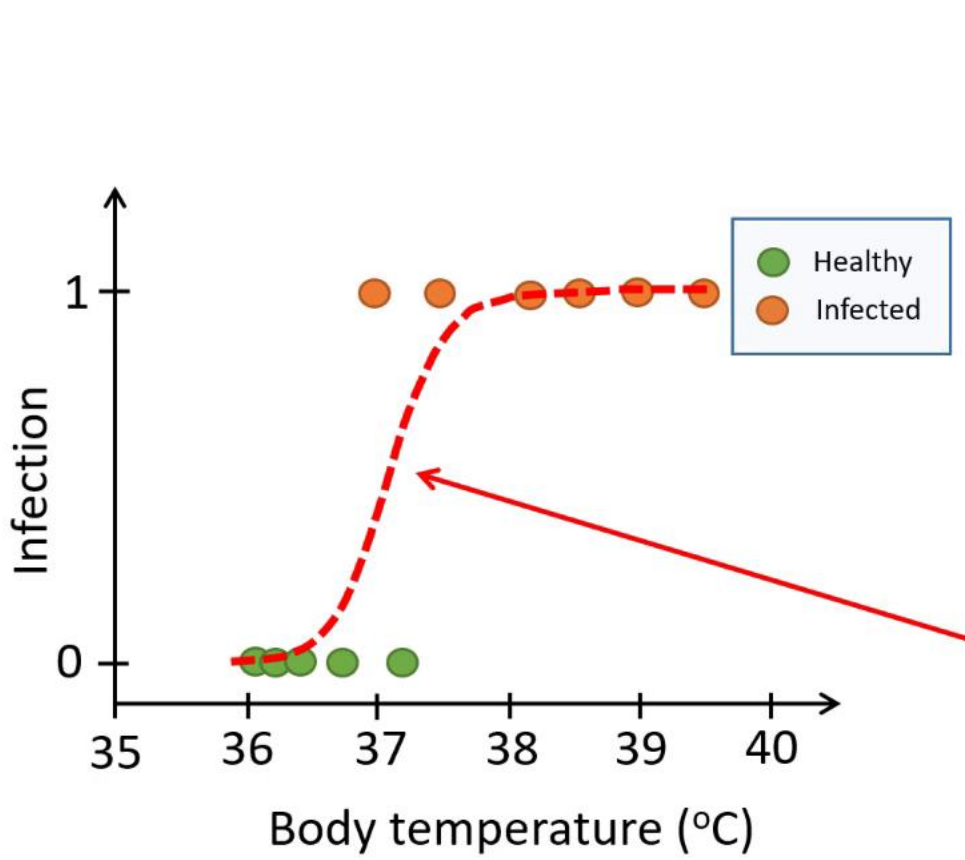


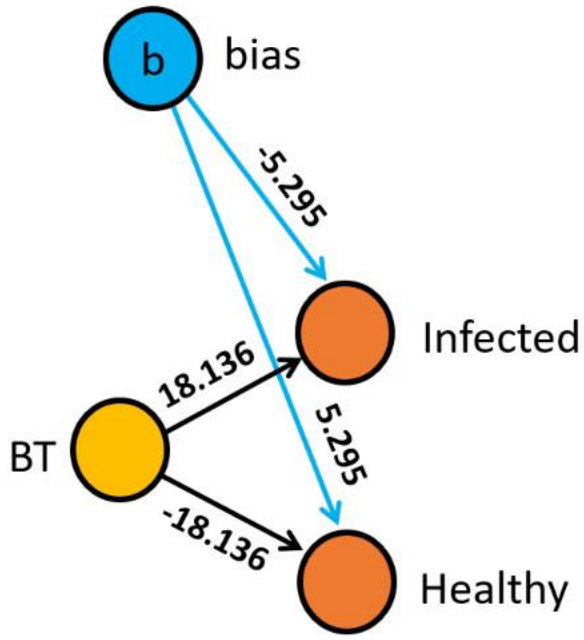
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295+18.136 \cdot x_{norm})}}$$

or right.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



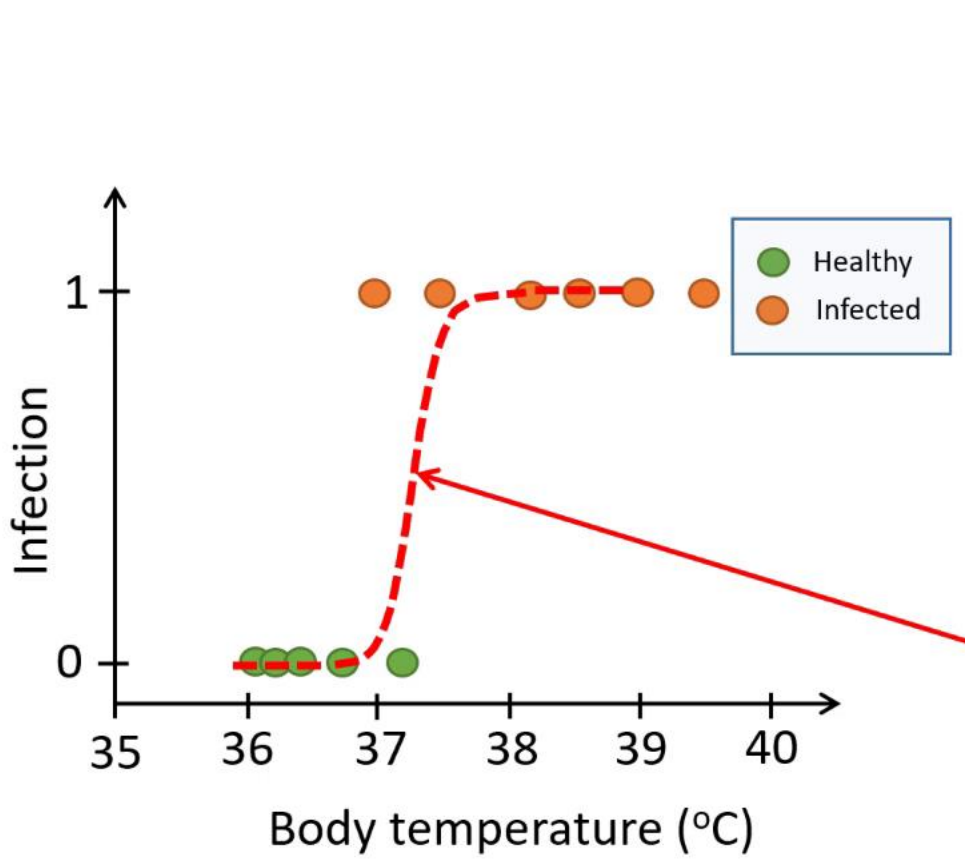


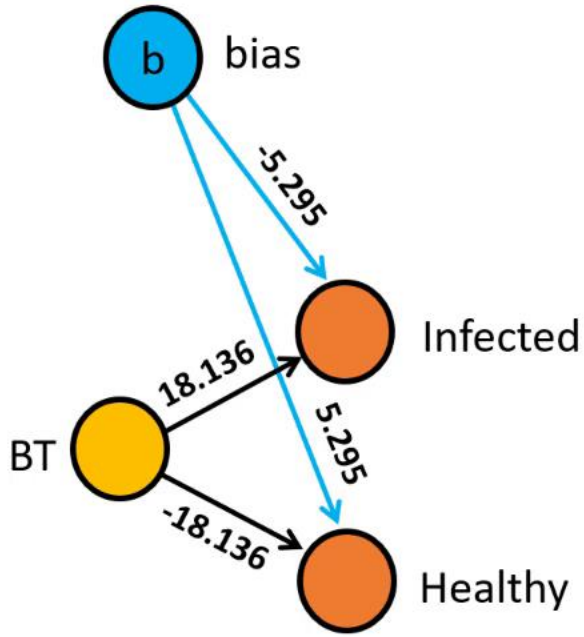
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 - 18.136\, x_{norm})}}$$

By instead changing the value of the weight that is multiplied by the value in the input node, we can change the slope of the curve. If we increase the value of this weight,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



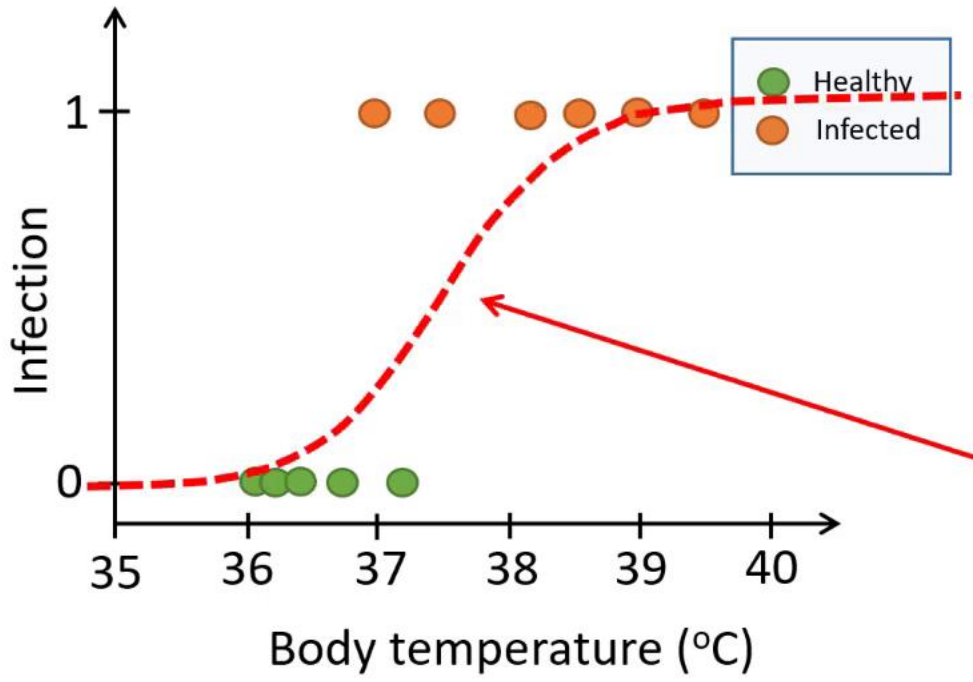


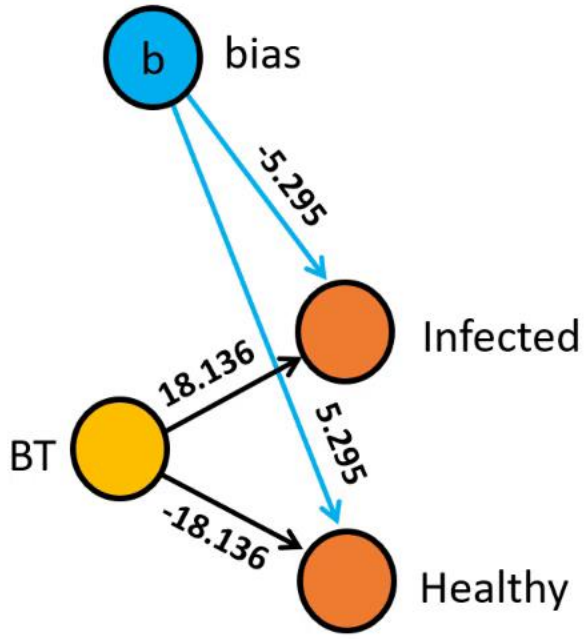
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 - 18.136\, x_{norm})}}$$

the curve becomes steeper,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



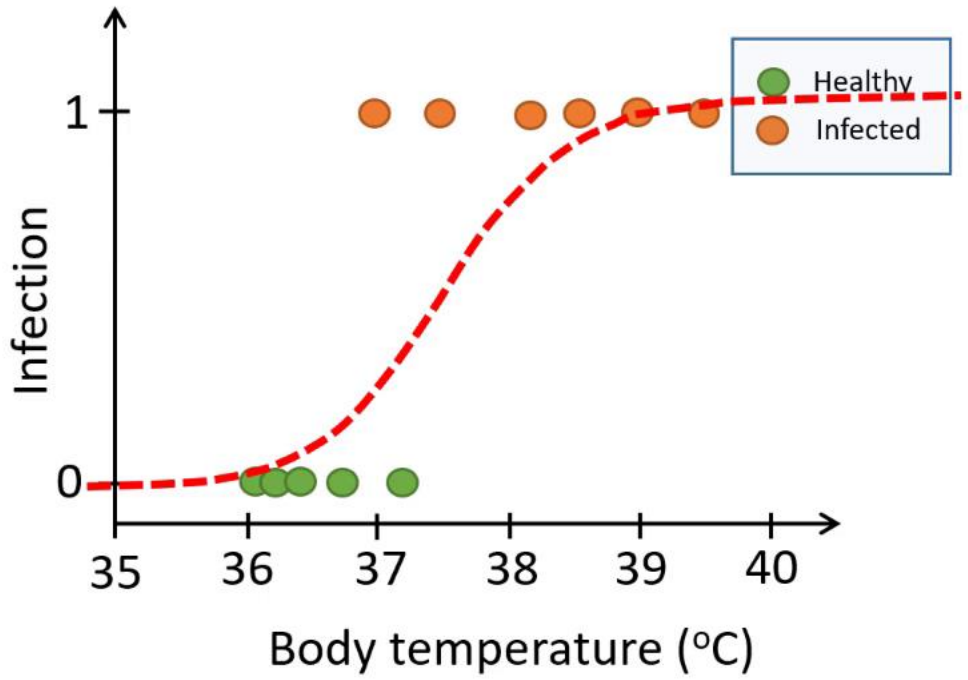


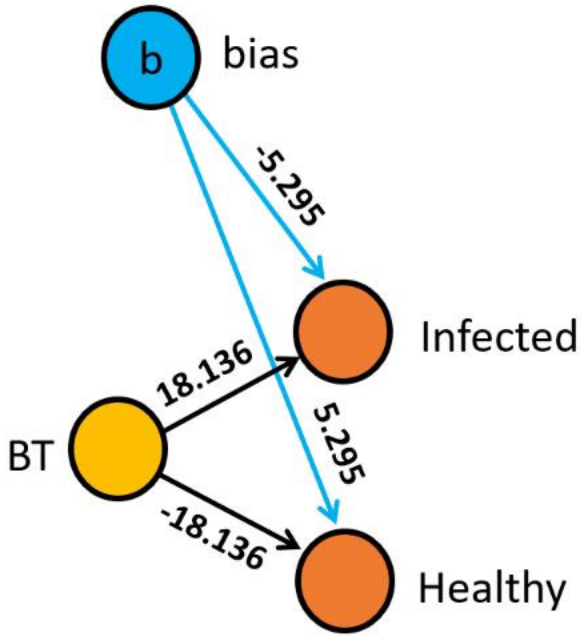
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 - 18.136\, x_{norm})}}$$

and if we reduce this value, the curve is less steep.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



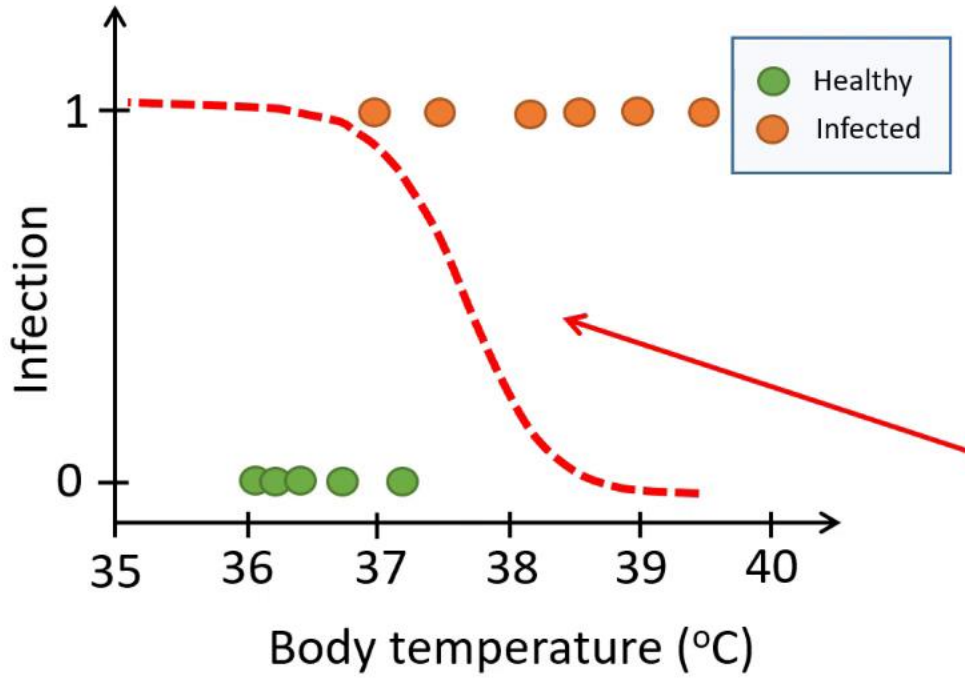


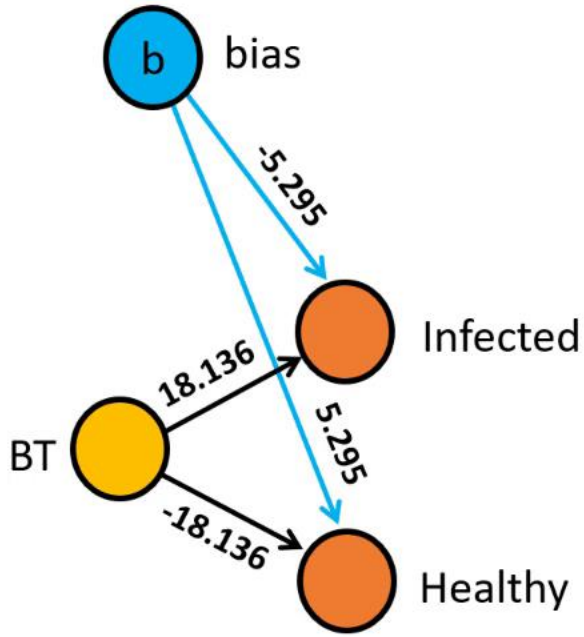
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 - 18.136\, x_{norm})}}$$

If the value is negative,

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



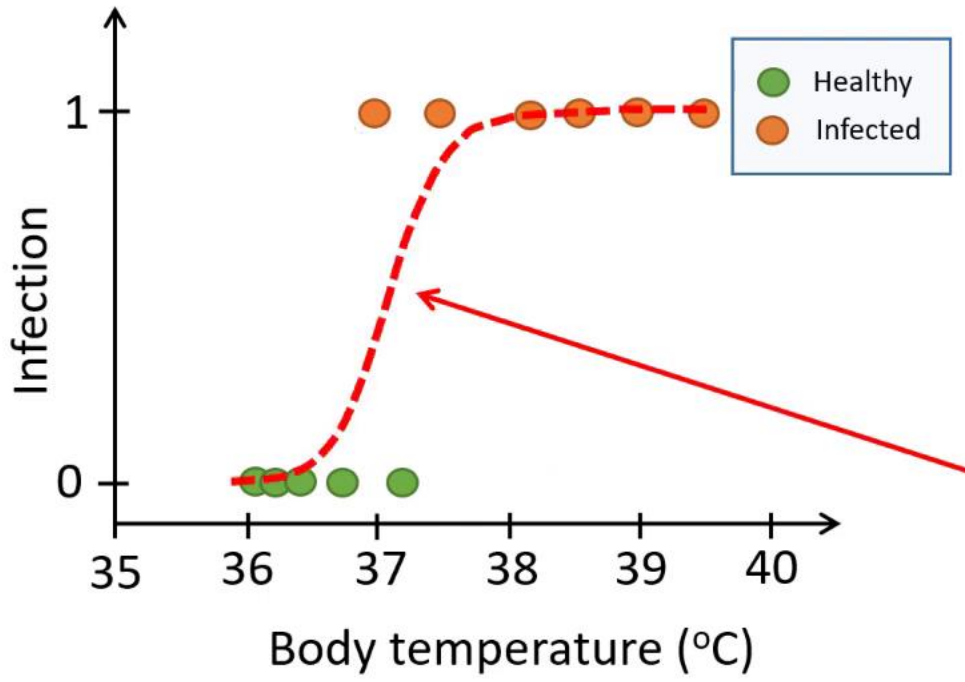


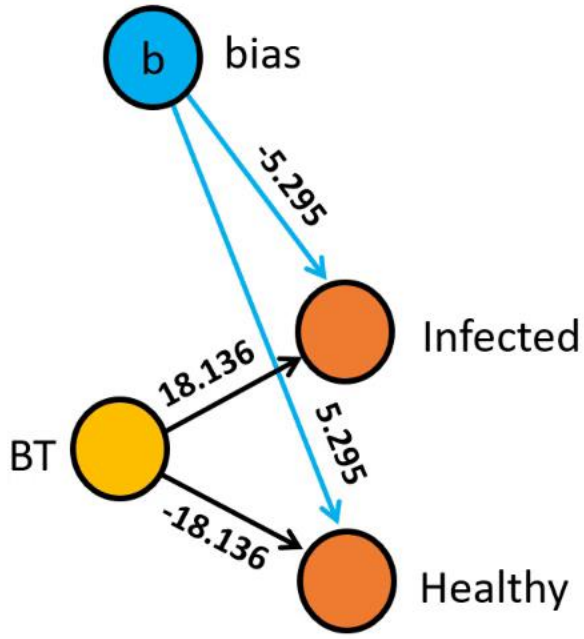
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 - 18.136\, x_{norm})}}$$

the curve may result in a negative slope.

A network without a hidden layer

Status	BT
Infected	37.0
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



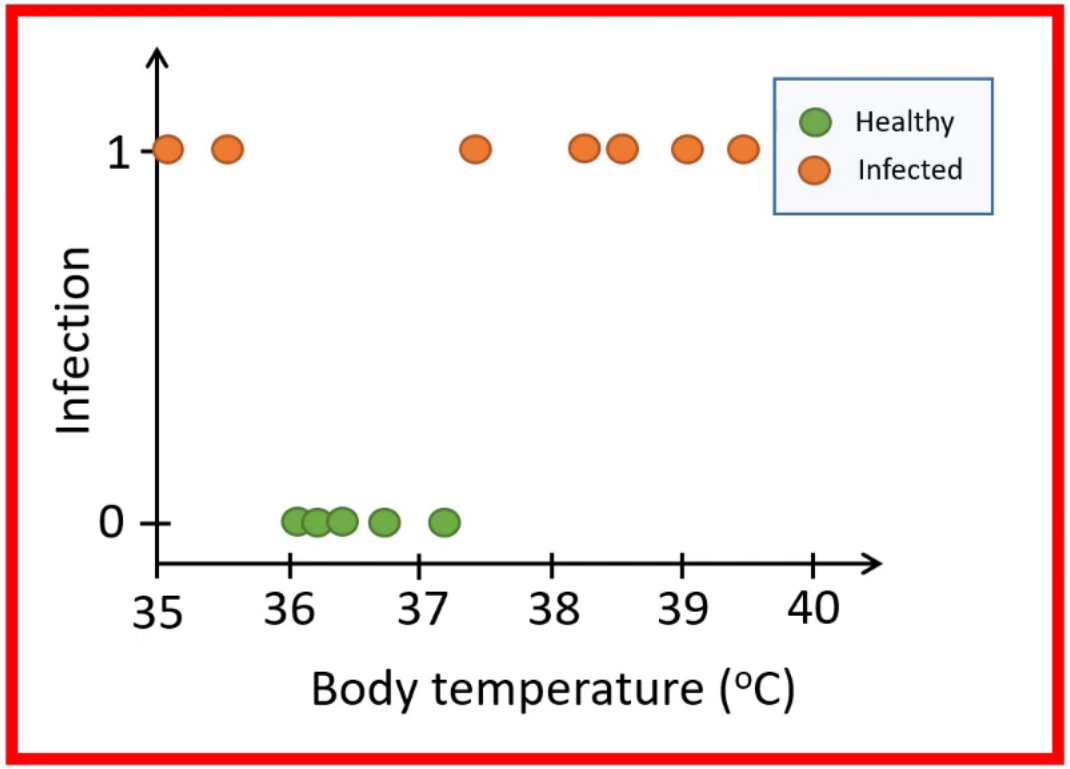


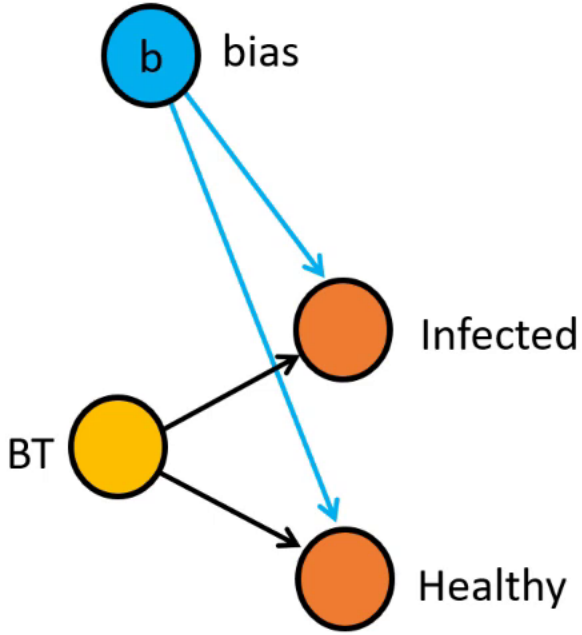
$$f(x_{norm}) = \frac{1}{1 + e^{-(-5.295 + 18.136 \cdot x_{norm})}}$$

A curve that makes the best prediction and fits best to the data is clearly a curve with the following values of the weights.

A network without a hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

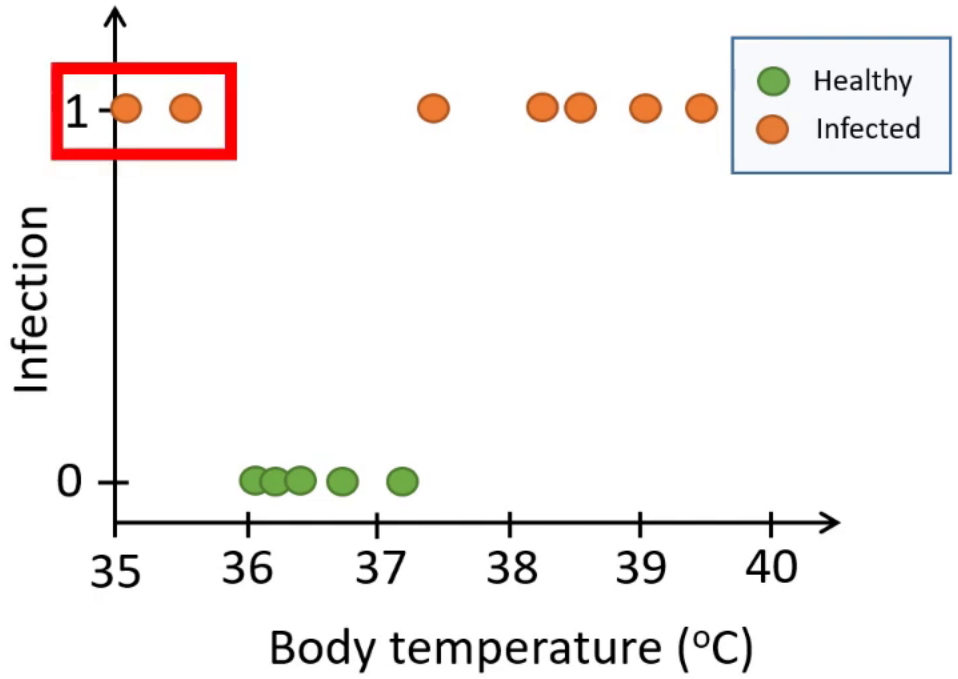


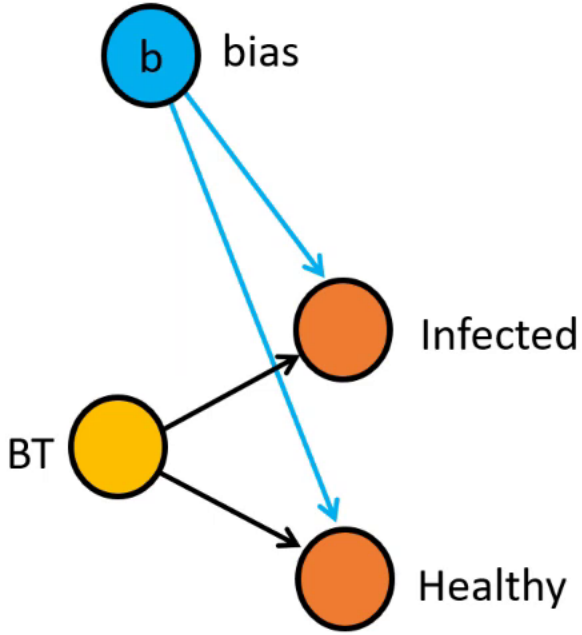


Now, suppose that we instead have the following data,

A network without a hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

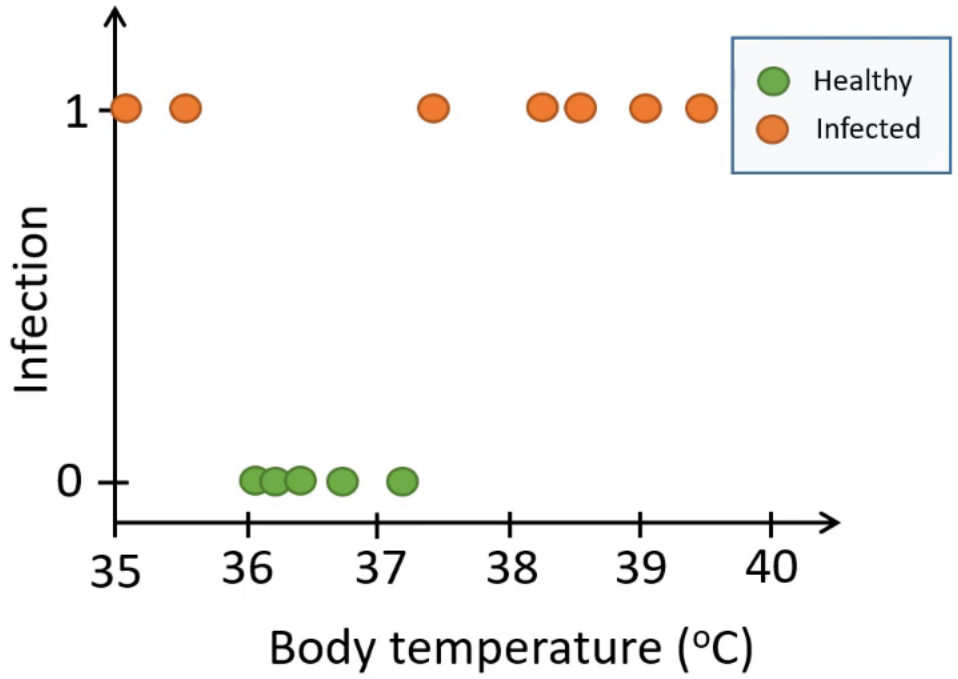


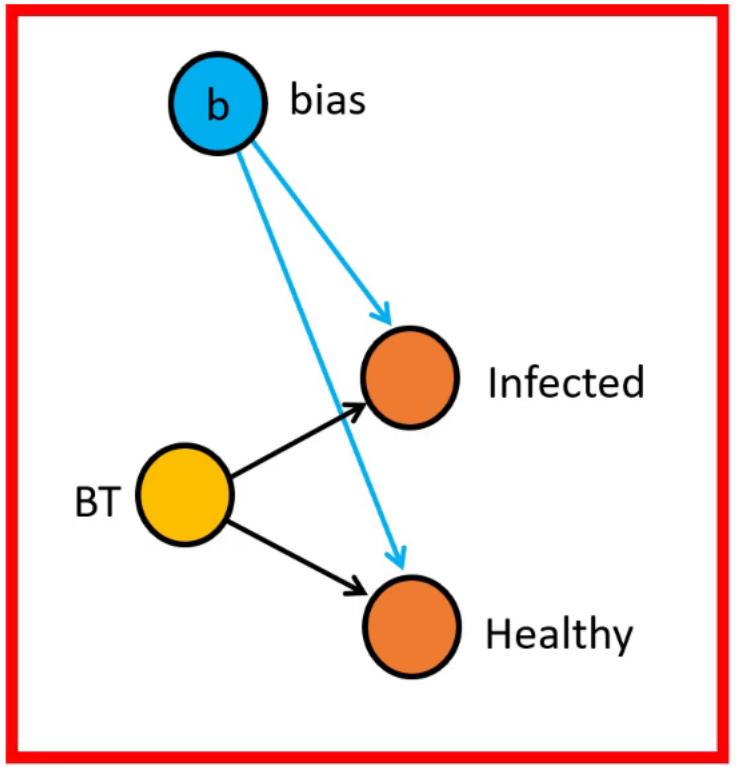


where two infected individuals now have very low body temperature.

A network without a hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

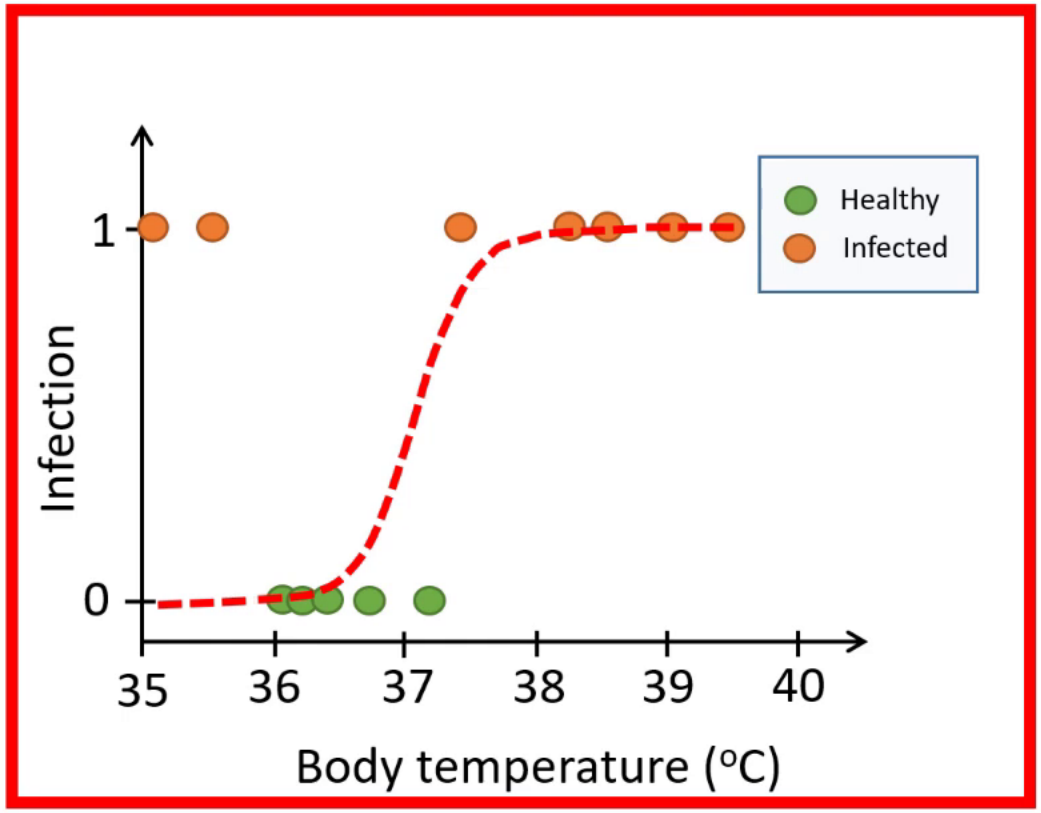


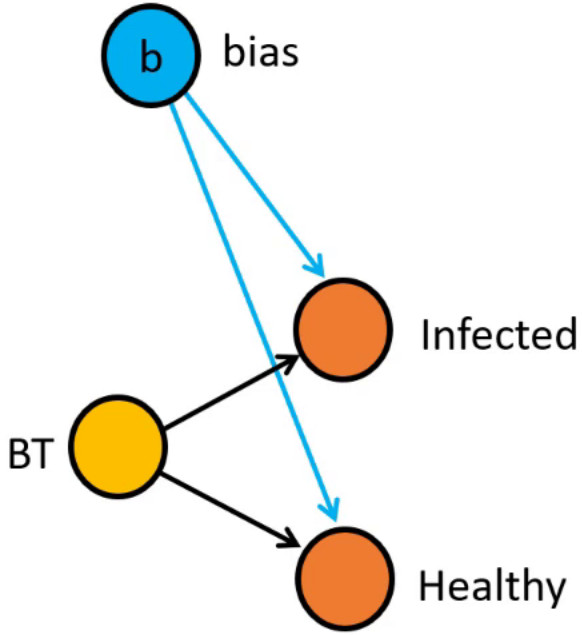


If we use this simple neural network,

A network without a hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

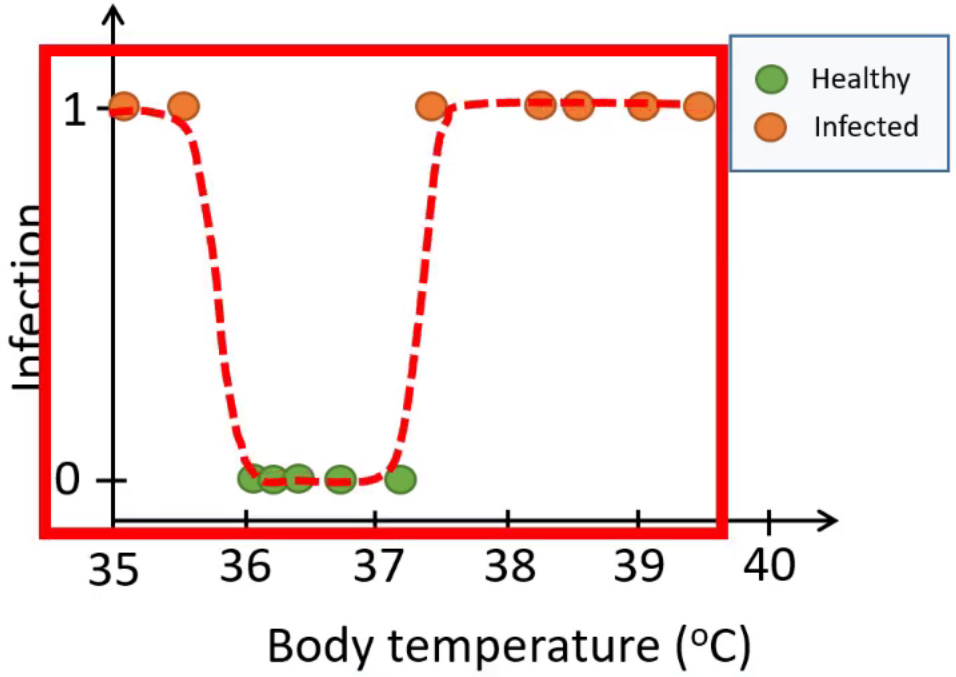


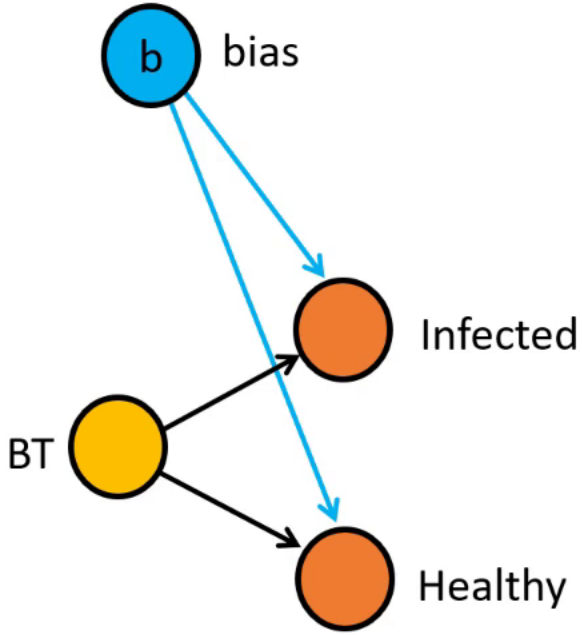


it will fail to discriminate healthy from infected individuals because it can only generate this s-shaped curve.

A network without a hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

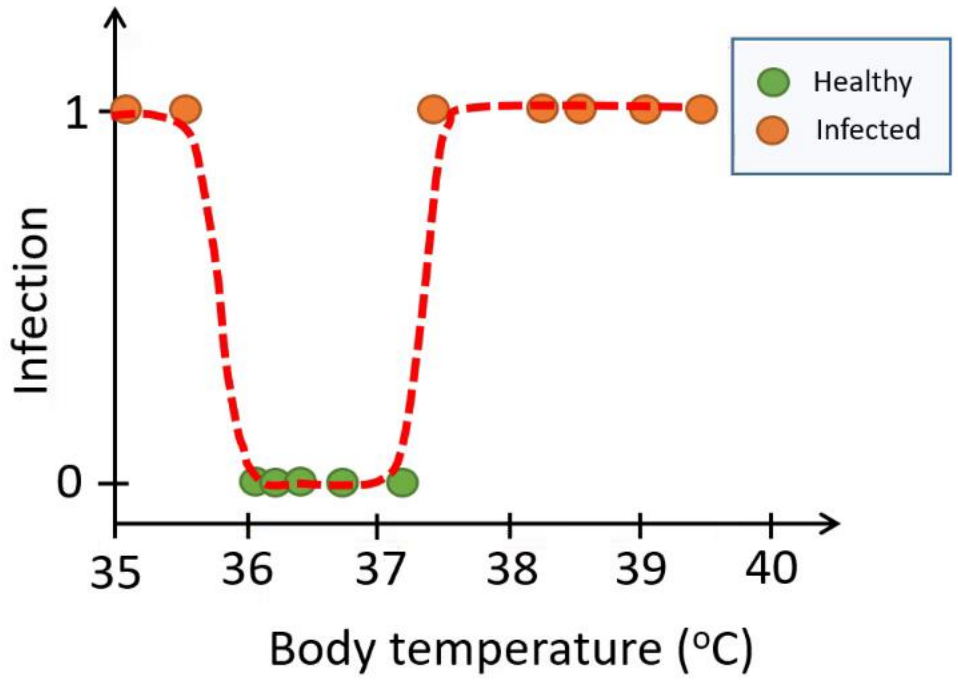


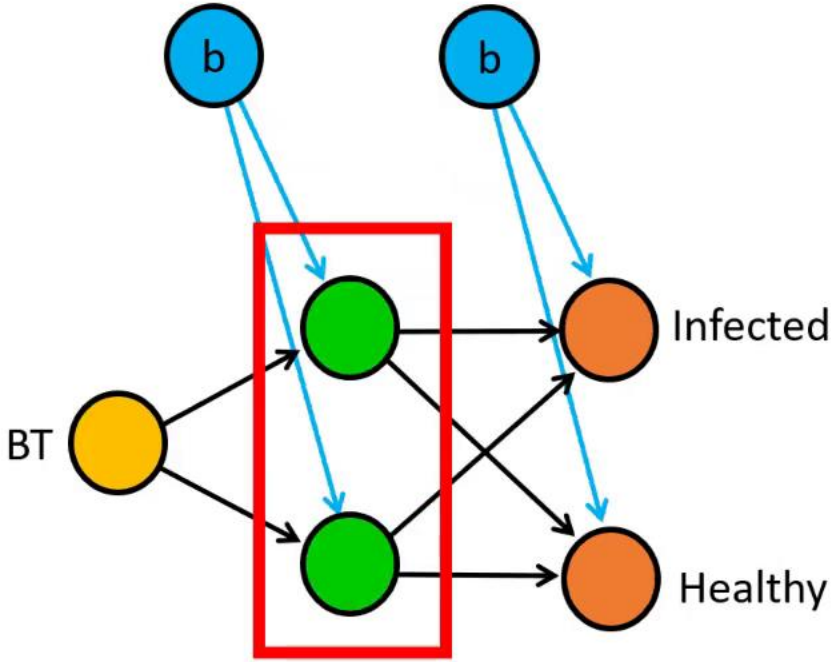


What we want is a curve that looks like this.

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

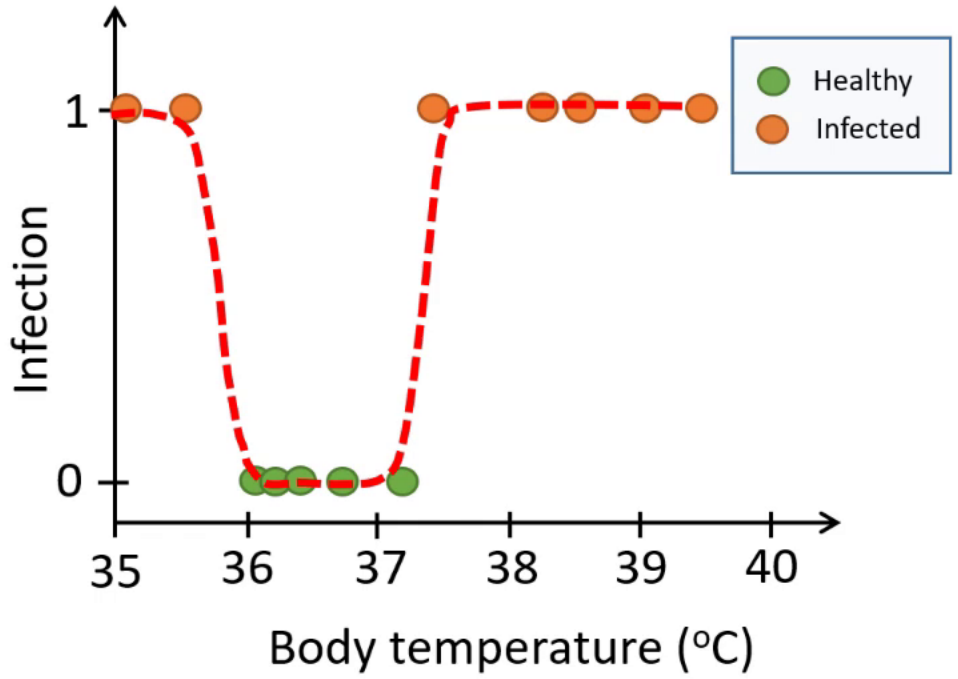


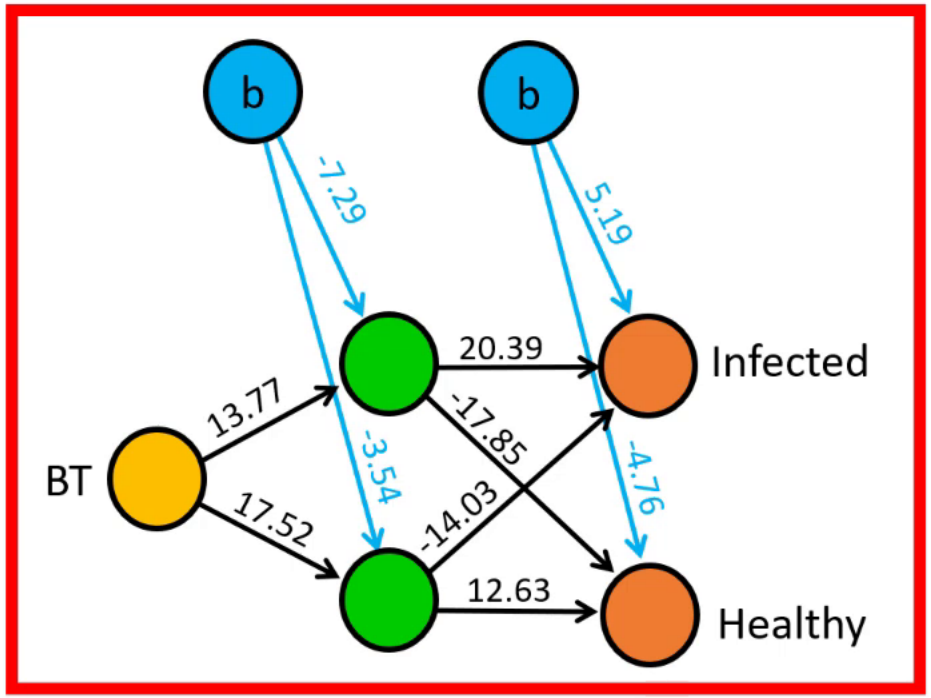


If we add a hidden layer with two hidden nodes,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

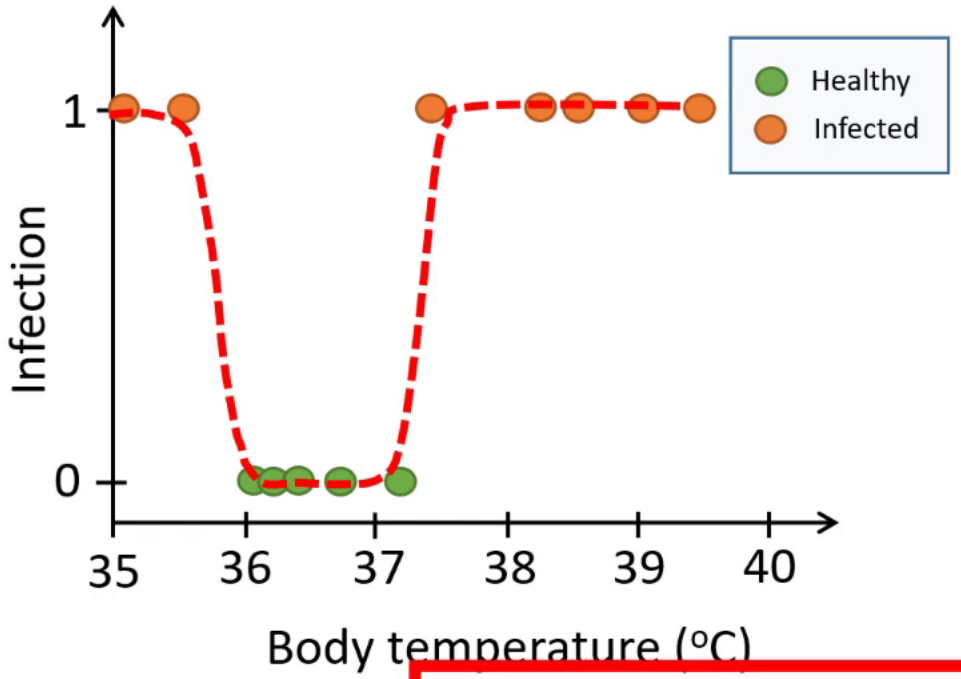


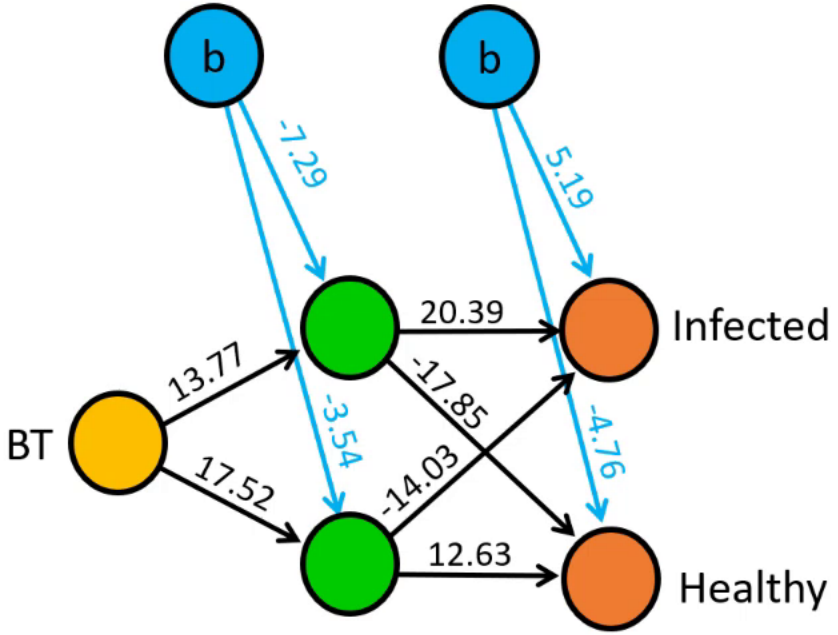


and train the network based on this data, we might obtain the following weights.

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



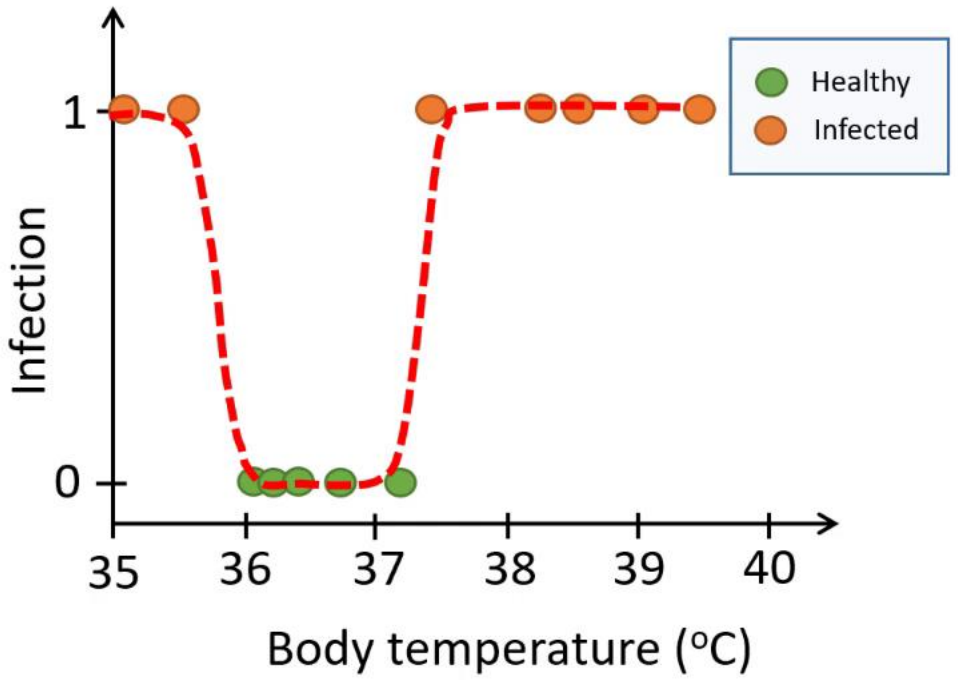


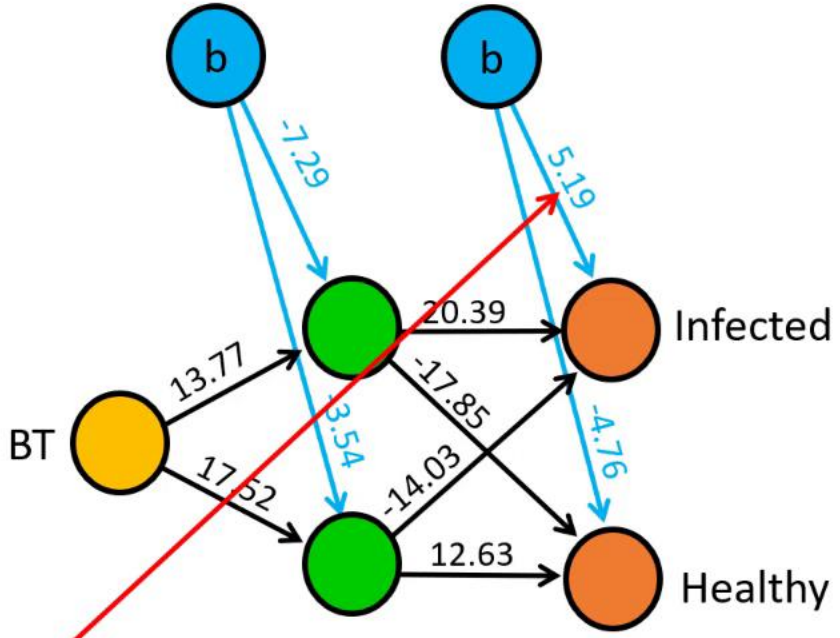
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19 + 20.39 \cdot \frac{1}{1+e^{-(-7.29+13.77 \cdot x_{norm})}} + (-14.03) \cdot \frac{1}{1+e^{-(-3.54+17.52 \cdot x_{norm})}})}}$$

The network now uses this complicated equation to compute the predicted probability of being infected,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



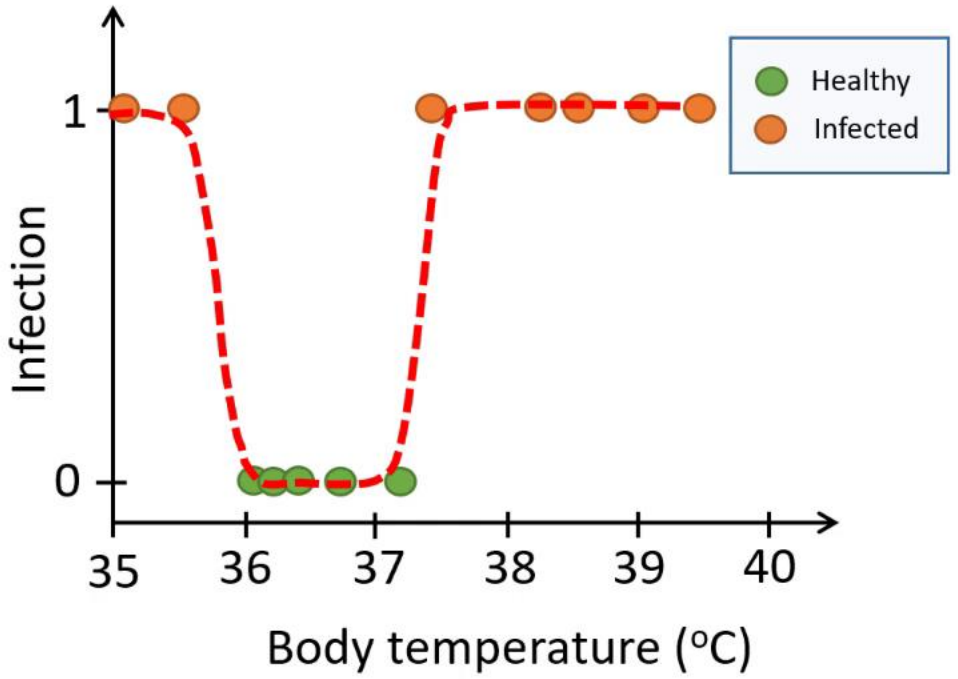


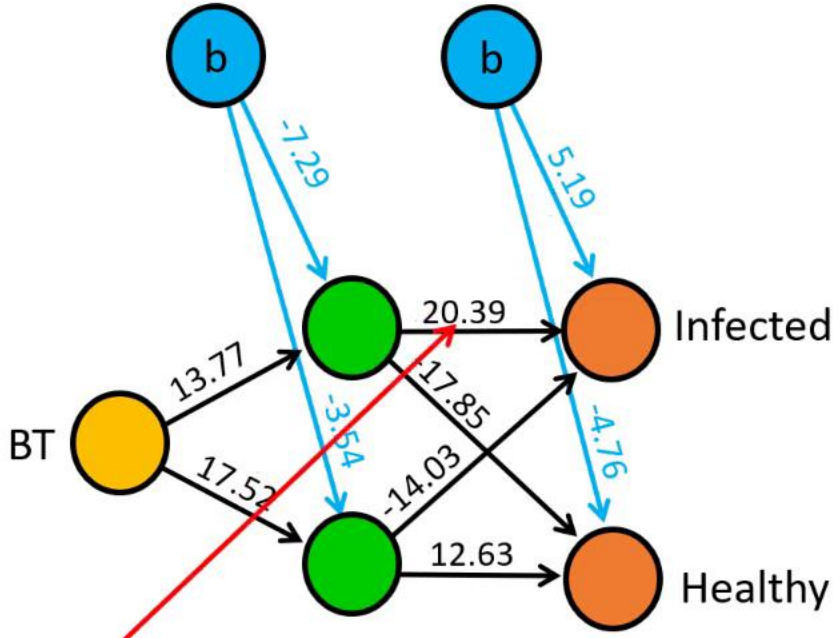
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19 + 20.39 \cdot \frac{1}{1+e^{-(-7.29+13.77 \cdot x_{norm})}} + (-14.03) \cdot \frac{1}{1+e^{-(-3.54+17.52 \cdot x_{norm})}})}}$$

where this value is the bias weight for this output node,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



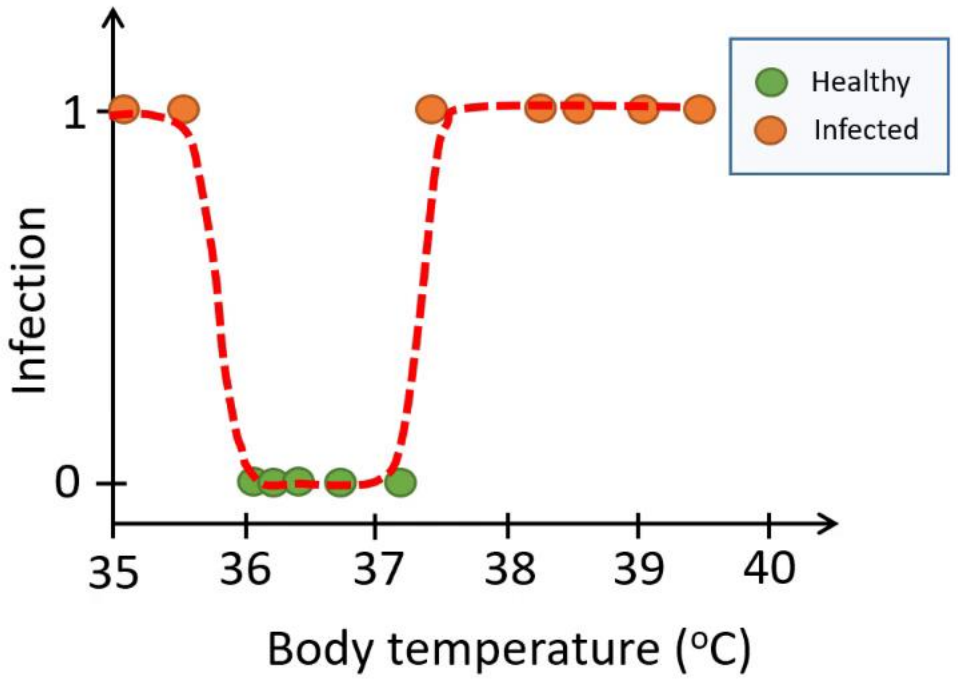


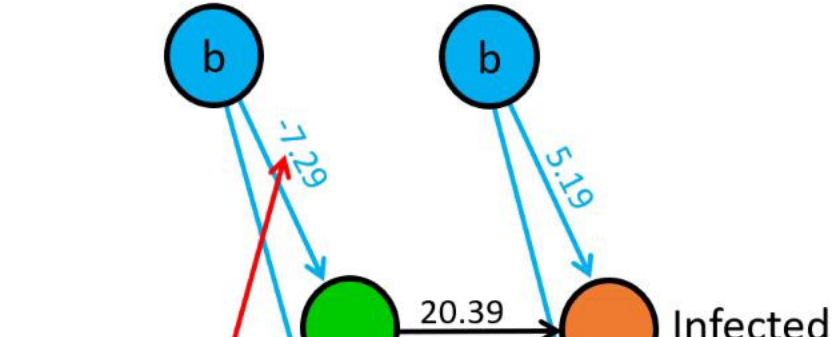
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19 + 20.39 \cdot \frac{1}{1+e^{-(-7.29+13.77 \cdot x_{norm})}} + (-14.03) \cdot \frac{1}{1+e^{-(-3.54+17.52 \cdot x_{norm})}})}}$$

and this is the weight,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



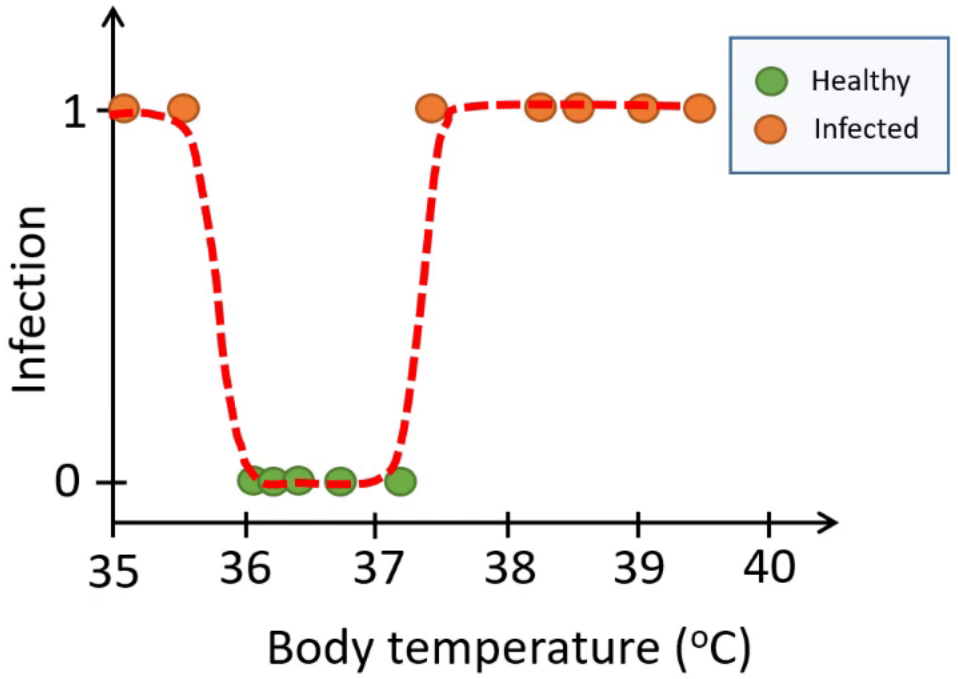


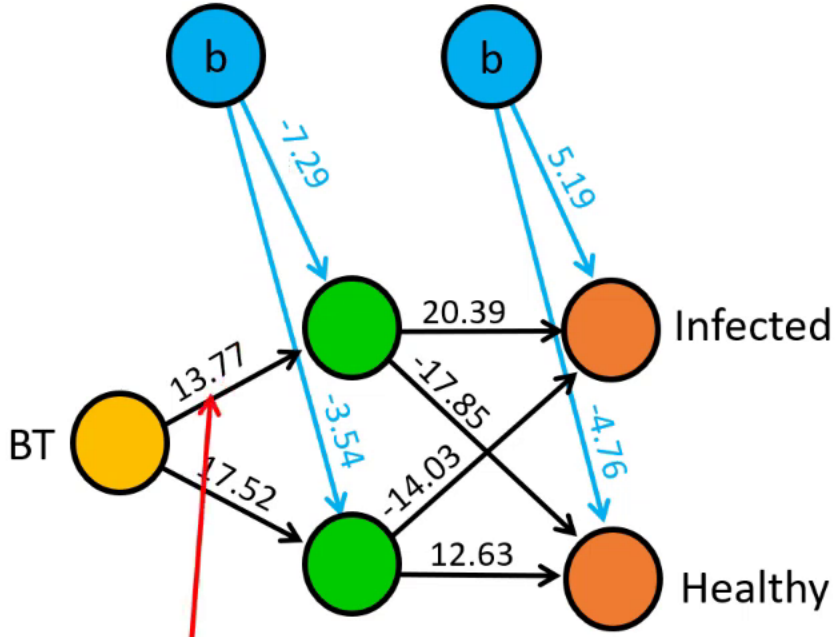
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19 + 20.39 \cdot \frac{1}{1+e^{-(-7.29+13.77 \cdot x_{norm})}} + (-14.03) \cdot \frac{1}{1+e^{-(-3.54+17.52 \cdot x_{norm})}})}}$$

The value inside this hidden node is determined by this bias weight,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



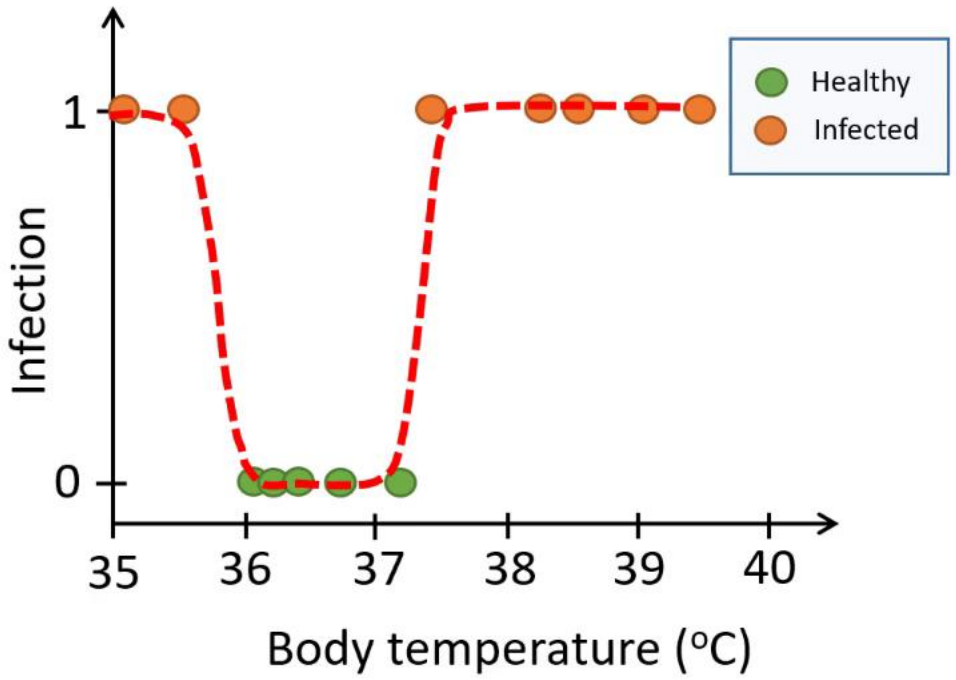


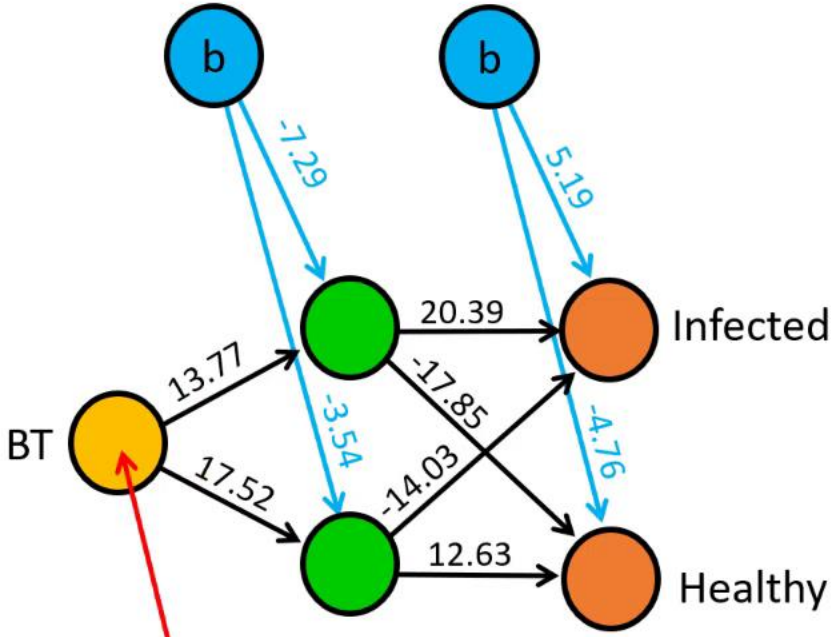
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19+20.39\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}}+(-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}})}}$$

and the weight,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



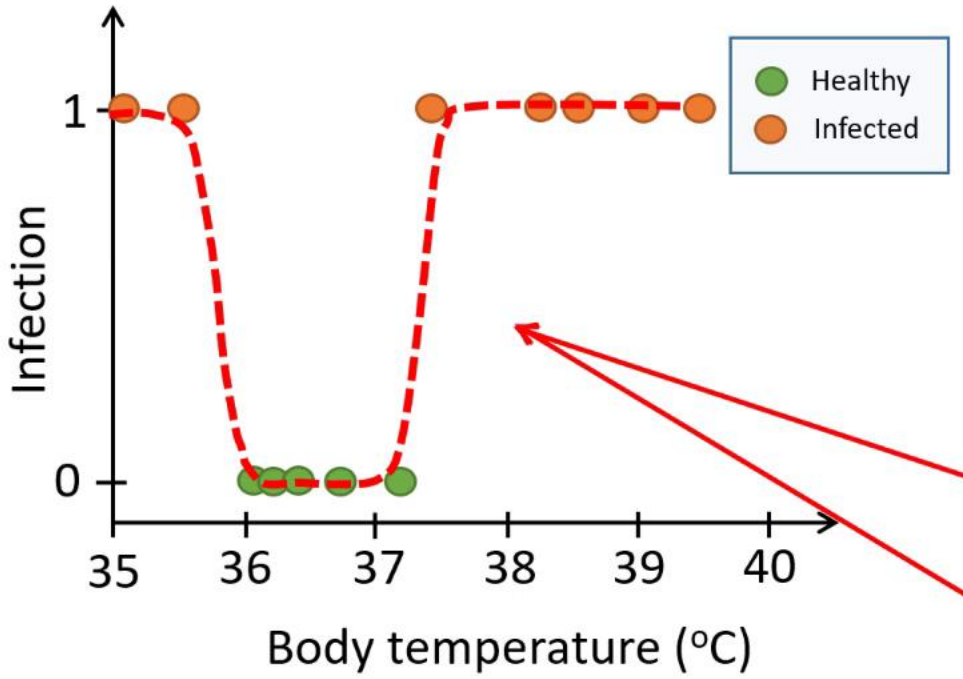


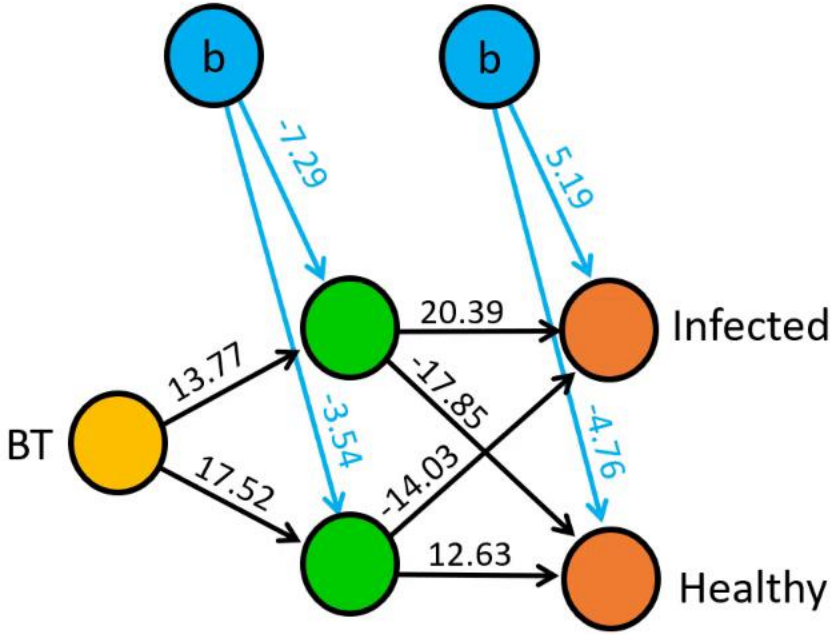
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19+20.39\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}}+(-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}})}}$$

which is multiplied by the normalized body temperature in the input node.

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



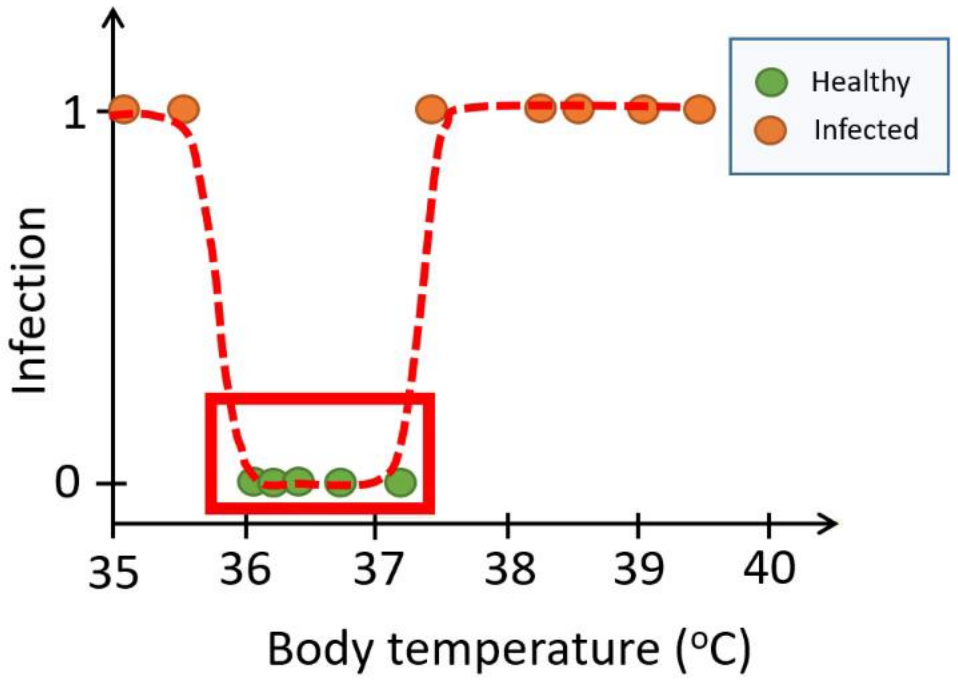


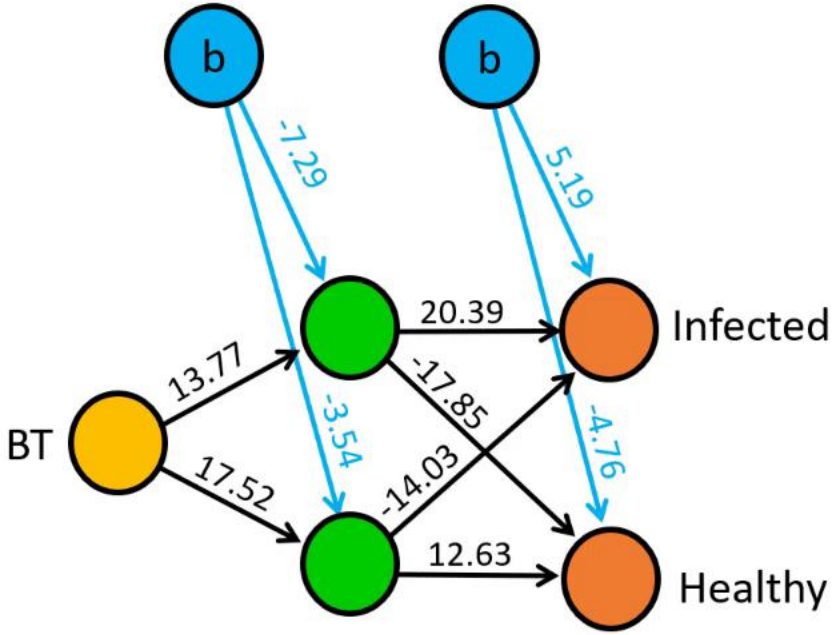
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19 + 20.39 \cdot \frac{1}{1+e^{-(-7.29+13.77 \cdot x_{norm})}} + (-14.03) \cdot \frac{1}{1+e^{-(-3.54+17.52 \cdot x_{norm})}})}}$$

If we change the value of xnorm between zero and one, we can generate the following curve that fits the data.

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



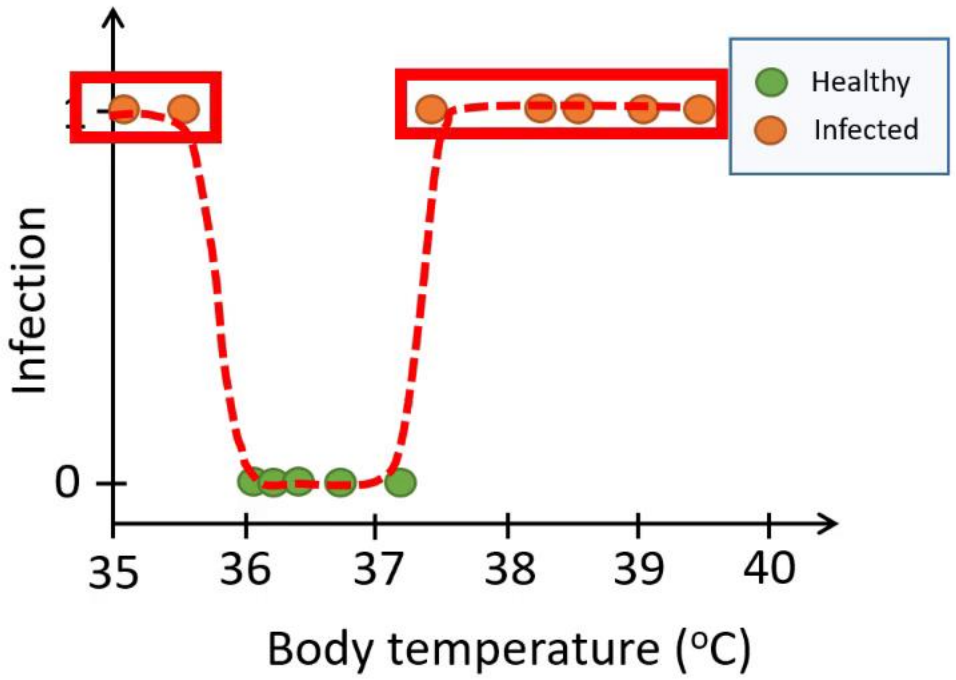


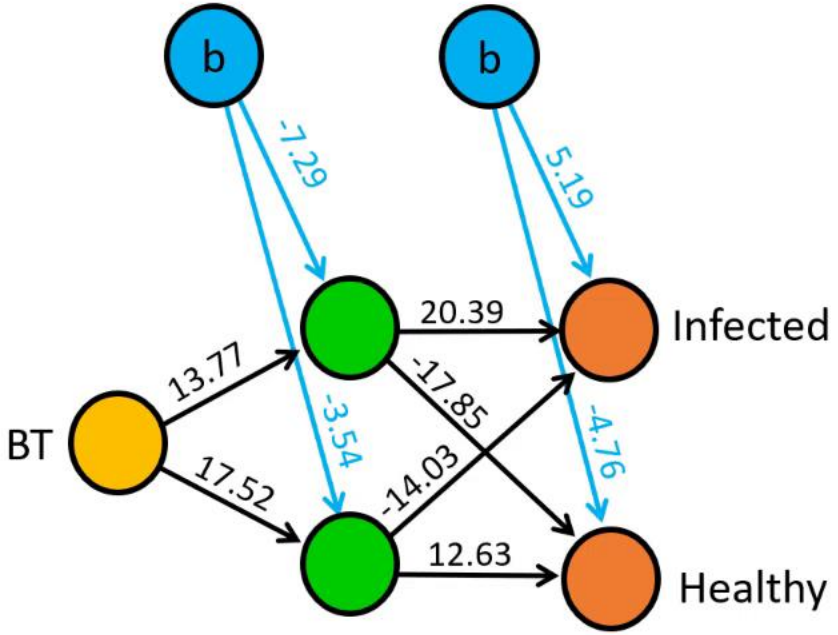
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19 + 20.39 \cdot \frac{1}{1+e^{-(-7.29+13.77 \cdot x_{norm})}} + (-14.03) \cdot \frac{1}{1+e^{-(-3.54+17.52 \cdot x_{norm})}})}}$$

We have therefore created a model that correctly predicts the individuals as being healthy if they have a body temperature between 36 and 37,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2



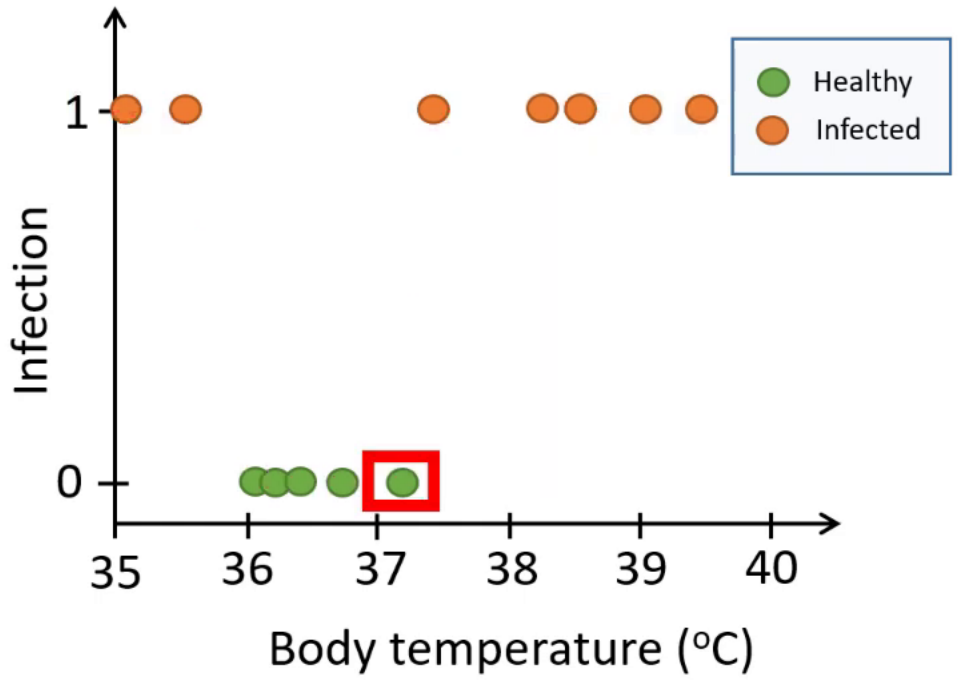


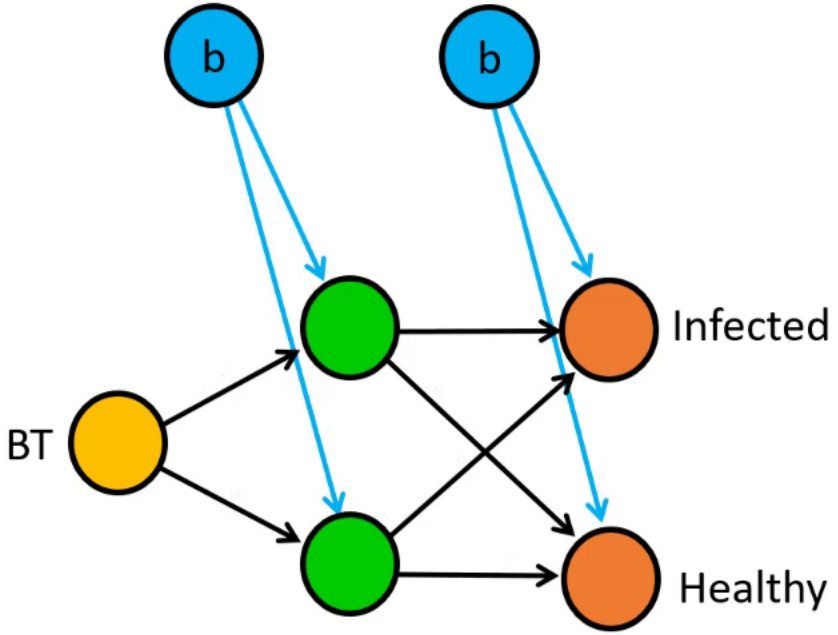
$$f(x_{norm}) = \frac{1}{1 + e^{-(5.19+20.39\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}}+(-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}})}}$$

and correctly predicts people that are outside this range as being infected.

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	37.2

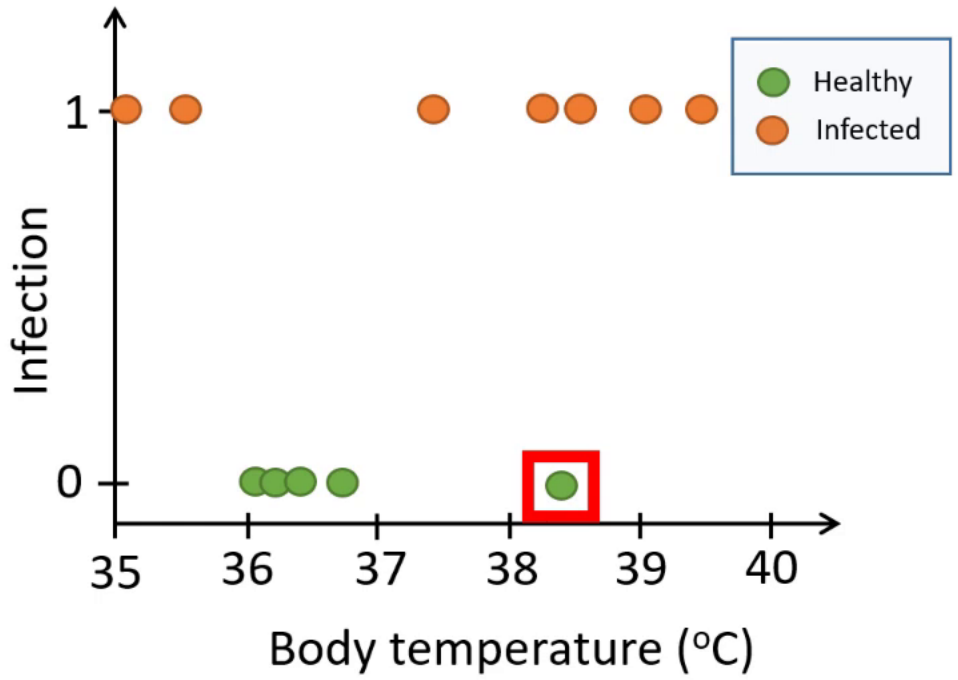


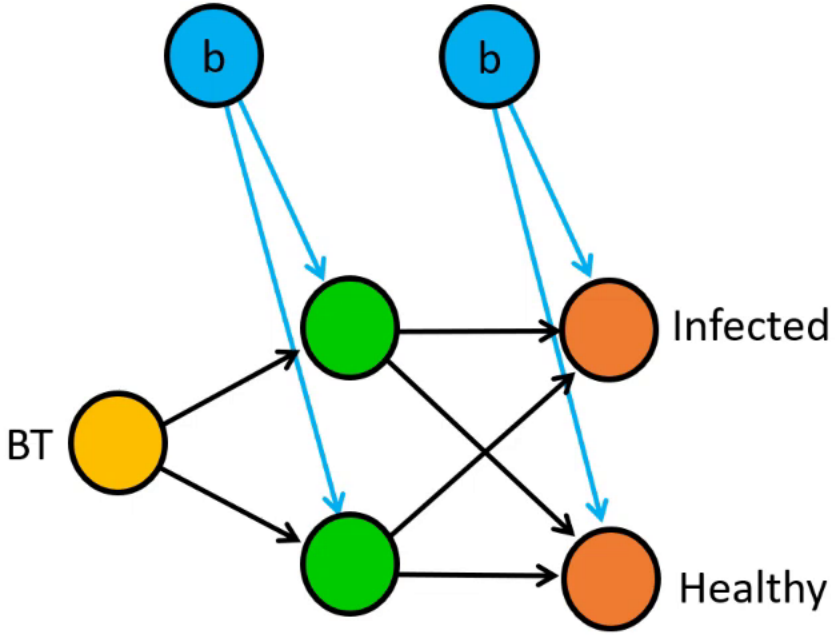


Now, suppose that we would move this data point to the right,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

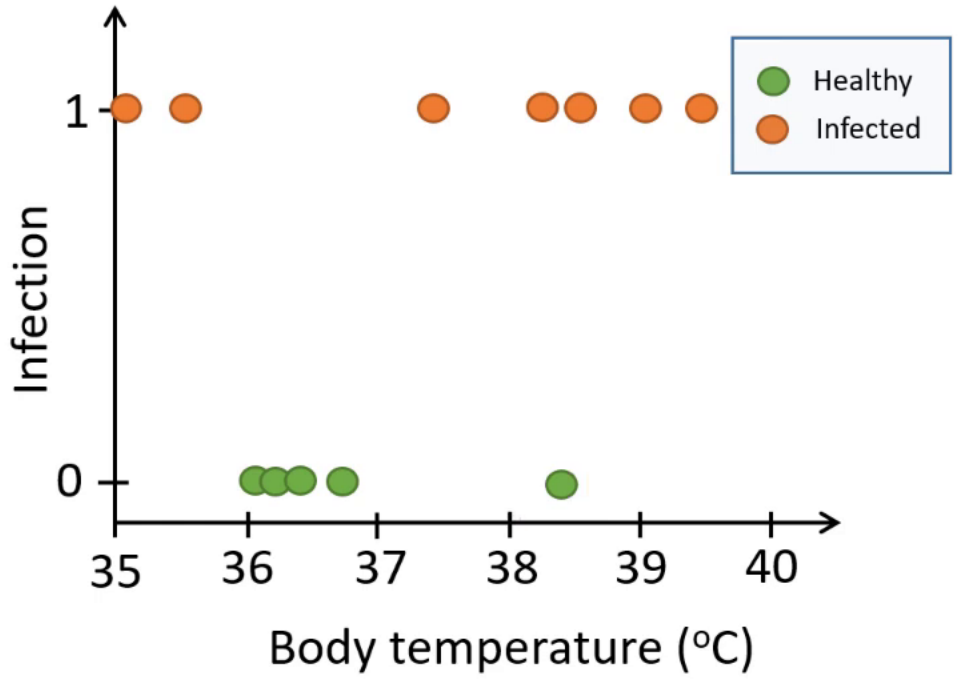


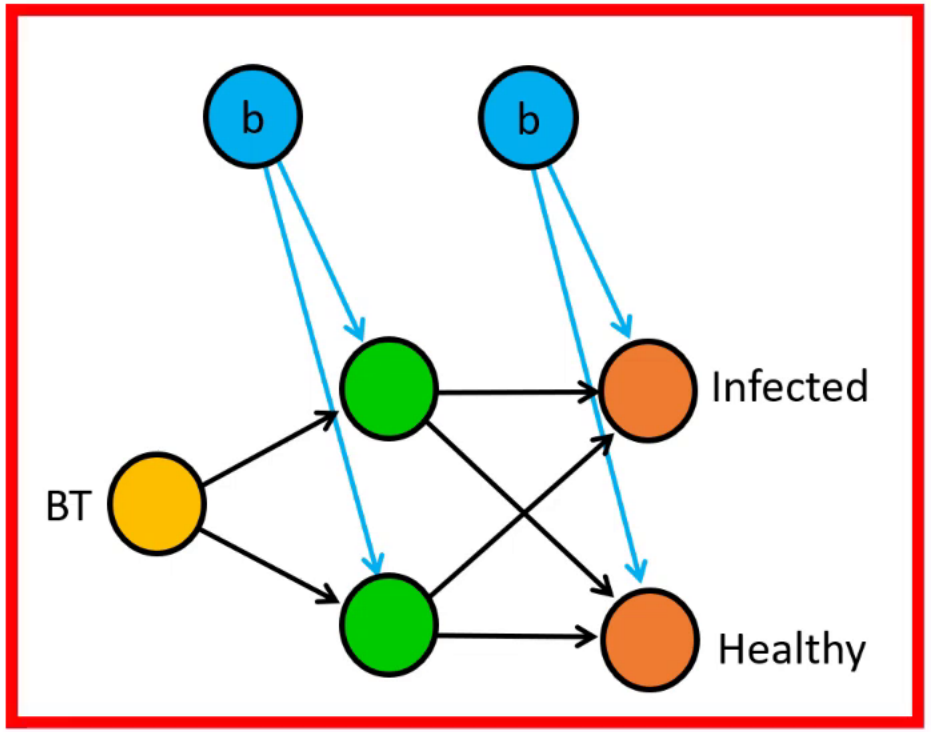


like this.

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

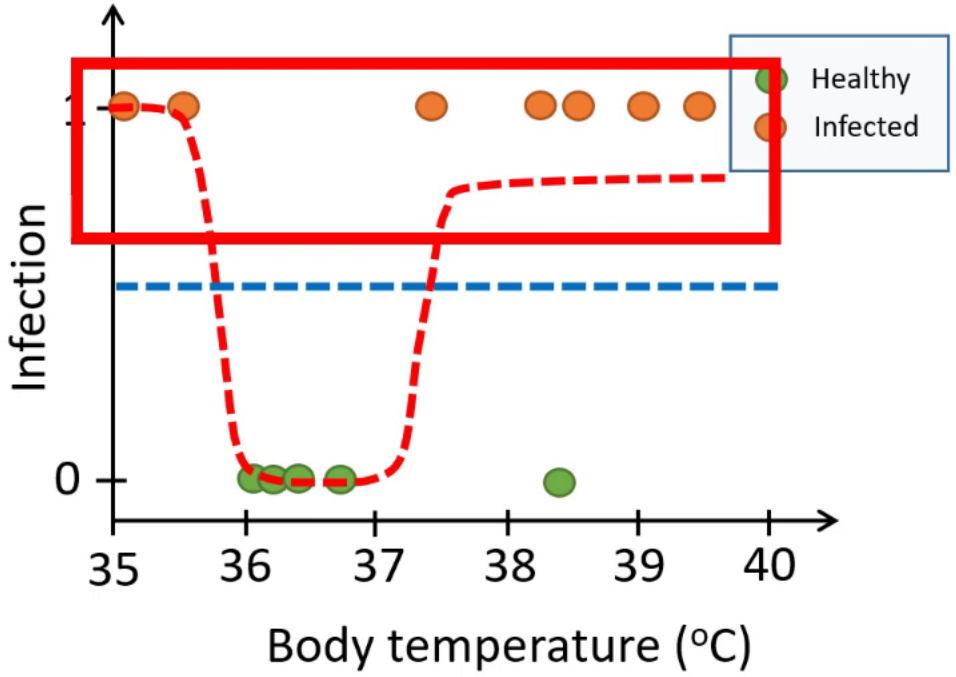


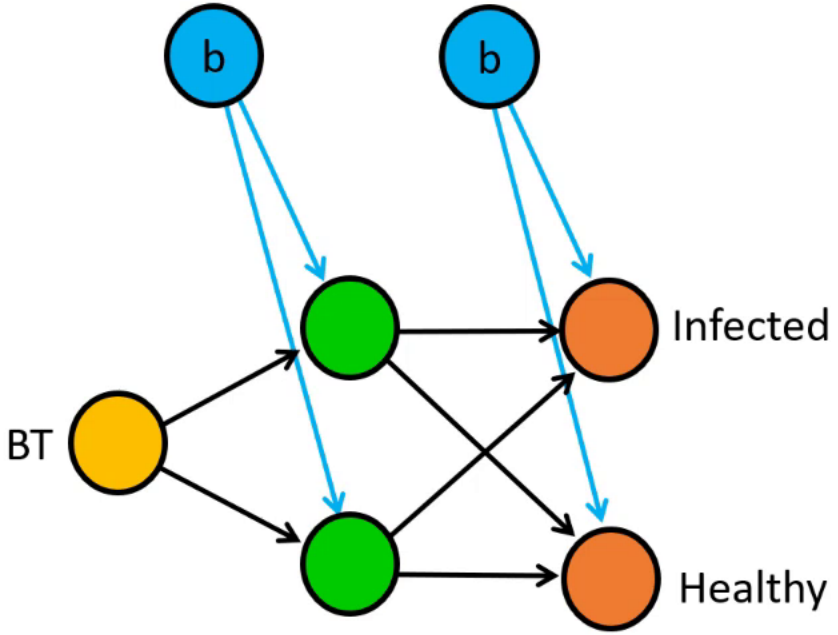


If we train this network on the new data,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

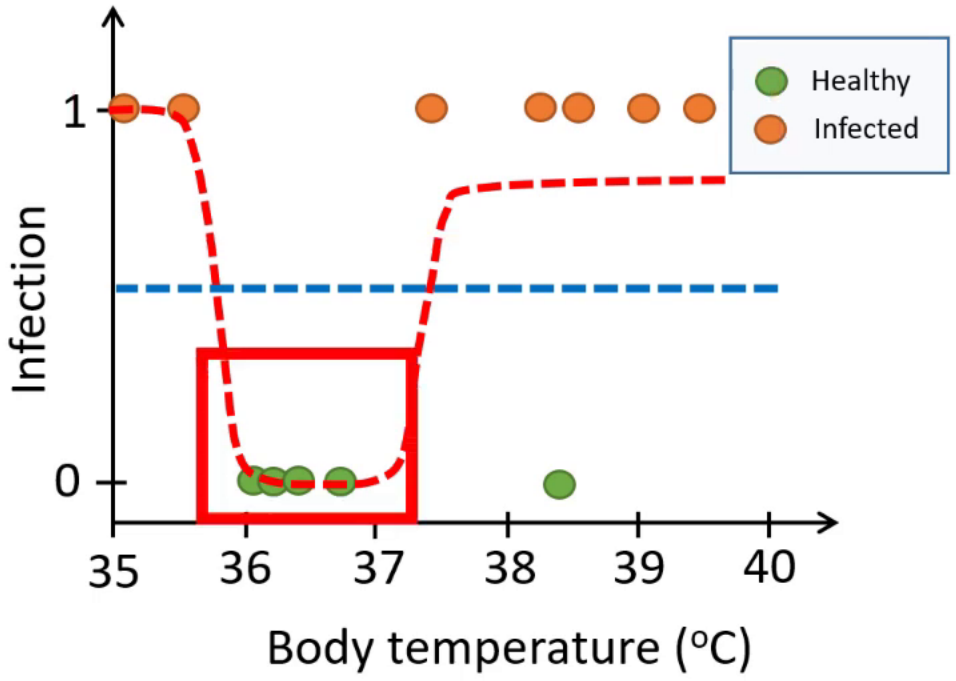


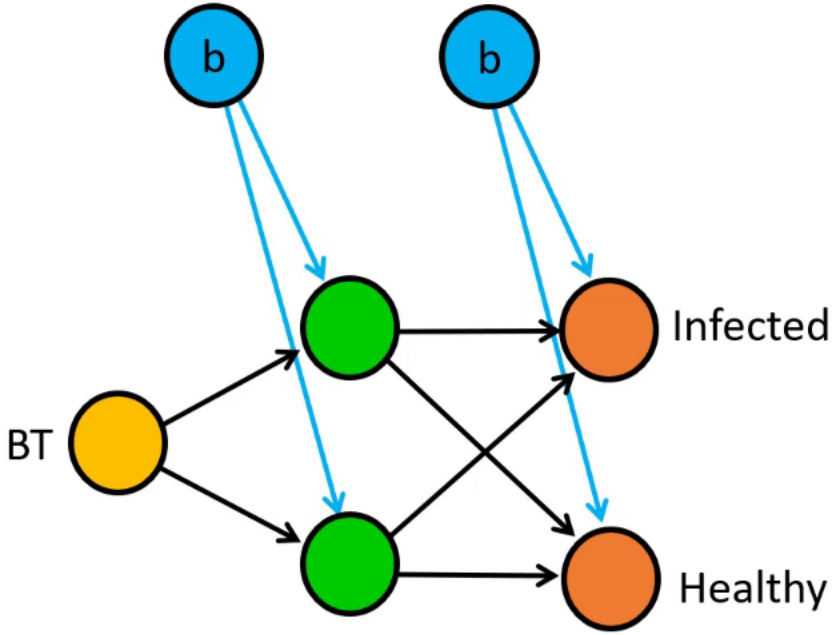


we see that it correctly predicts all infected individuals as being infected,

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

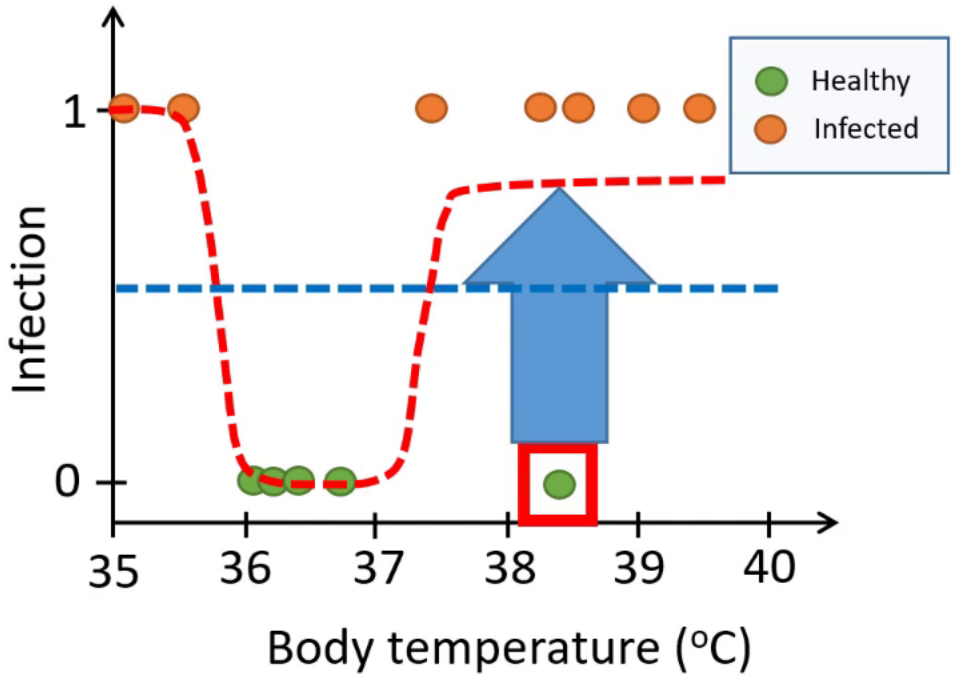


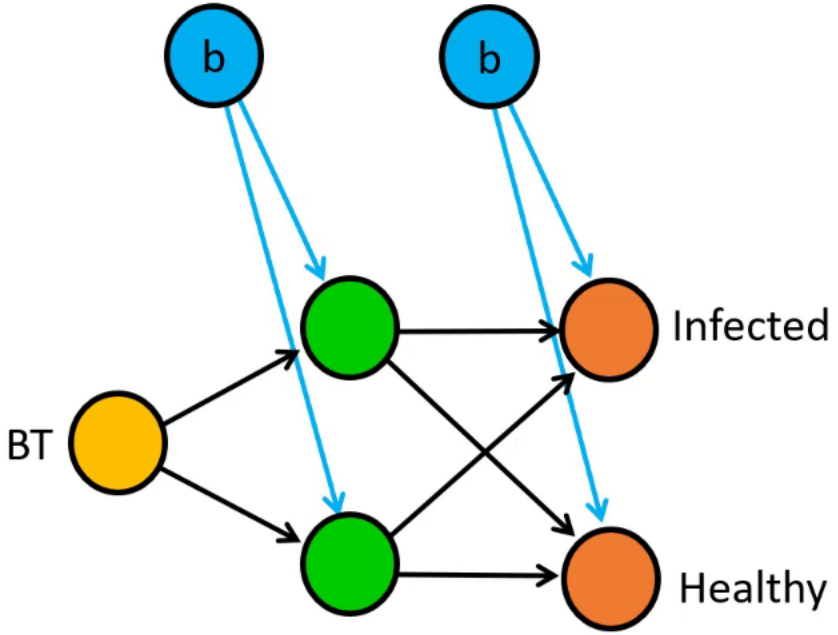


and almost all healthy individuals as being healthy.

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

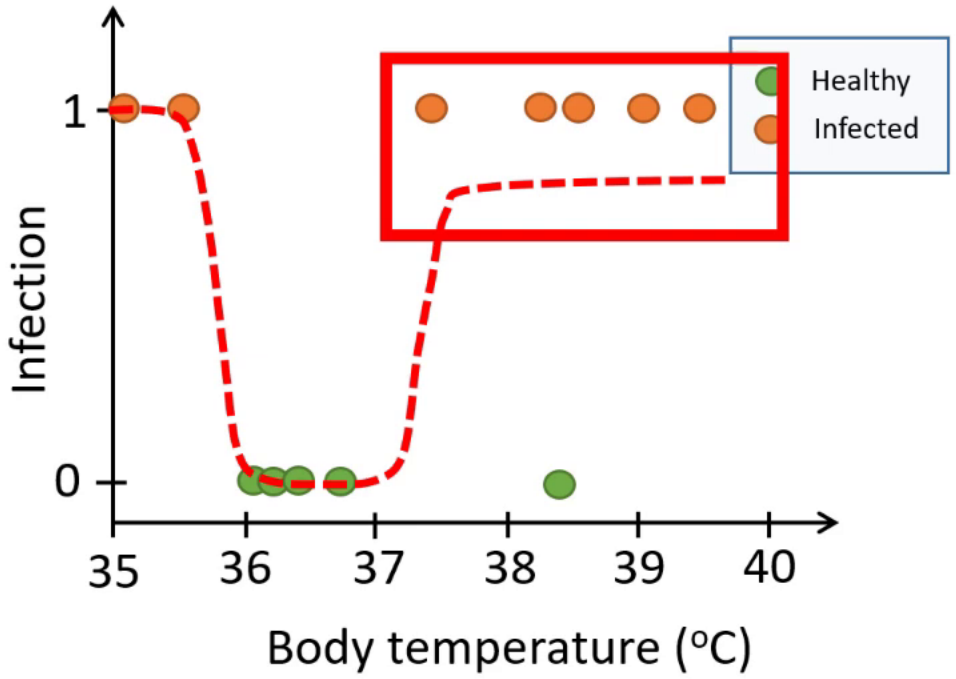


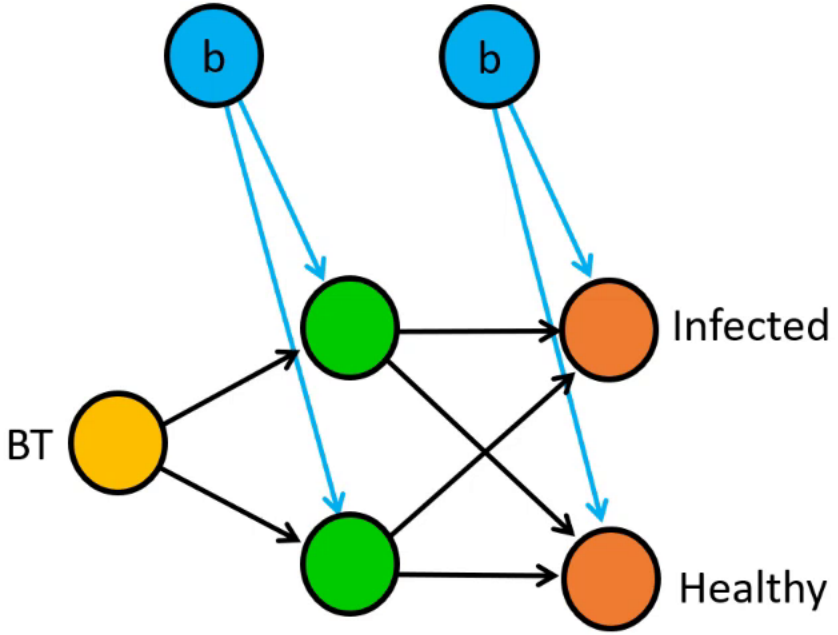


This healthy individual will be incorrectly predicted as infected. However, this data point seems to be due to a measurement error because no healthy person should have a body temperature of 38.4.

A network with one hidden layer

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

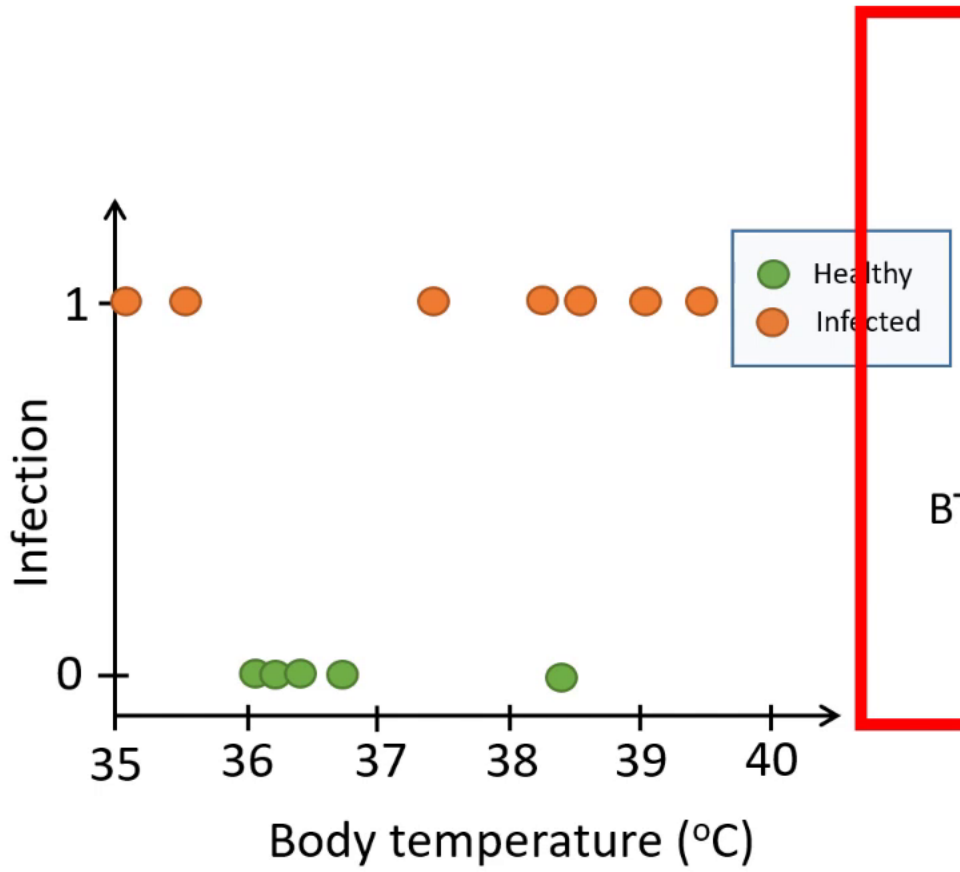


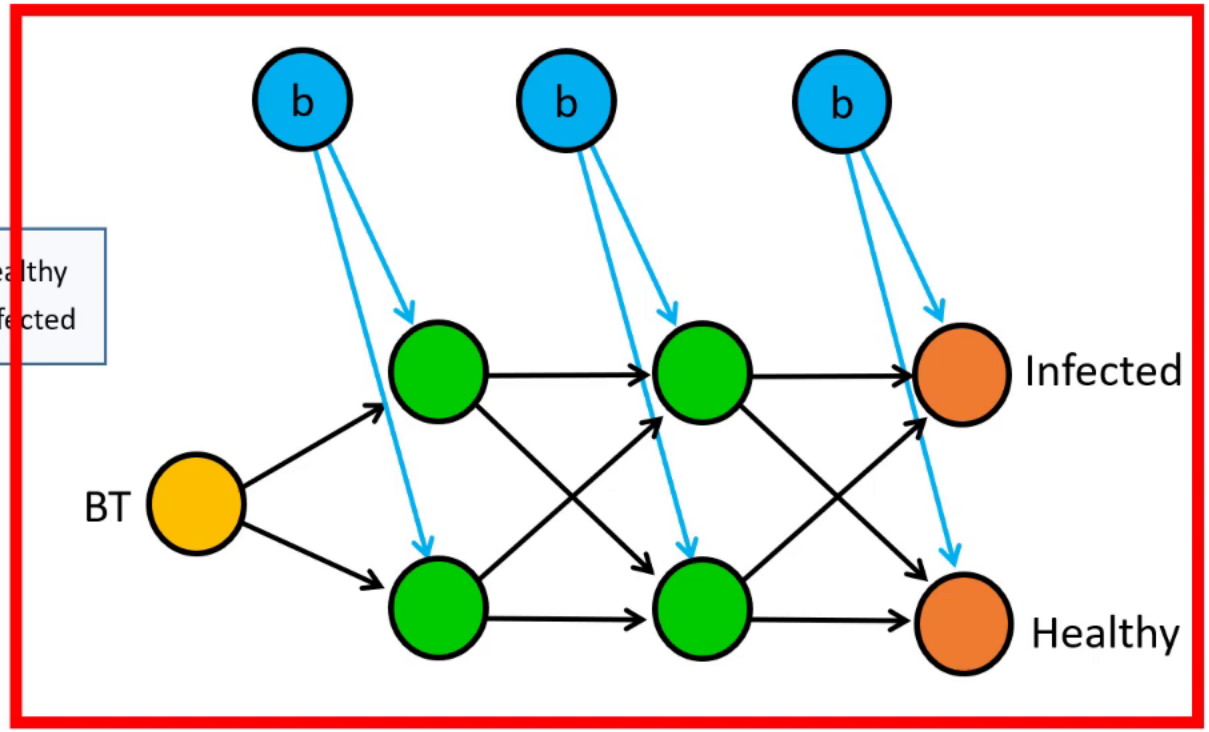


The network therefore makes a great job of ignoring this data point and predicting it as being infected.

A deep neural network

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

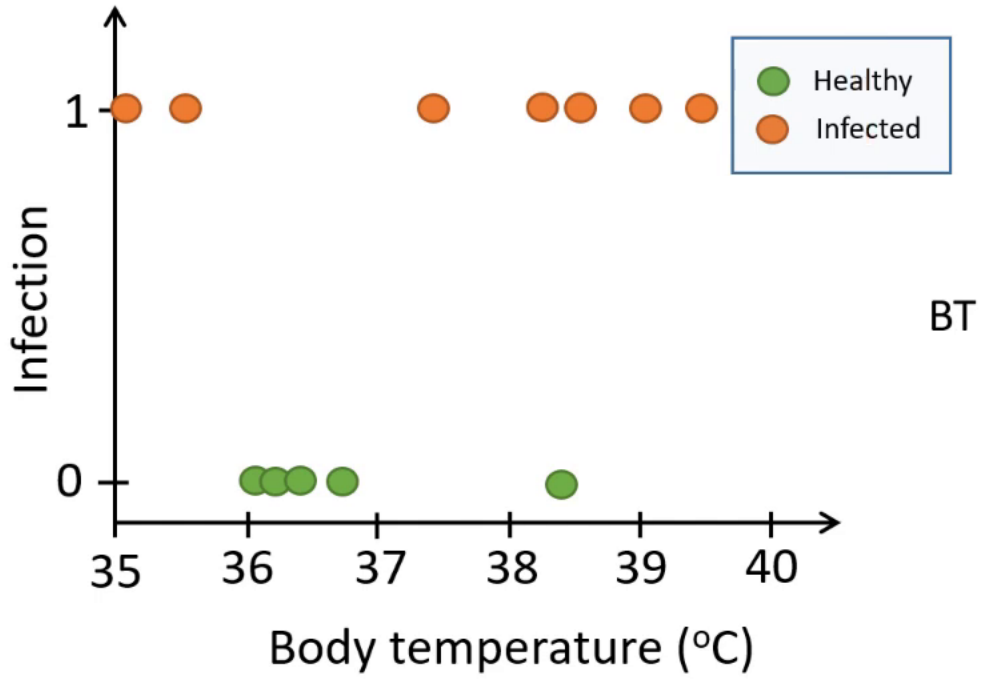


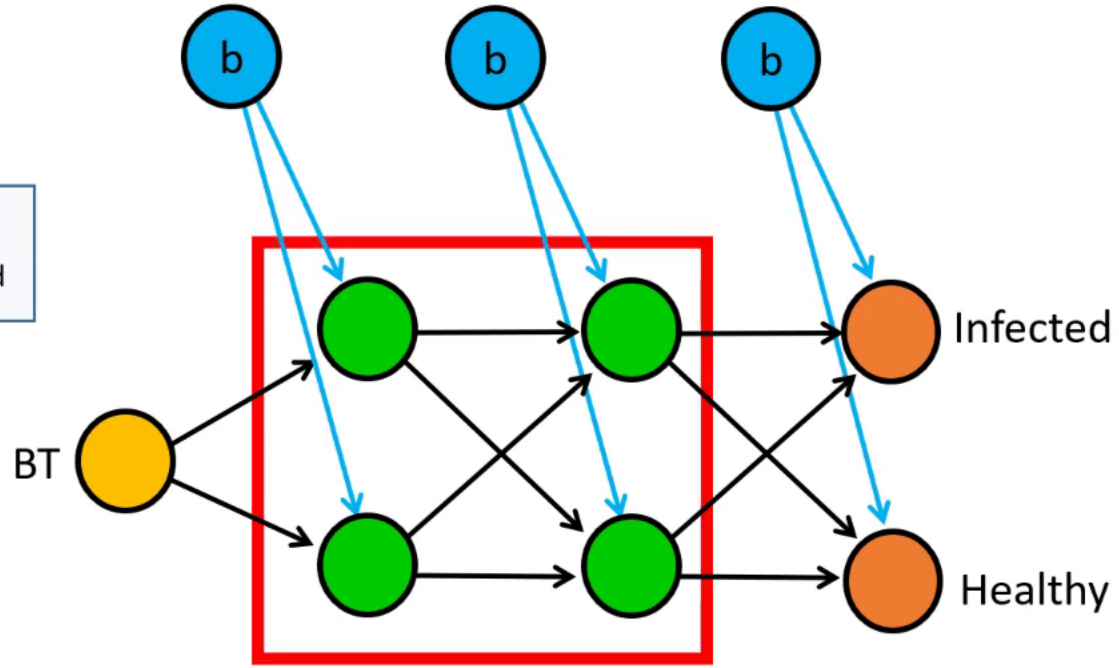


But what would happen if we use a deep neural network,

A deep neural network

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

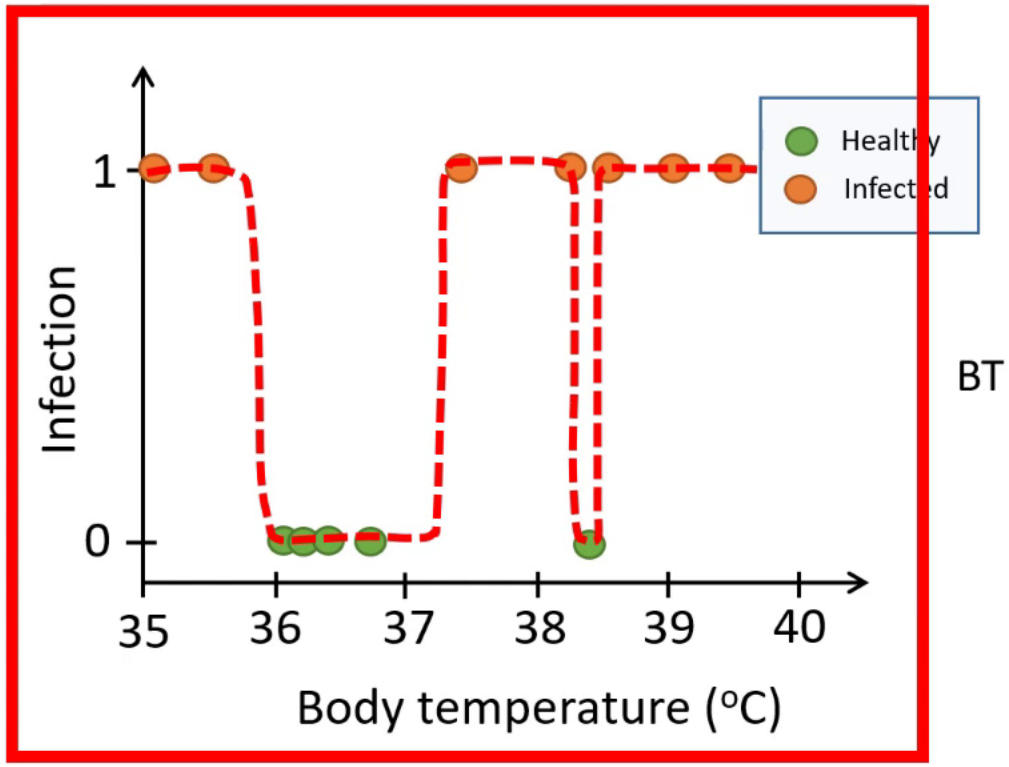


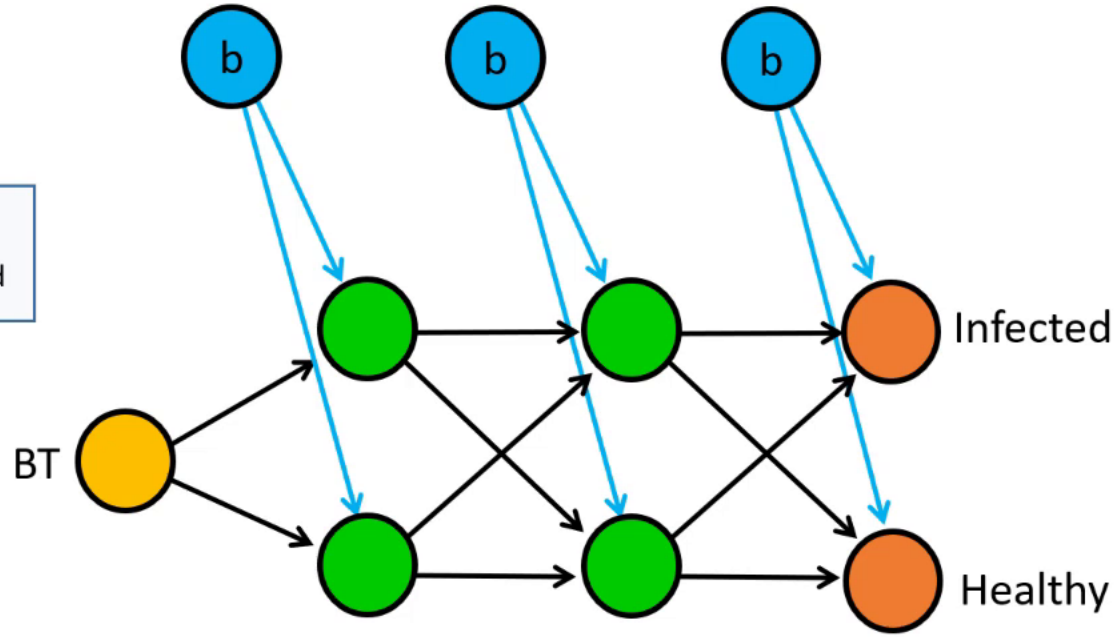


with two hidden layers?

A deep neural network

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

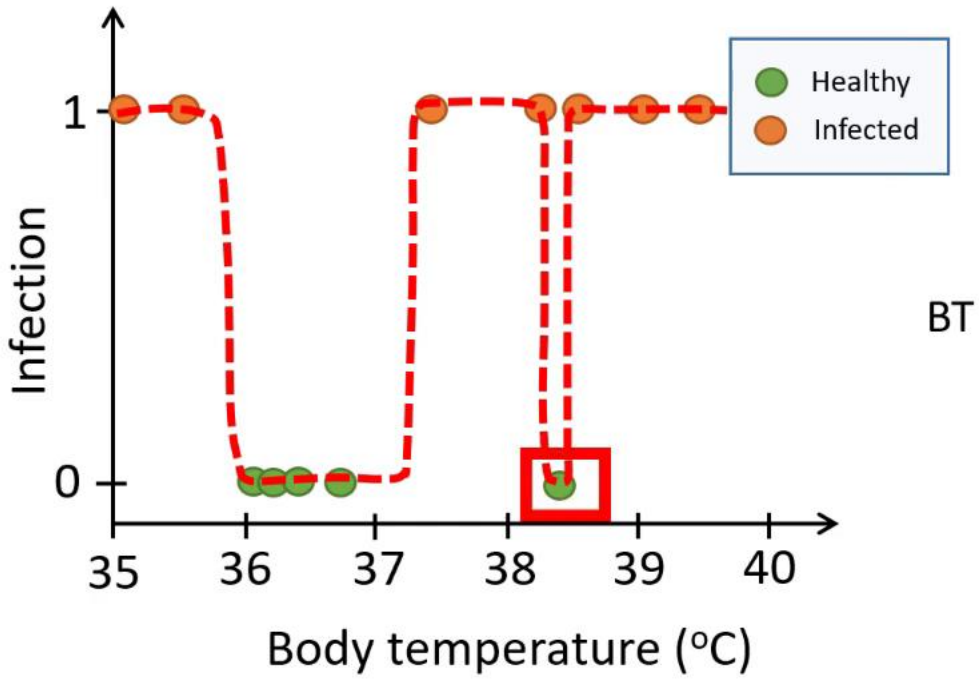


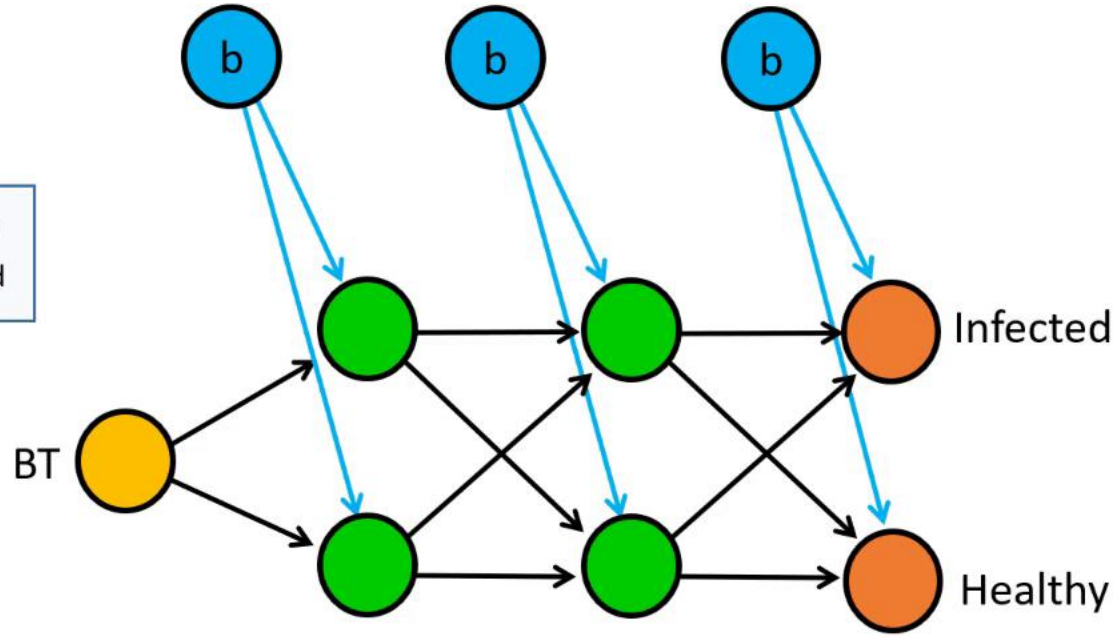


Well, such a deep network may actually generate the following strange curve,

A deep neural network

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

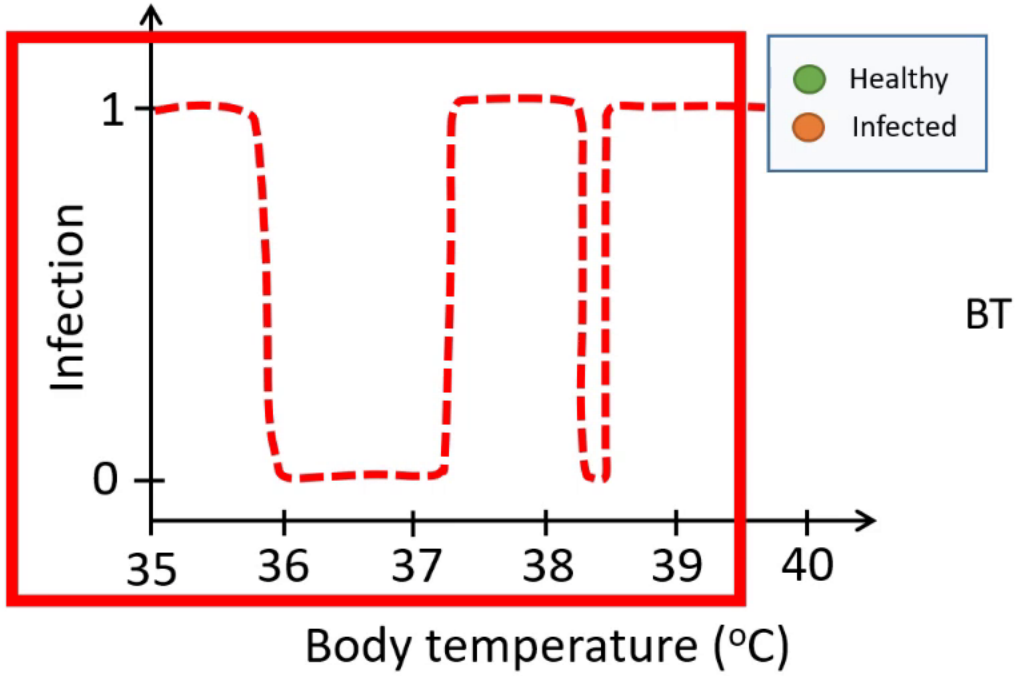


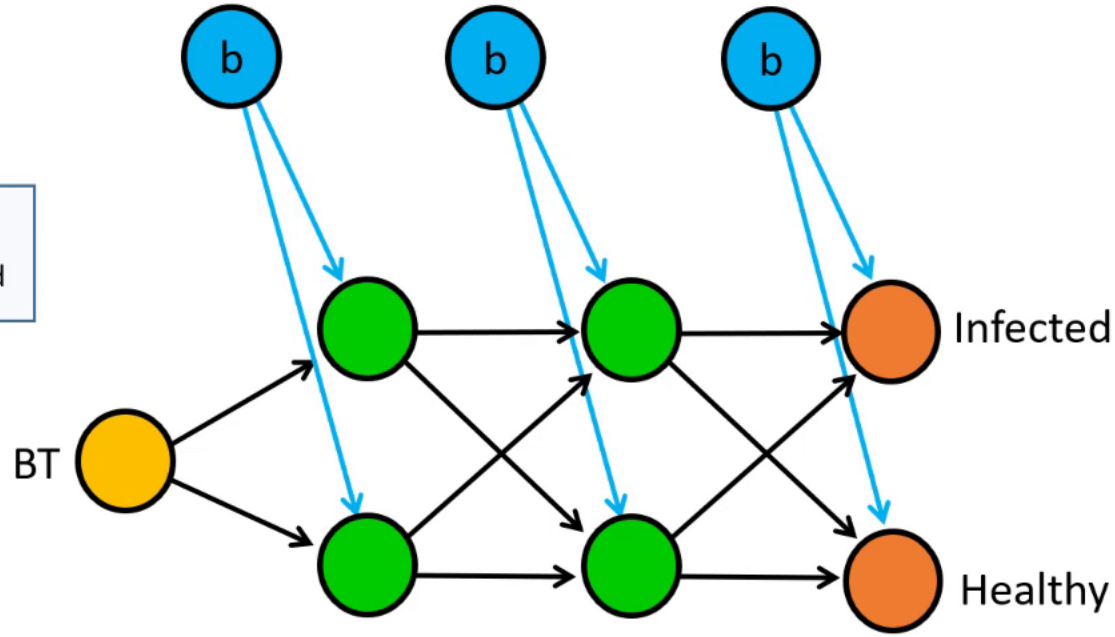


where this healthy individual is now correctly predicted as being healthy. This is the cool thing with deep neural networks, because they can learn almost any pattern in the data.

A deep neural network

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

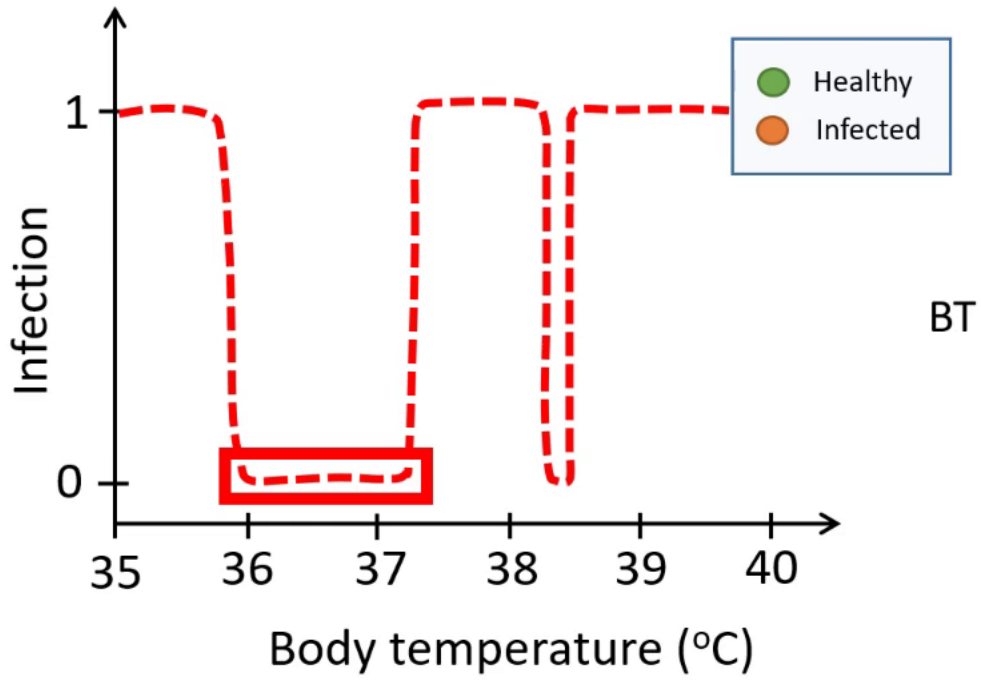


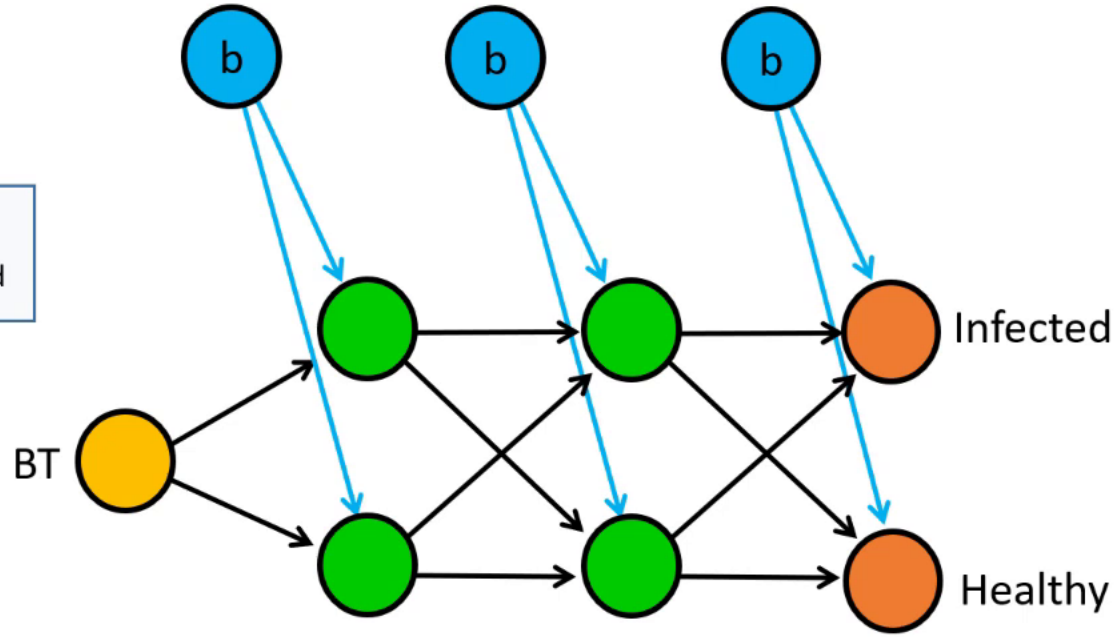


However, is this network really useful? Suppose that we would use the following network to predict if someone is infected or not.

A deep neural network

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

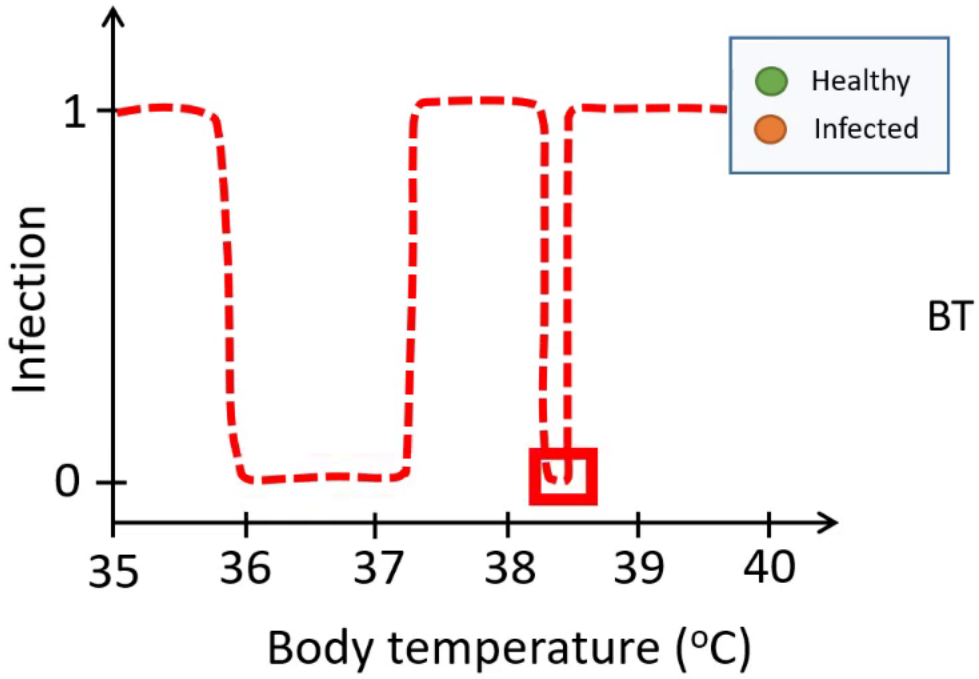


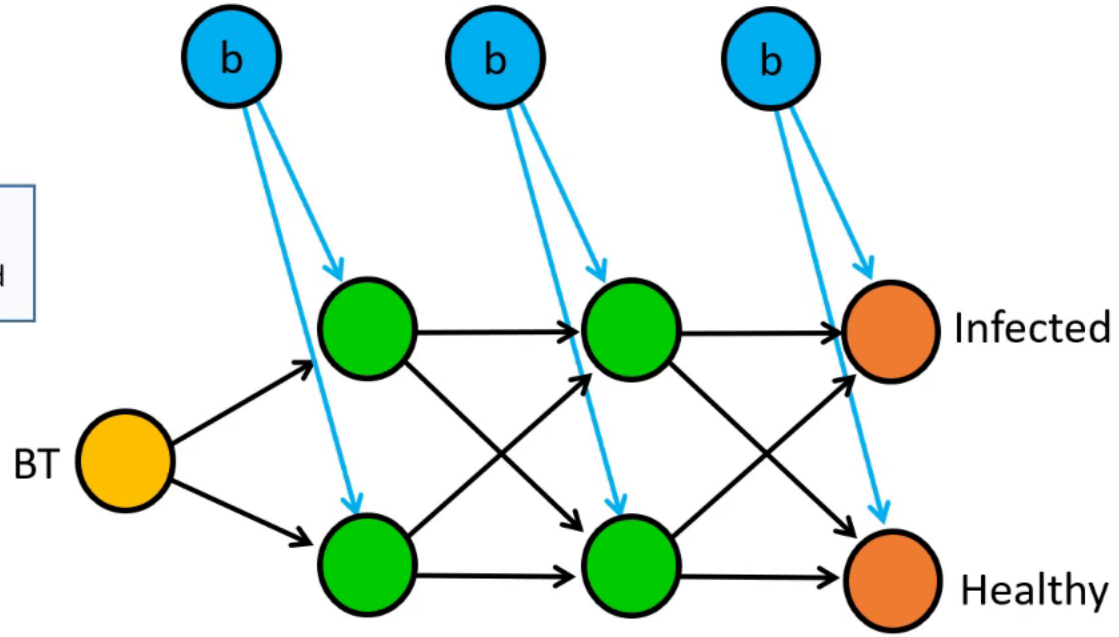


It makes sense that people with a body temperature between 36 and 37 are predicted to be healthy,

A deep neural network

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

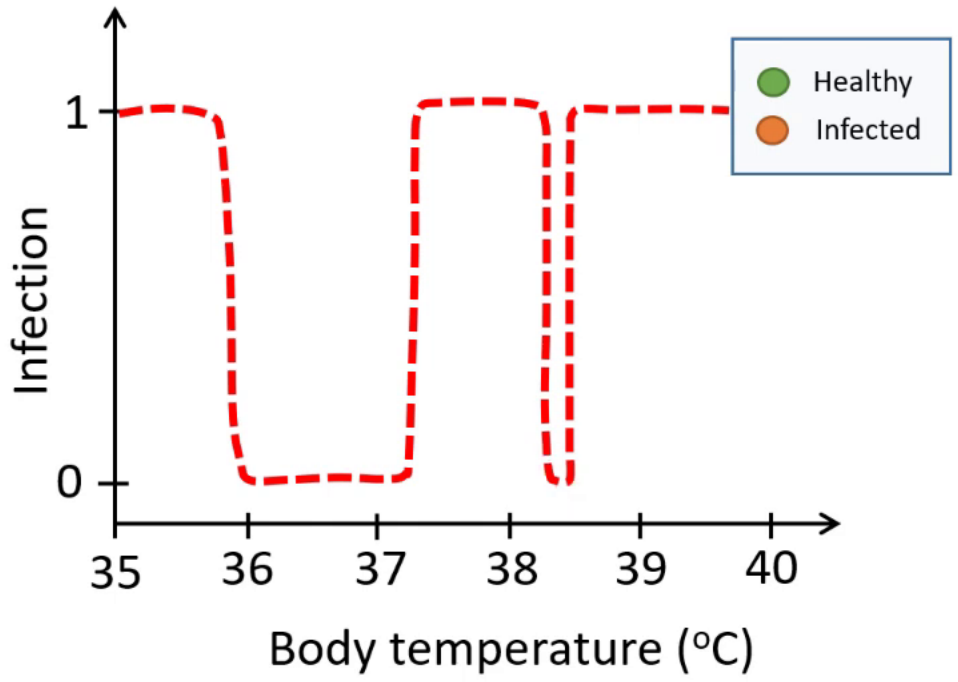




but it makes no sense that a person with a body temperature of exactly 38.4 is predicted to be healthy.

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

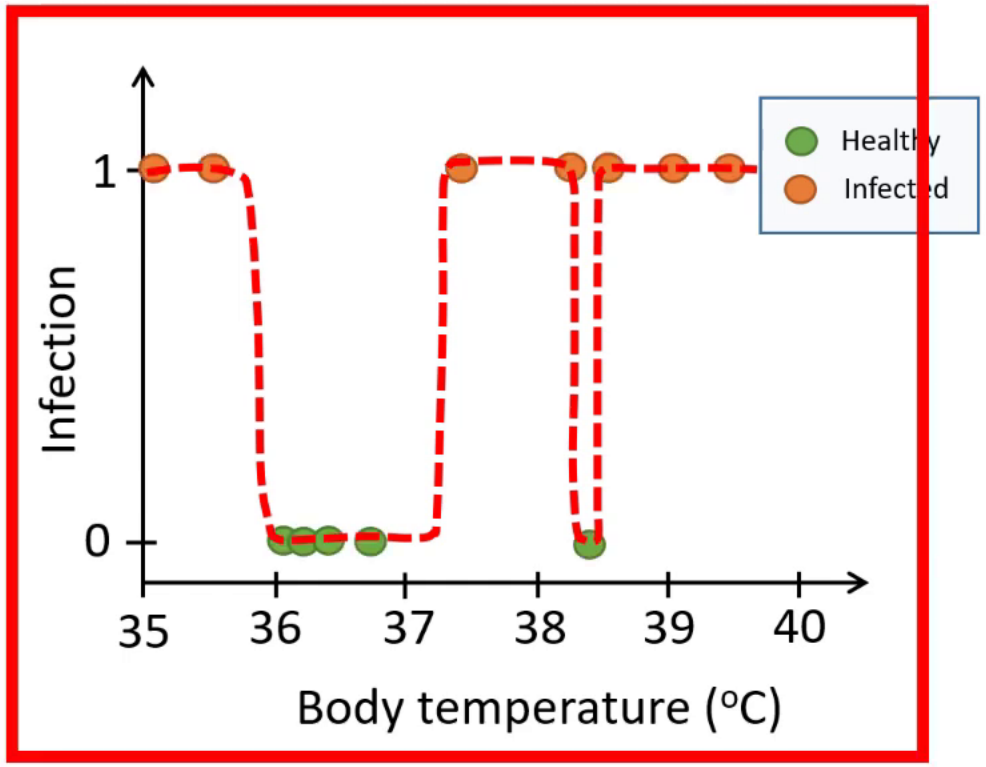


Overfitting

By using a too deep network with many hidden nodes usually results in what we call overfitting,

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

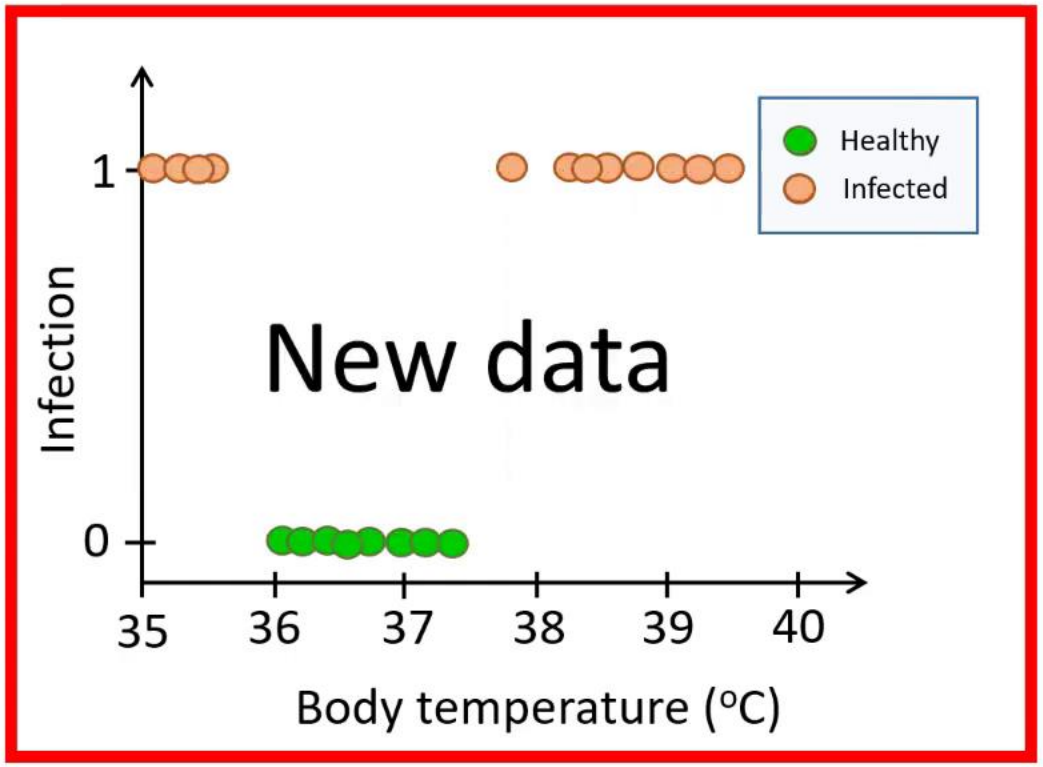


Overfitting

which occurs when the network fits very well with the training data,

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

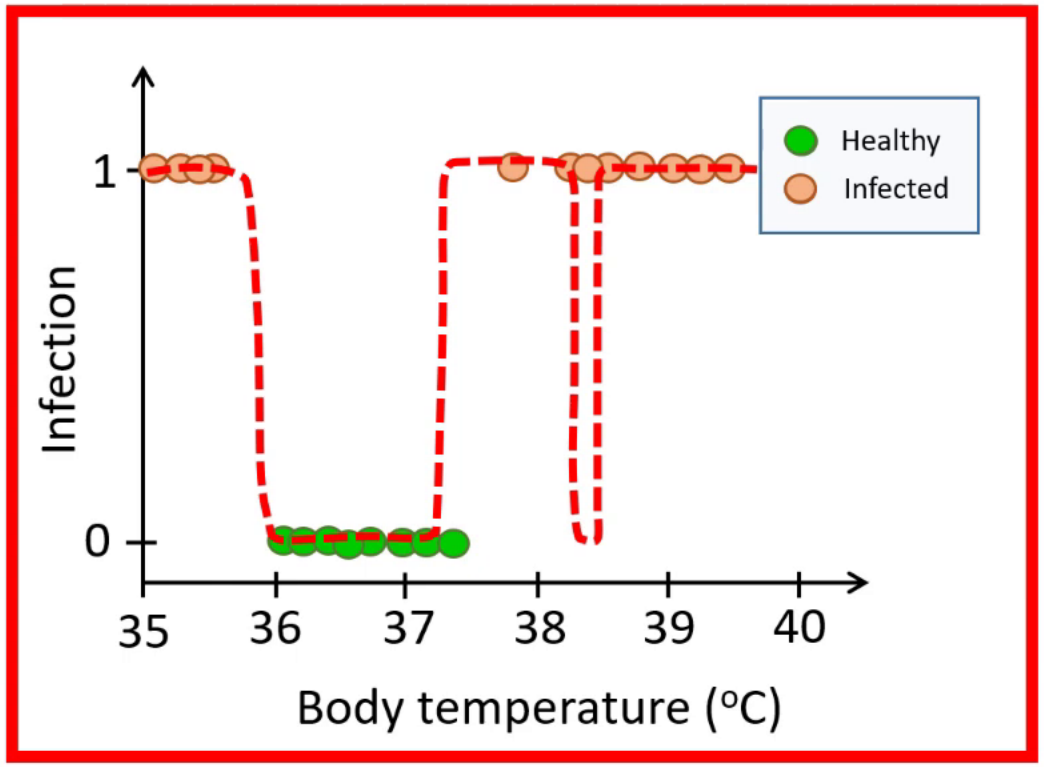


Overfitting

but does not fit well to a new data set.

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

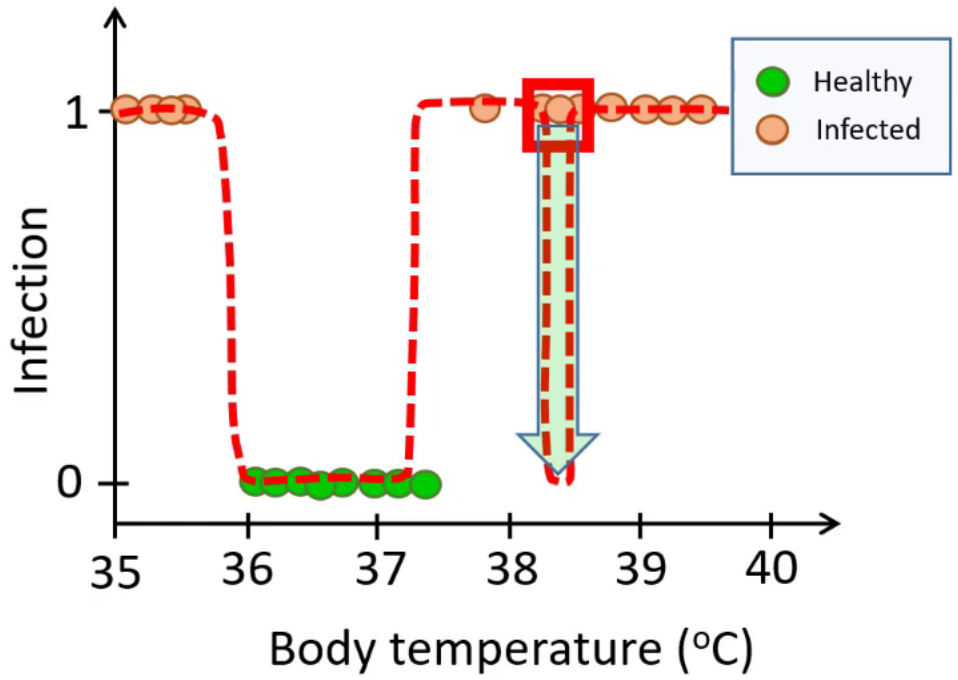


Overfitting

The reason why this network does not fit well with the new data set,

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

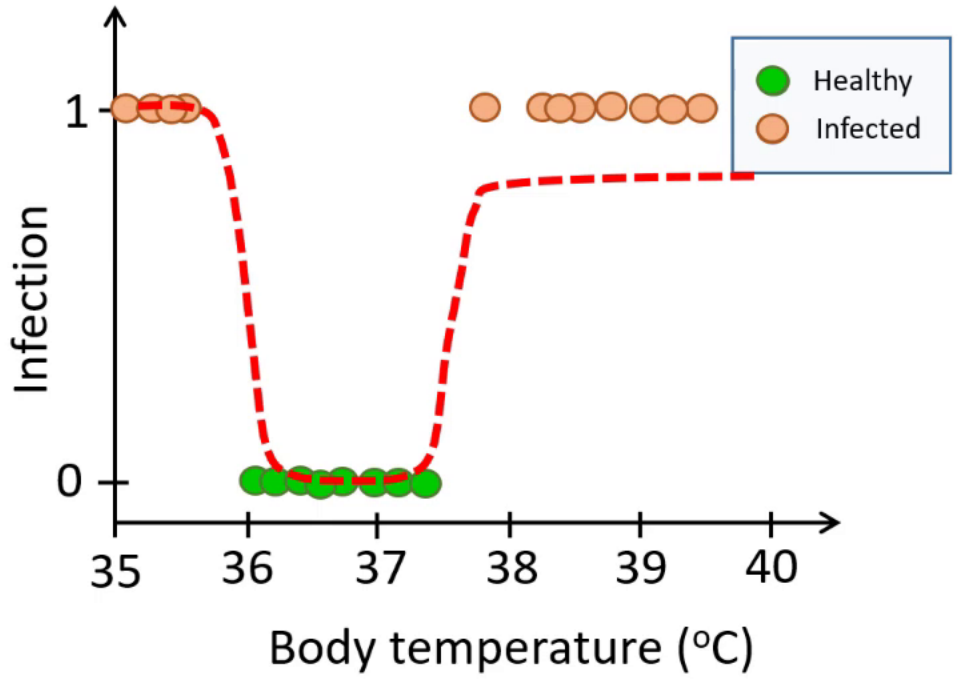


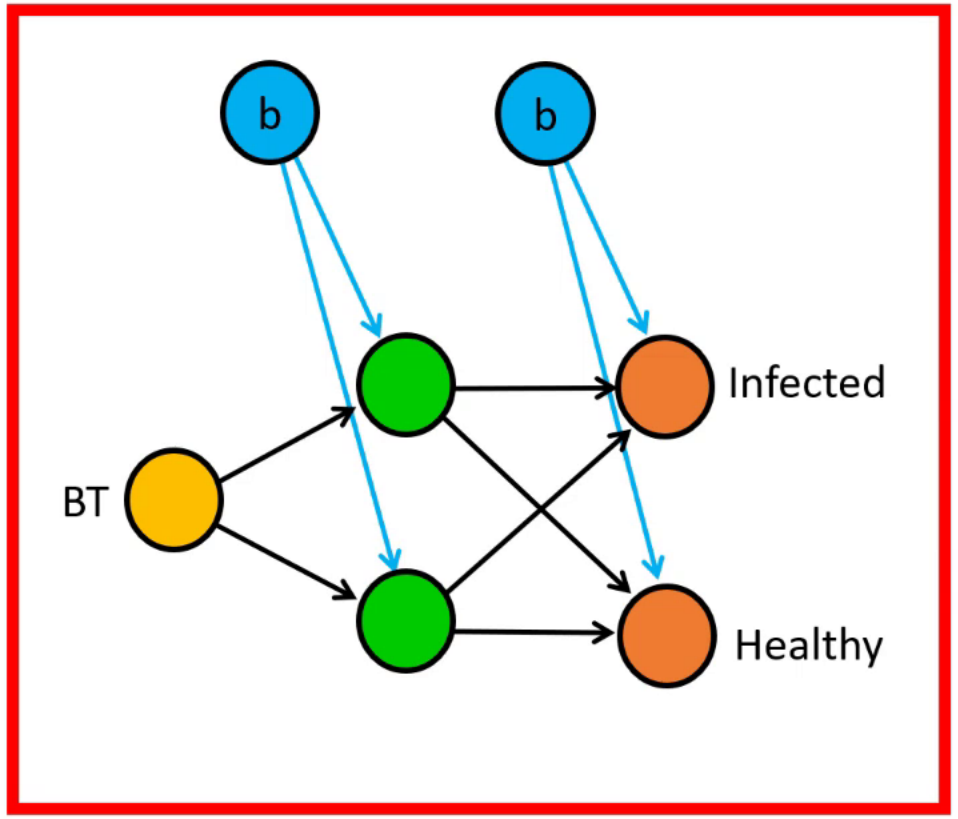
Overfitting

is because this person, who is infected, is now incorrectly predicted as being healthy because its body temperature is 38.4.

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

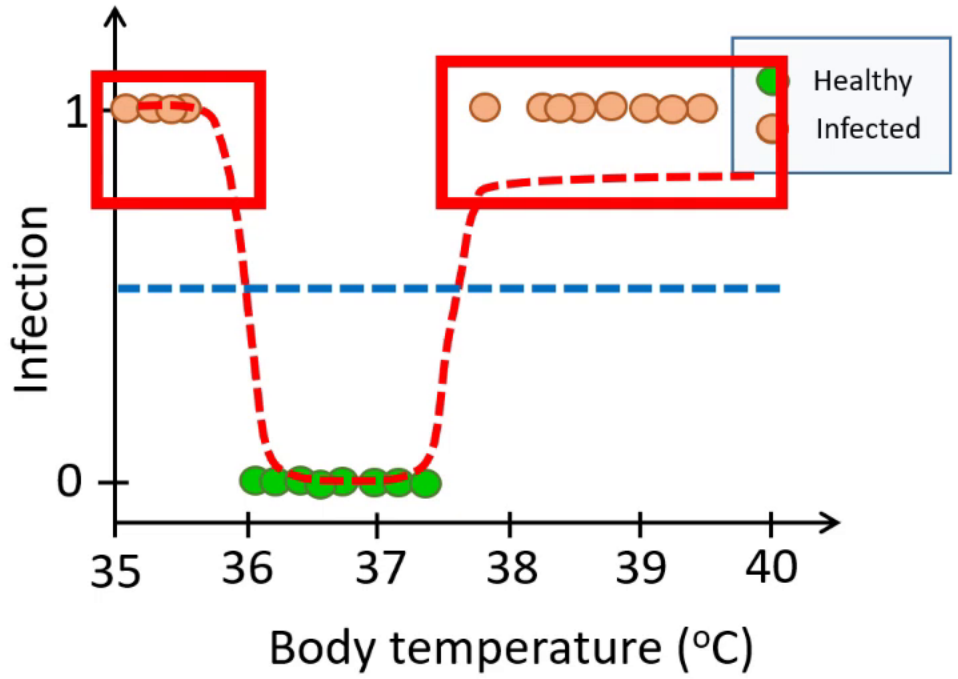


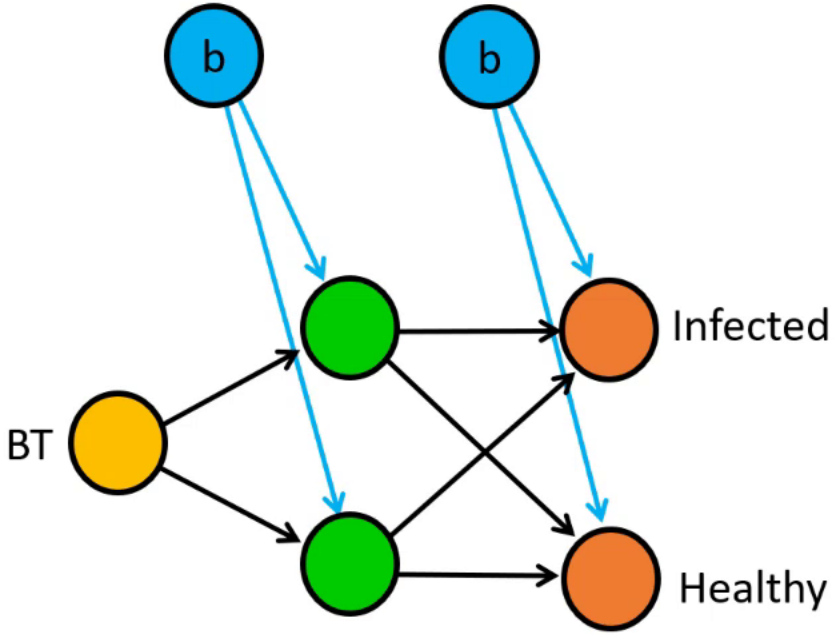


A network with just one hidden layer will do a better job of predicting the new data,

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

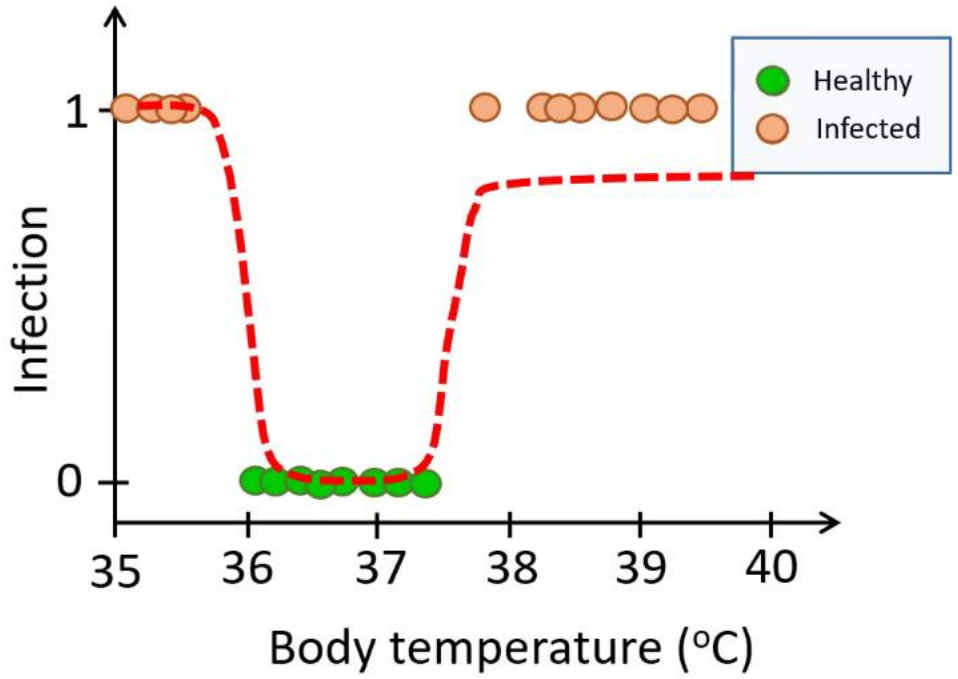


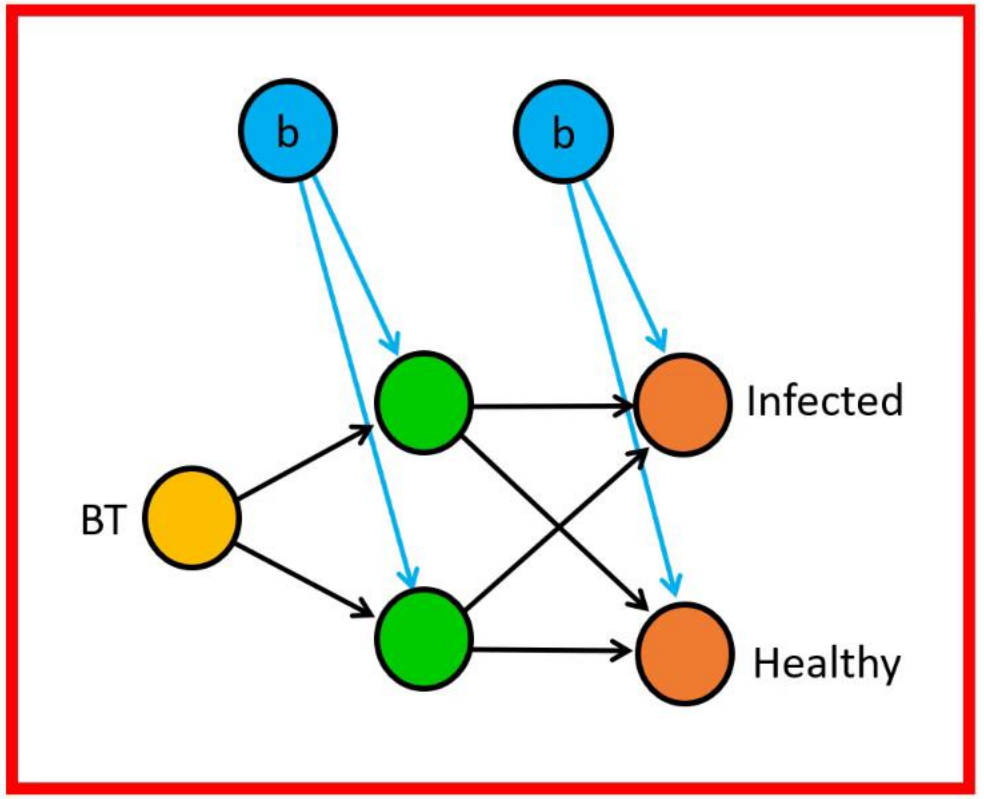


whereas people outside this range are predicted as being infected.

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

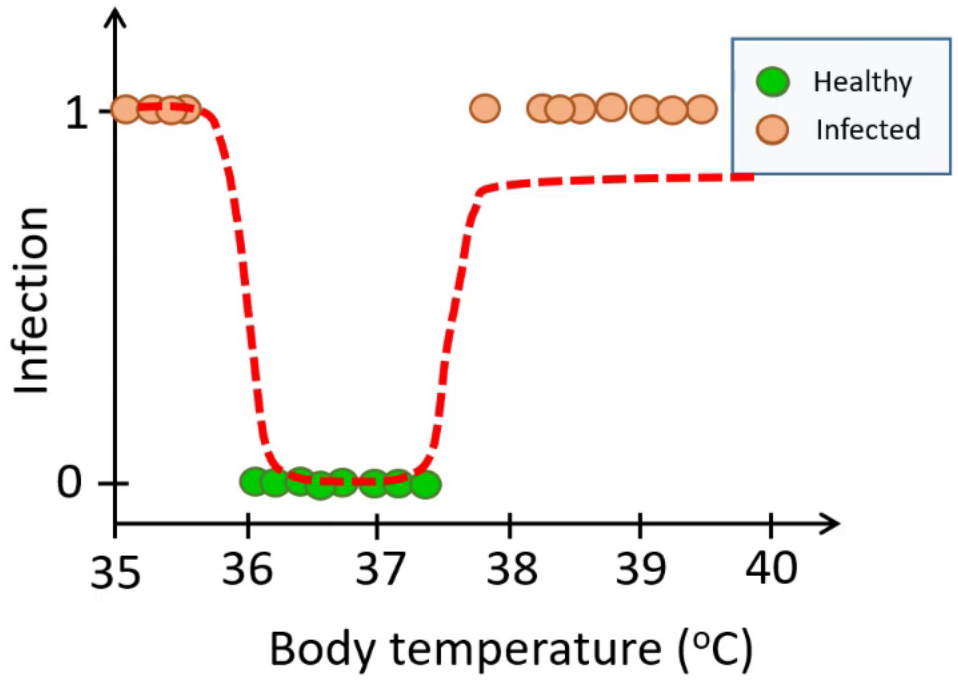


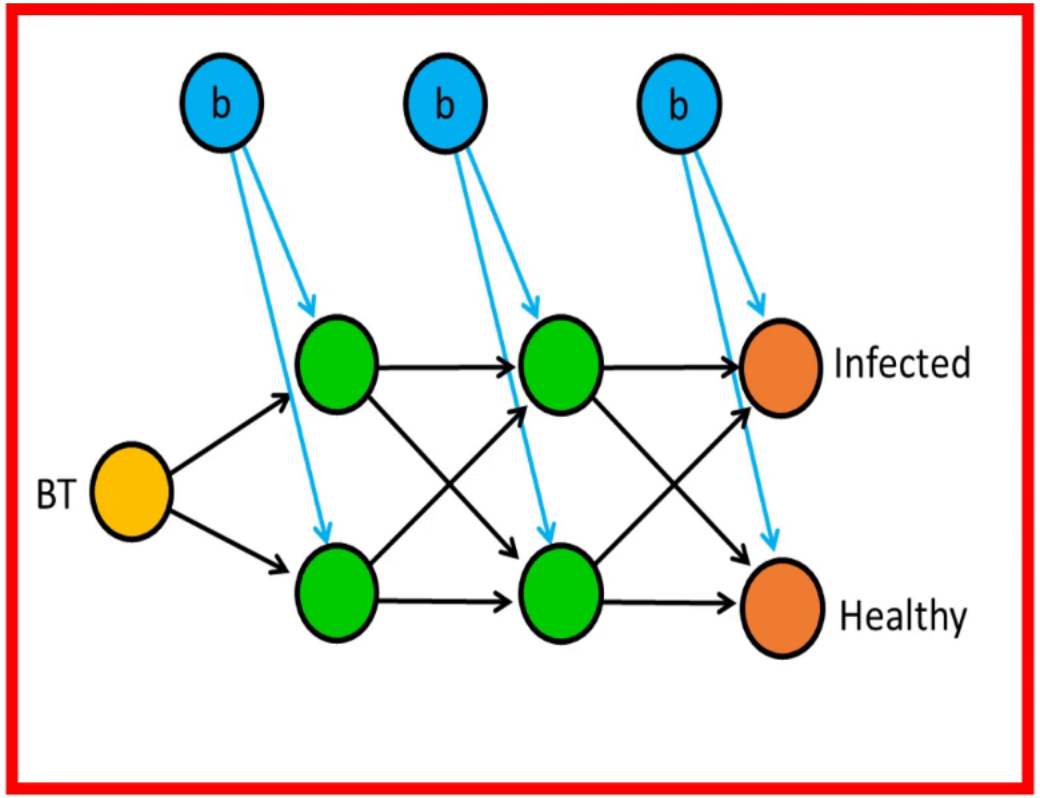


So, for this data, it would make sense to use a simpler network,

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4

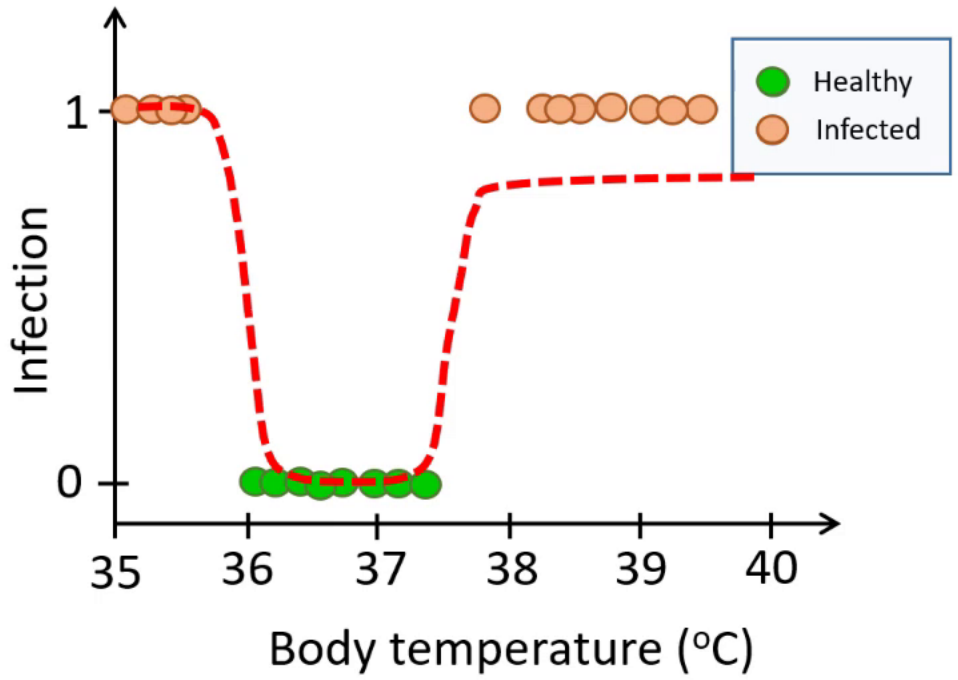




or to use a deep neural network where we use a method to prevent overfitting.

Overfitting

Status	BT
Infected	35.0
Infected	35.5
Infected	37.4
Infected	38.2
Infected	38.5
Infected	39.0
Infected	39.5
Healthy	36.1
Healthy	36.2
Healthy	36.4
Healthy	36.7
Healthy	38.4



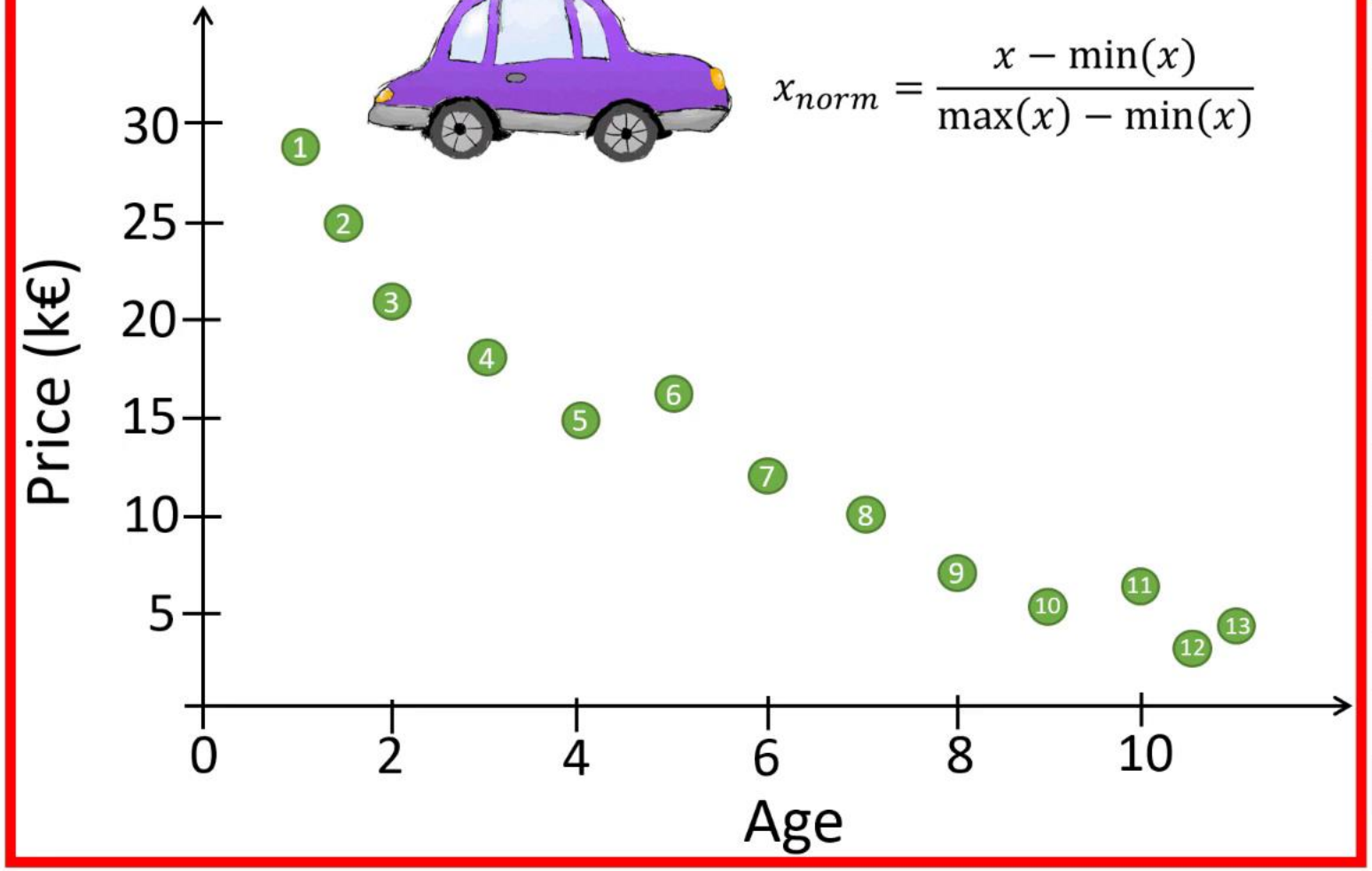
- Early stopping
- Dropout

We will discuss two such methods in a few minutes.

A second example



$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$



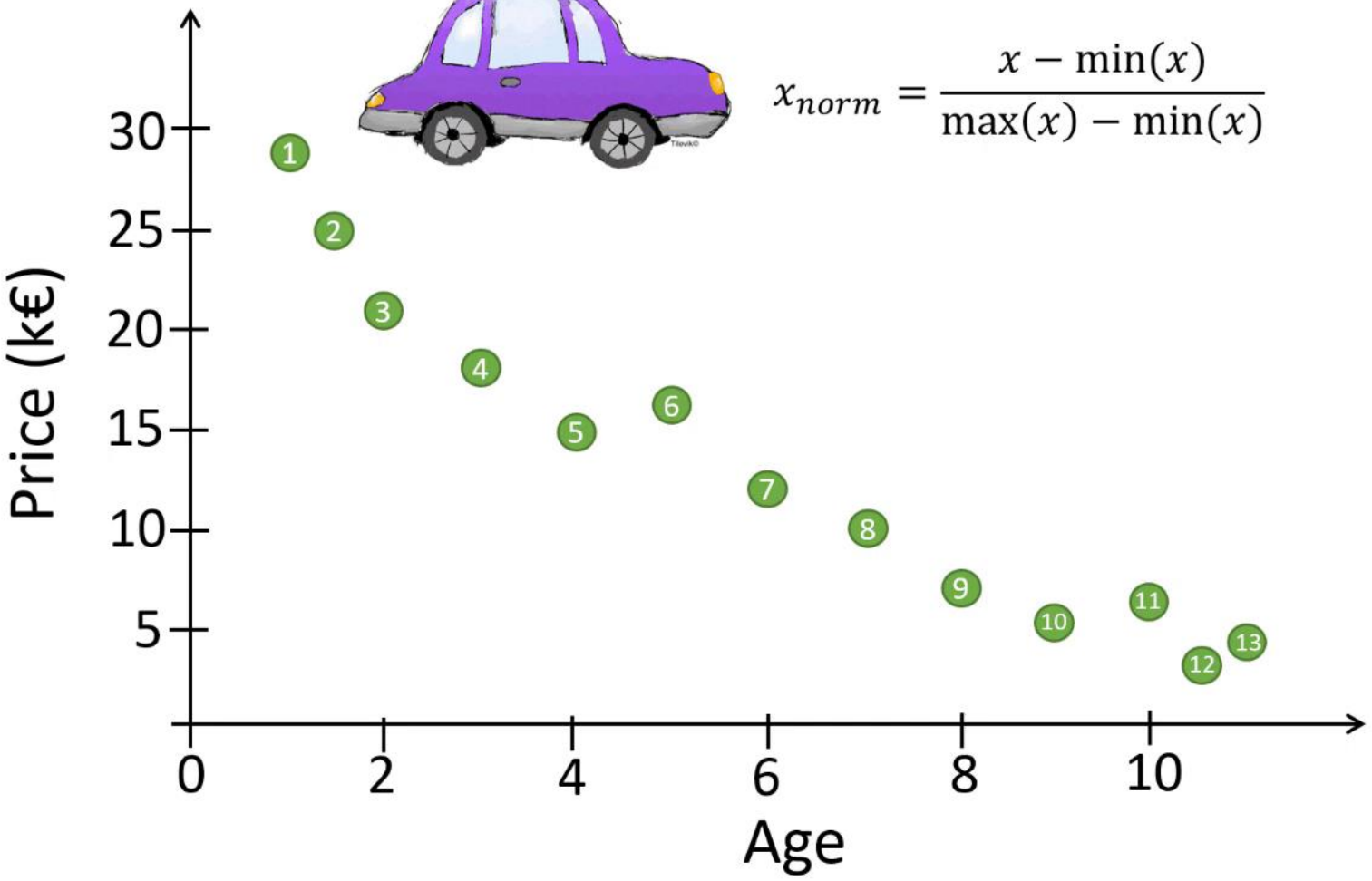
Car	Price (k€)	Normalized Price	Age (years)	Normalized Age
1	29	1.00	1	0.00
2	25	0.85	1.5	0.05
3	21	0.69	2	0.10
4	18	0.58	3	0.20
5	15	0.46	4	0.30
6	16	0.50	5	0.40
7	12	0.35	6	0.50
8	10	0.27	7	0.60
9	7	0.15	8	0.70
10	5	0.08	9	0.80
11	6	0.12	10	0.90
12	3	0.00	10.5	0.95
13	4	0.04	11	1.00

But let's first take another example where we like to train a network based on some car sales data so that we can use the network to predict an appropriate price of a used car based on its age.

A second example



$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$



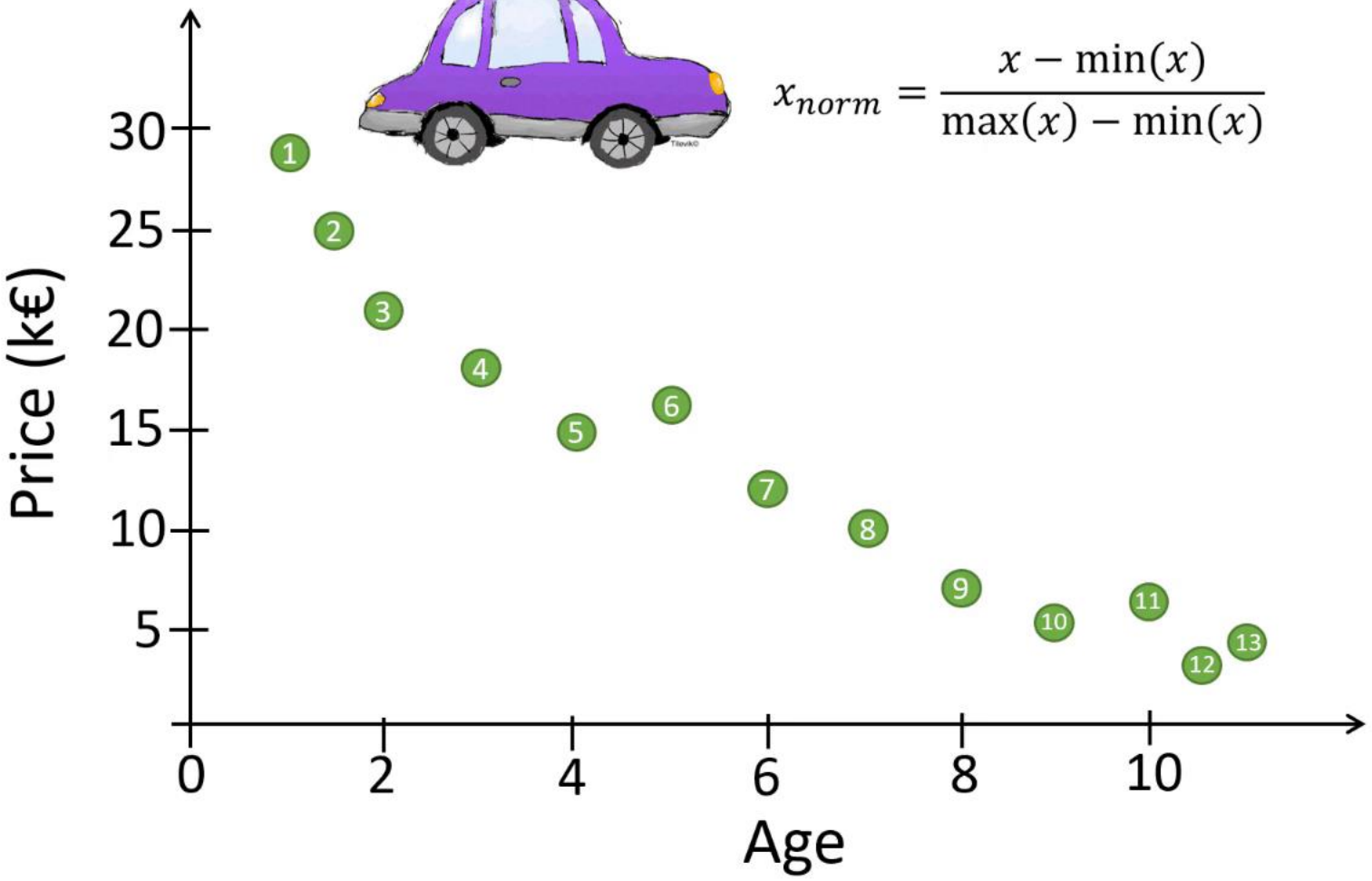
Car	Price (k€)	Normalized Price	Age (years)	Normalized Age
1	29	1.00	1	0.00
2	25	0.85	1.5	0.05
3	21	0.69	2	0.10
4	18	0.58	3	0.20
5	15	0.46	4	0.30
6	16	0.50	5	0.40
7	12	0.35	6	0.50
8	10	0.27	7	0.60
9	7	0.15	8	0.70
10	5	0.08	9	0.80
11	6	0.12	10	0.90
12	3	0.00	10.5	0.95
13	4	0.04	11	1.00

The input for our network will be the normalized ages of the cars,

A second example



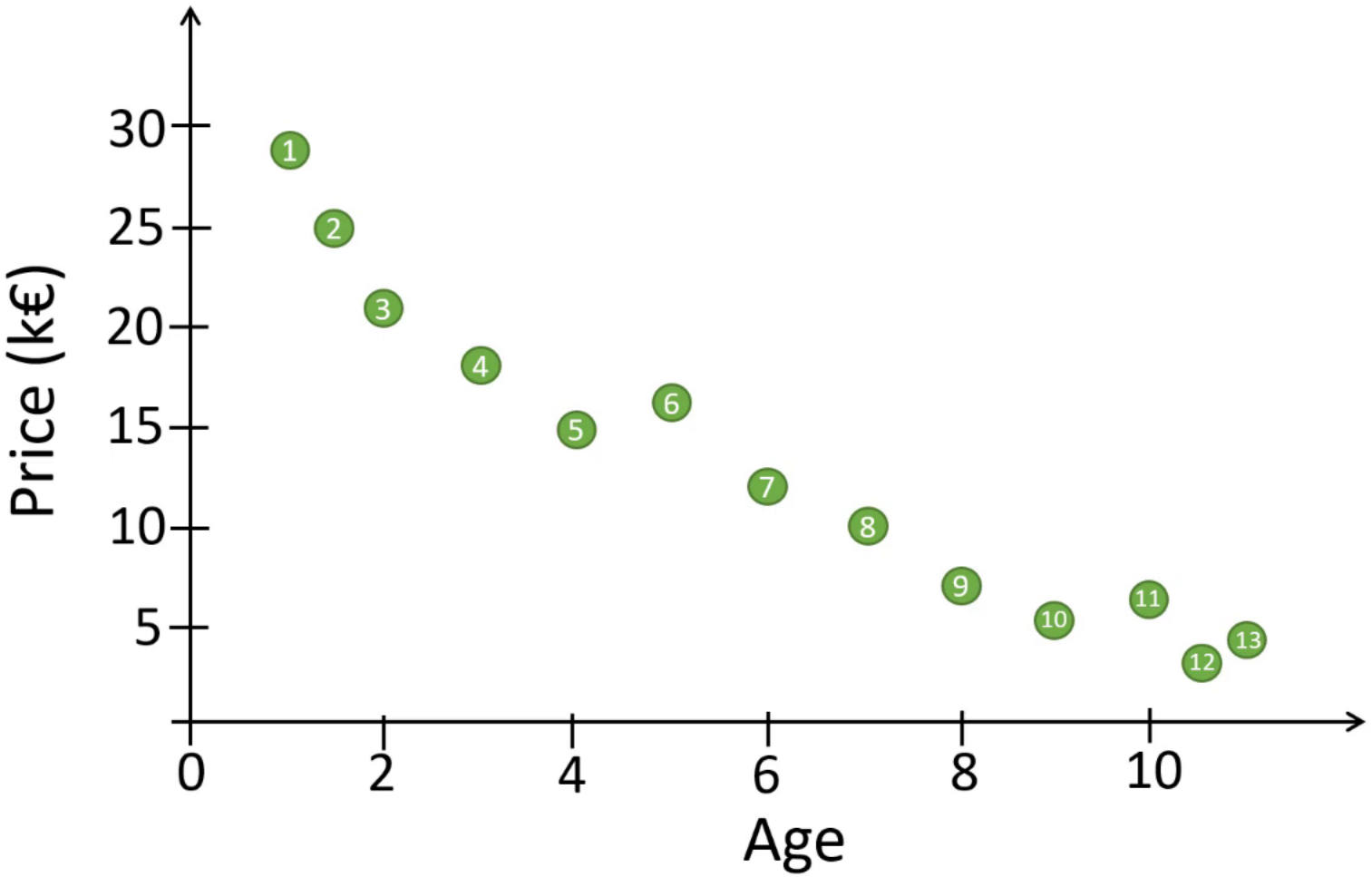
$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

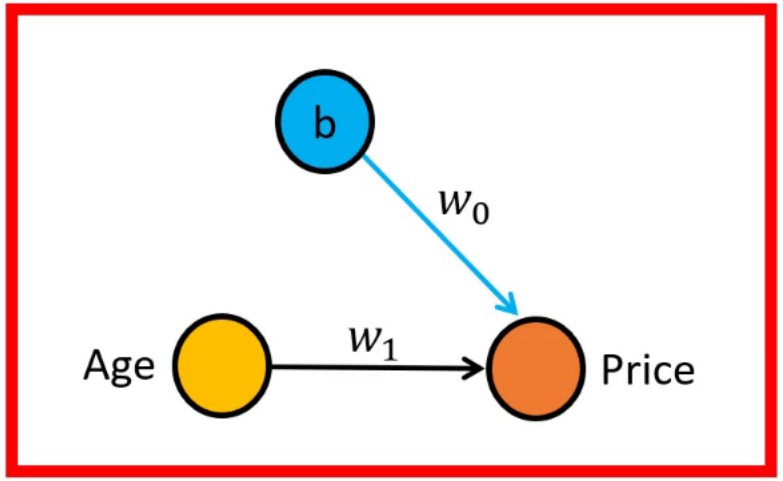


Car	Price (k€)	Normalized Price	Age (years)	Normalized Age
1	29	1.00	1	0.00
2	25	0.85	1.5	0.05
3	21	0.69	2	0.10
4	18	0.58	3	0.20
5	15	0.46	4	0.30
6	16	0.50	5	0.40
7	12	0.35	6	0.50
8	10	0.27	7	0.60
9	7	0.15	8	0.70
10	5	0.08	9	0.80
11	6	0.12	10	0.90
12	3	0.00	10.5	0.95
13	4	0.04	11	1.00

whereas the target values will be the normalized prices.

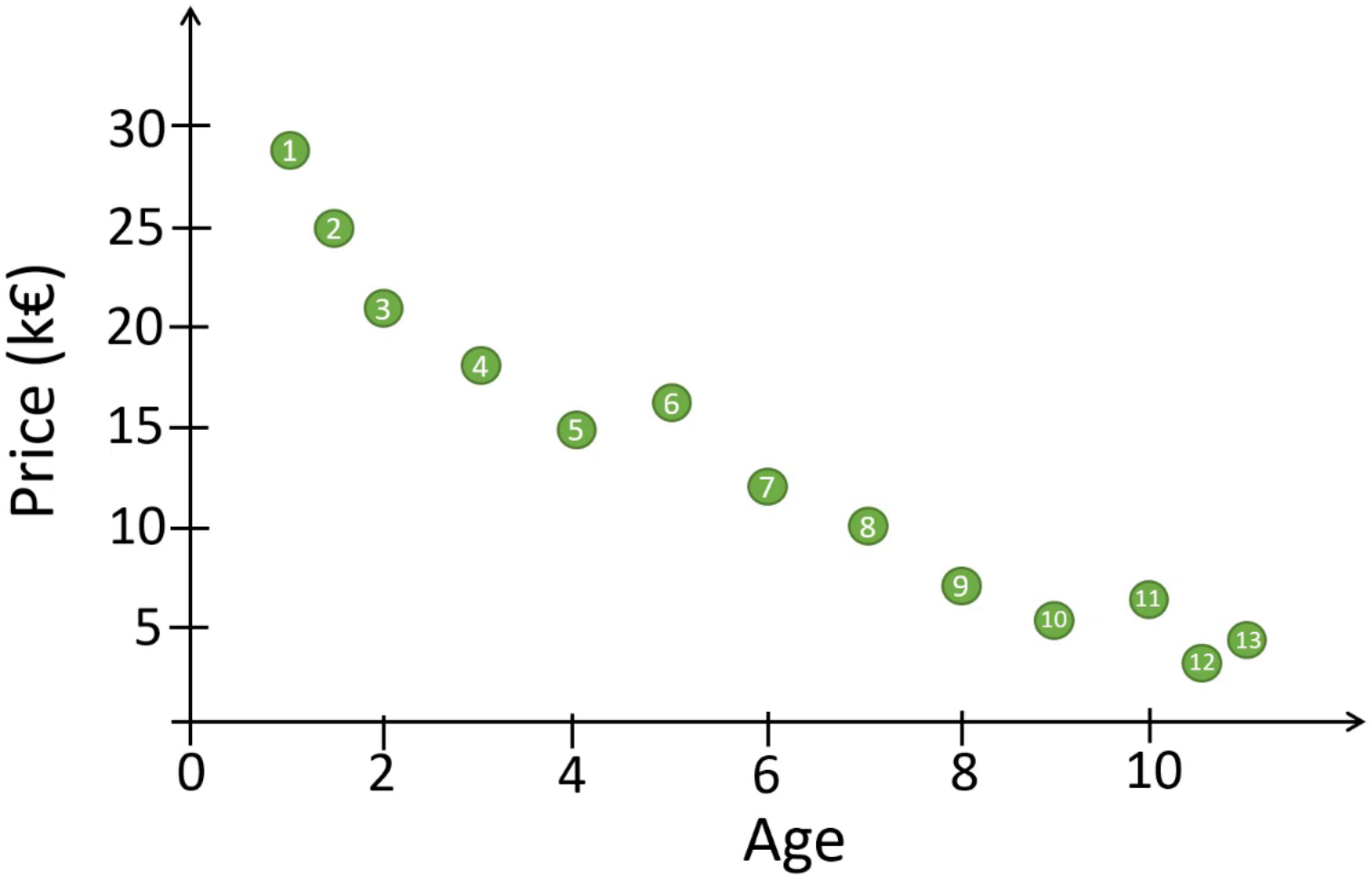
A second example

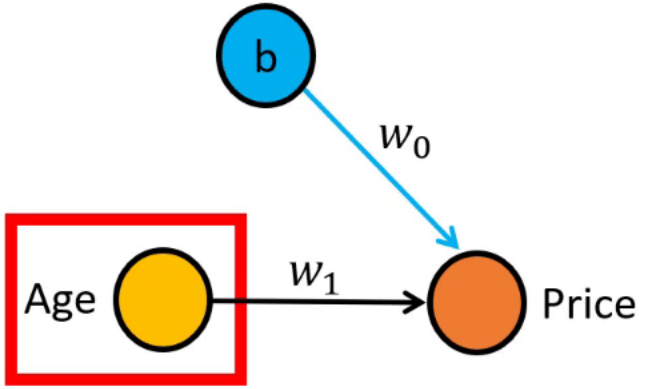




We will first start to use the following network, without any hidden layer,

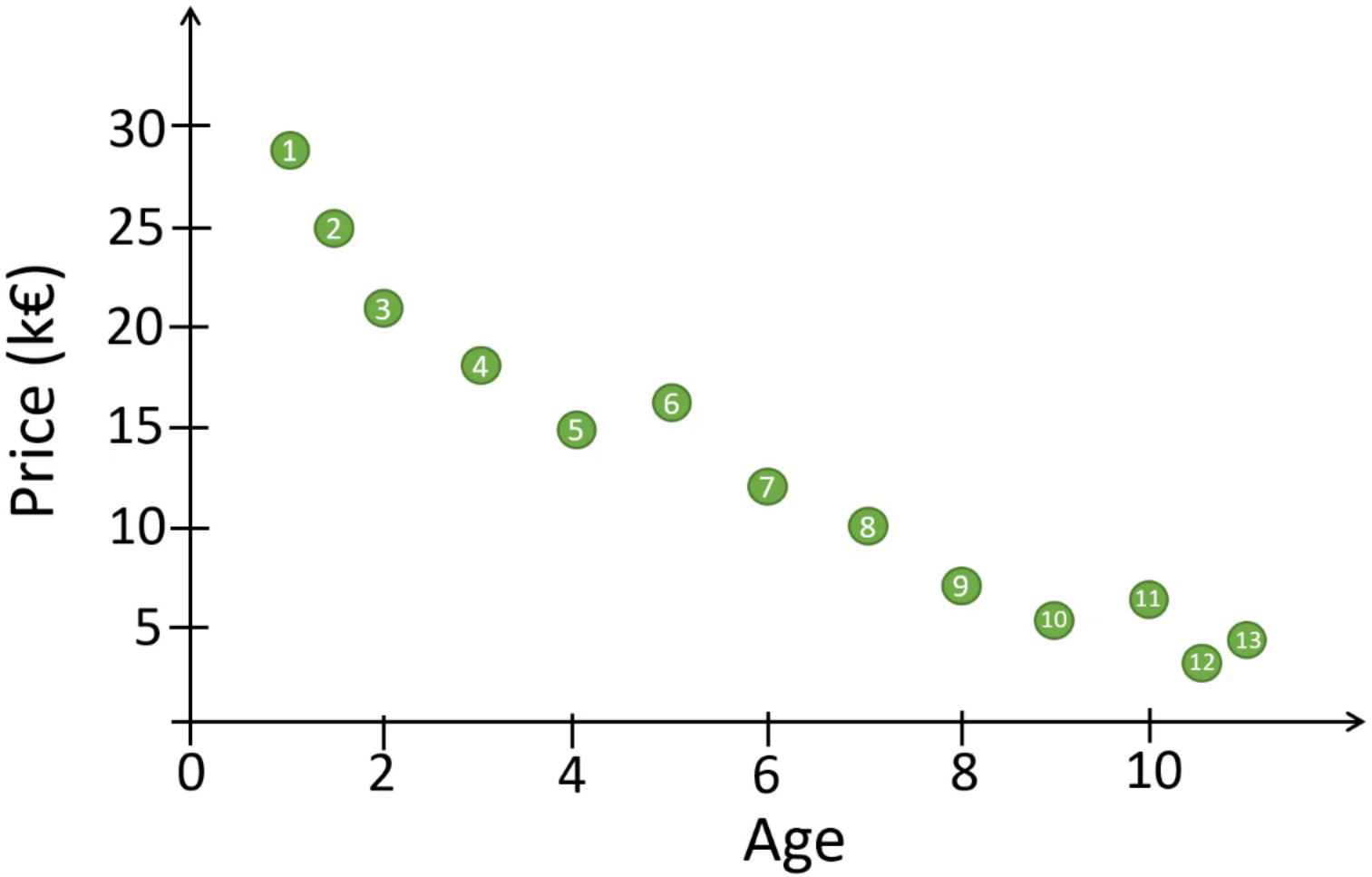
A second example

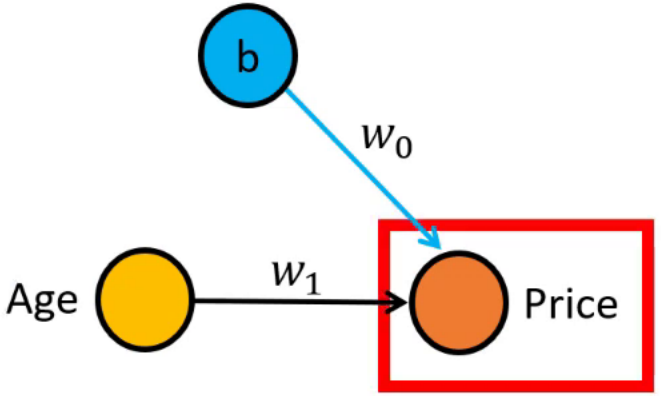




with just one input node that takes the normalized ages of the cars,

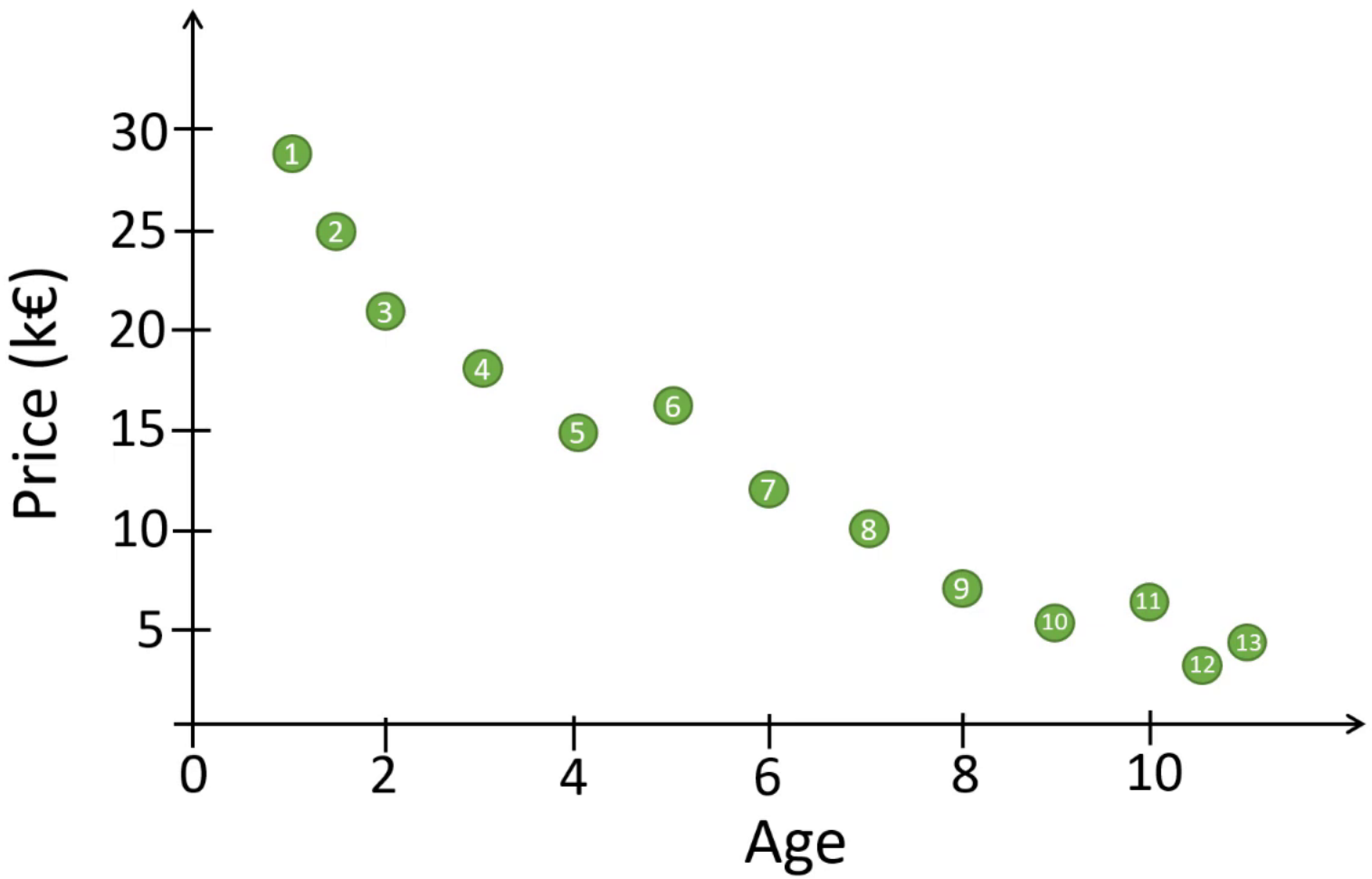
A second example

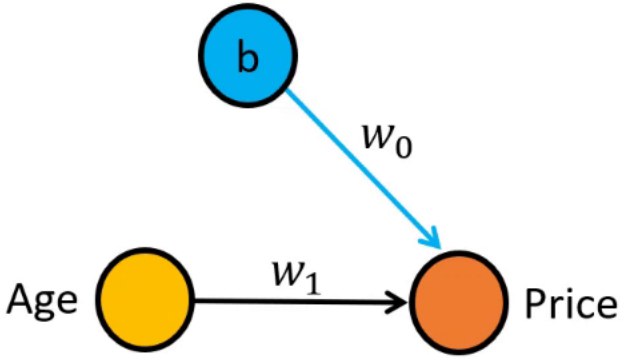




and one output node that generates the predicted normalized price.

A second example



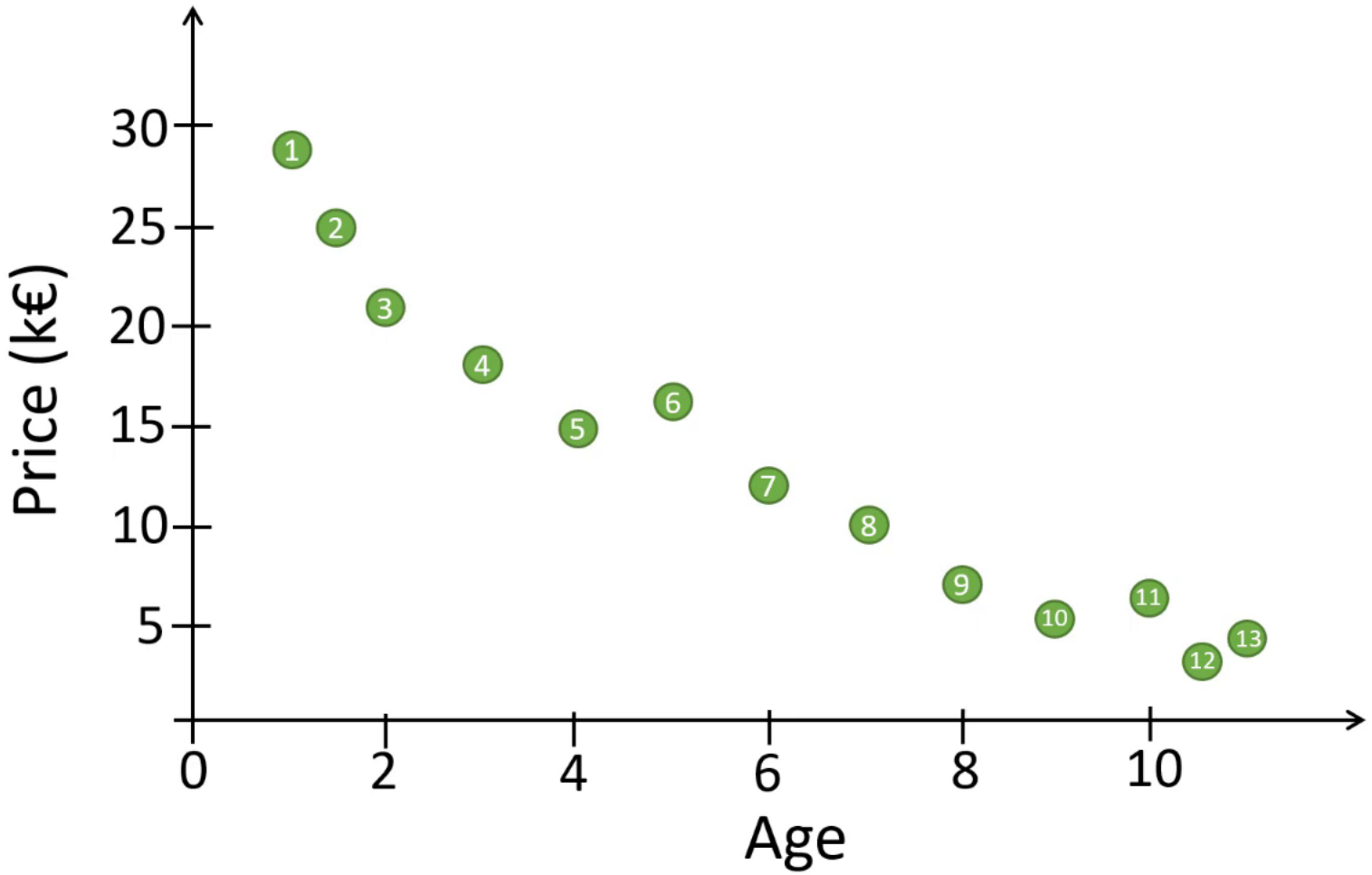


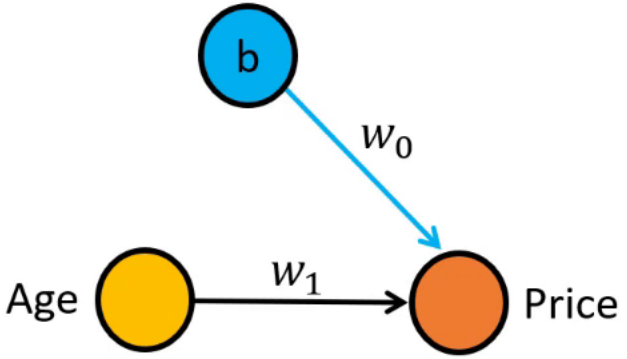
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

We will here use the linear activation function for the output node,

A second example



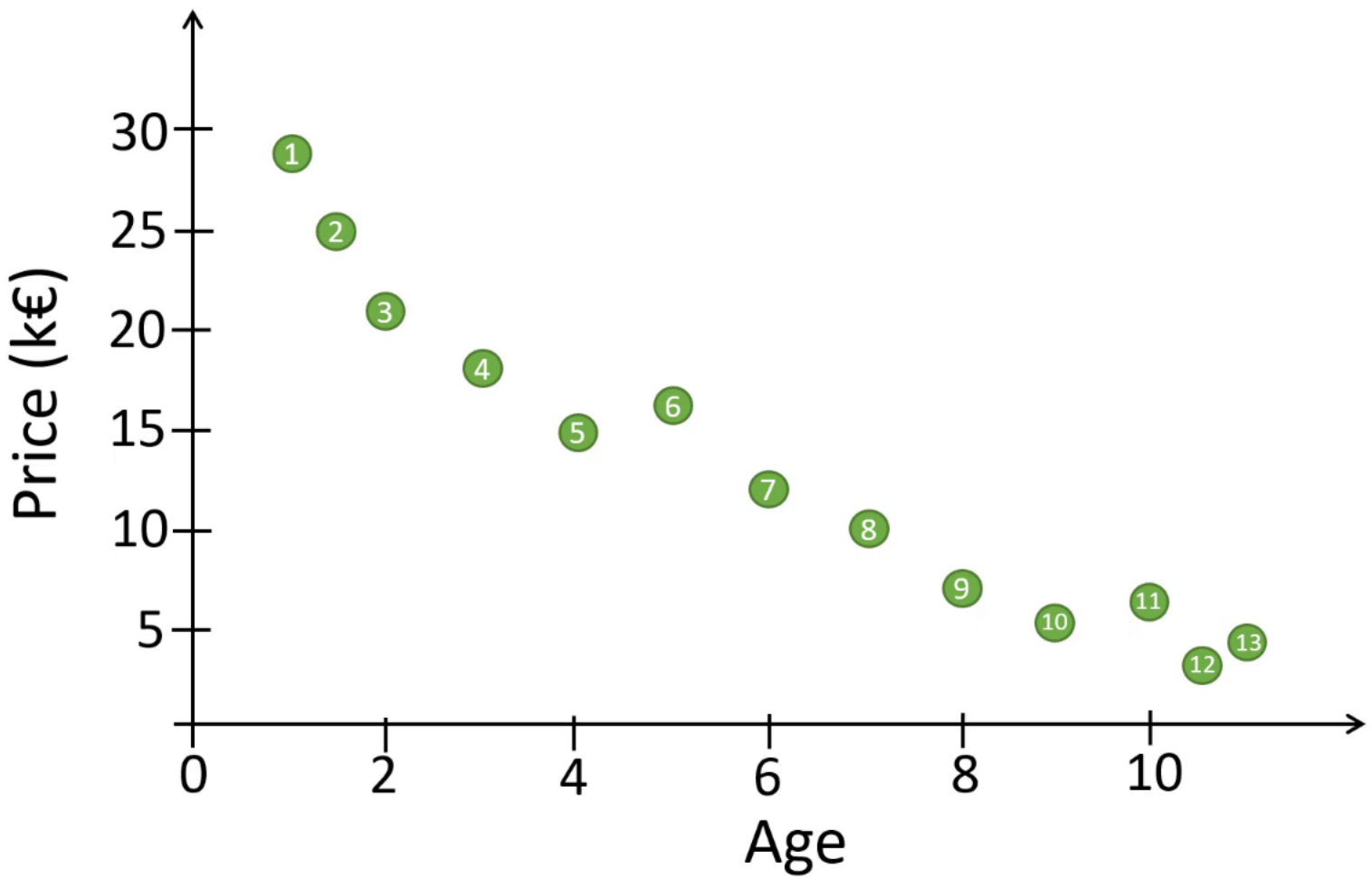


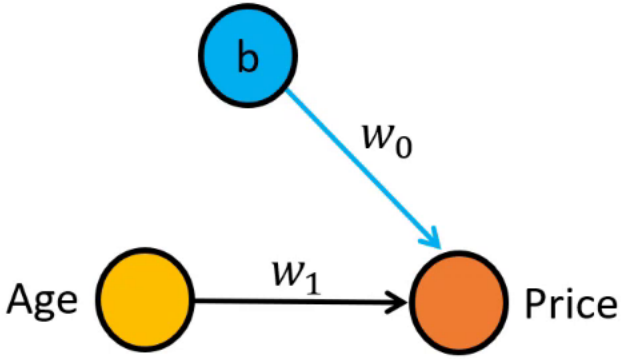
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

which means that the output node is simply equal to z.

A second example



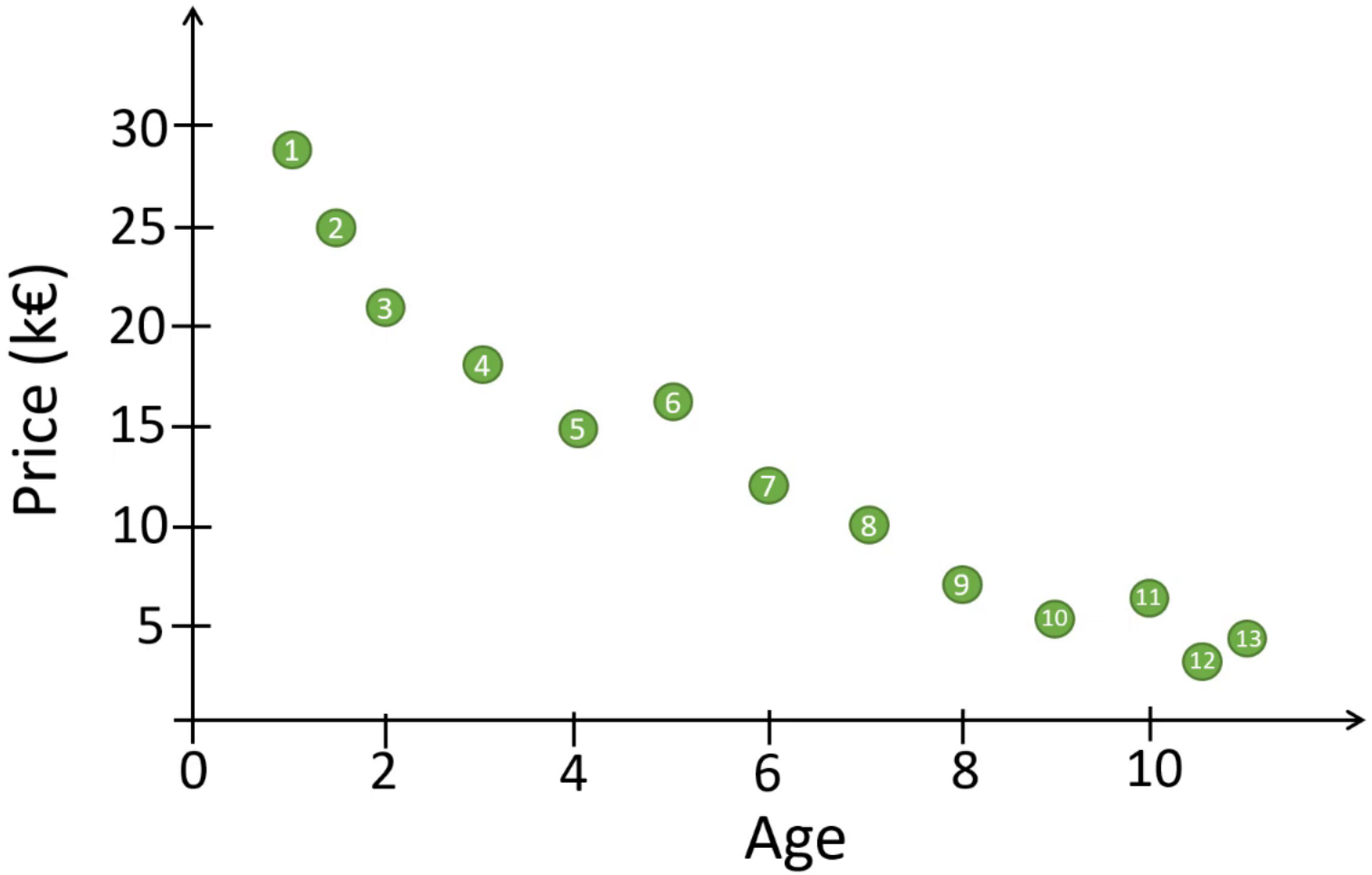


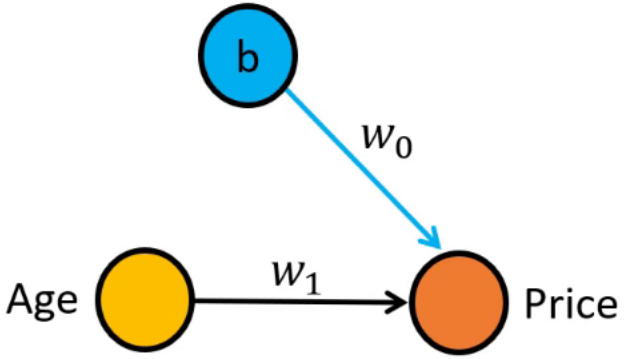
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

Since we only have one input node in this example,

A second example



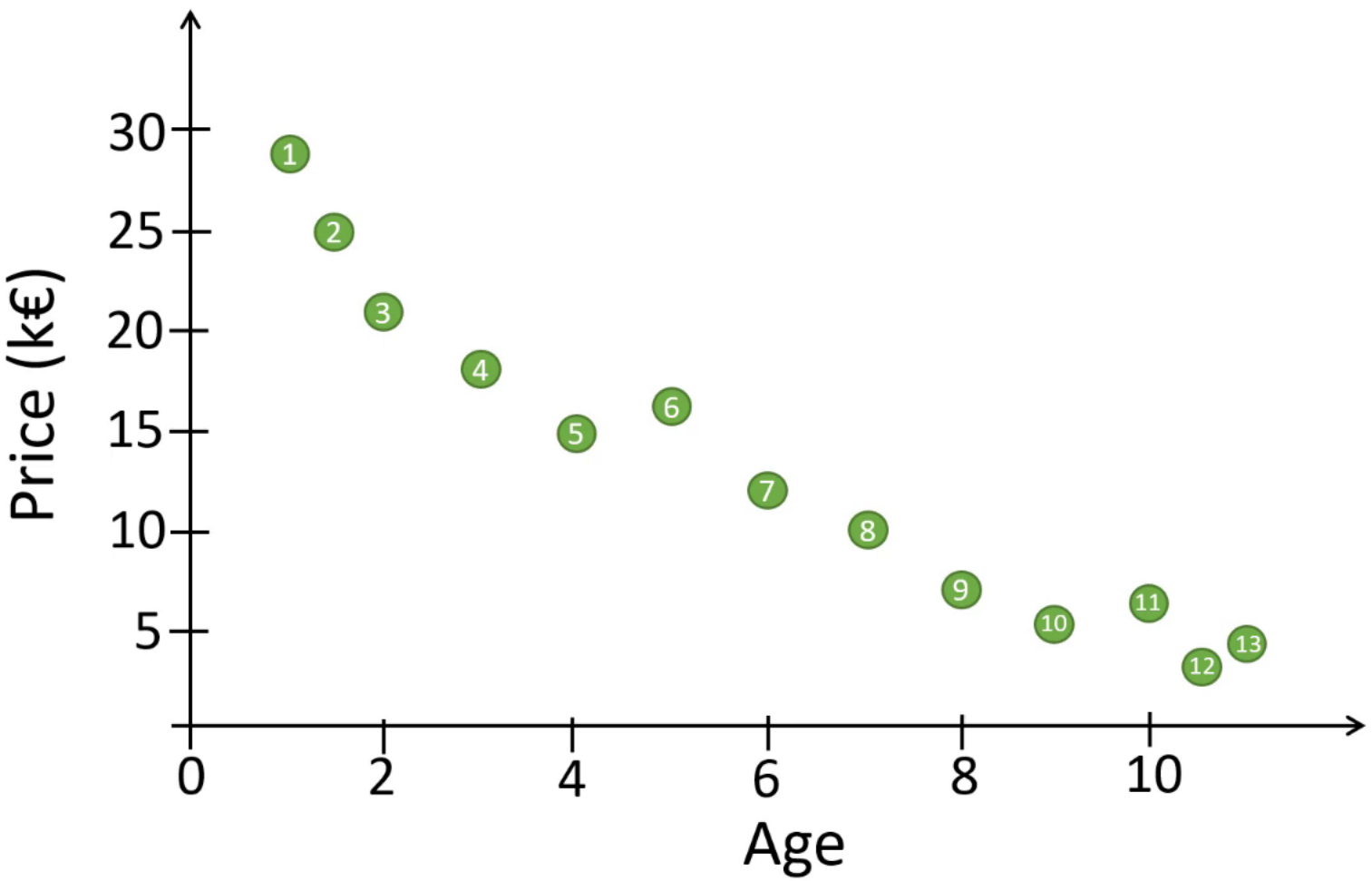


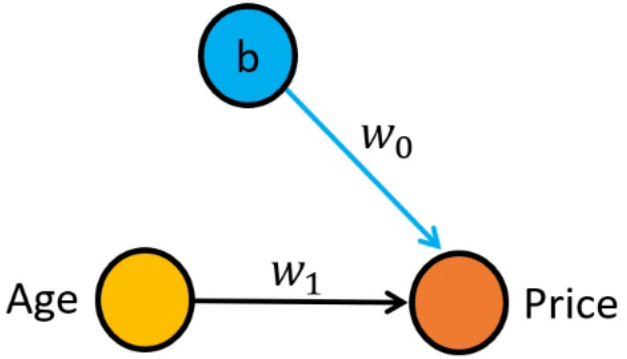
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

we can remove this summation notation,

A second example



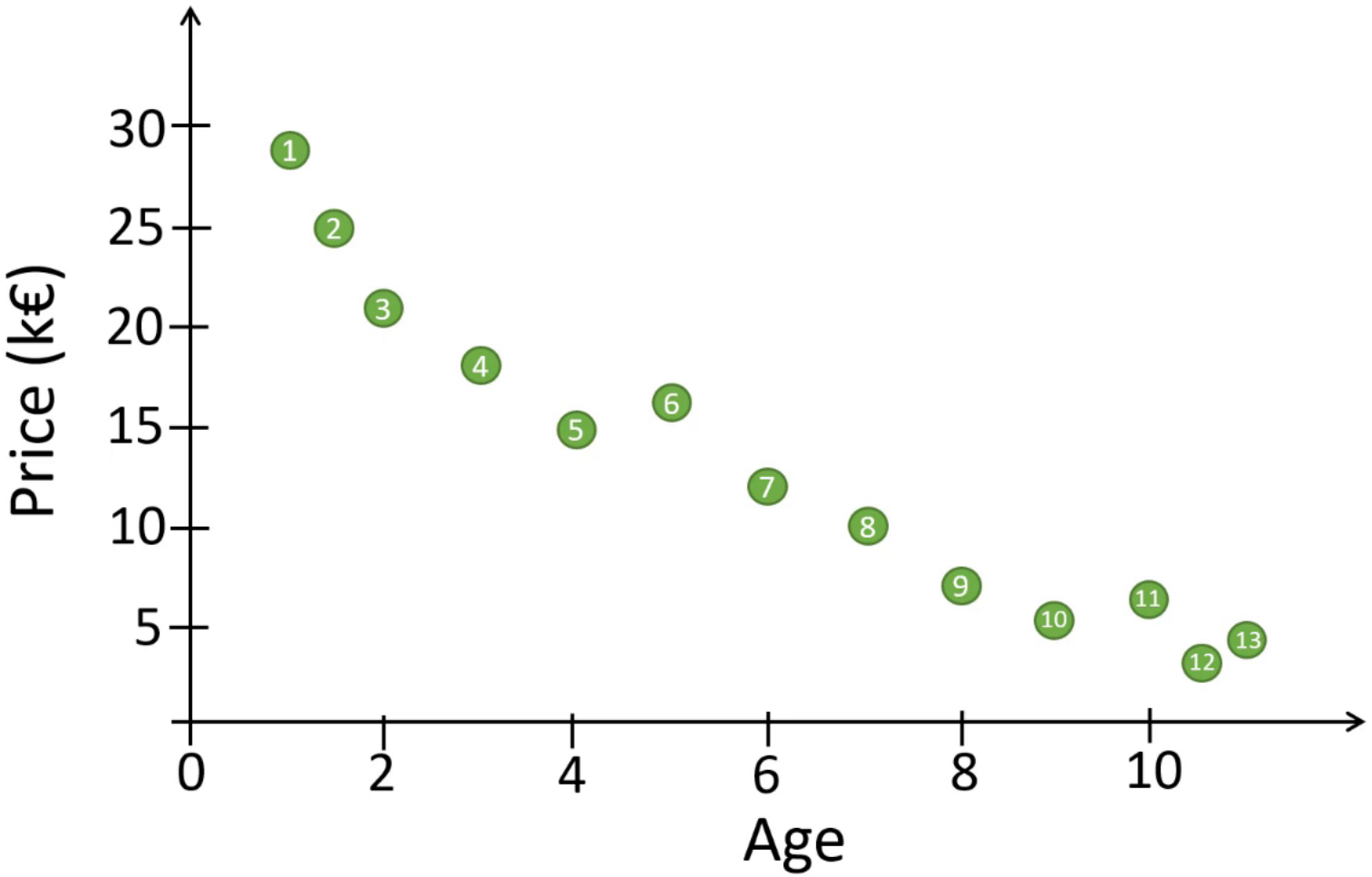


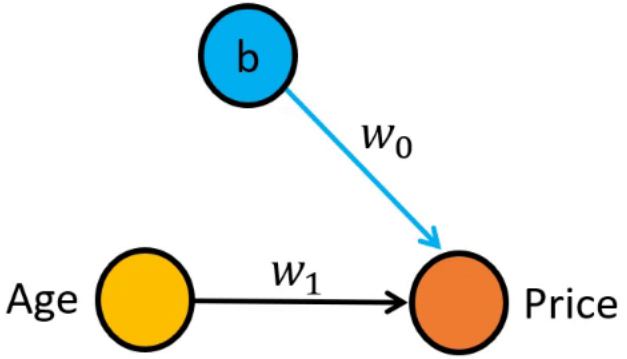
$$z = w_0 + w_1 \cdot Age_{norm}$$

$$f(z) = z$$

and write z like this.

A second example



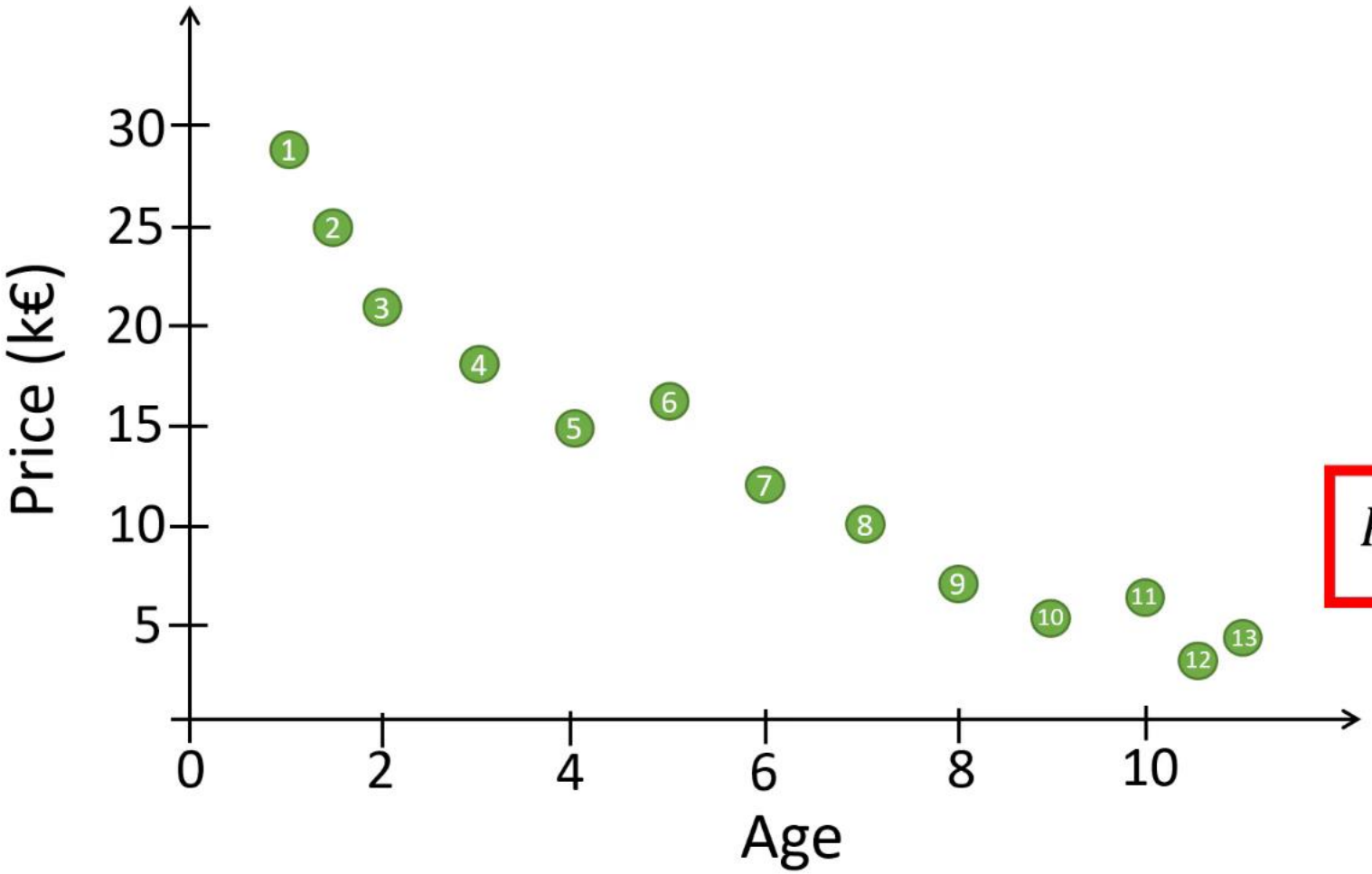


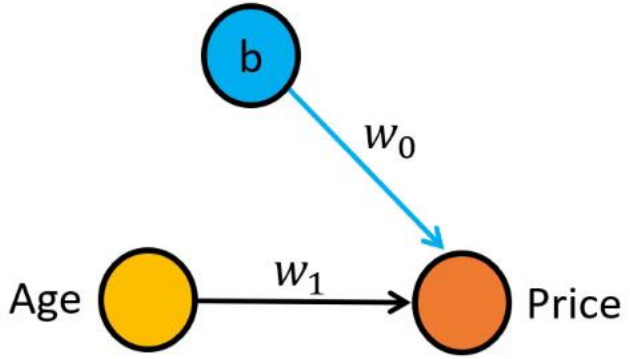
$$z = w_0 + w_1 \cdot Age_{norm}$$

$$f(z) = z$$

Let's substitute z with the terms on the right hand side,

A second example

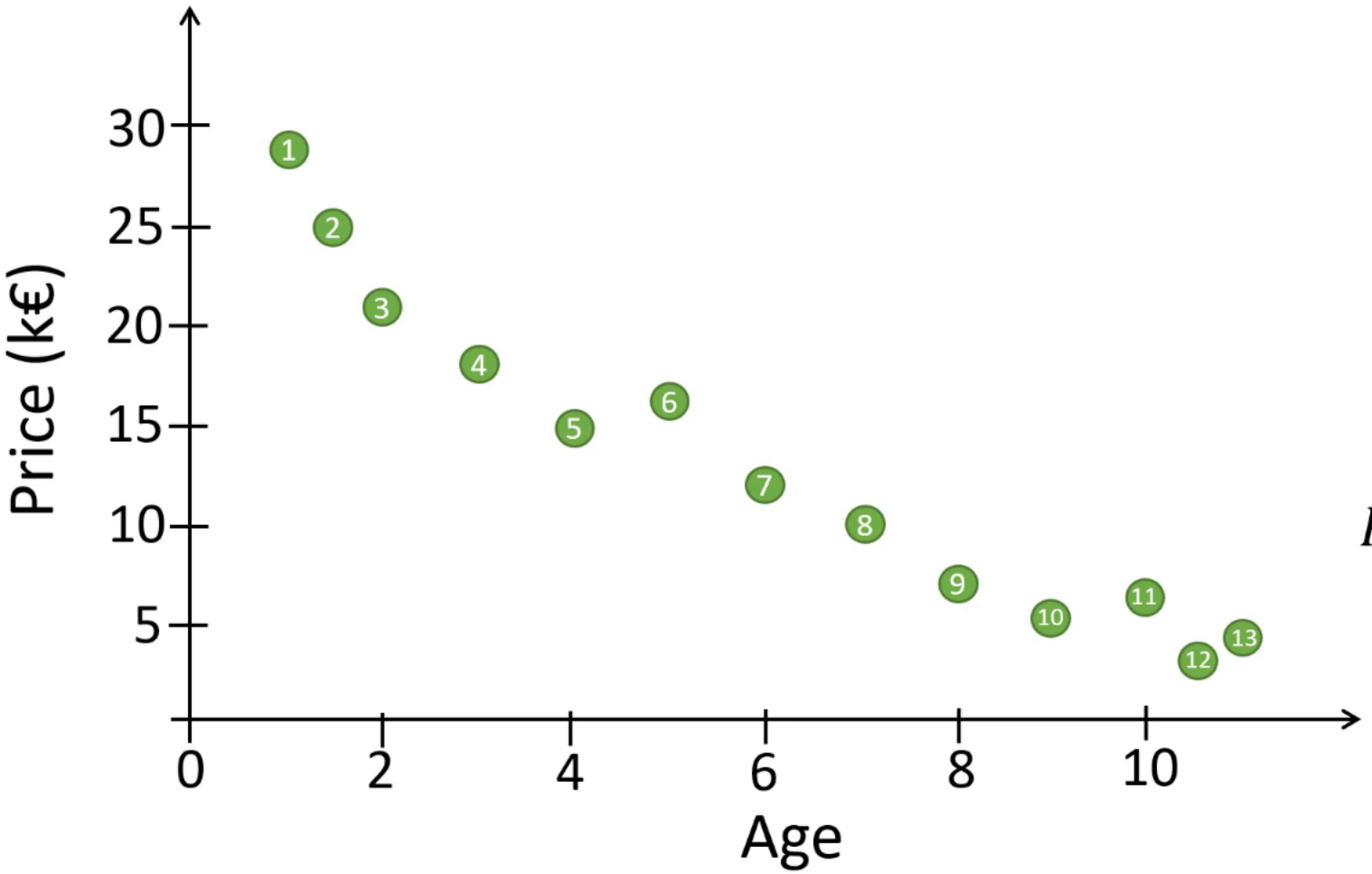


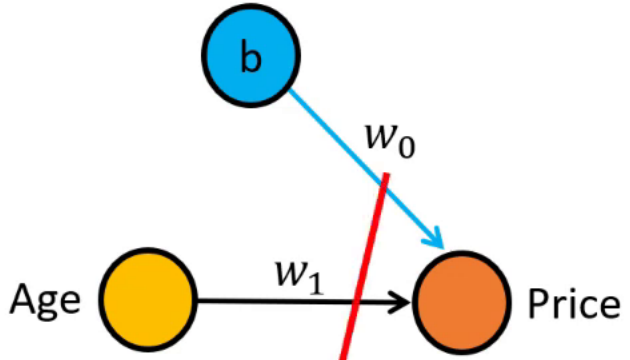


$$Price_{norm} = w_0 + w_1 \cdot Age_{norm}$$

so that we get the following function that the network uses to compute the predicted normalized price based on the normalized age.

A second example

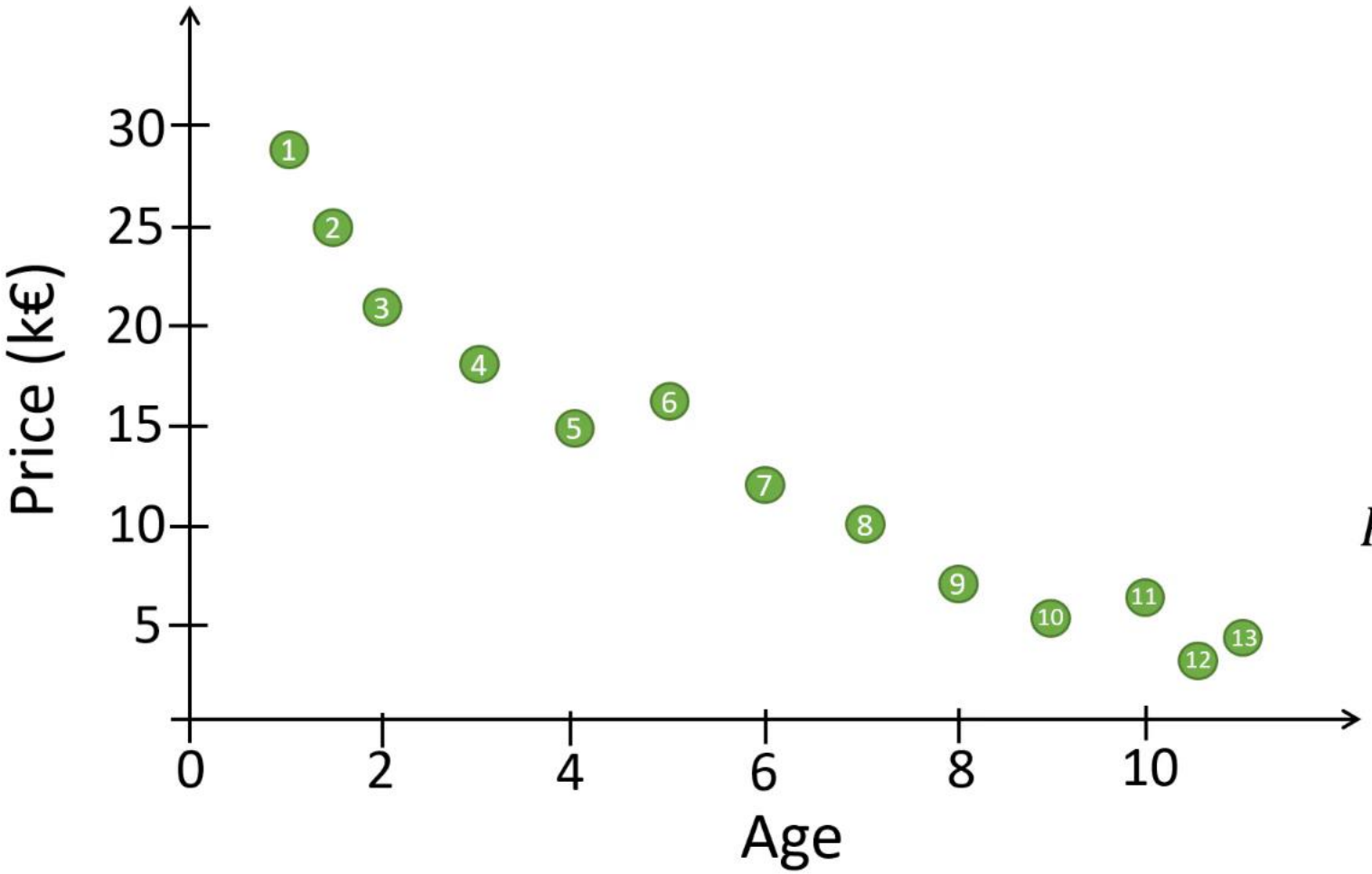


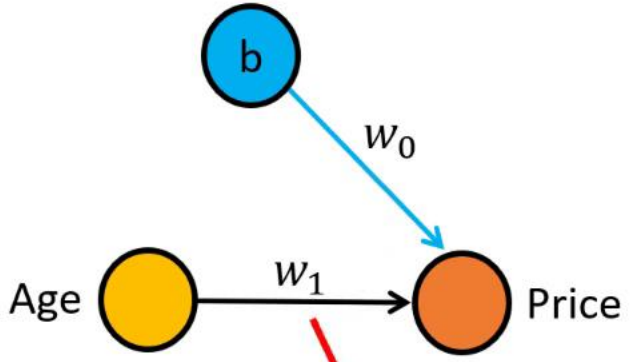


$$Price_{norm} = 0.83 + w_1 \cdot Age_{norm}$$

If we train this simple network on the normalized data, we see that the bias weight is equal to about 0.83,

A second example

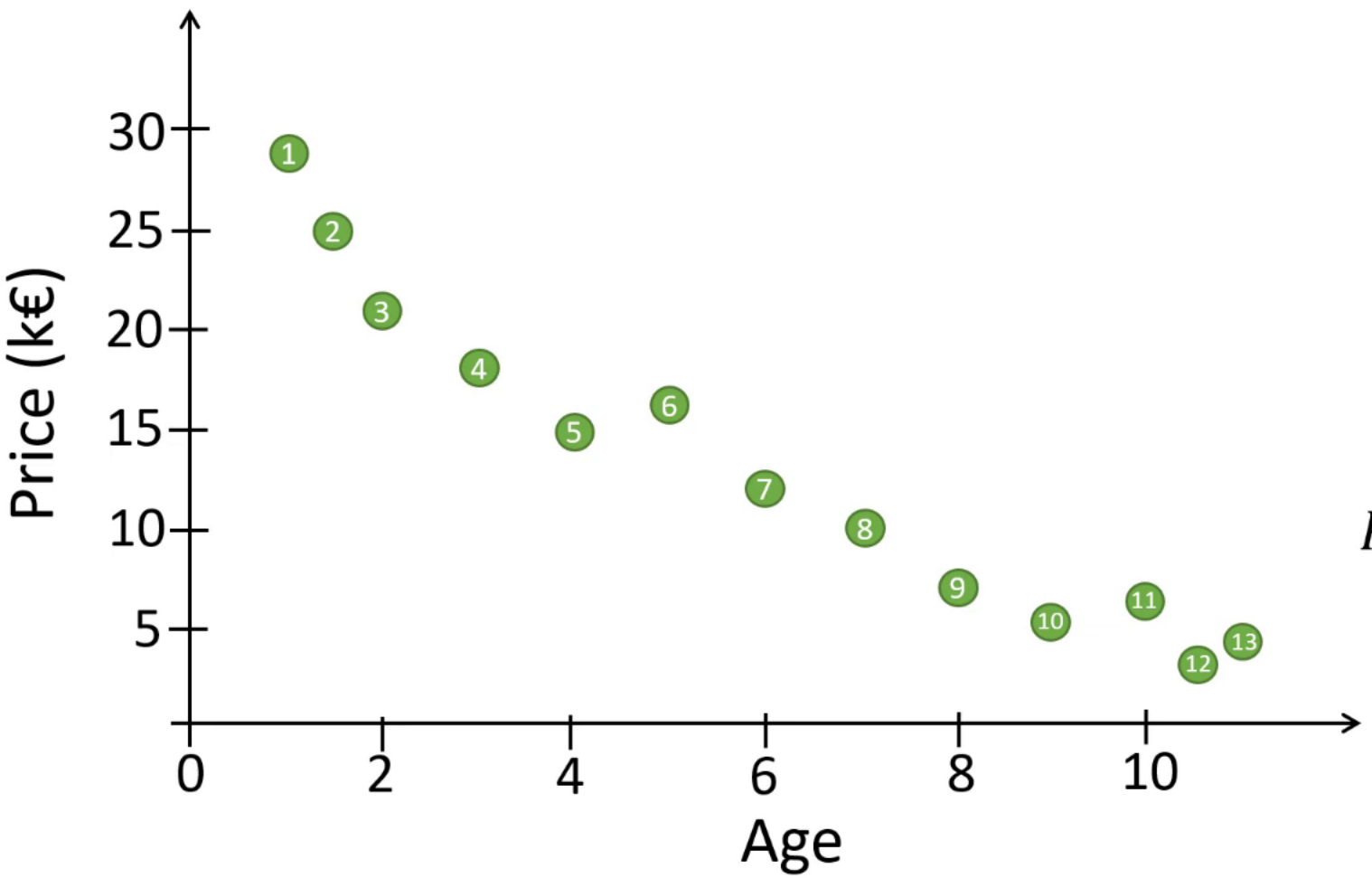


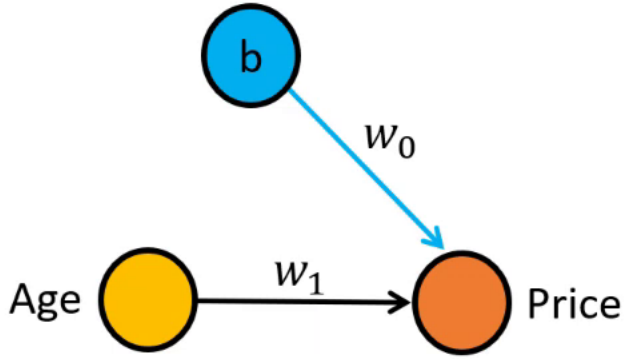


$$Price_{norm} = 0.83 - 0.88 \cdot Age_{norm}$$

and that the weight associated with the input node is about negative 0.88.

A second example

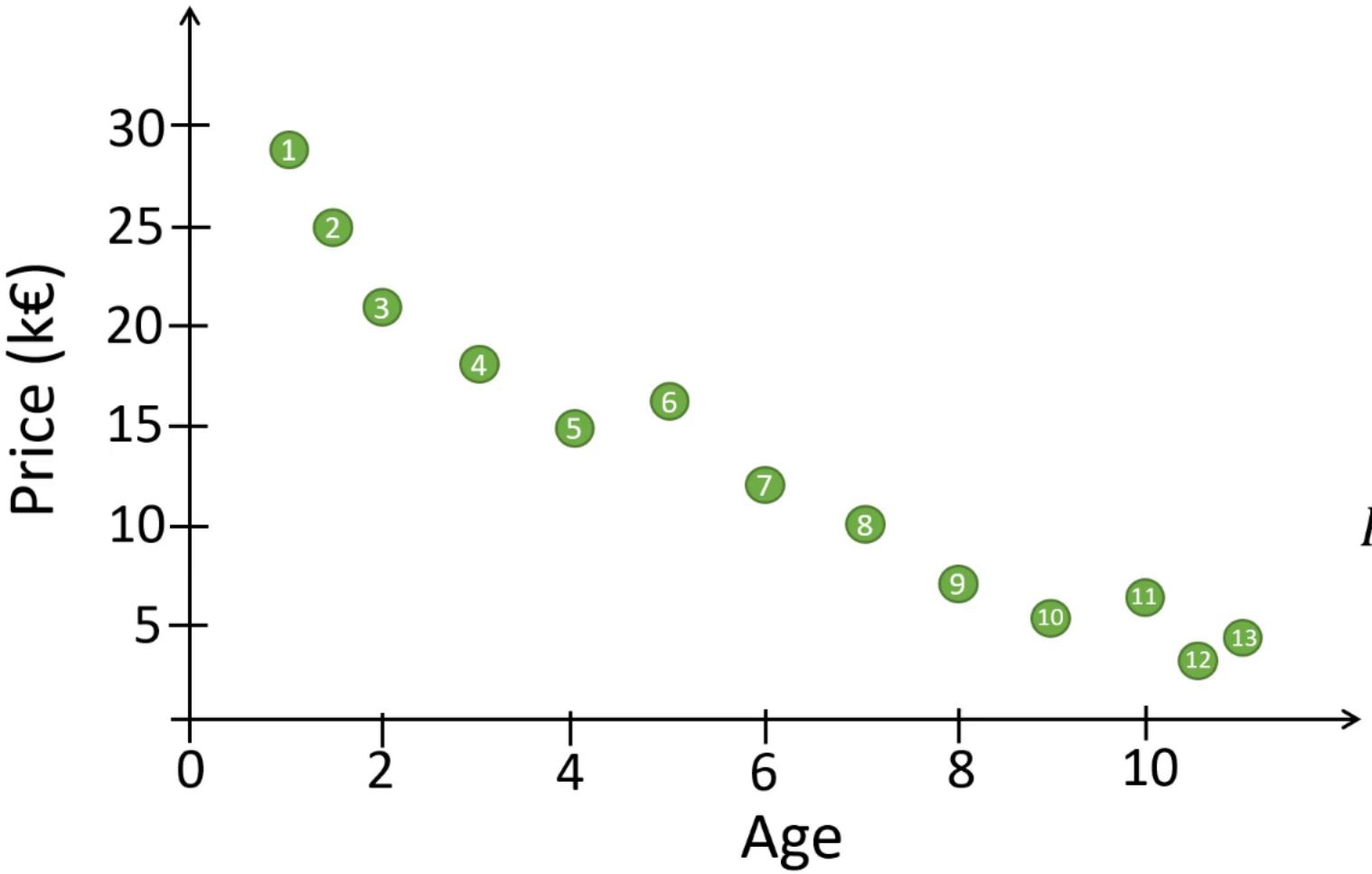


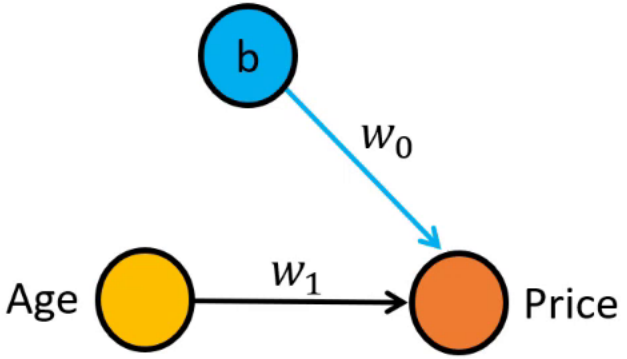


$$Price_{norm} = 0.83 - 0.88 \cdot Age_{norm}$$

These are the exact same estimated parameters we will get if we would use linear regression on the normalized data.

A second example

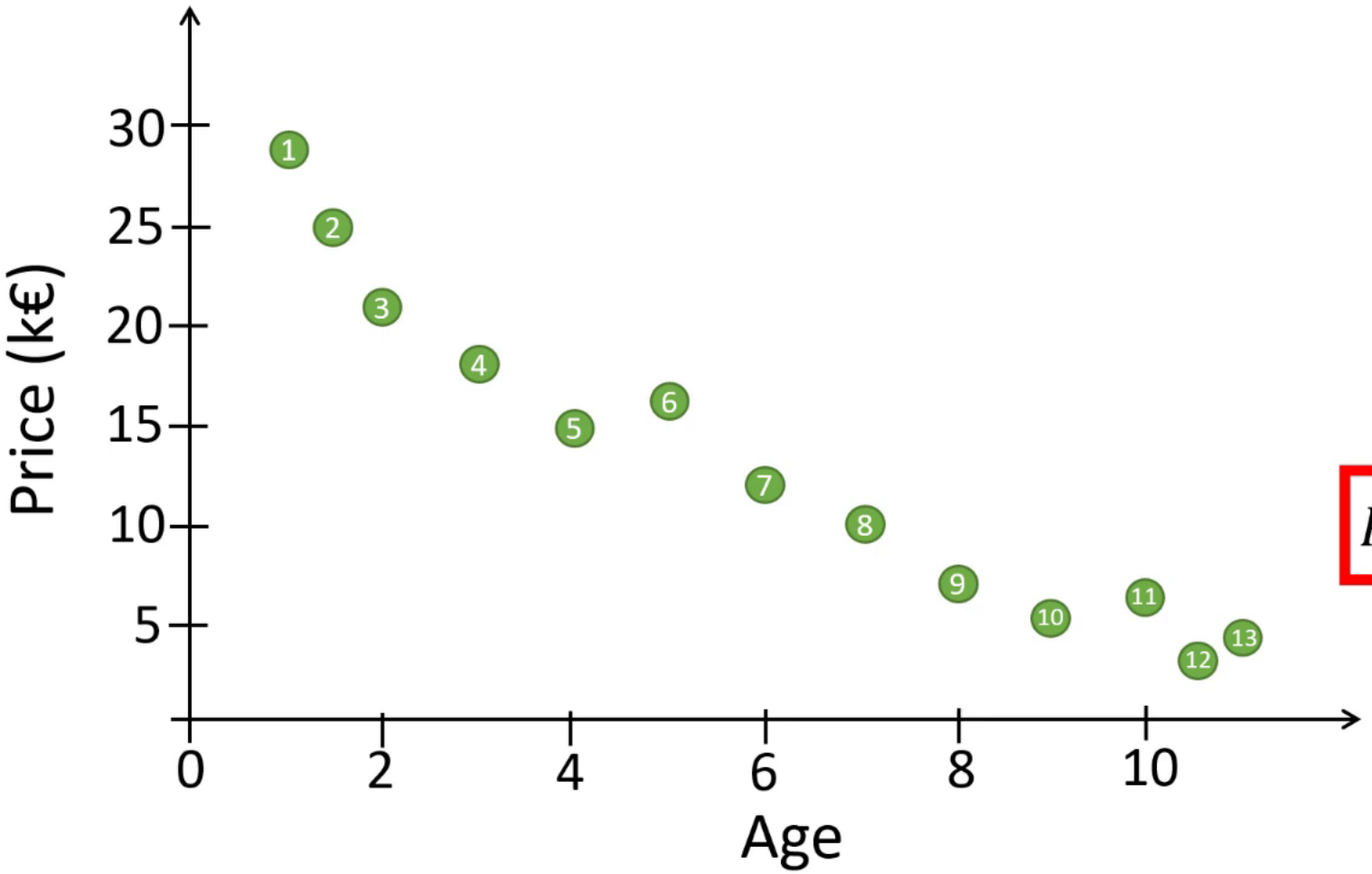


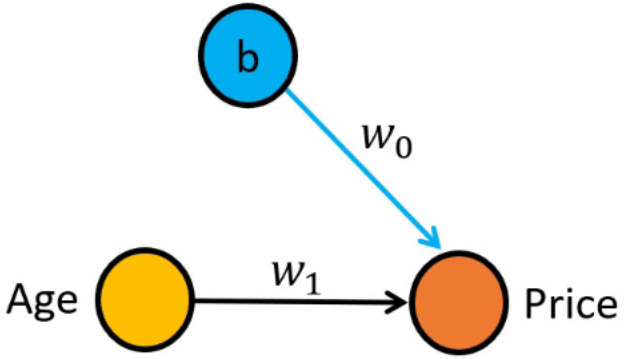


$$Price_{norm} = 0.83 - 0.88 \cdot Age_{norm}$$

By changing the value of the input node between zero and one,

A second example

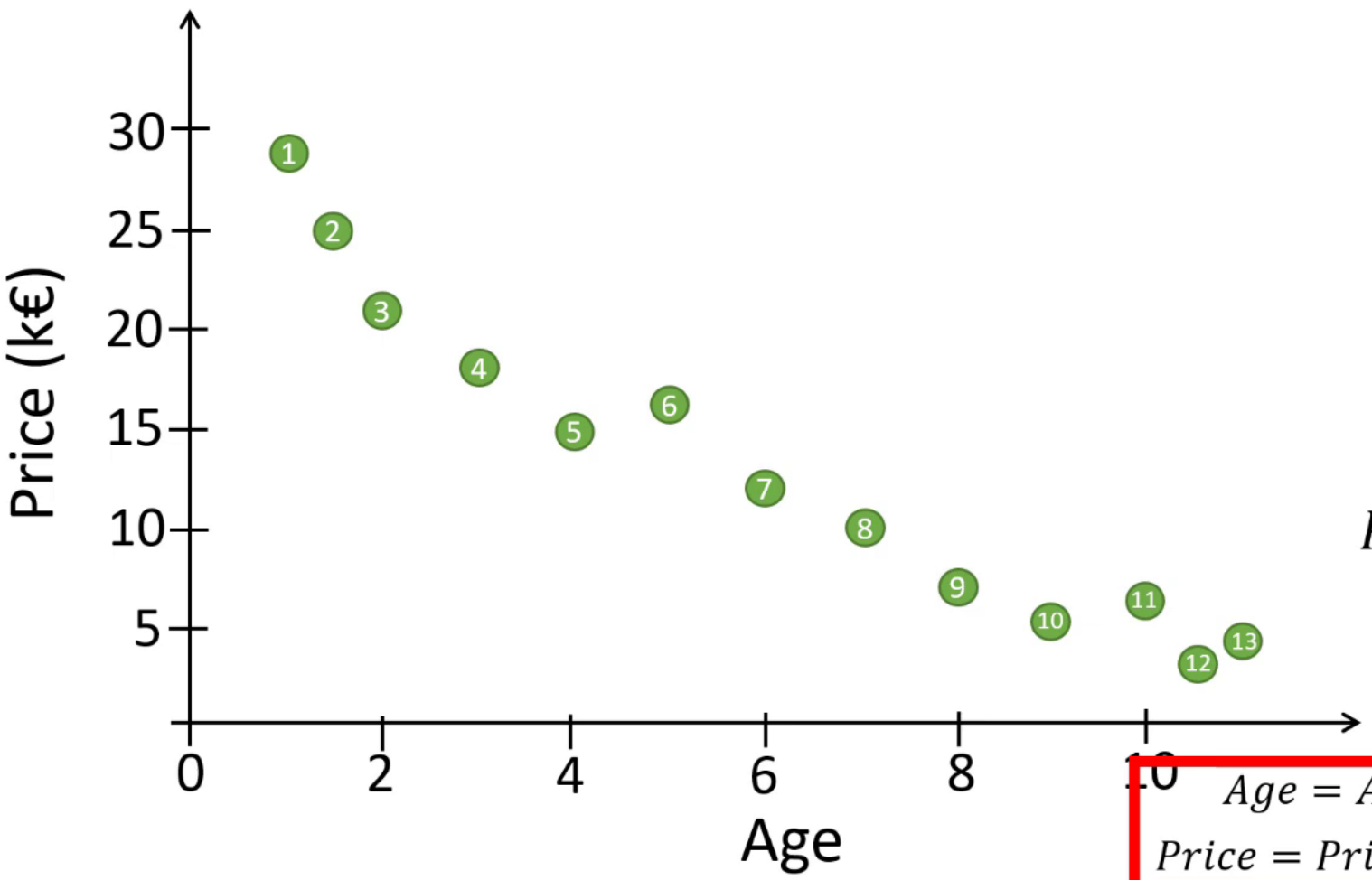


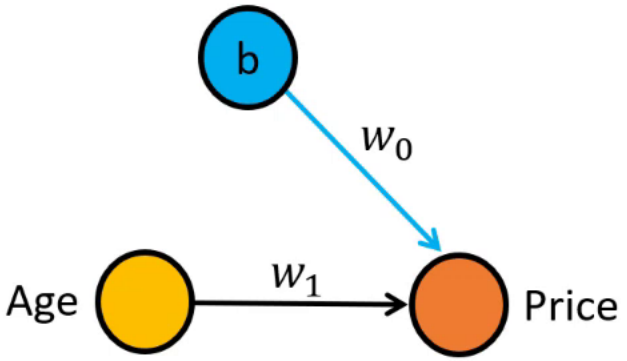


$$Price_{norm} = 0.83 - 0.88 \cdot Age_{norm}$$

we can compute the predicted normalized price.

A second example





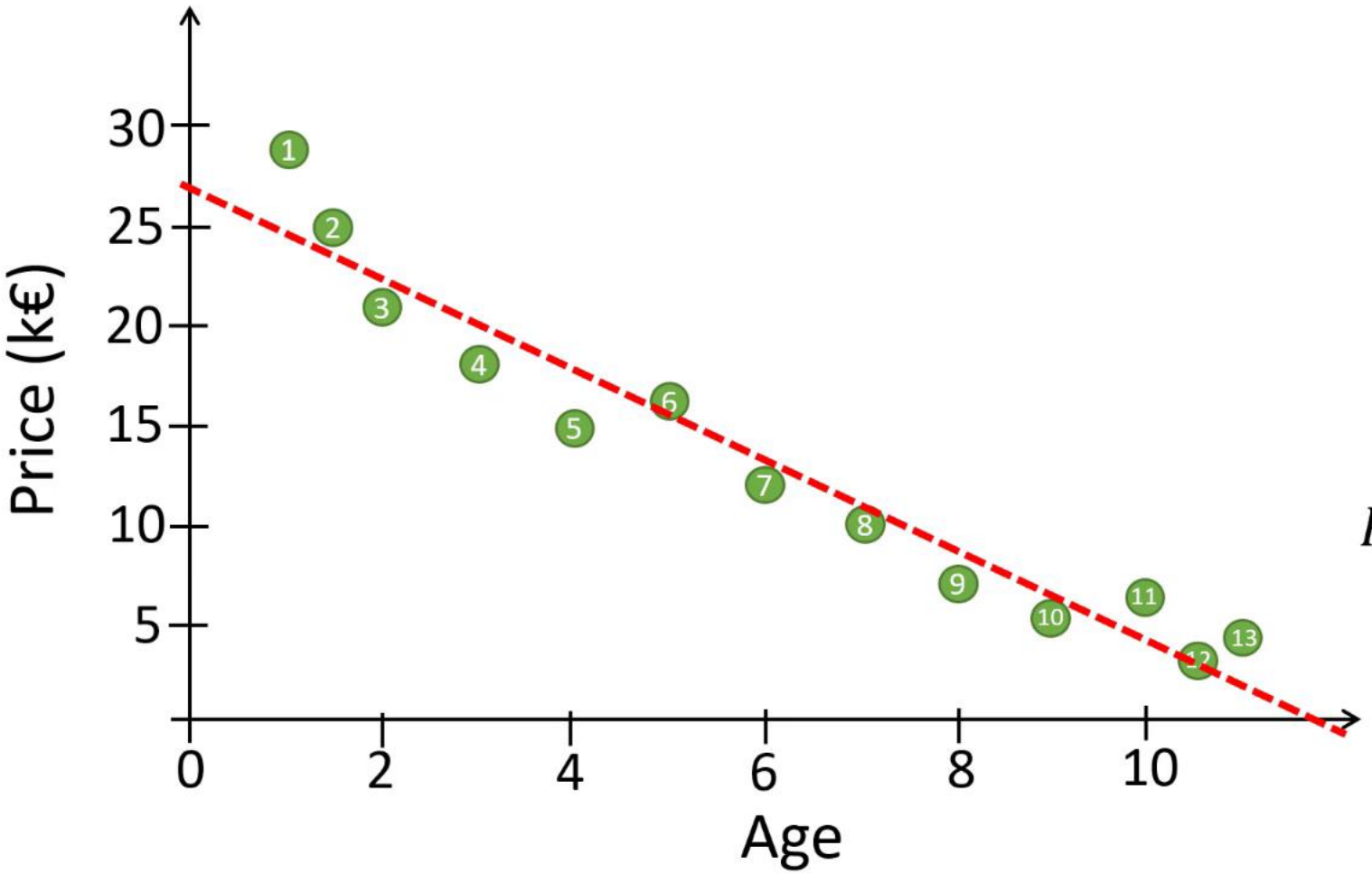
$$Price_{norm} = 0.83 - 0.88 \cdot Age_{norm}$$

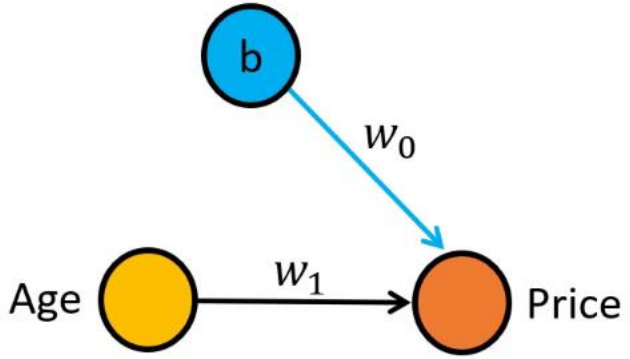
$$Age = Age_{norm}(\max(Age) - \min(Age)) + \min(Age)$$

$$Price = Price_{norm}(\max(Price) - \min(Price)) + \min(Price)$$

If we then convert the normalized age and price back to the original scales,

A second example

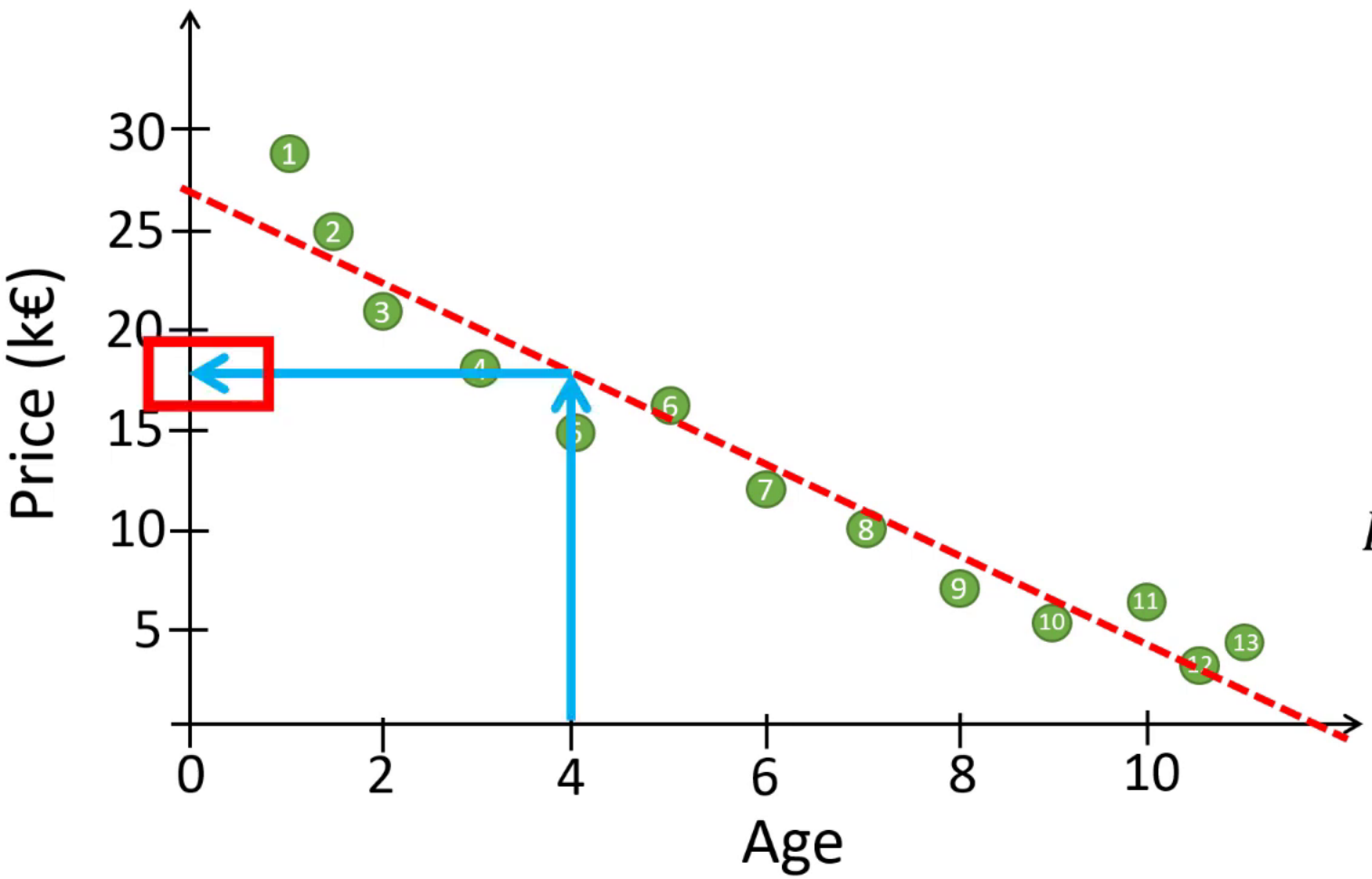


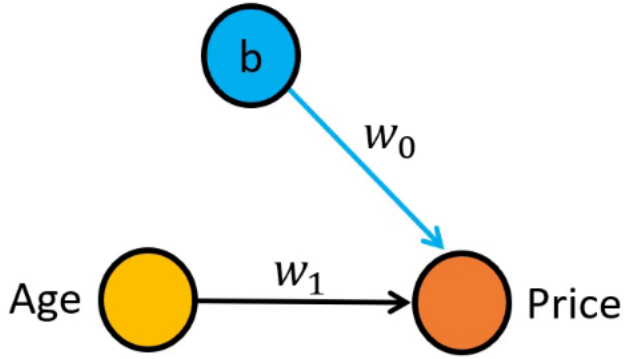


$$Price_{norm} = 0.83 - 0.88 \cdot Age_{norm}$$

we will be able to add the following line, which shows the predicted car prices by the network. This is the exact same line as we would get if we would add a regression line to the data.

A second example

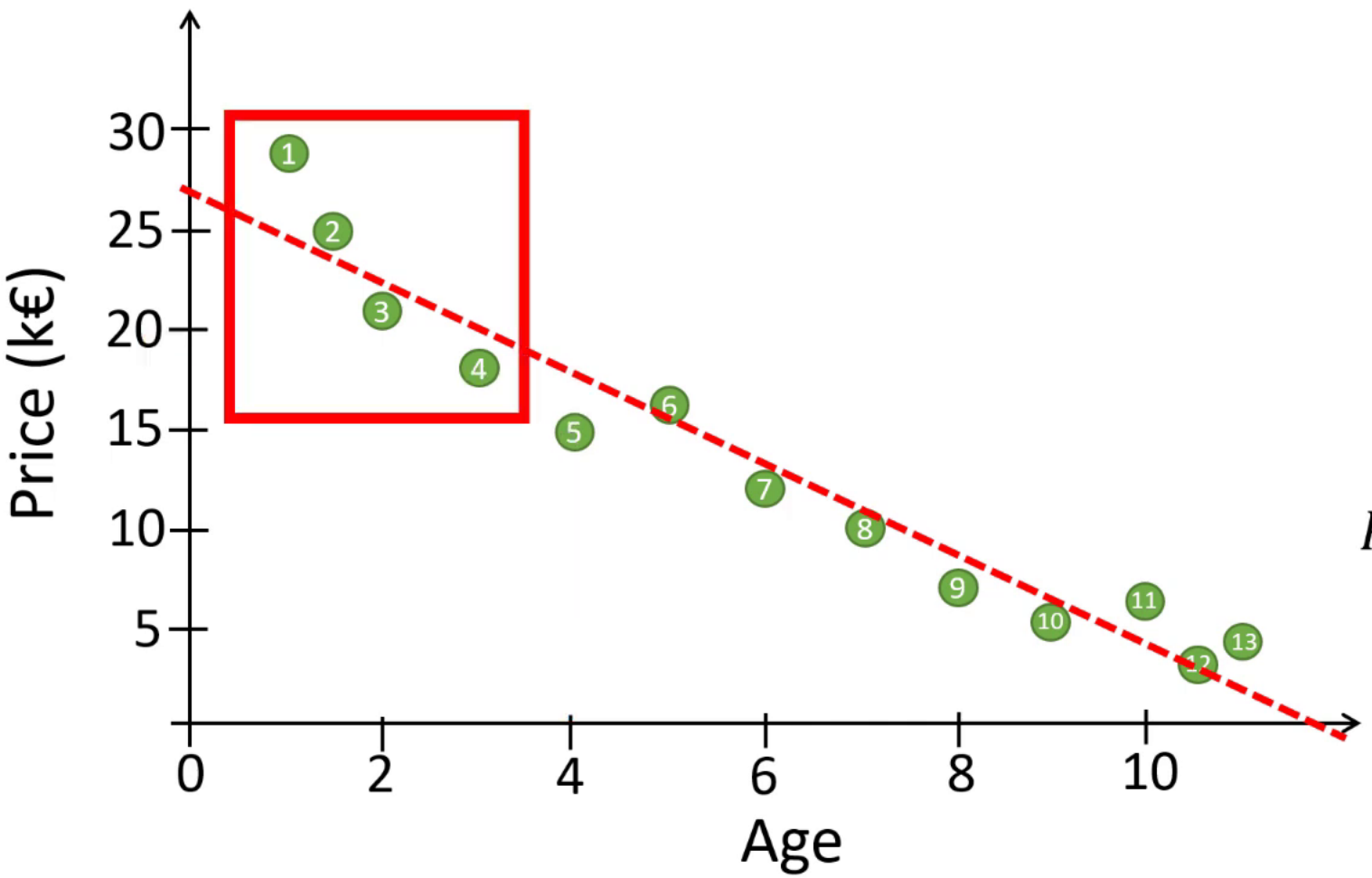


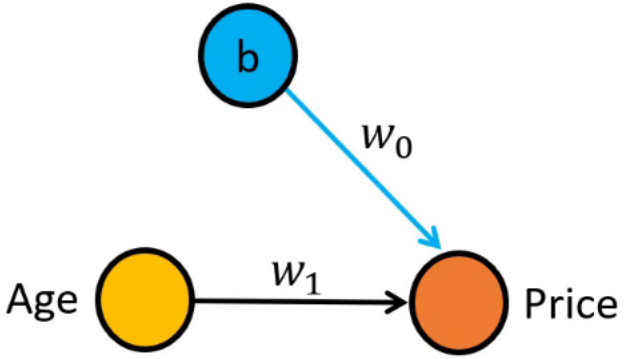


$$Price_{norm} = 0.83 - 0.88 \cdot Age_{norm}$$

For example, the network predicts that a car that is four years old will have a price of about 18.000 euros.

A second example

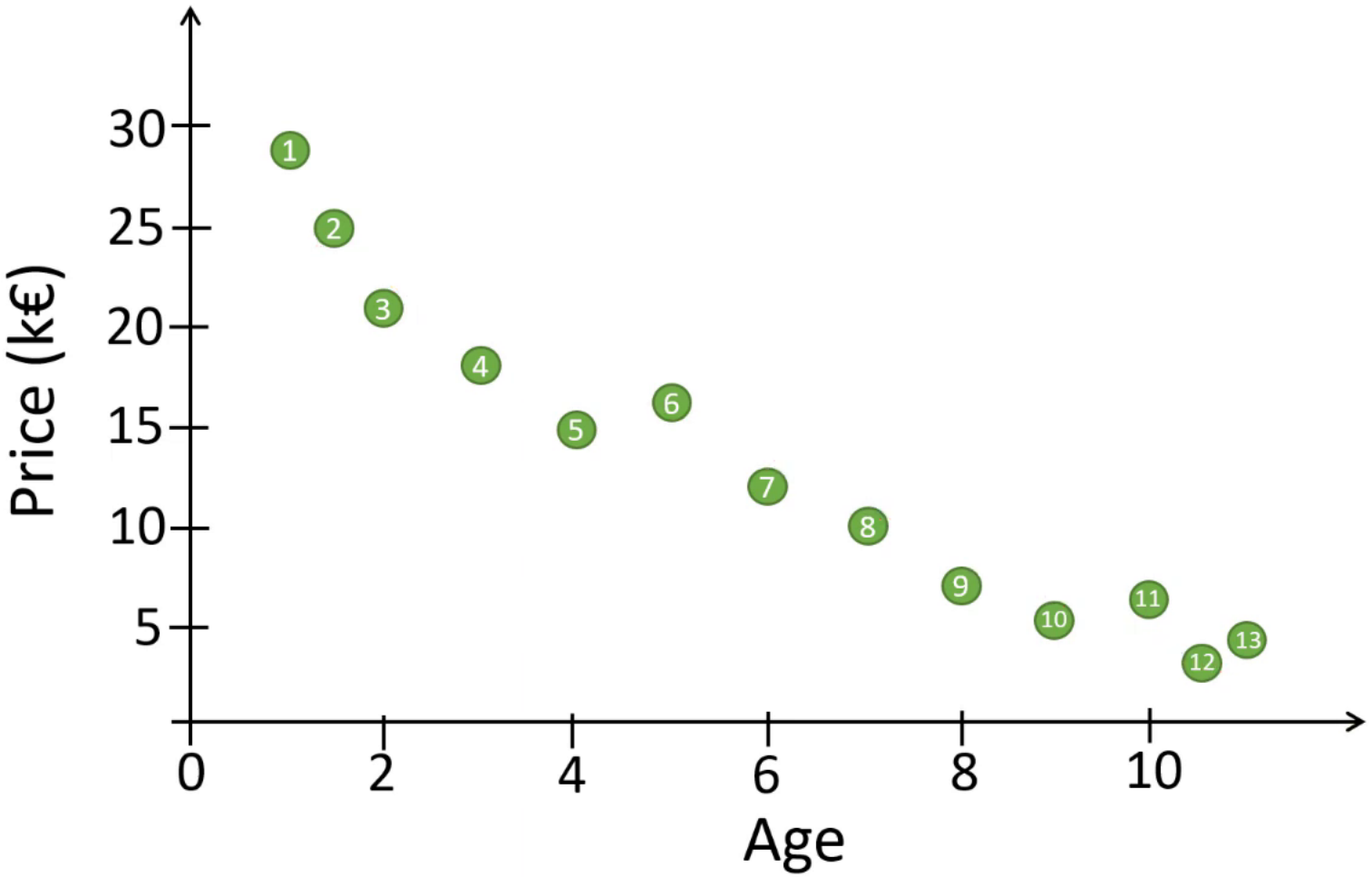


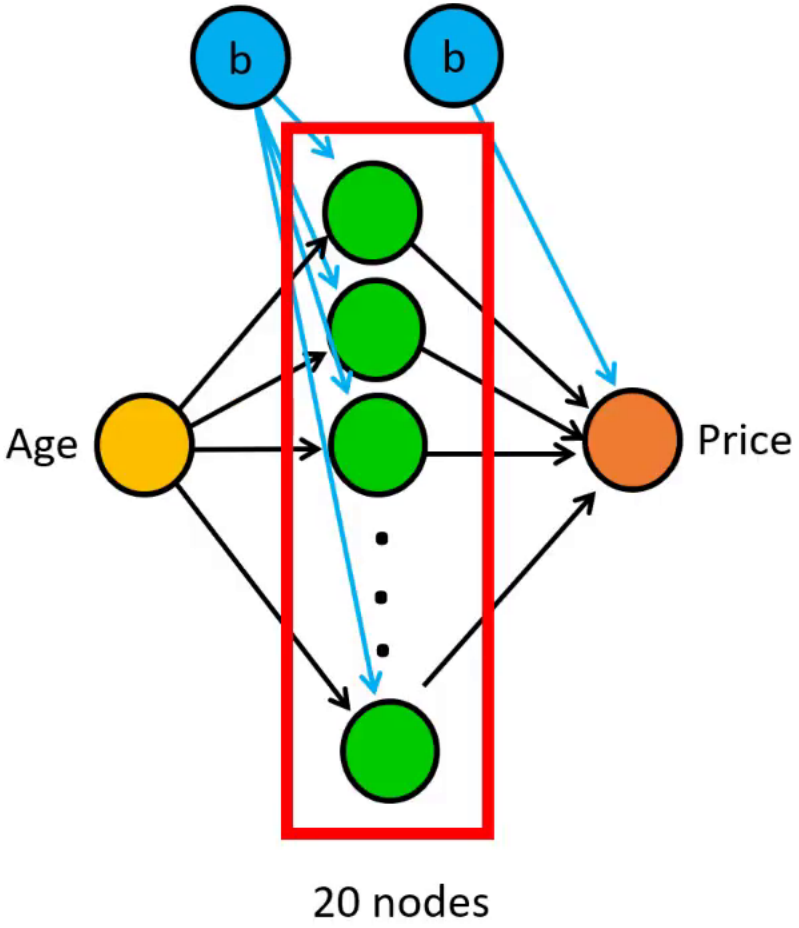


$$Price_{norm} = 0.83 - 0.88 \cdot Age_{norm}$$

Although a straight line fits quite well with the data, it does not account for the fact that the price declines faster for newer cars,

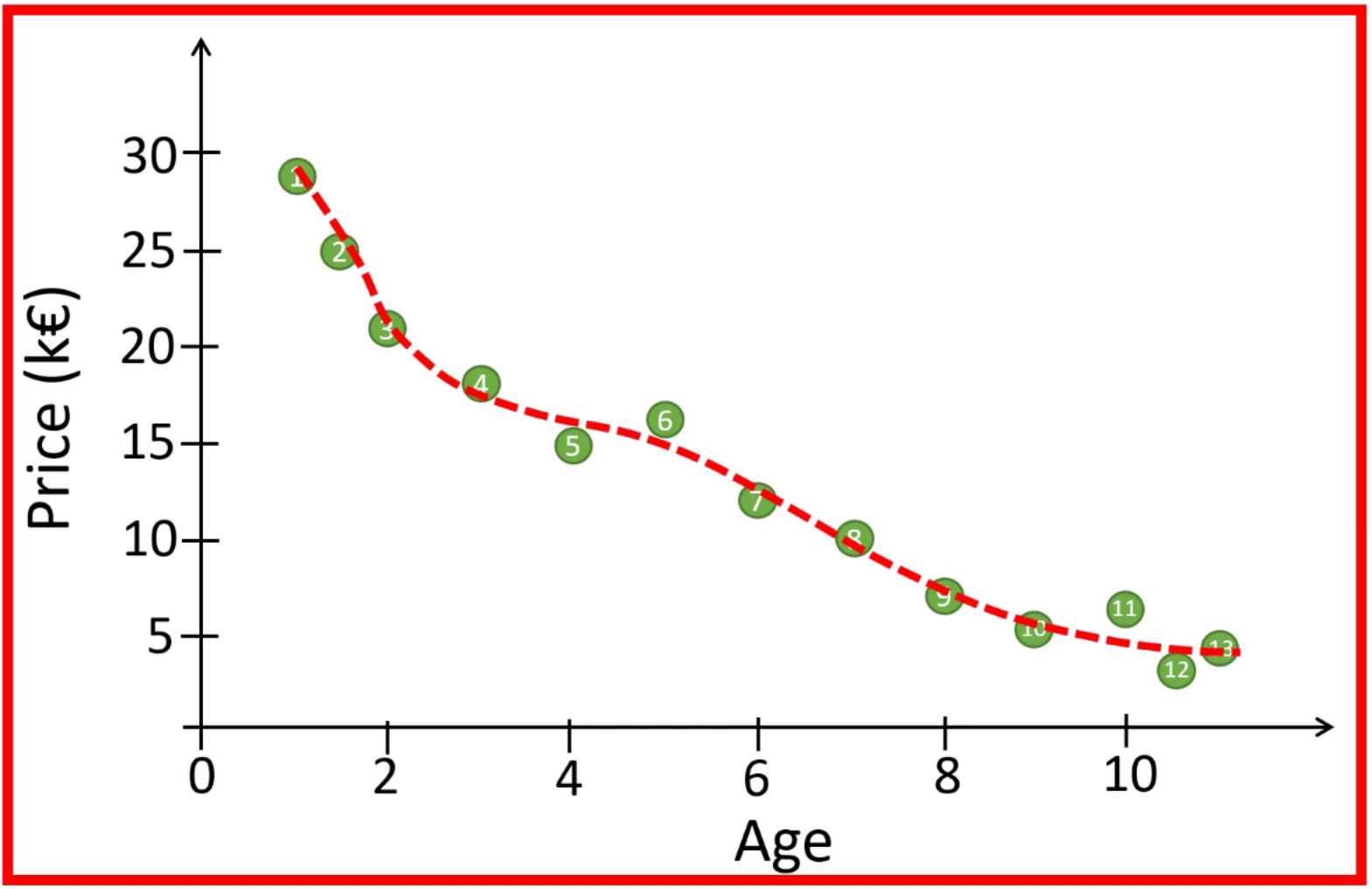
A network with one hidden layer

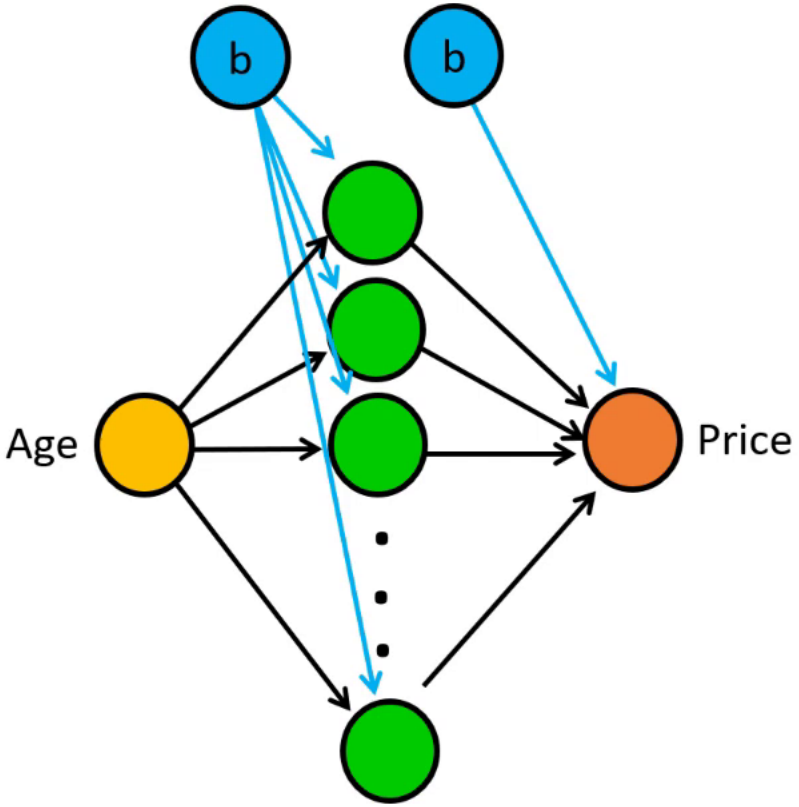




If we instead use a network with one hidden layer that uses the sigmoid activation function to compute the values in the hidden nodes,

A network with one hidden layer

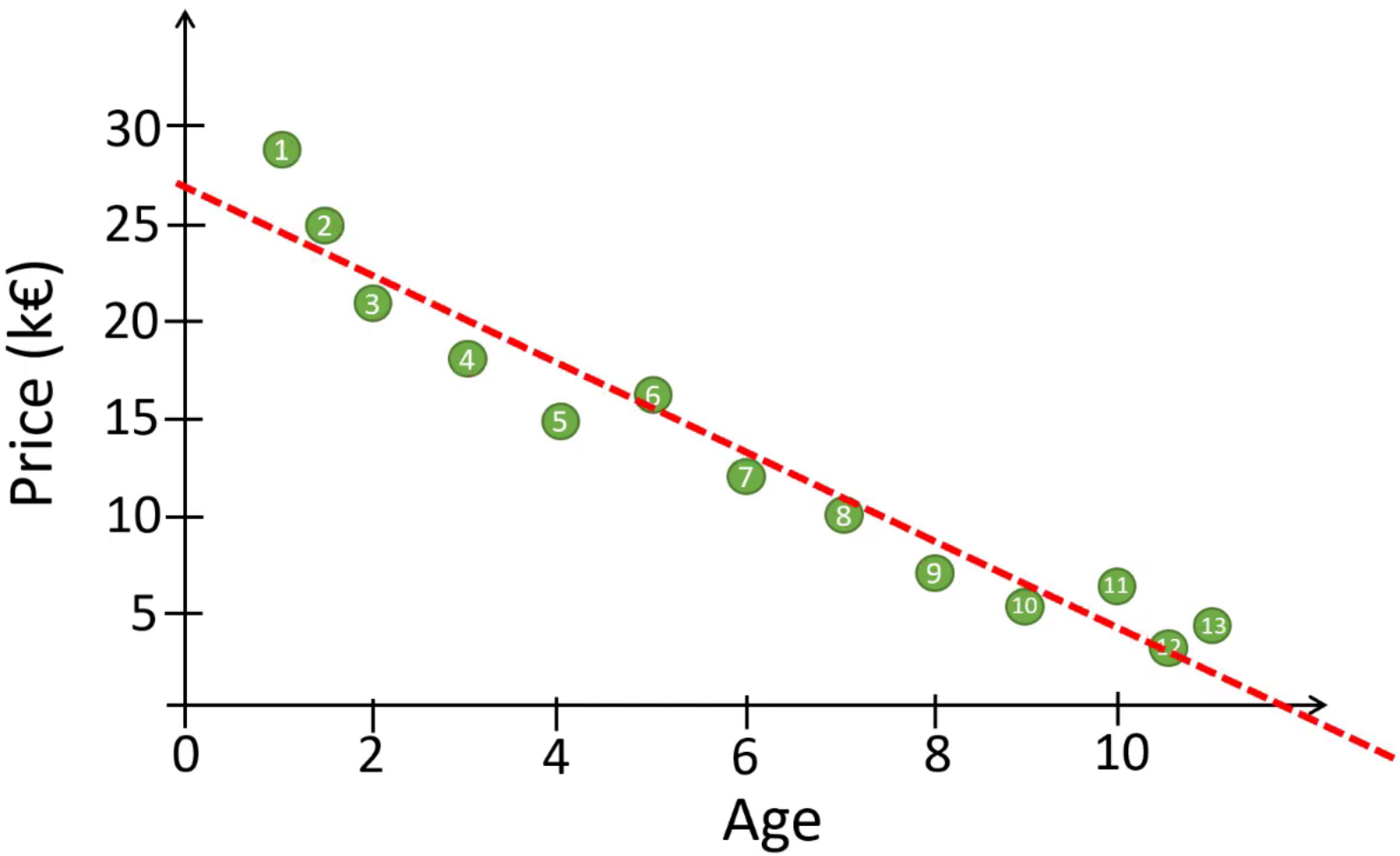


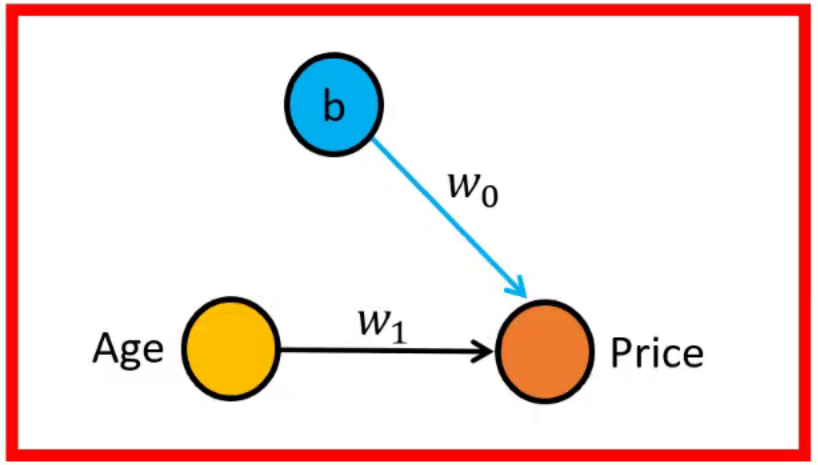


20 nodes

the curve now fits a lot better to the data. We see that the data points are now much closer to the curve,

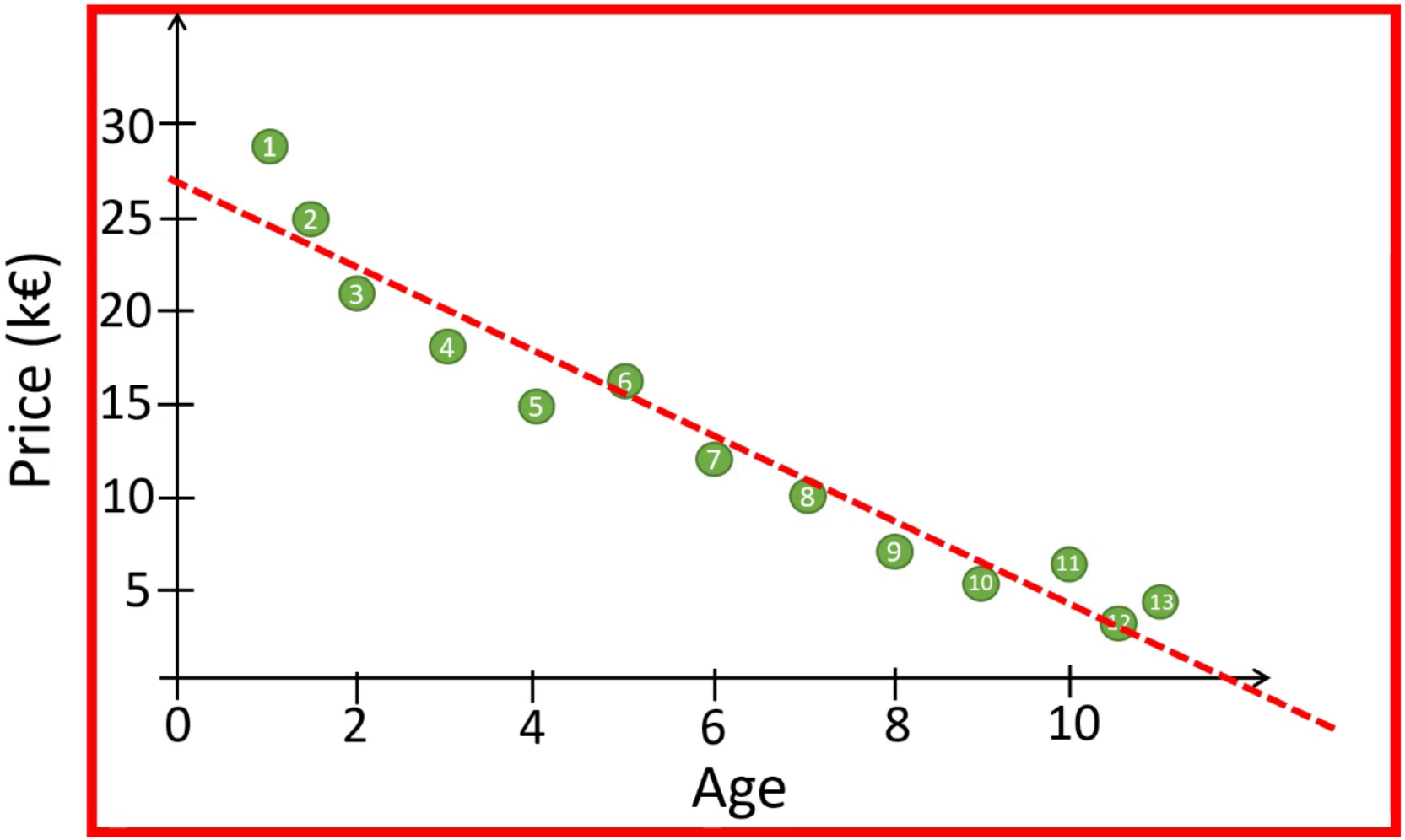
A network with no hidden layer

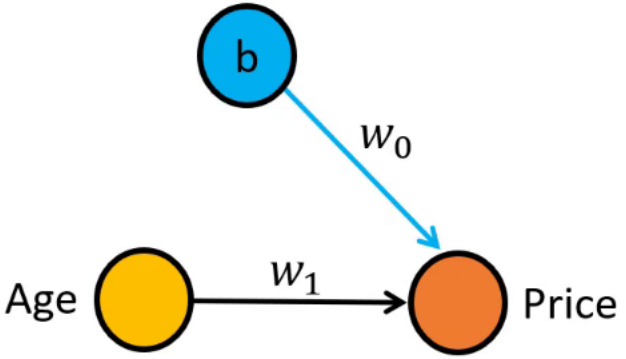




compared to if we use a network without any hidden layer,

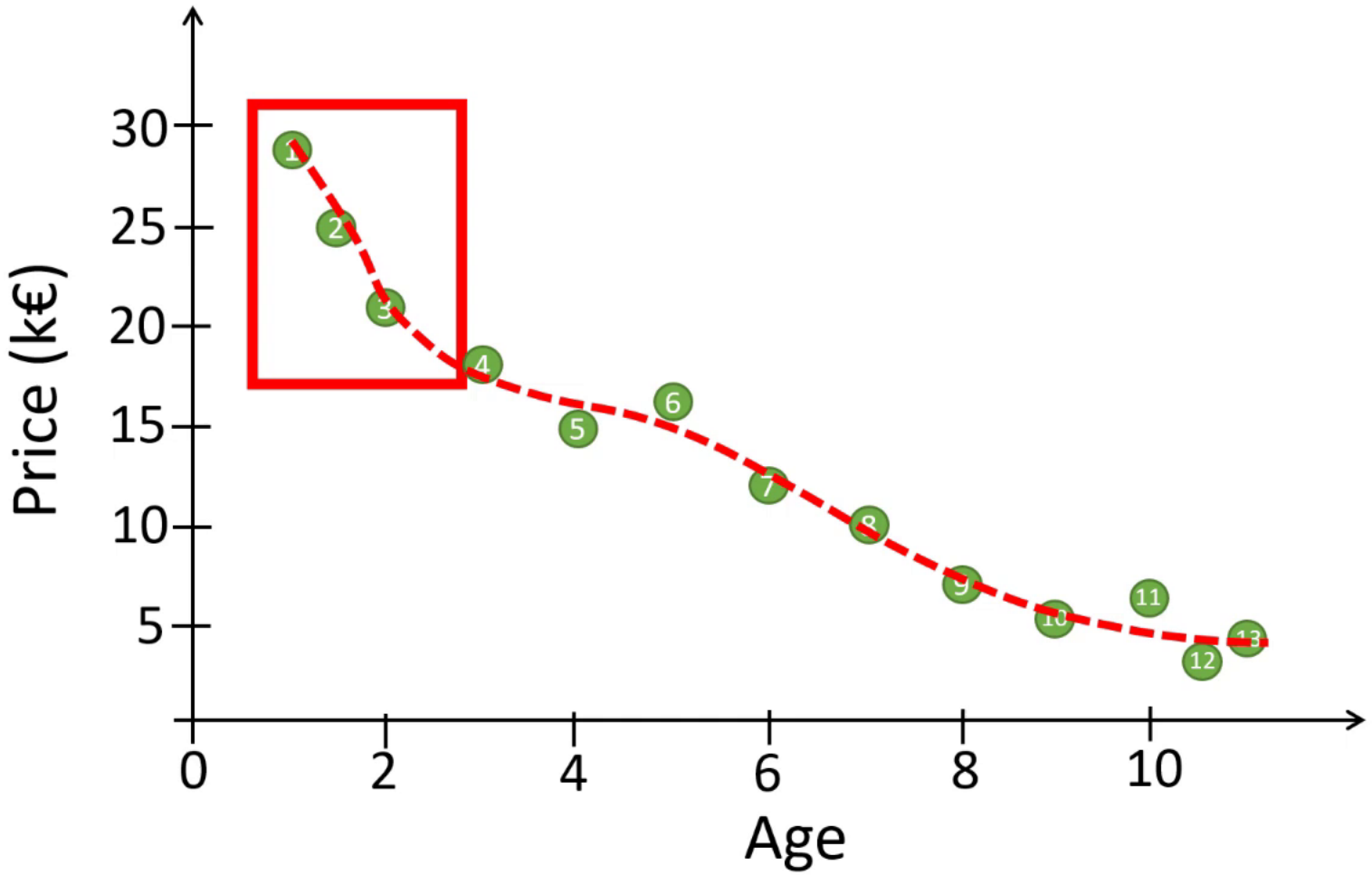
A network with no hidden layer

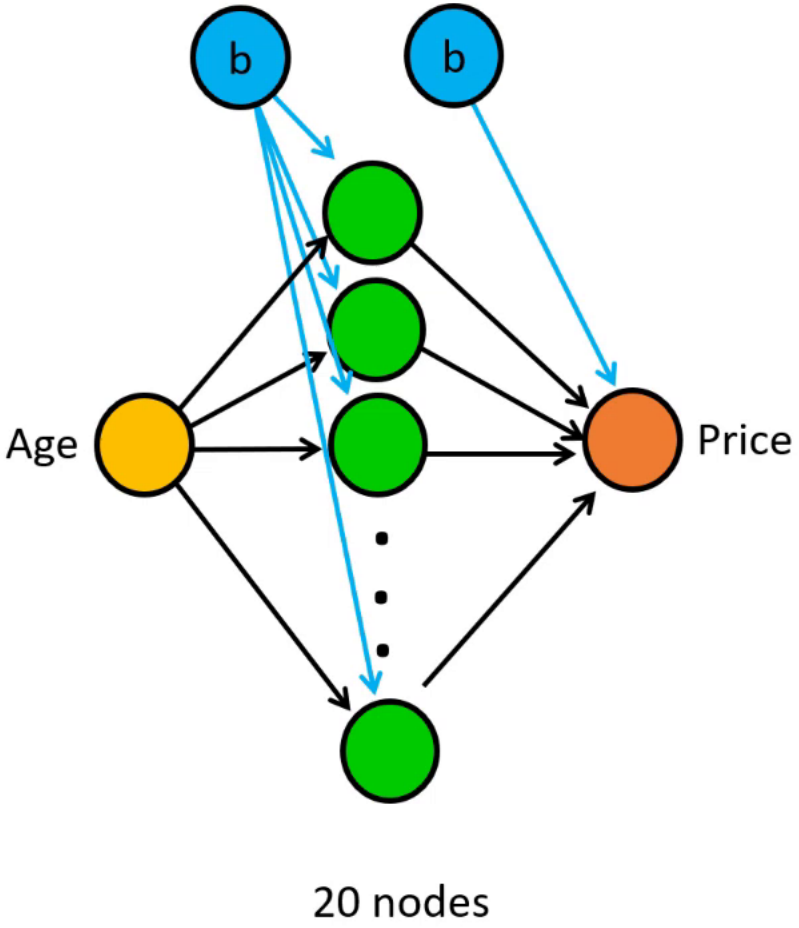




which can only generate a straight line.

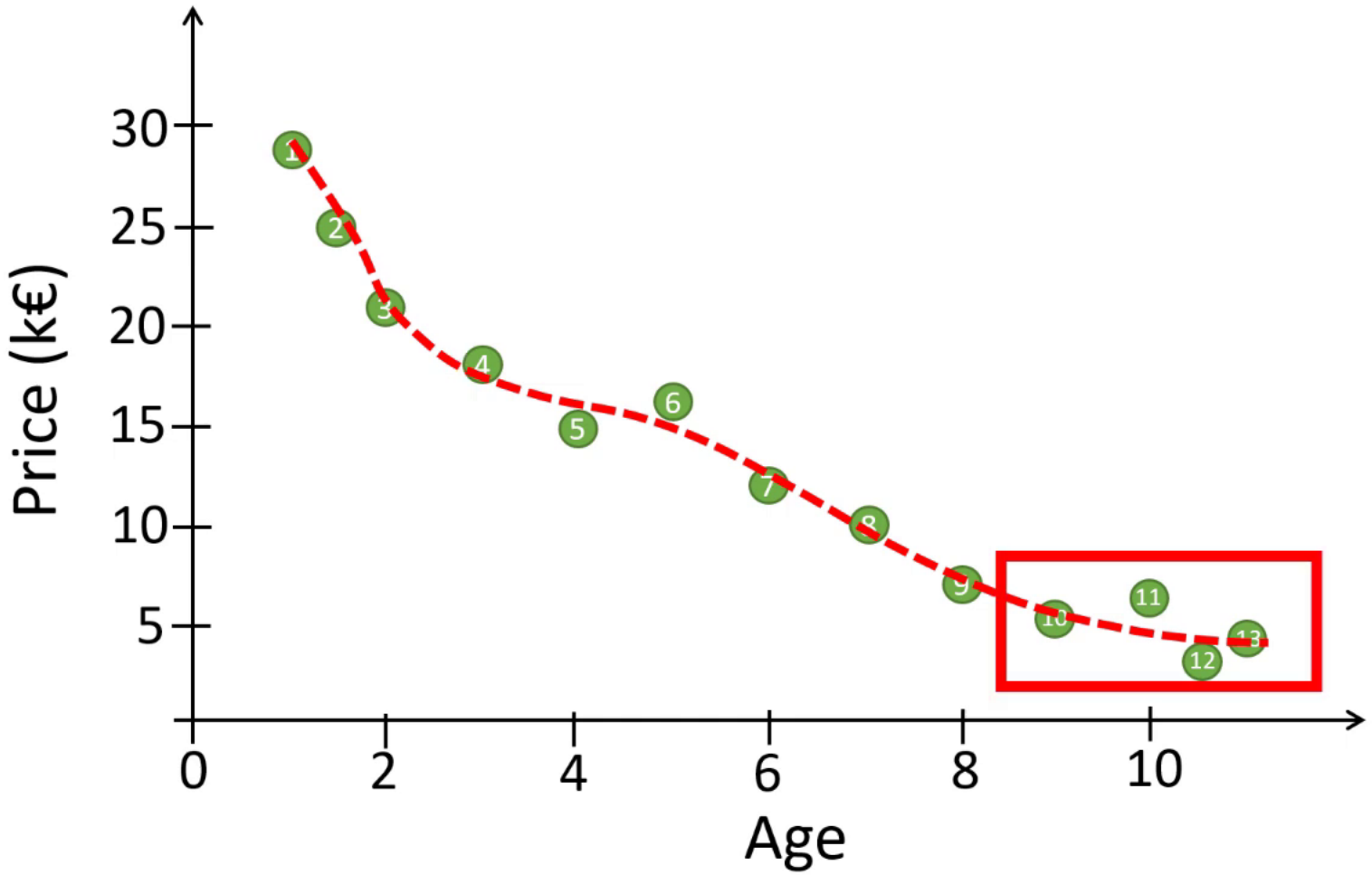
A network with one hidden layer

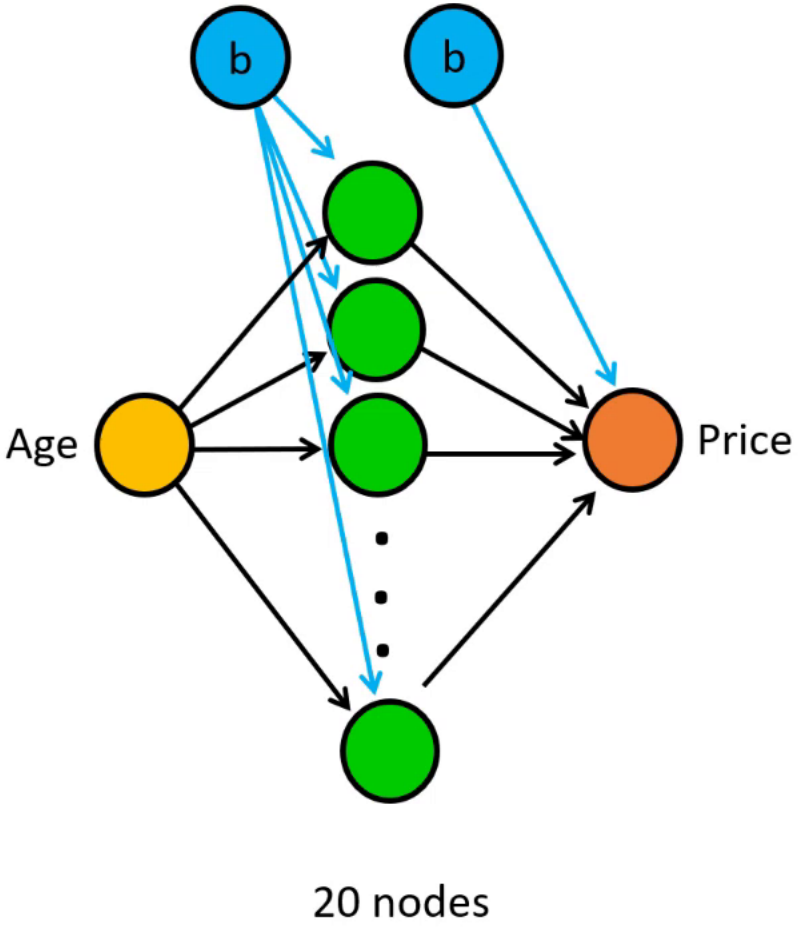




The network now predicts a fast drop in price for newer cars,

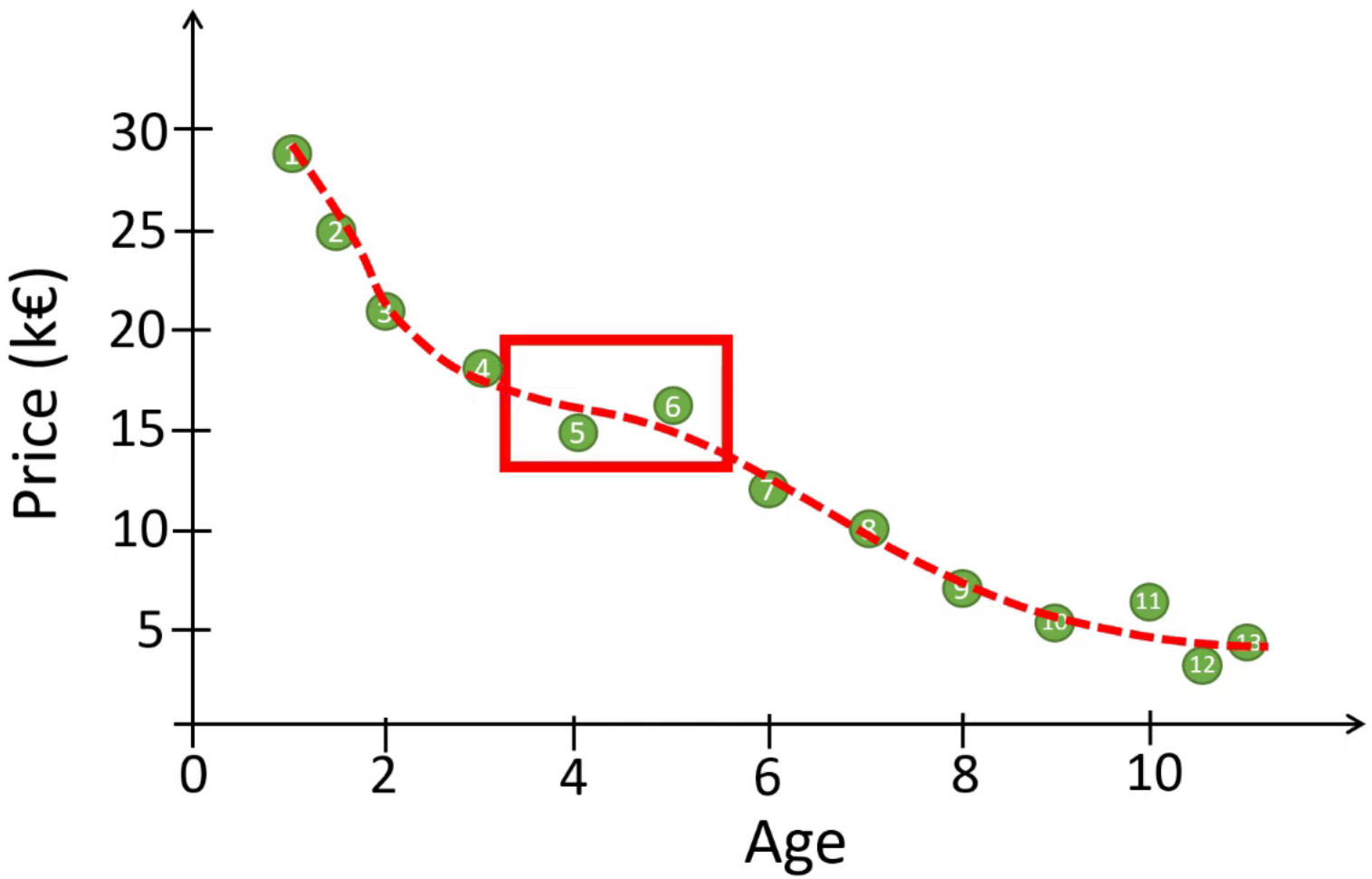
A network with one hidden layer

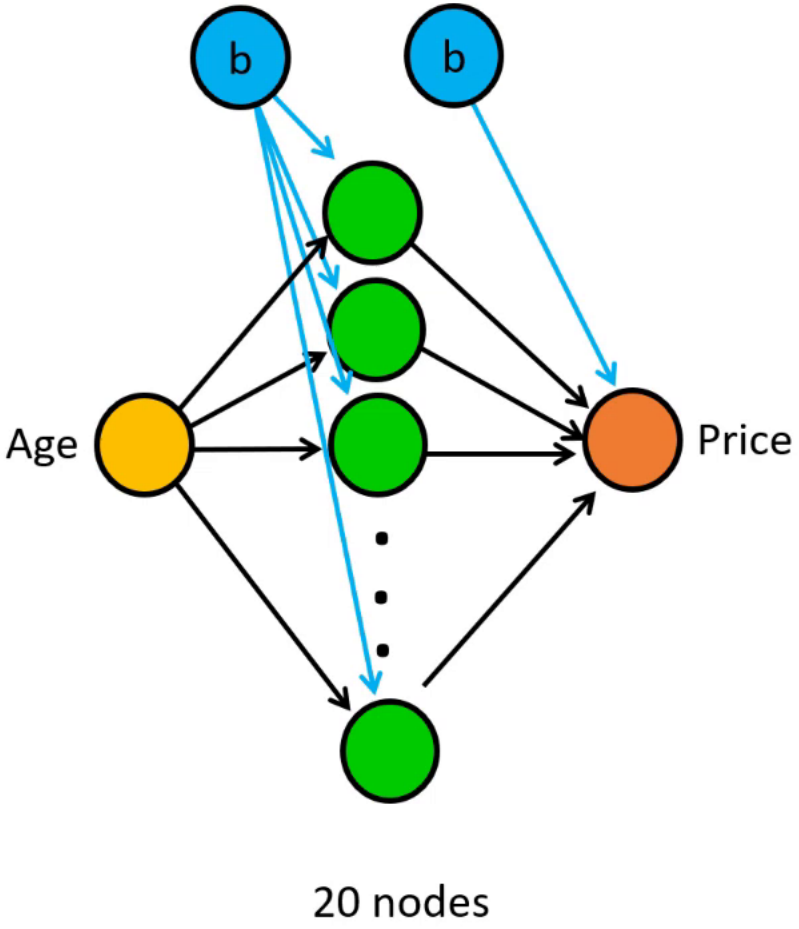




and a slower price fall for older cars.

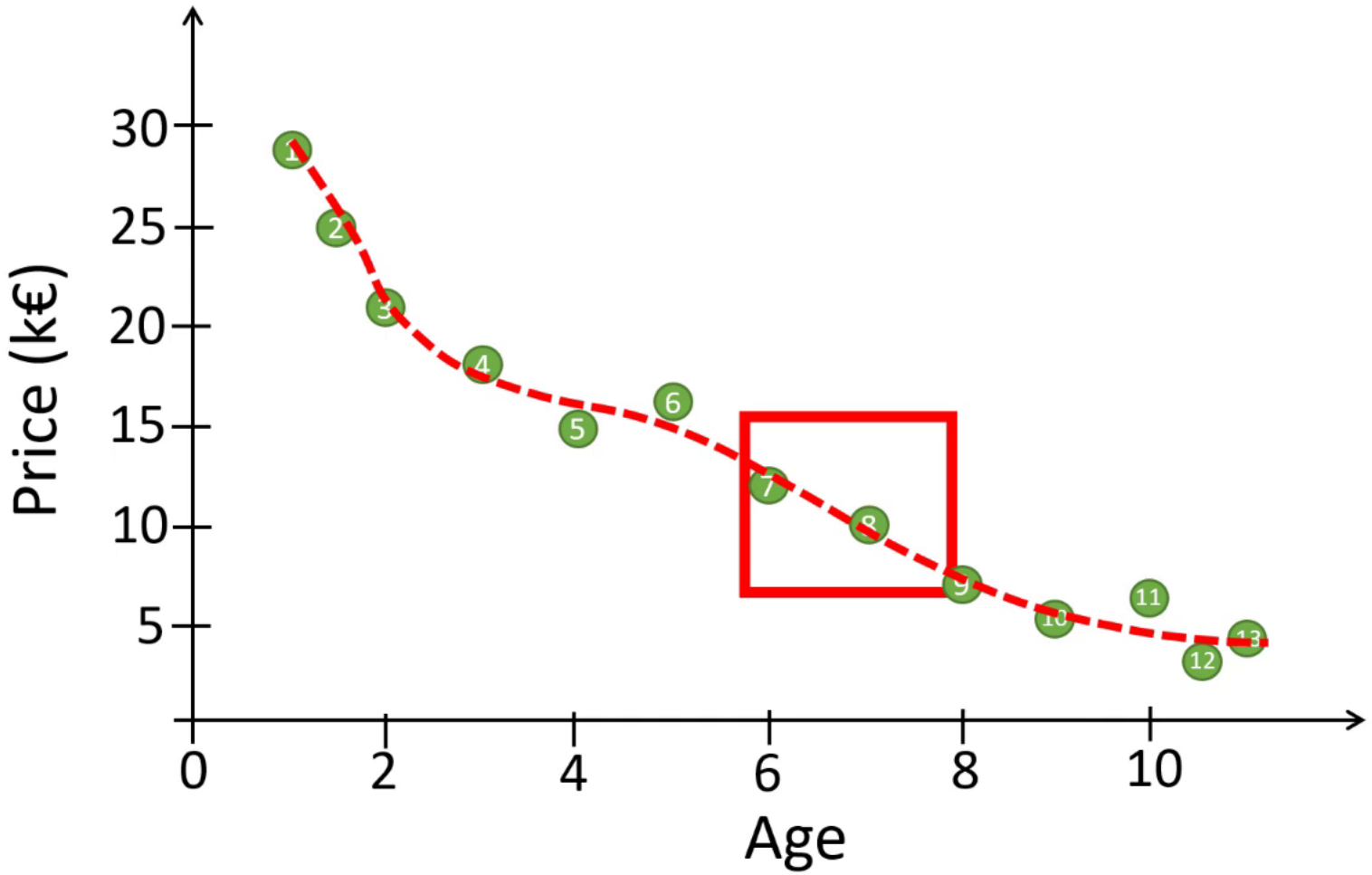
A network with one hidden layer

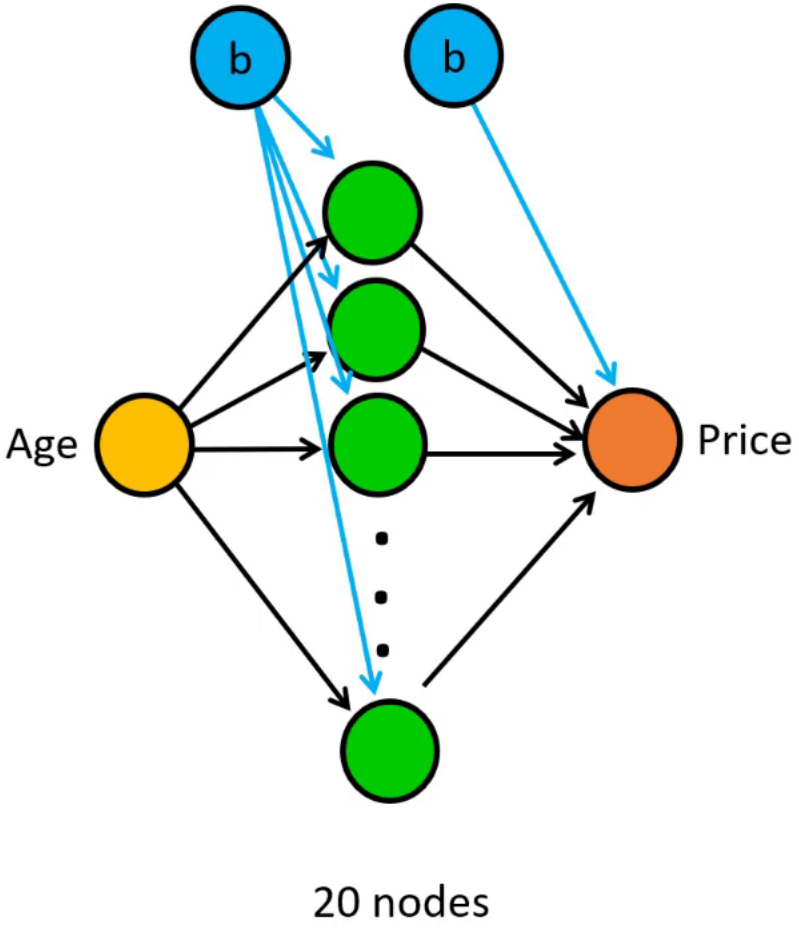




However, it is a bit strange that the price declines slower here,

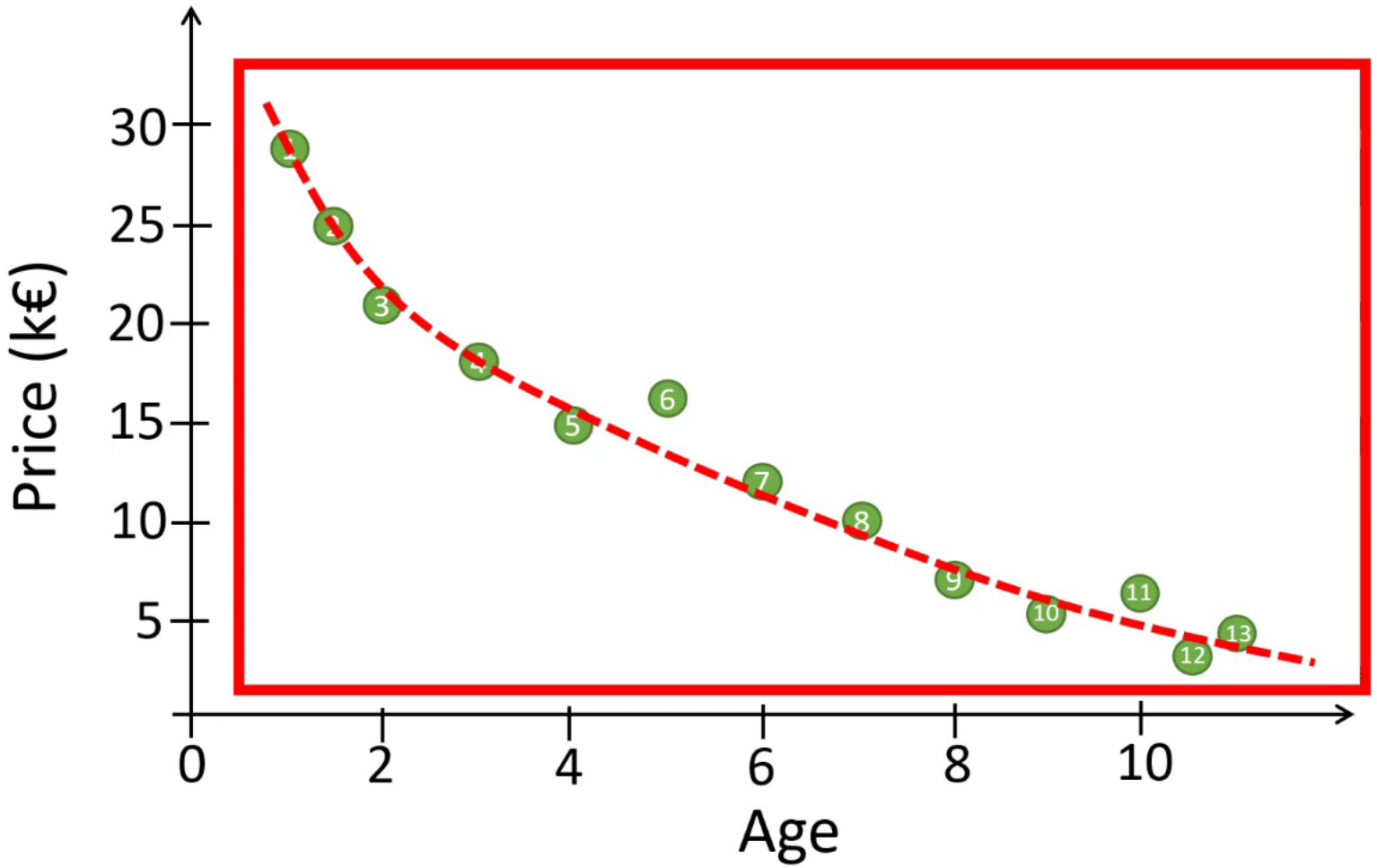
A network with one hidden layer

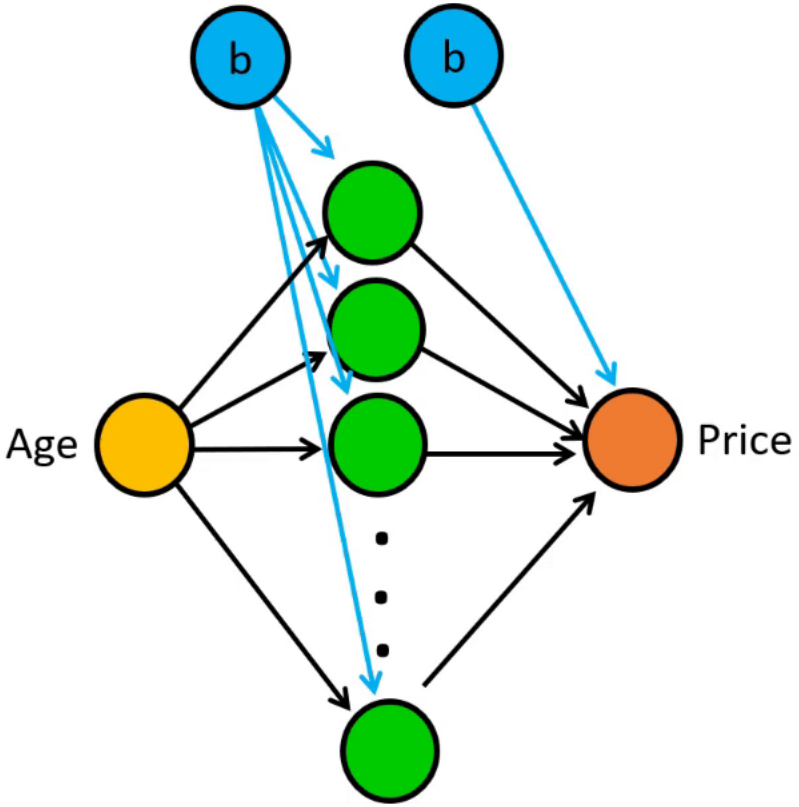




compared to here.

A network with one hidden layer

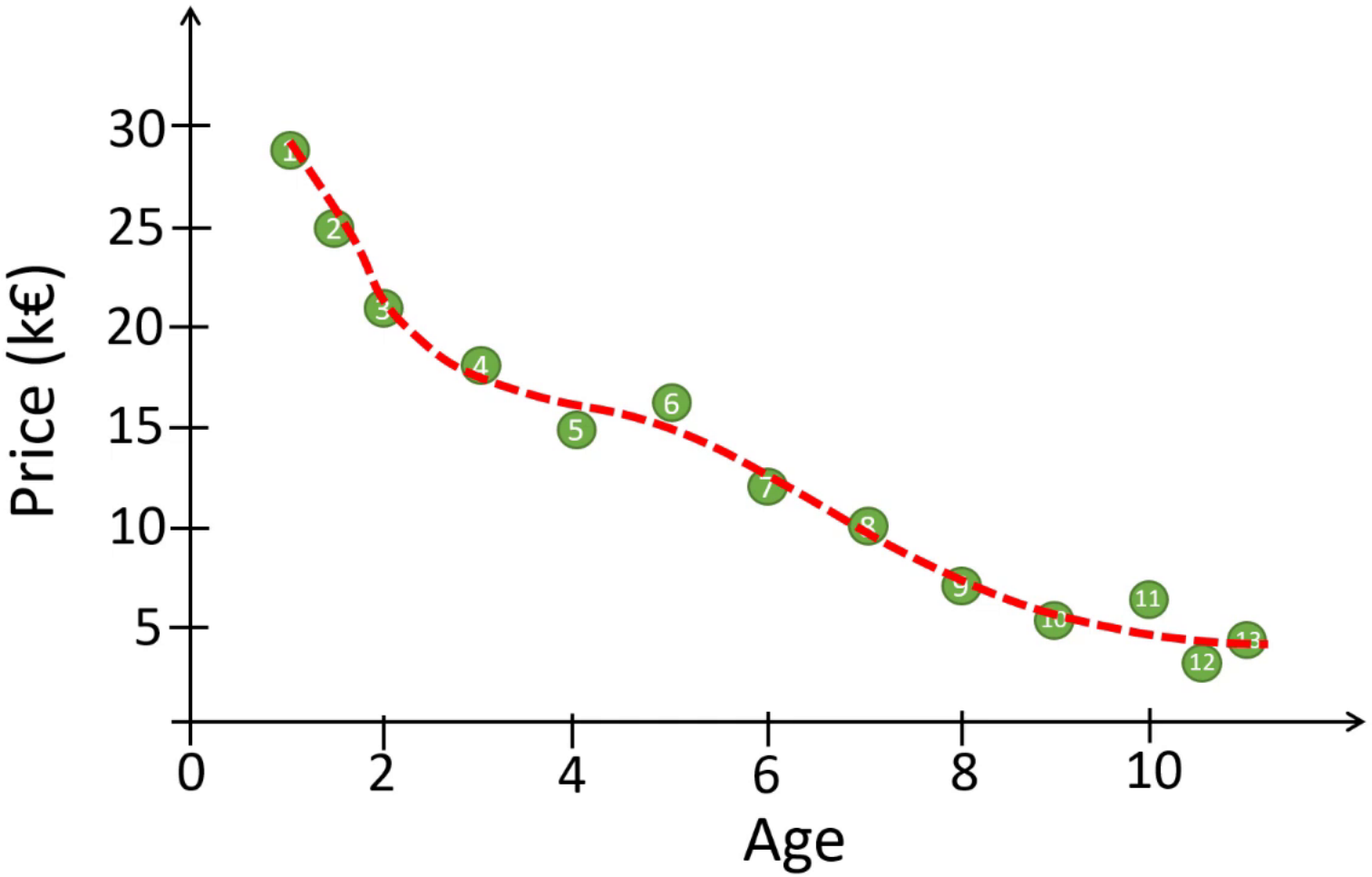




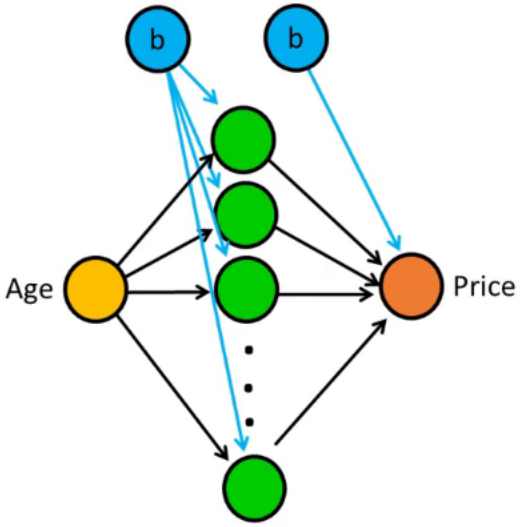
20 nodes

A curve like this seems more reasonable. We will come back to this kind of curve later on when we apply the method of early stopping.

A network with one hidden layer



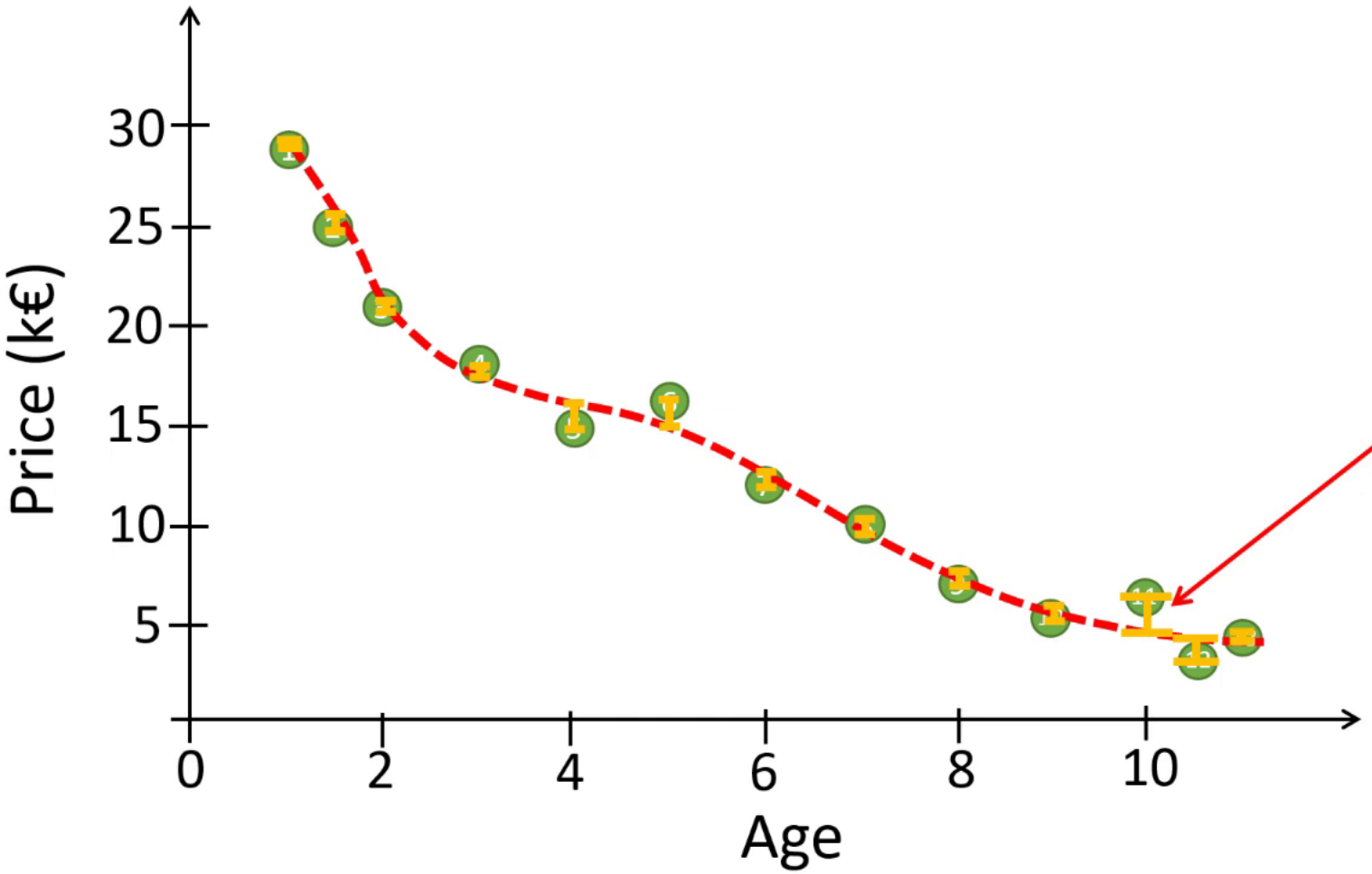
$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$



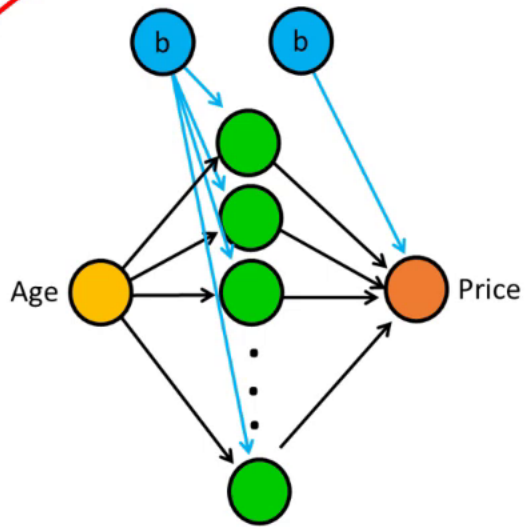
20 nodes

Anyway, to measure how well the network fits the training data, we can, for example, calculate the sum of the squared errors.

A network with one hidden layer



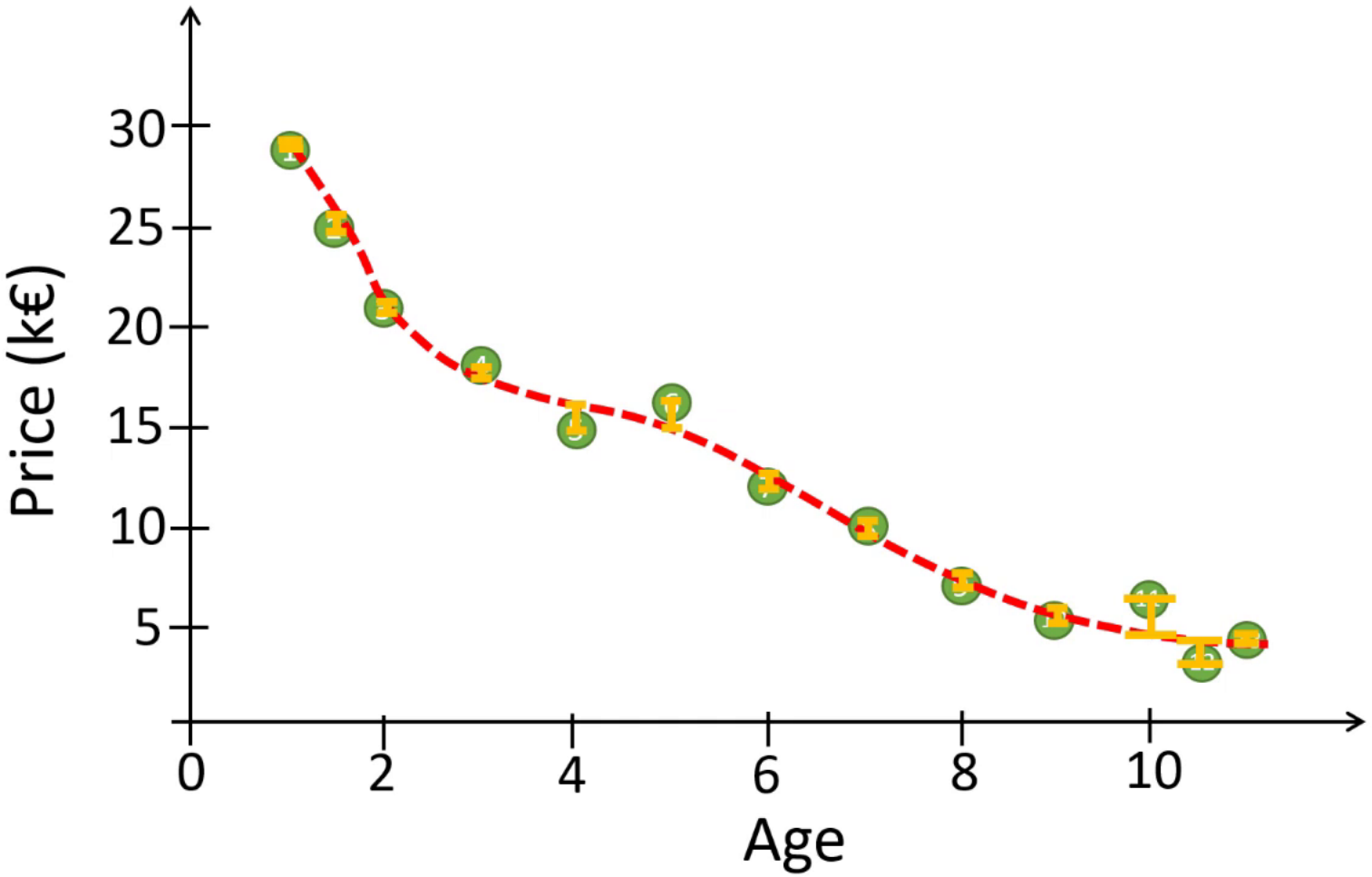
$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$



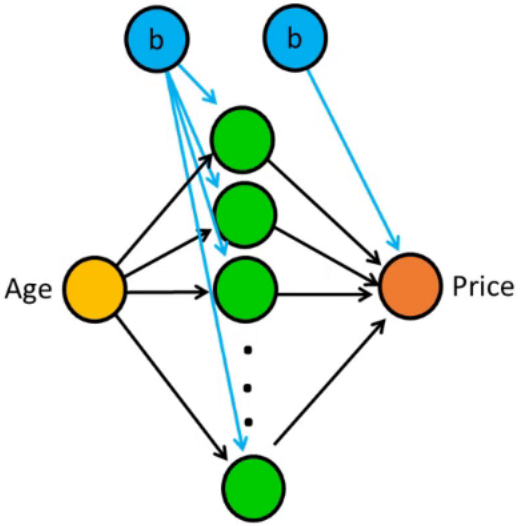
20 nodes

An error is the difference between the observed price and the predicted price by the network.

A network with one hidden layer



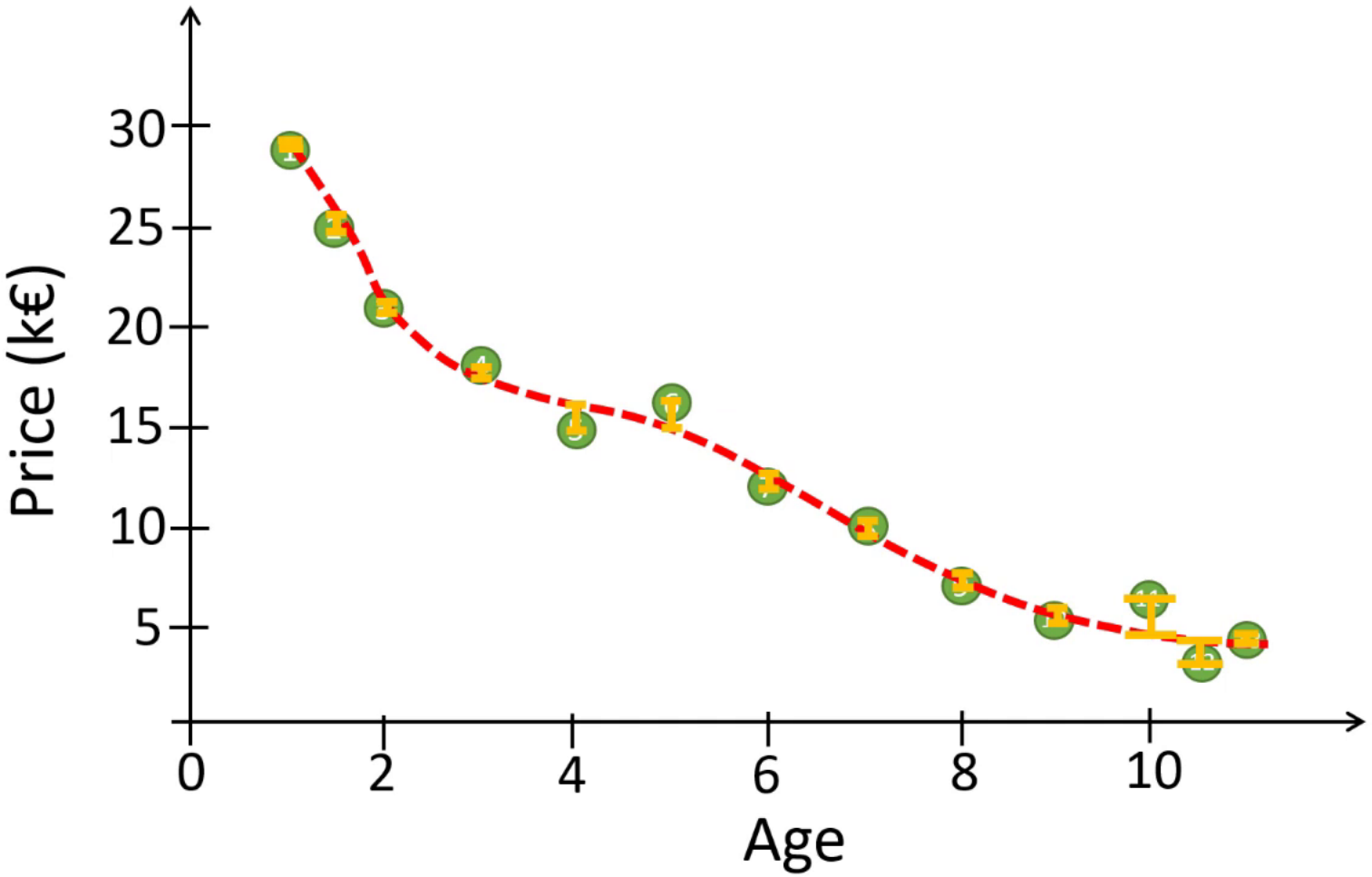
$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$



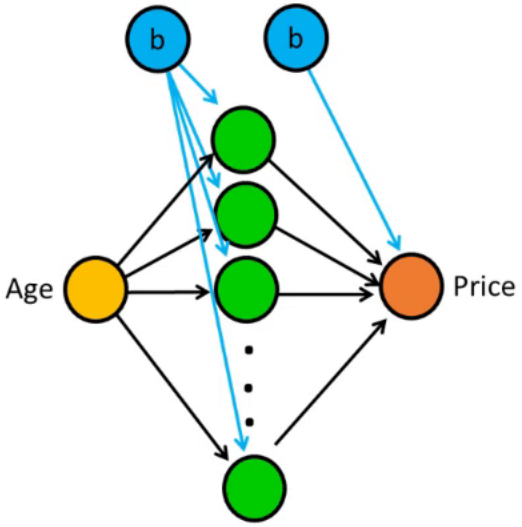
20 nodes

squared,

A network with one hidden layer



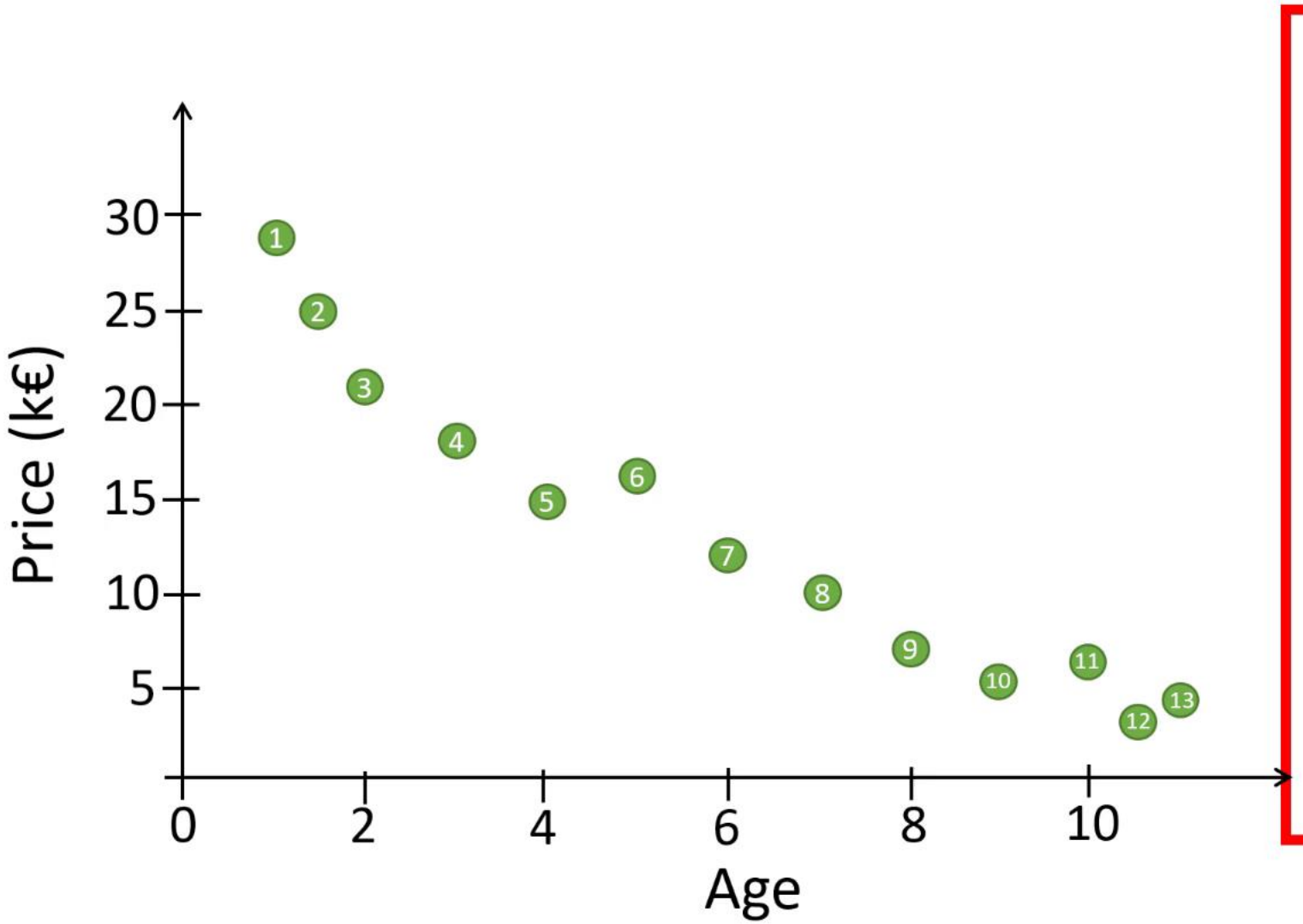
$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 7.7$$

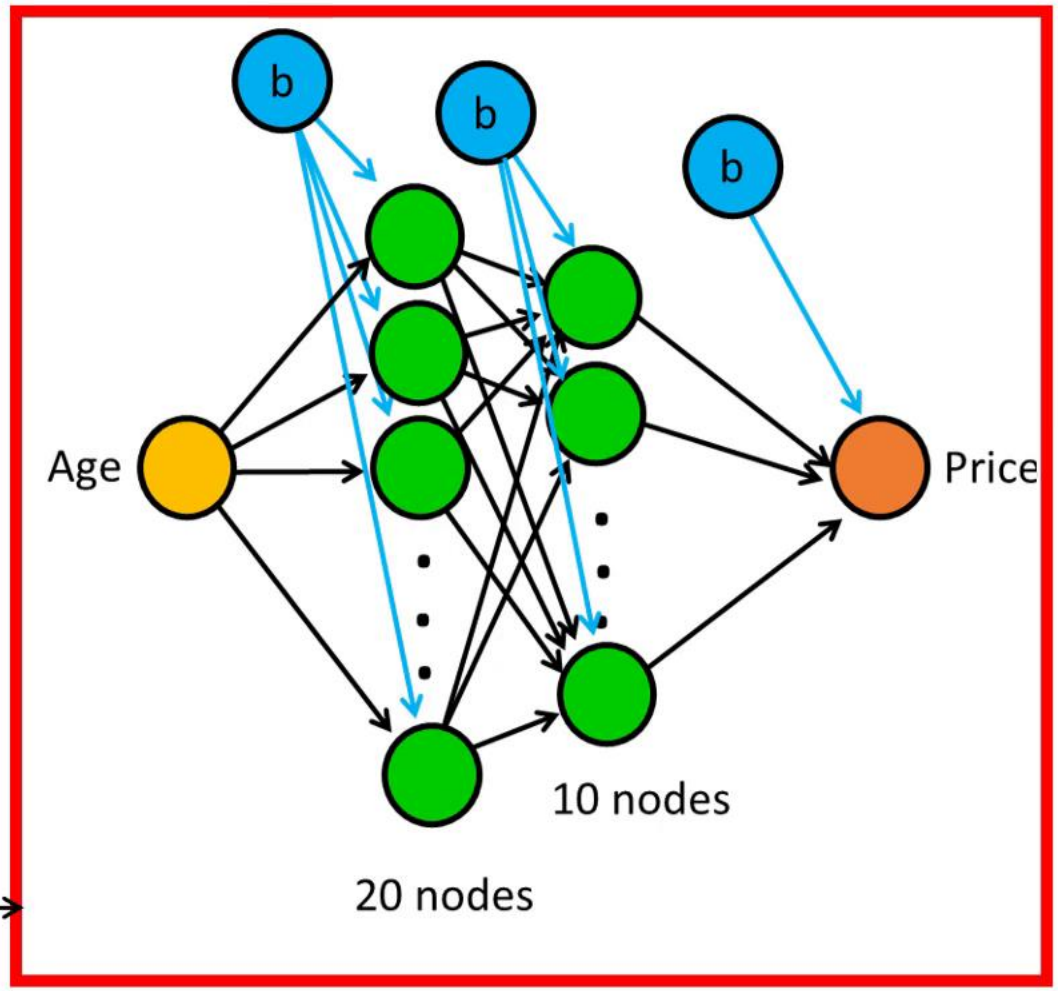


20 nodes

then we would get a value of about 7.7. The smaller this value is, the better the model fits the data.

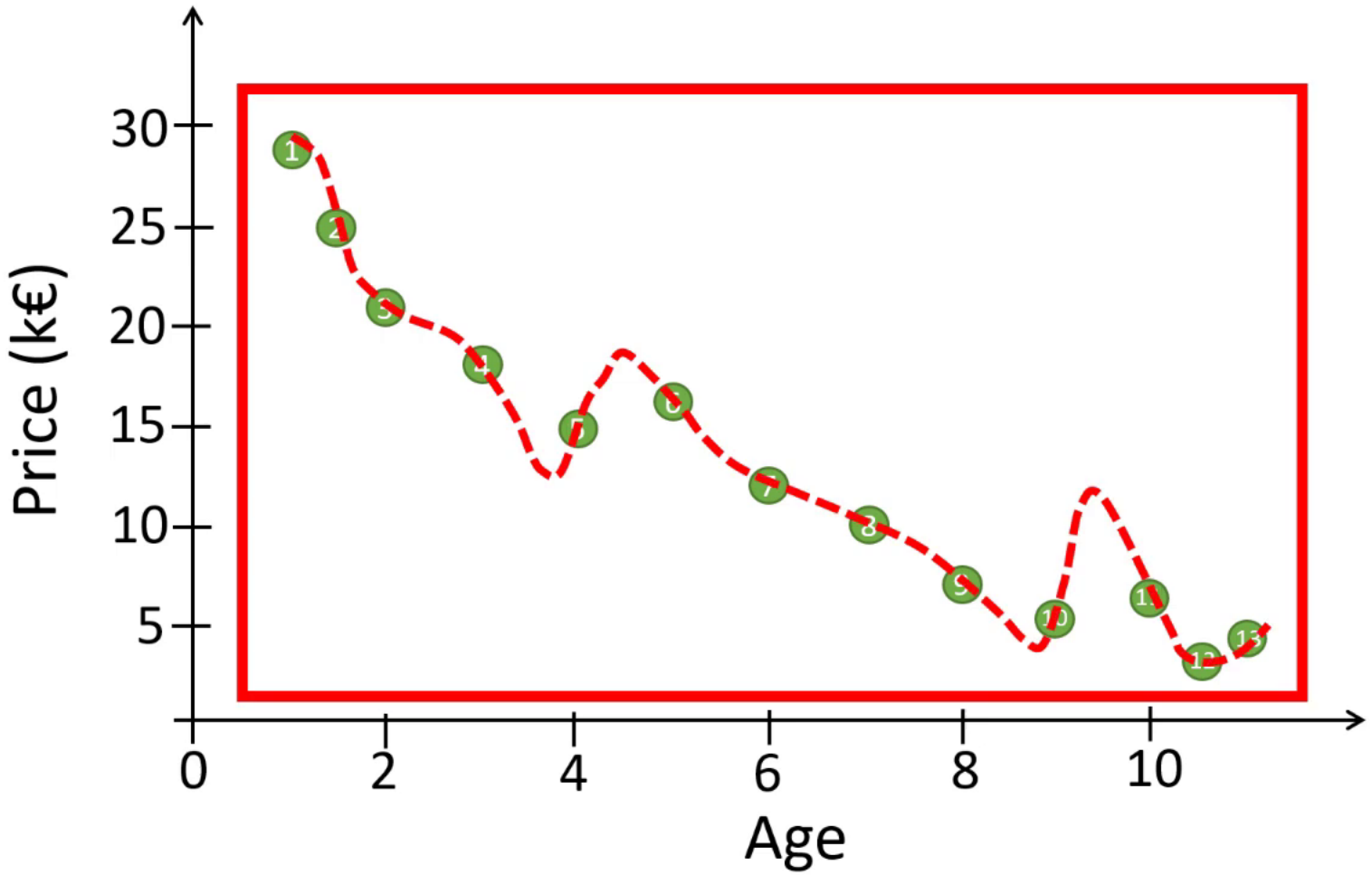
A deep neural network

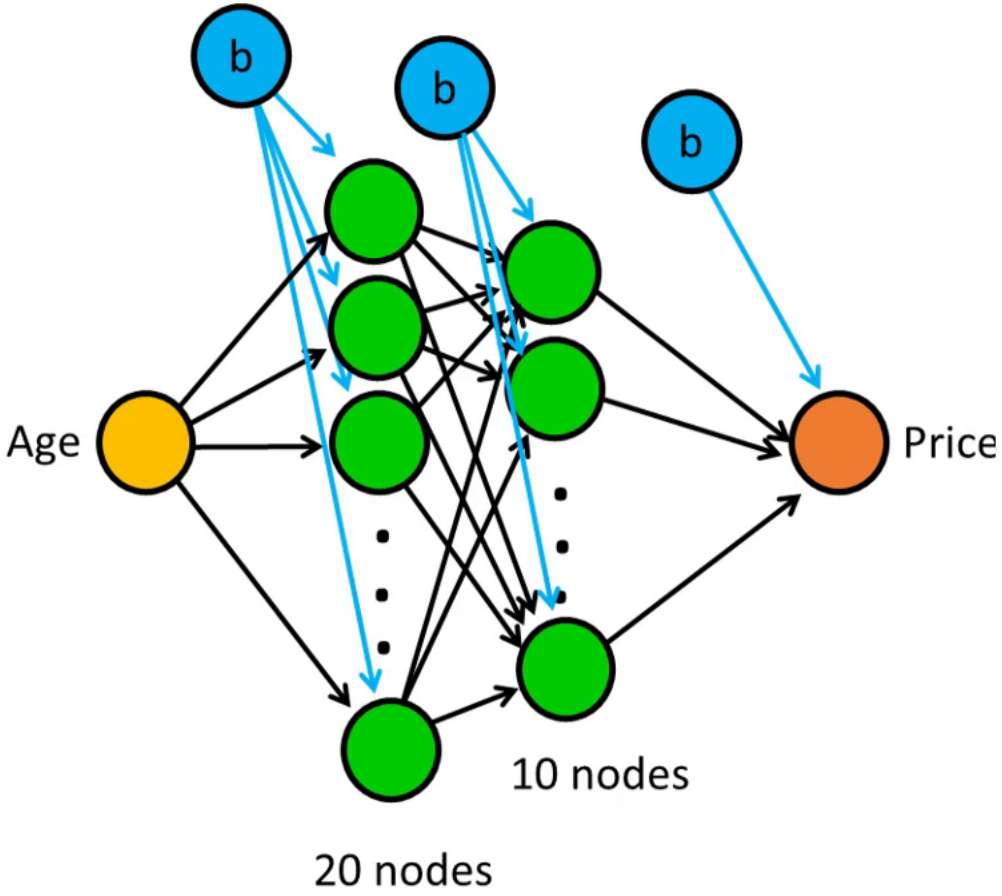




However, if we instead use a deep neural network with two hidden layers,

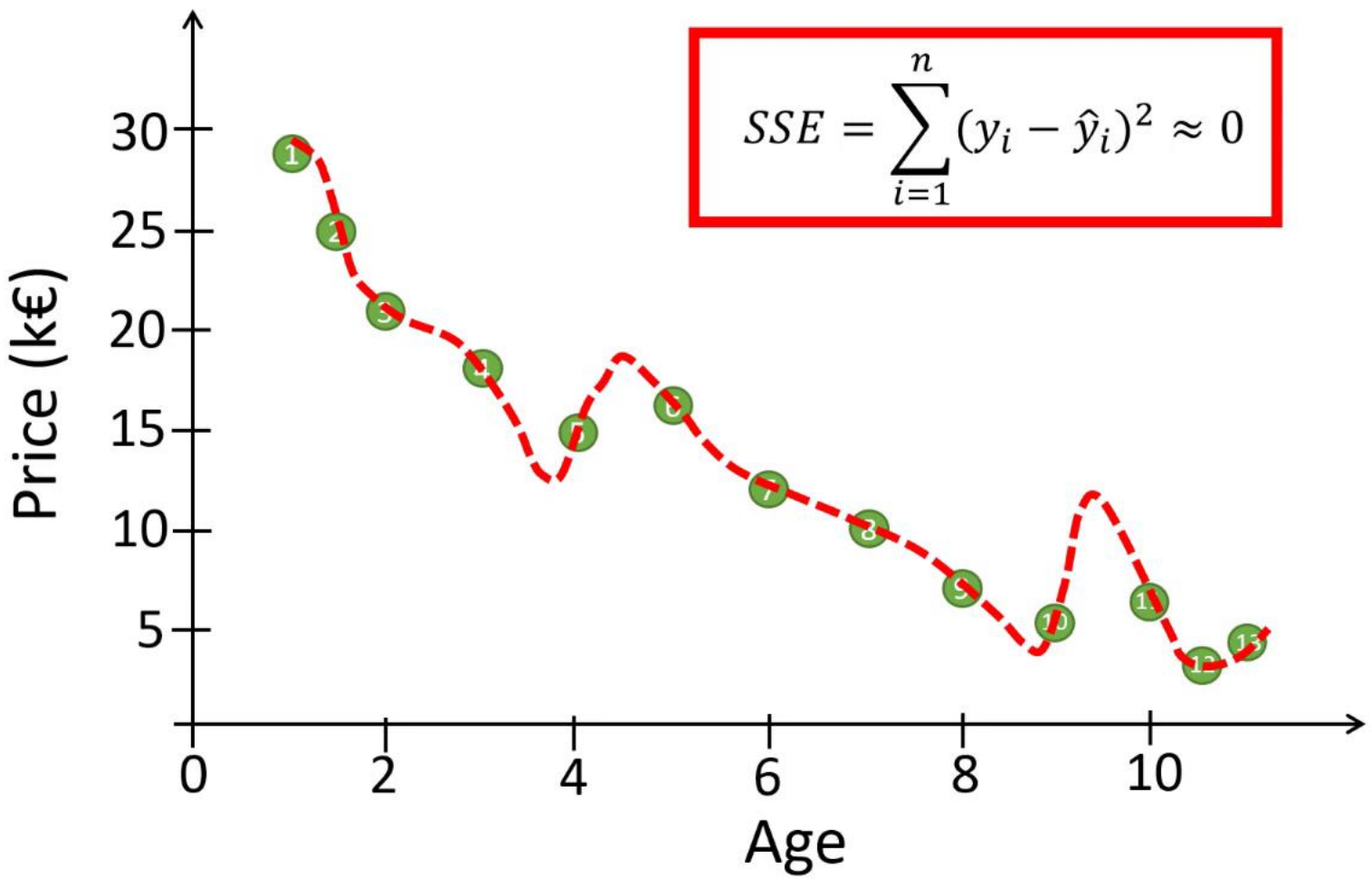
A deep neural network



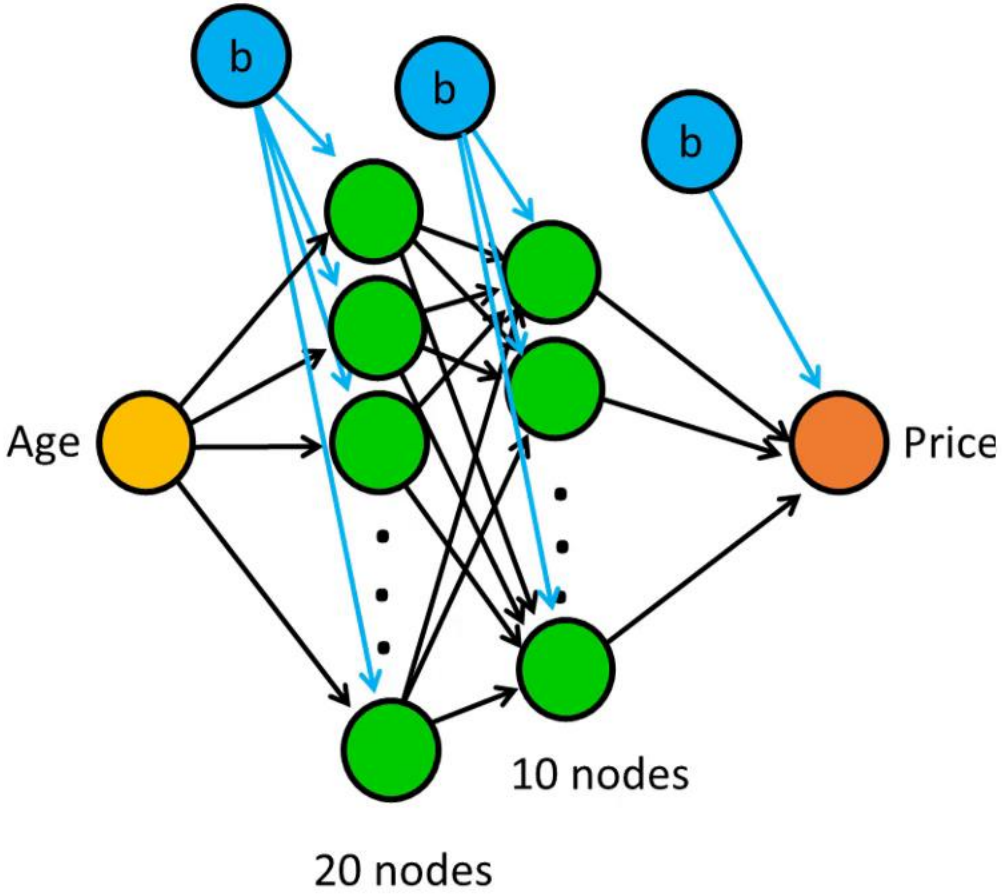


the curve now goes through every data point,

A deep neural network

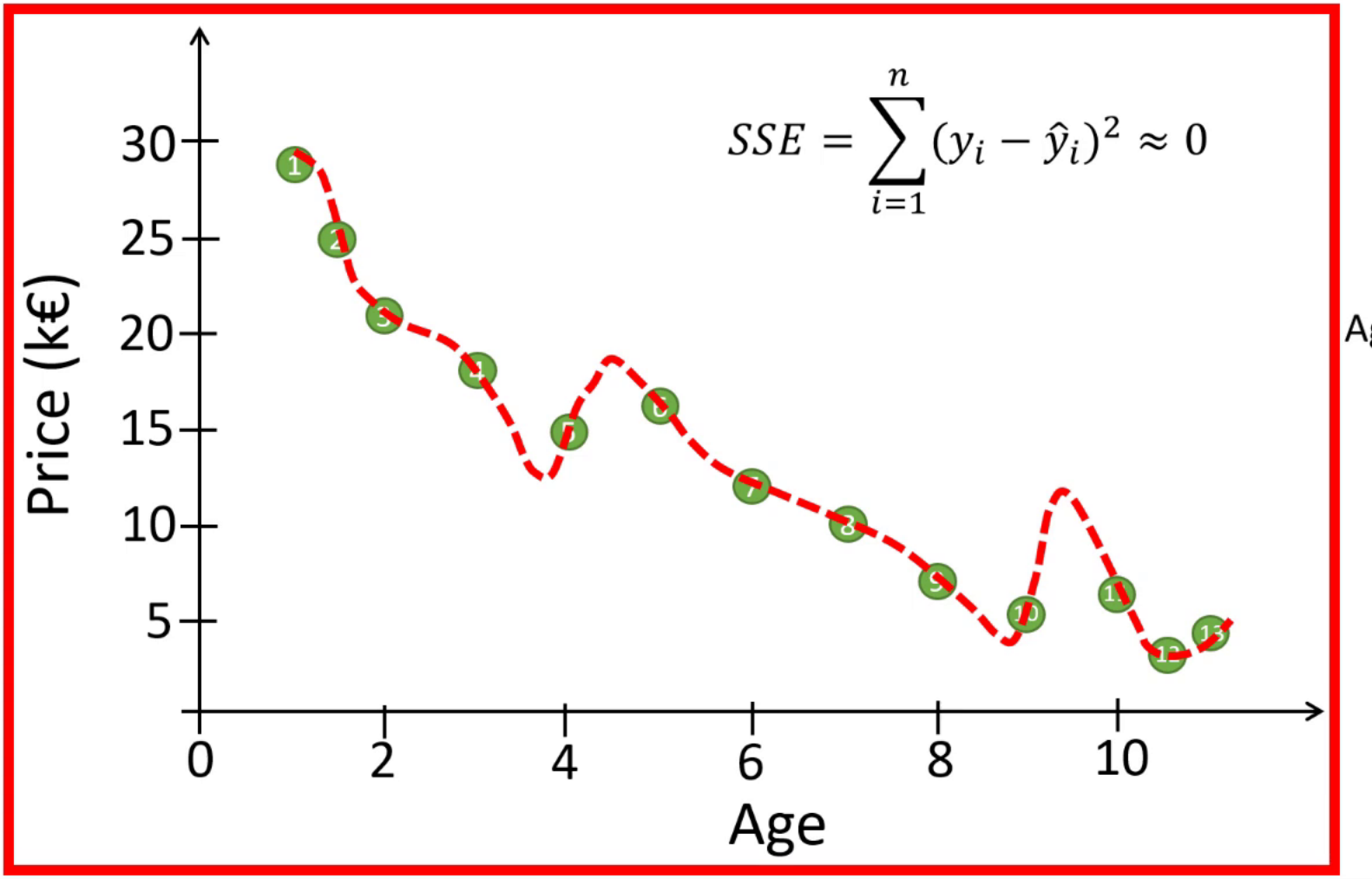


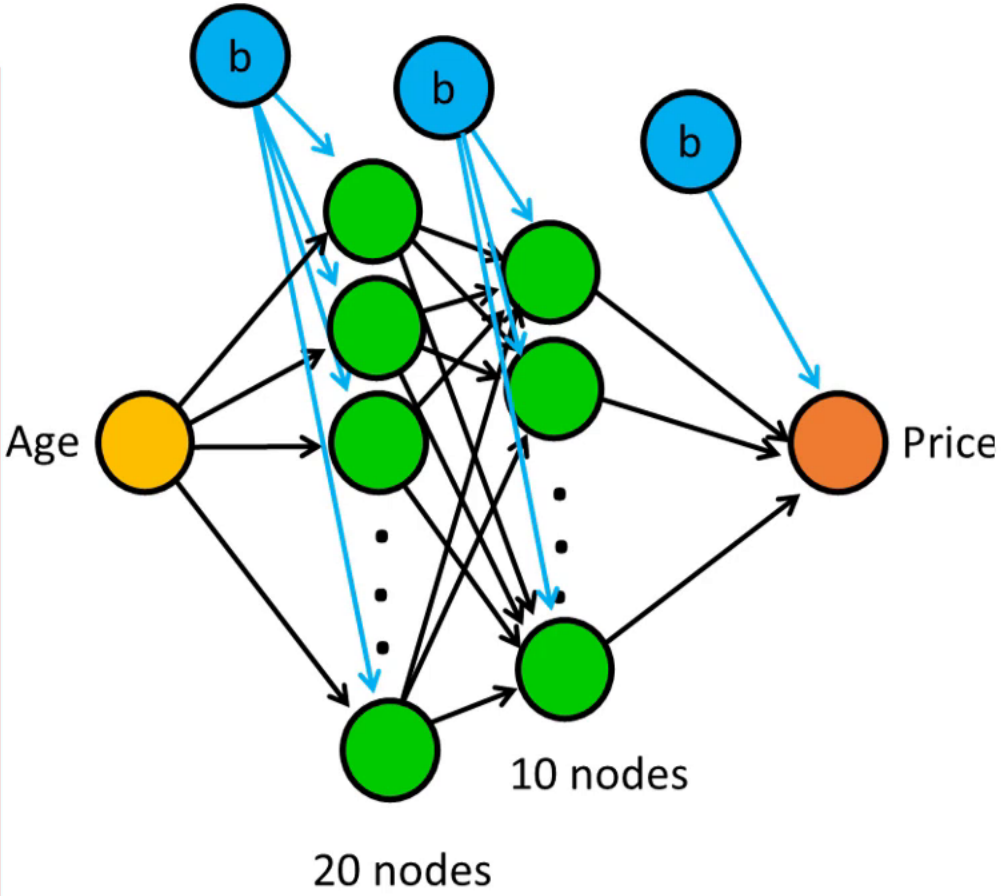
$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \approx 0$$



which will result in an error that is almost zero.

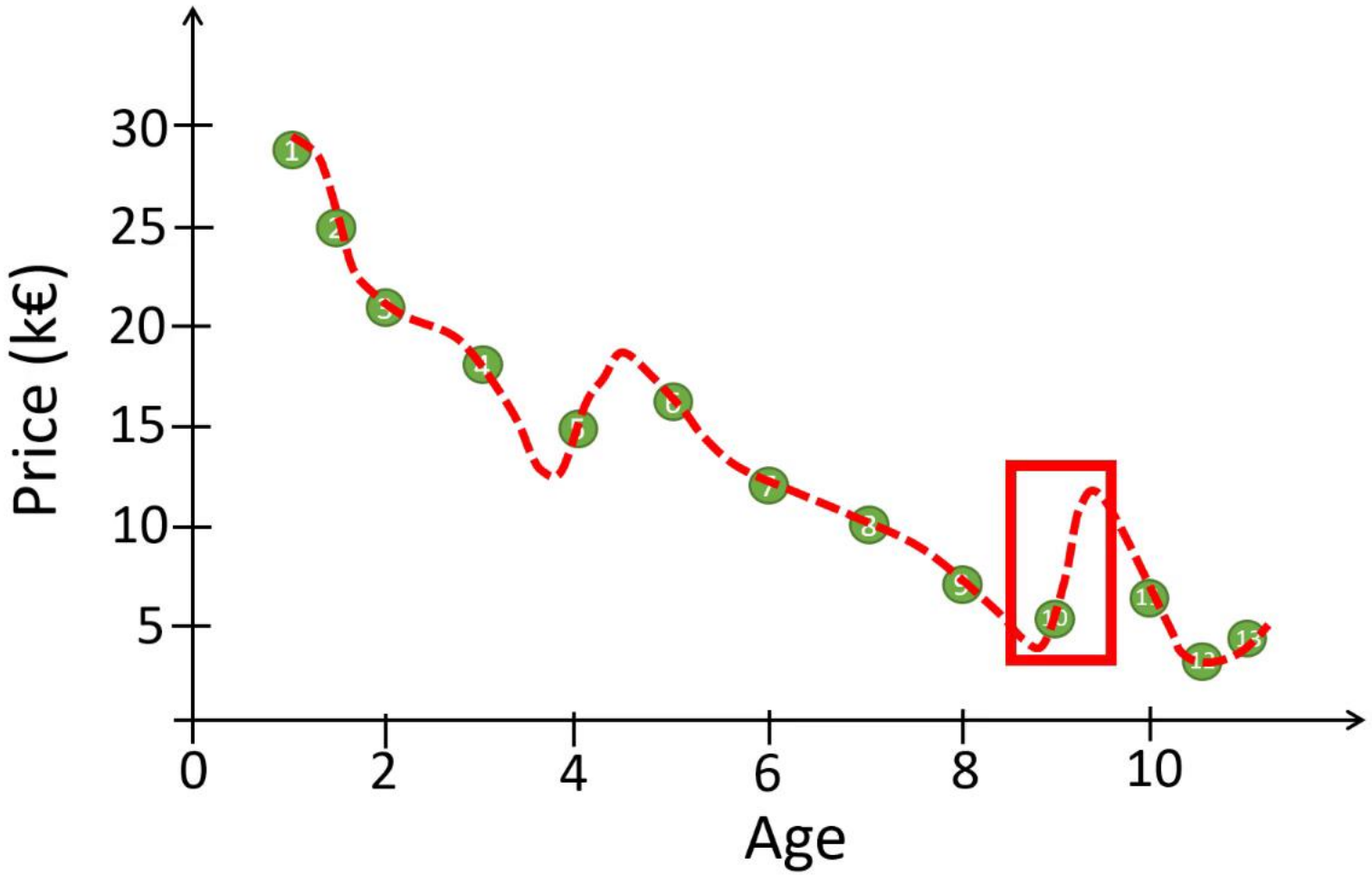
A deep neural network

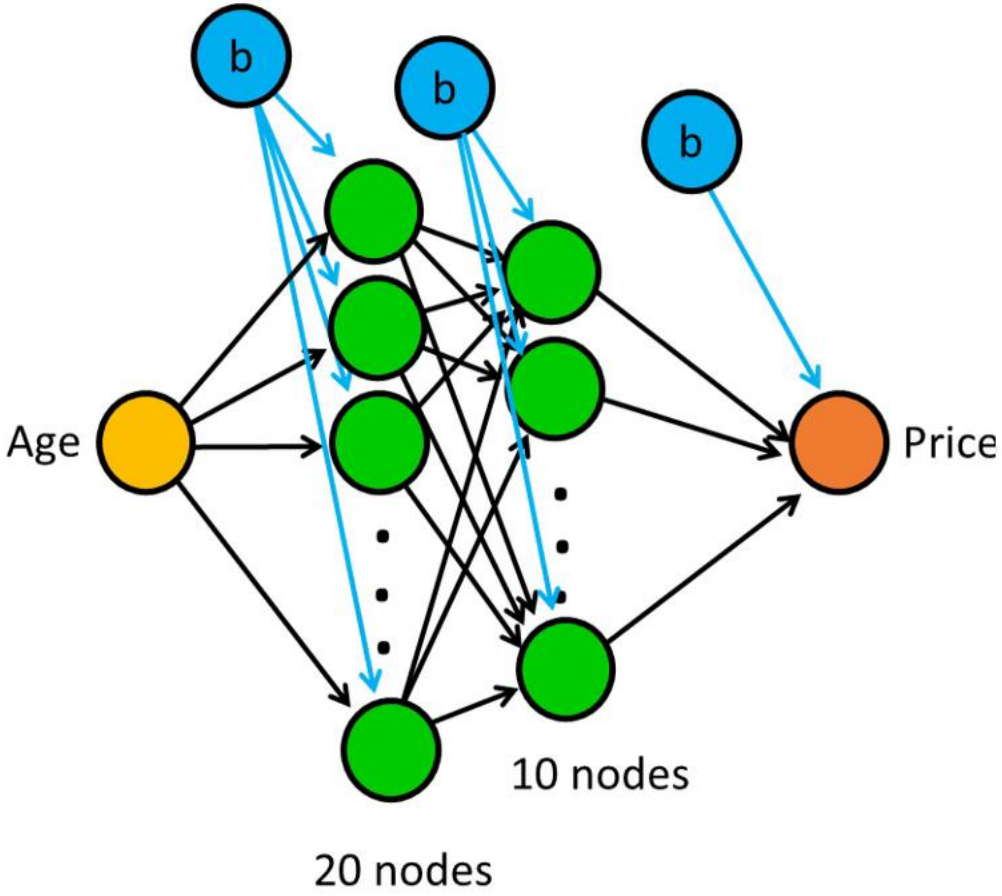




Although this model fits the data perfectly, it does not make sense,

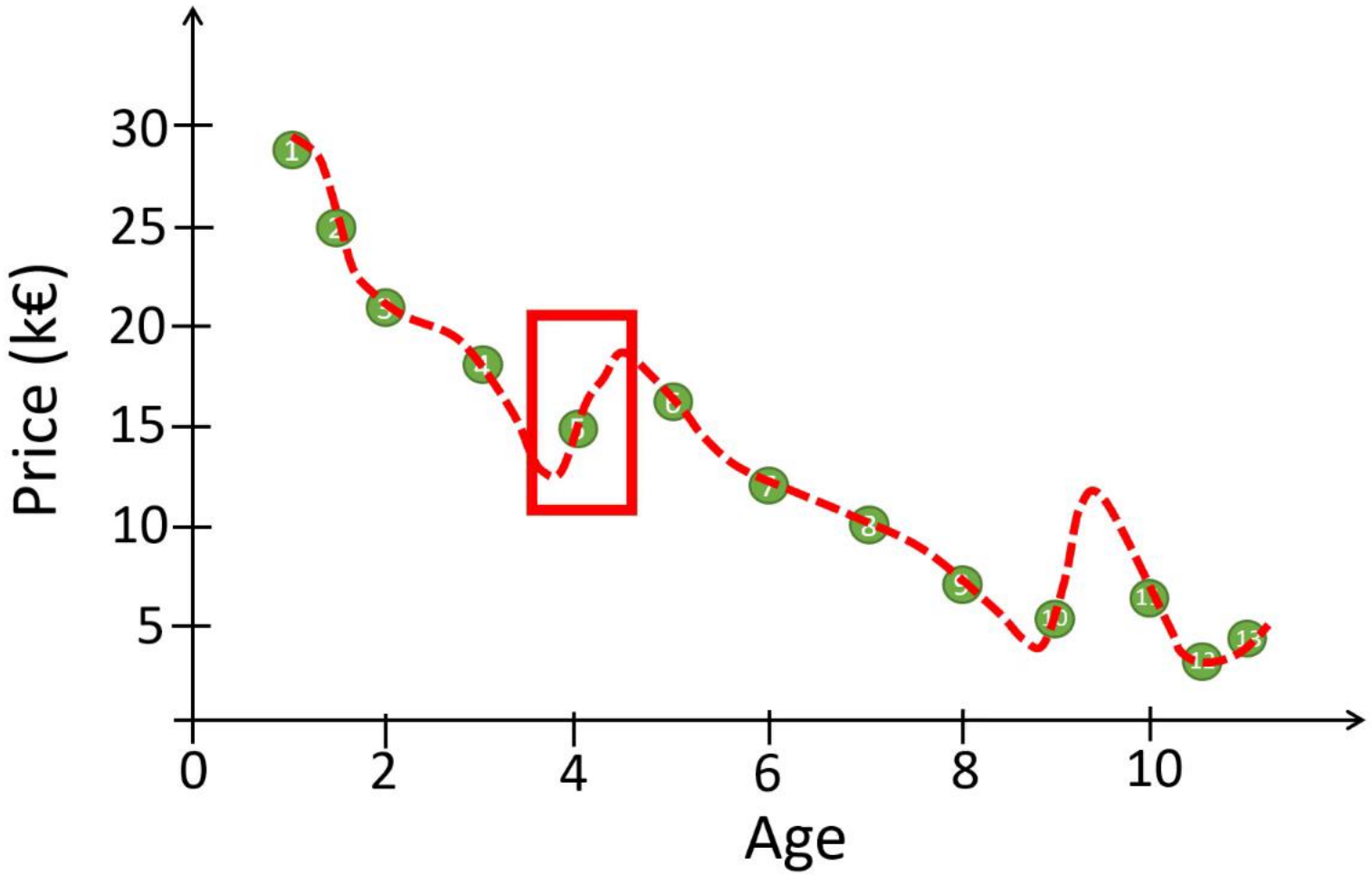
A deep neural network

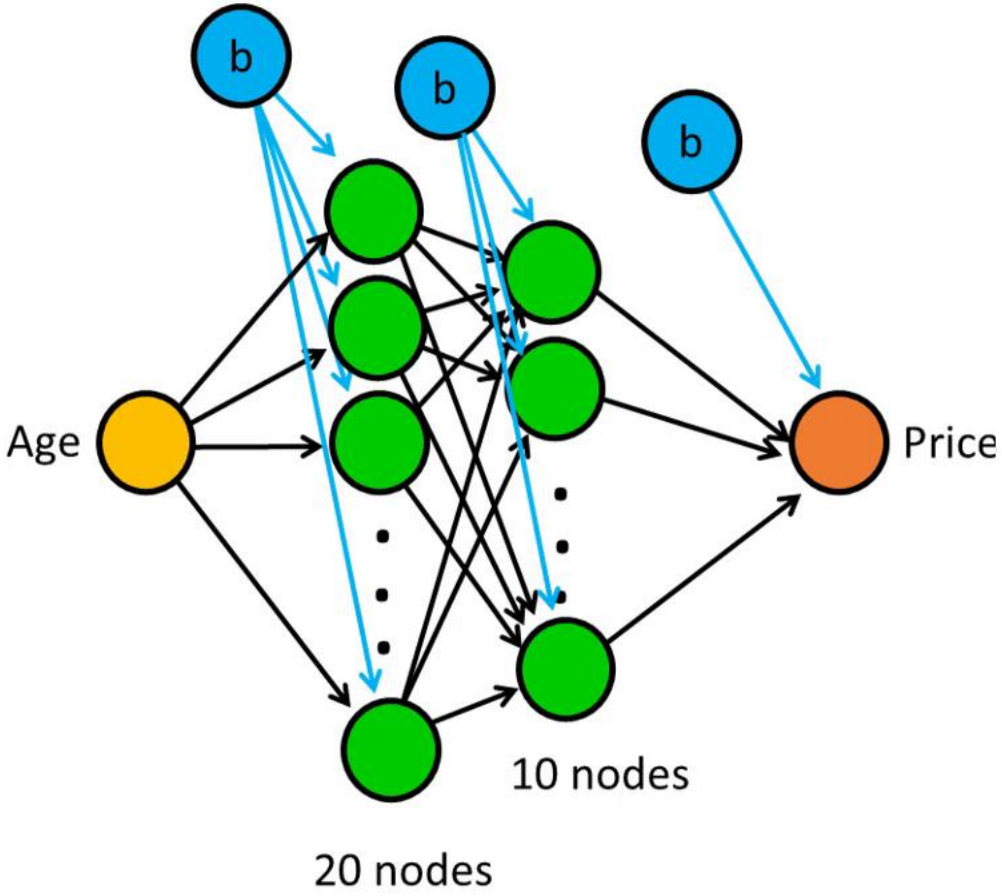




because, for example, this peak suggests that the price suddenly increases after nine years,

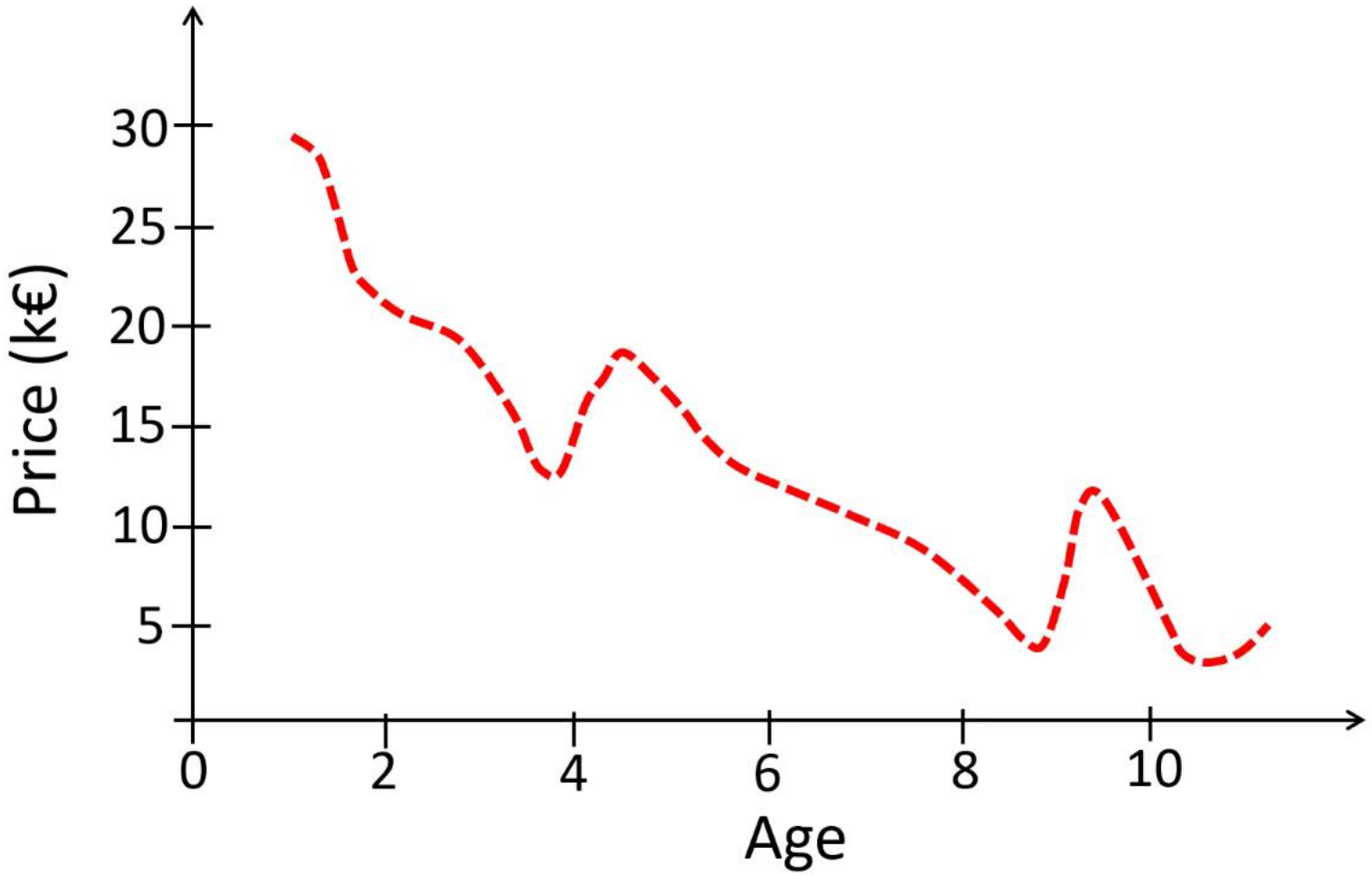
A deep neural network

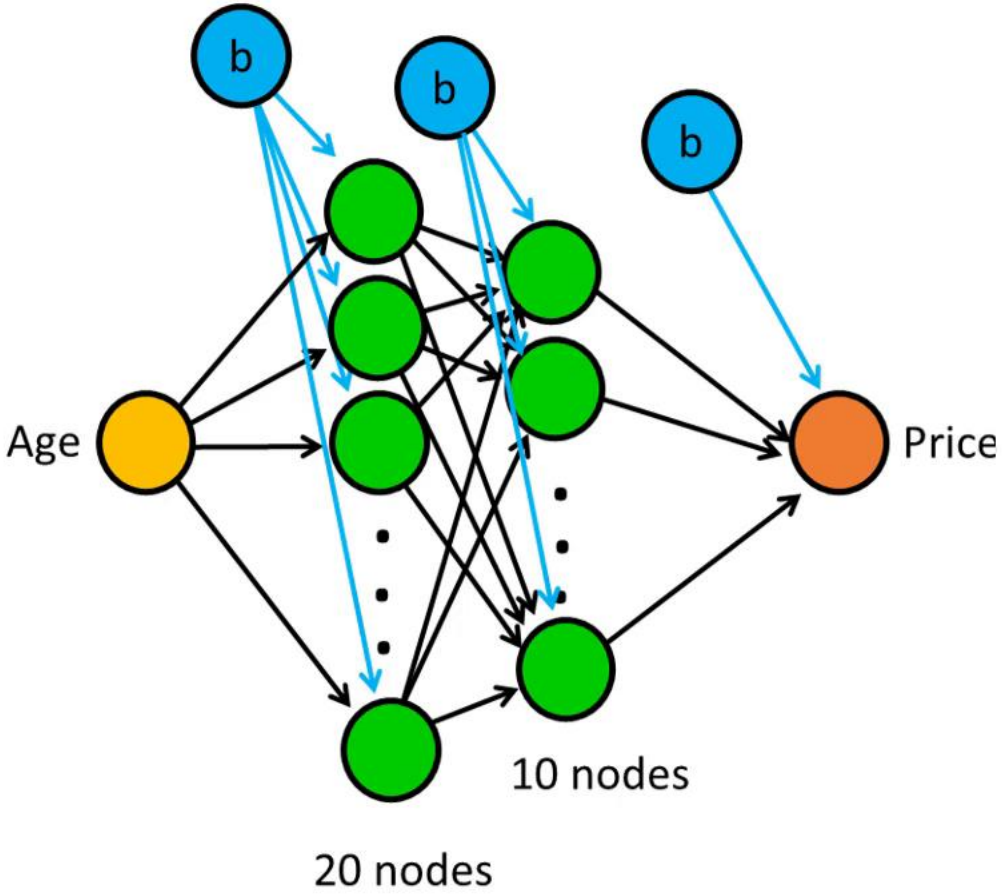




which happens also at around four years.

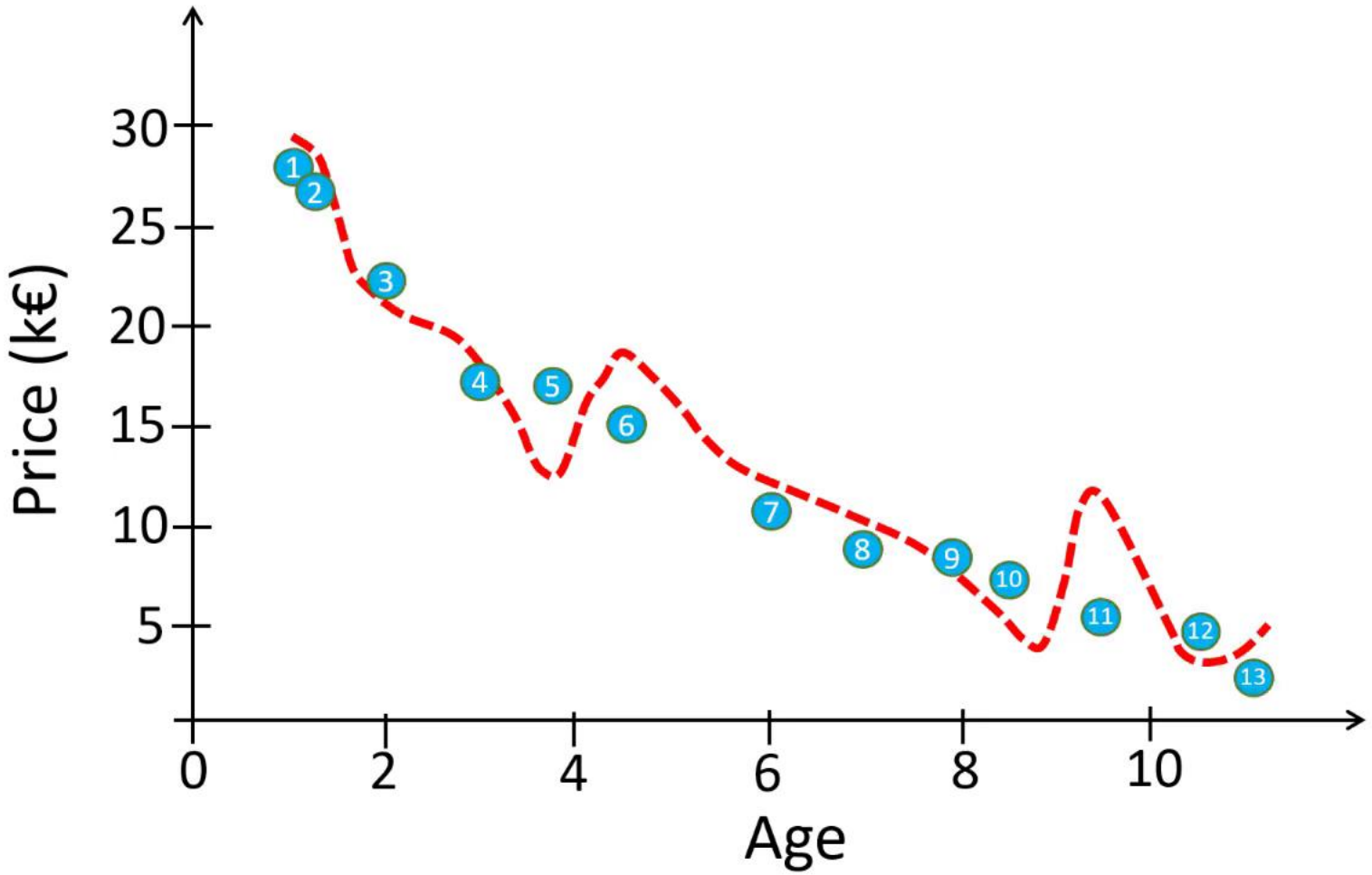
A deep neural network

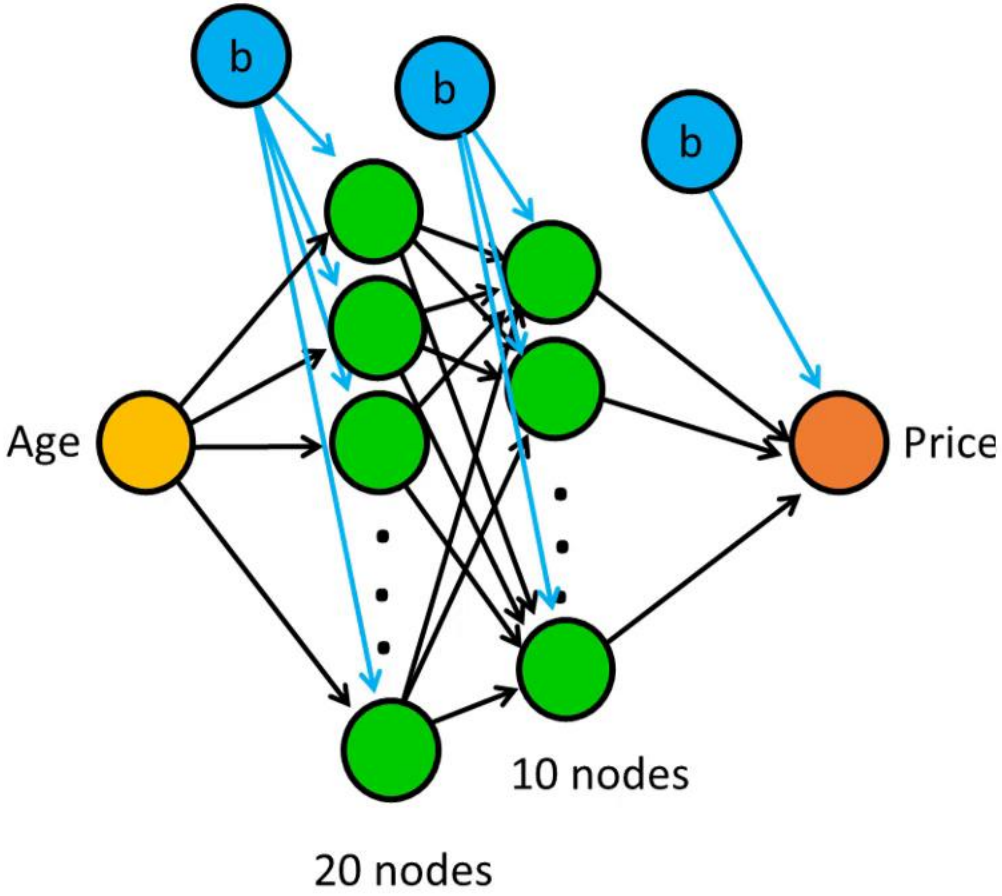




Suppose that we would remove the training data,

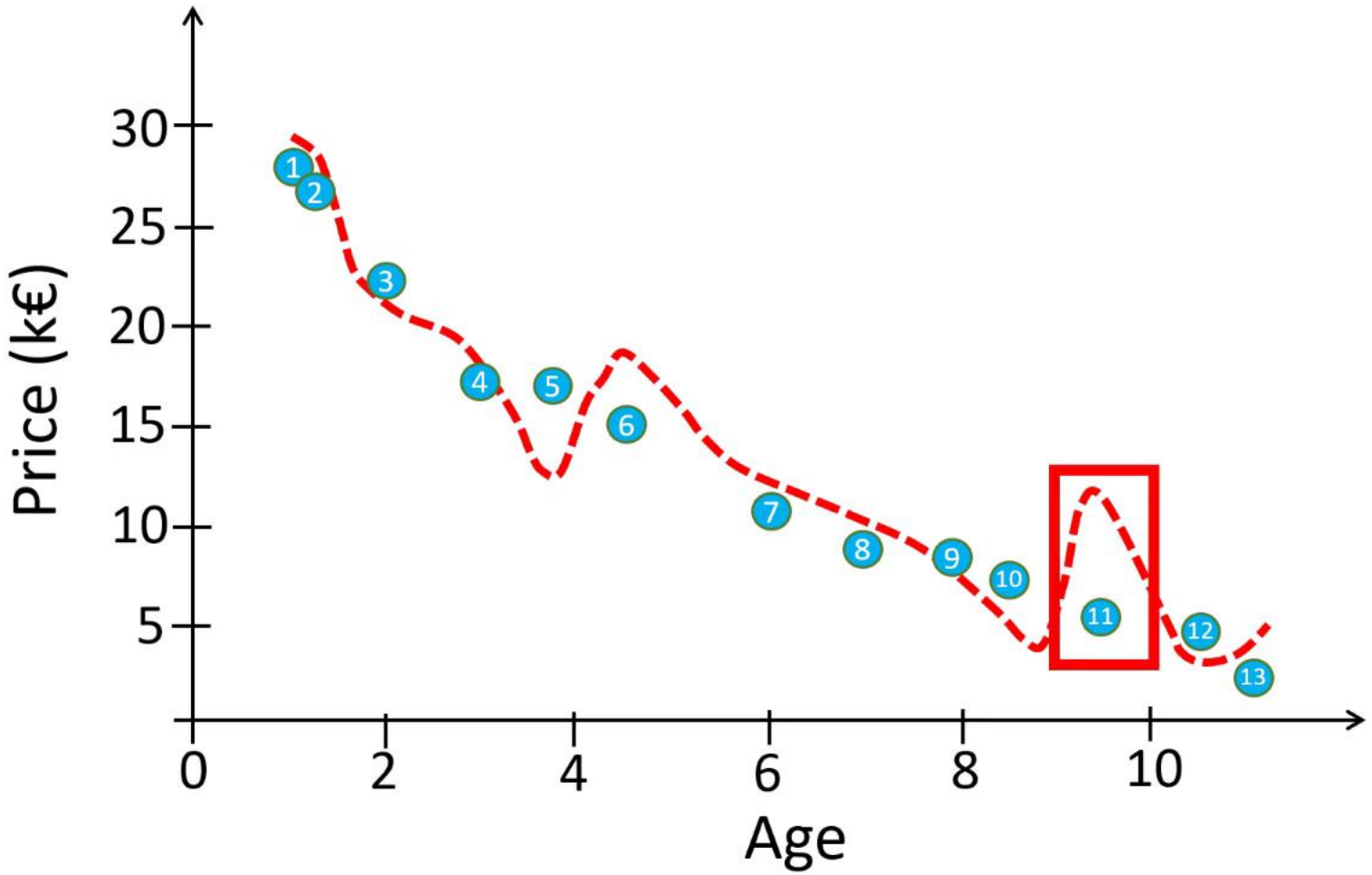
A deep neural network

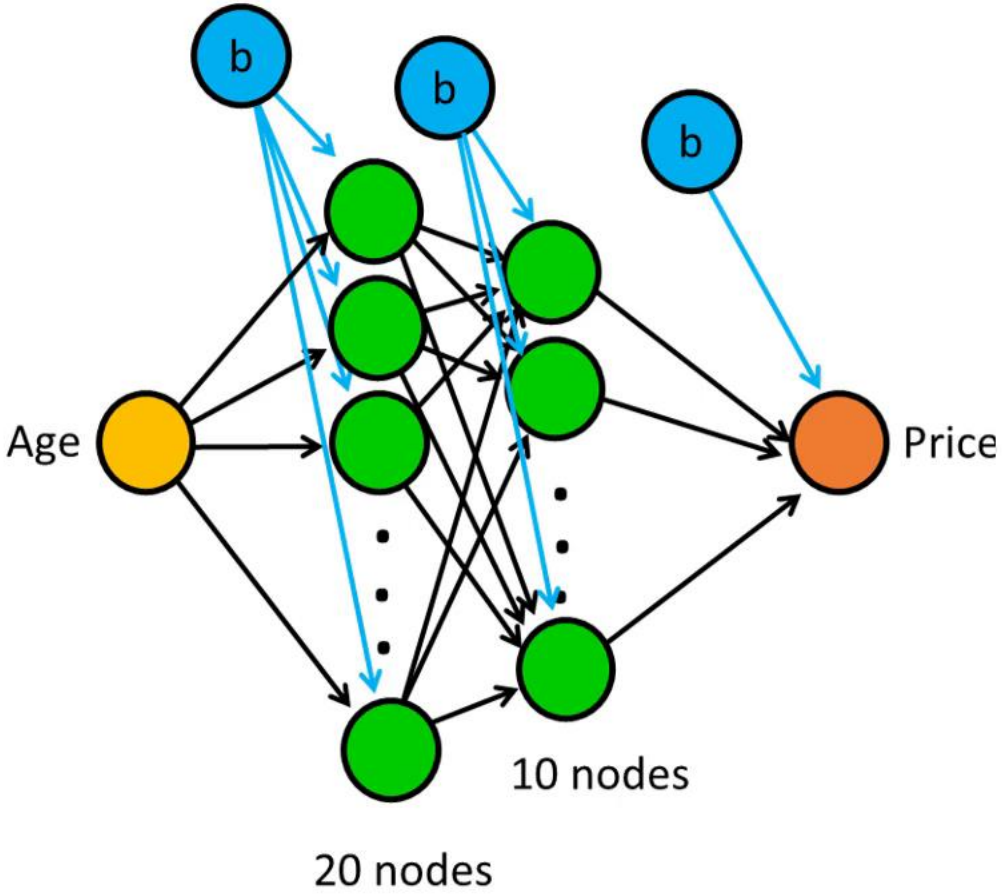




and collect a new data set that the network has not been trained on, then we see that the network predicts the new data quite badly,

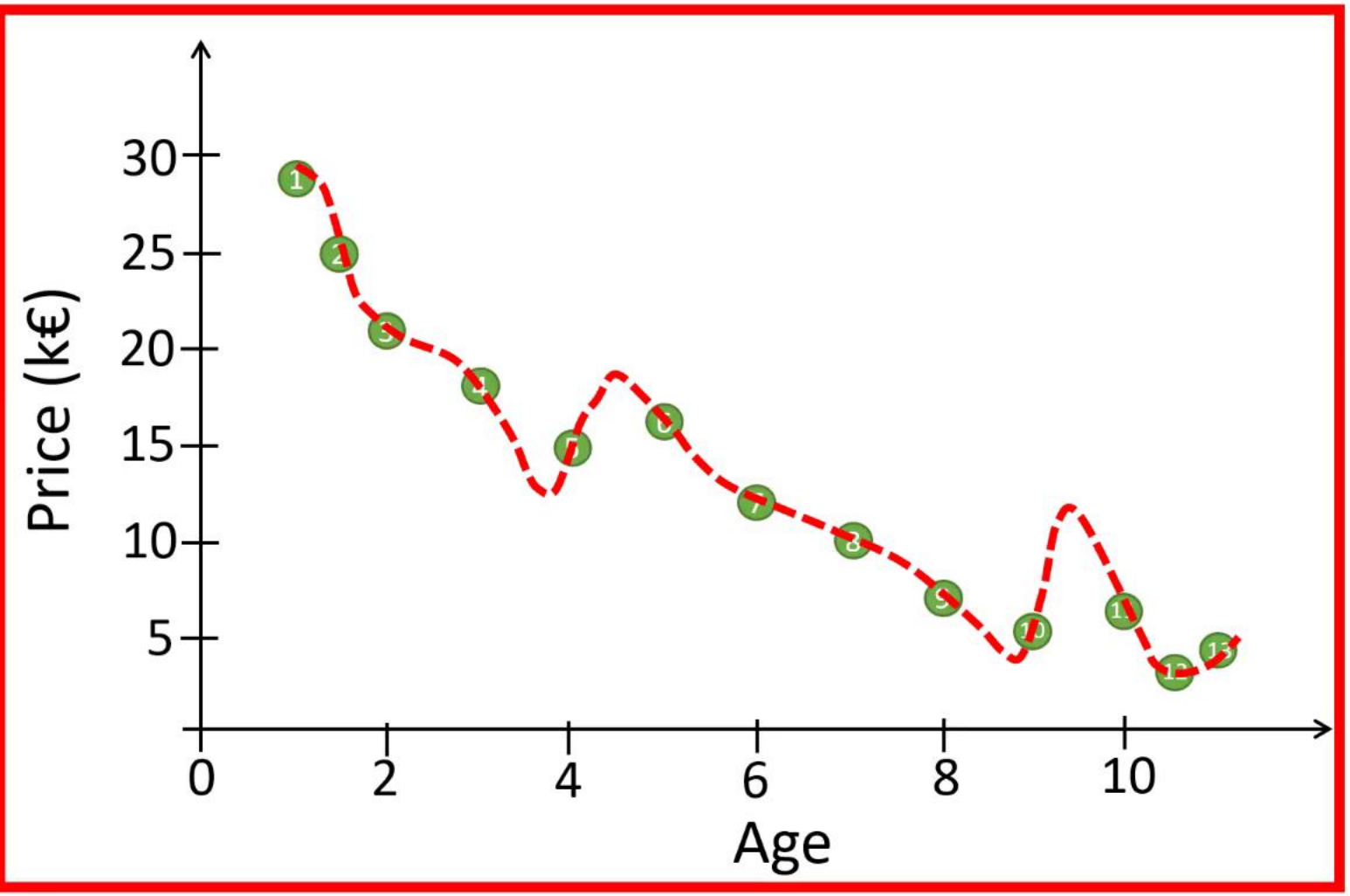
A deep neural network

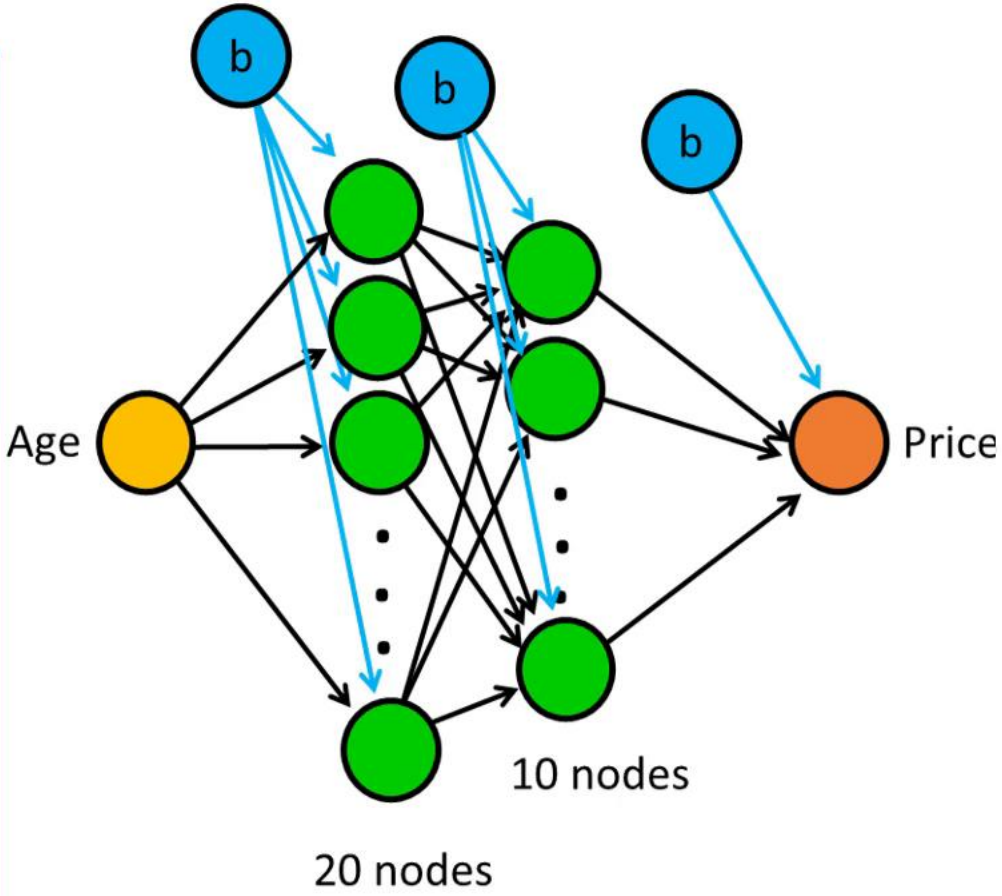




because, for example, the curve is quite far away from this data point. This is what we call overfitting,

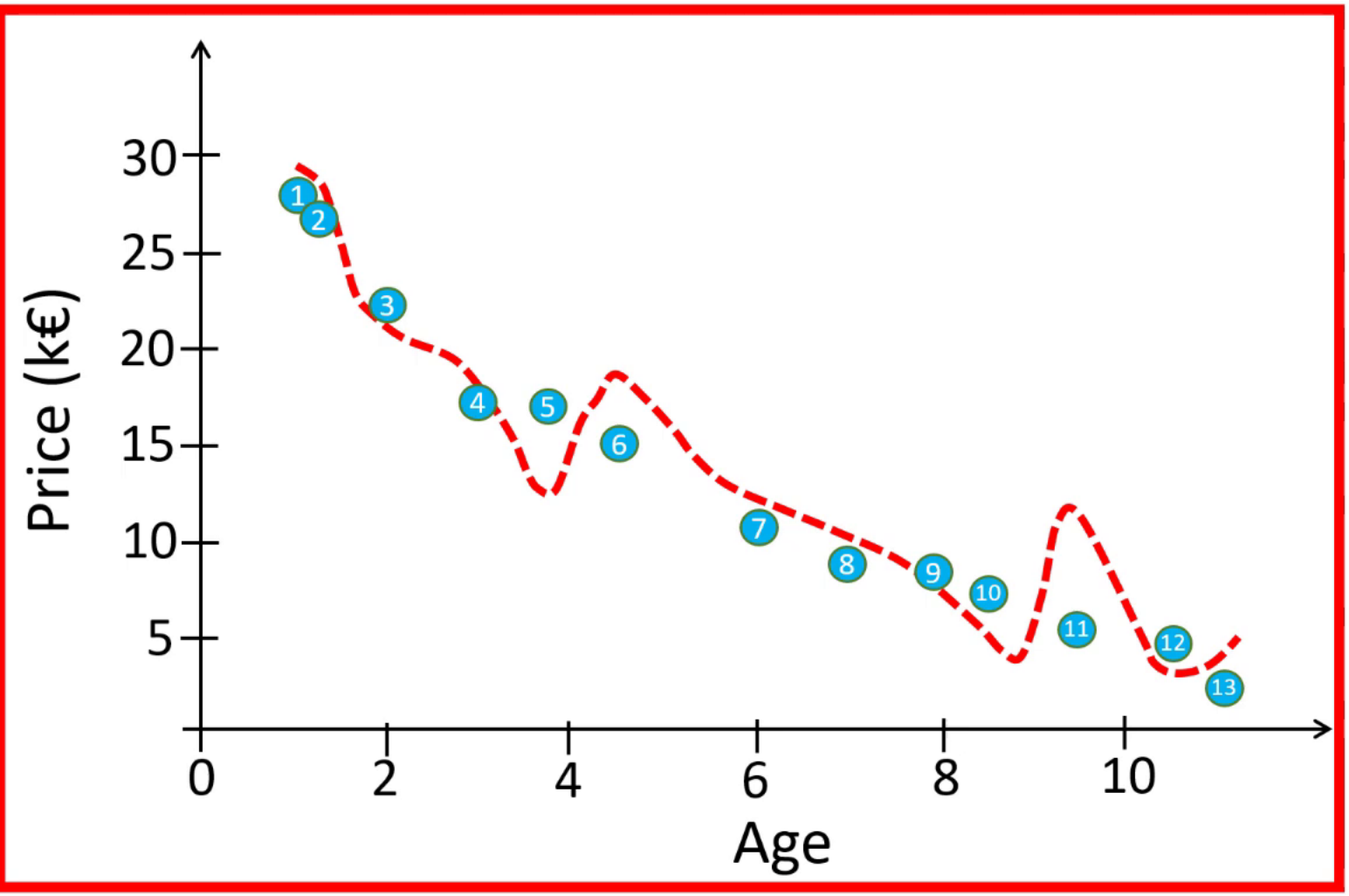
A deep neural network

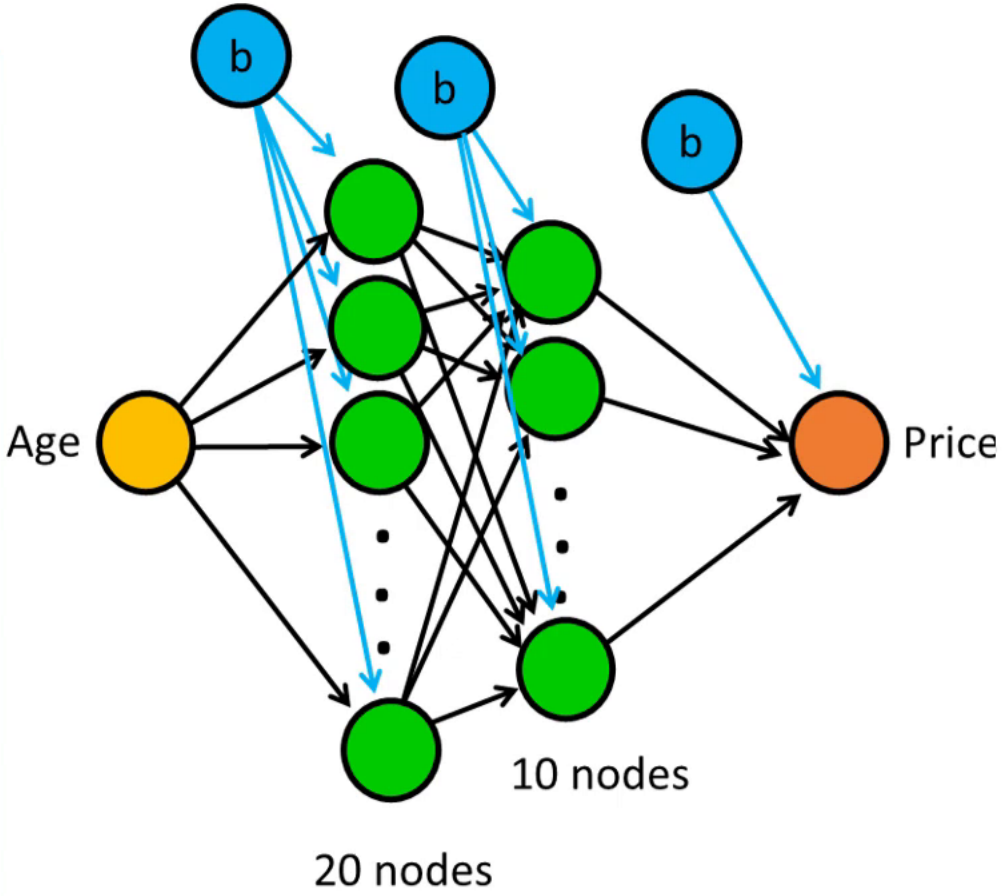




which means that the model fits very good to the training data,

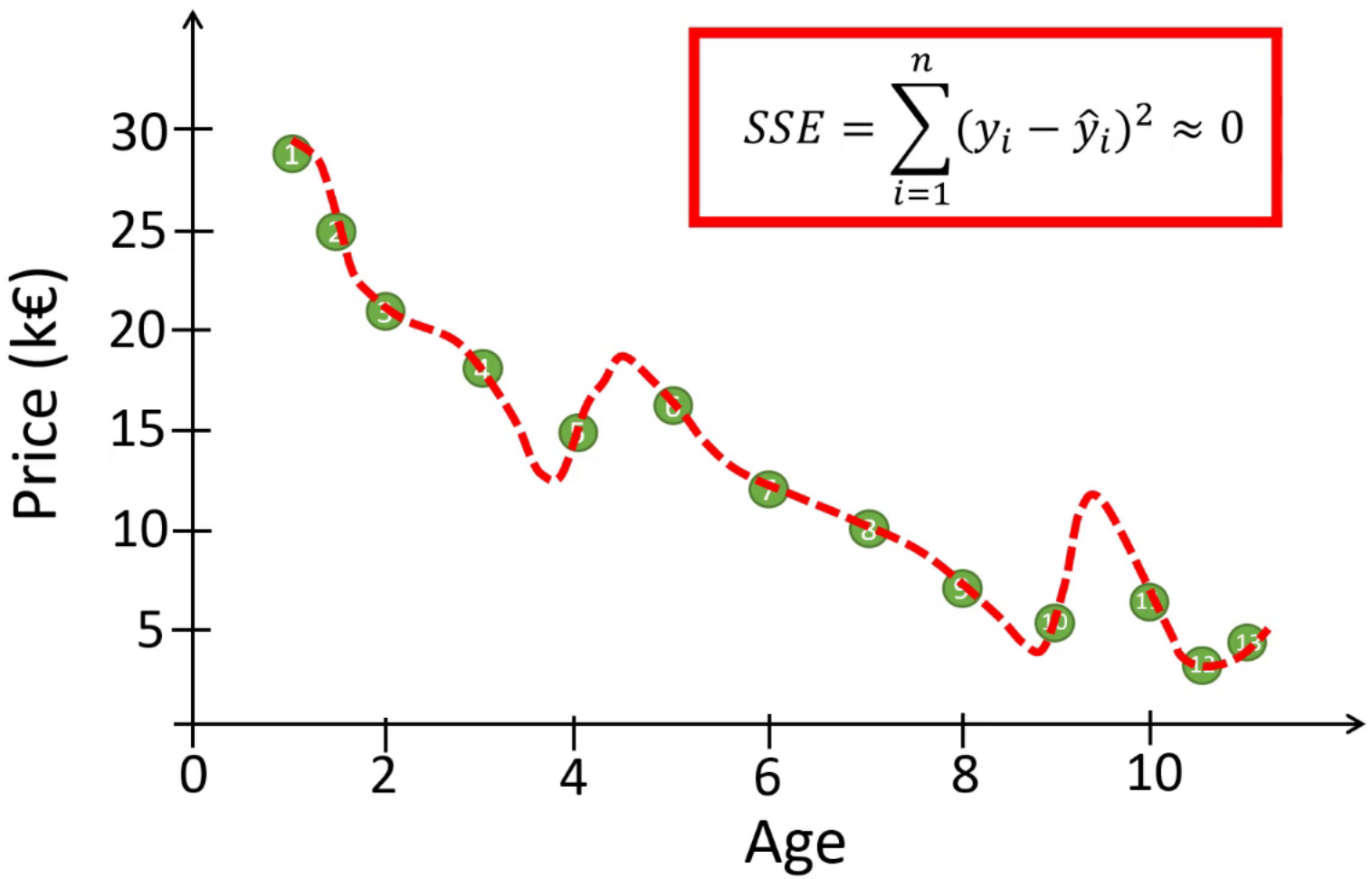
A deep neural network



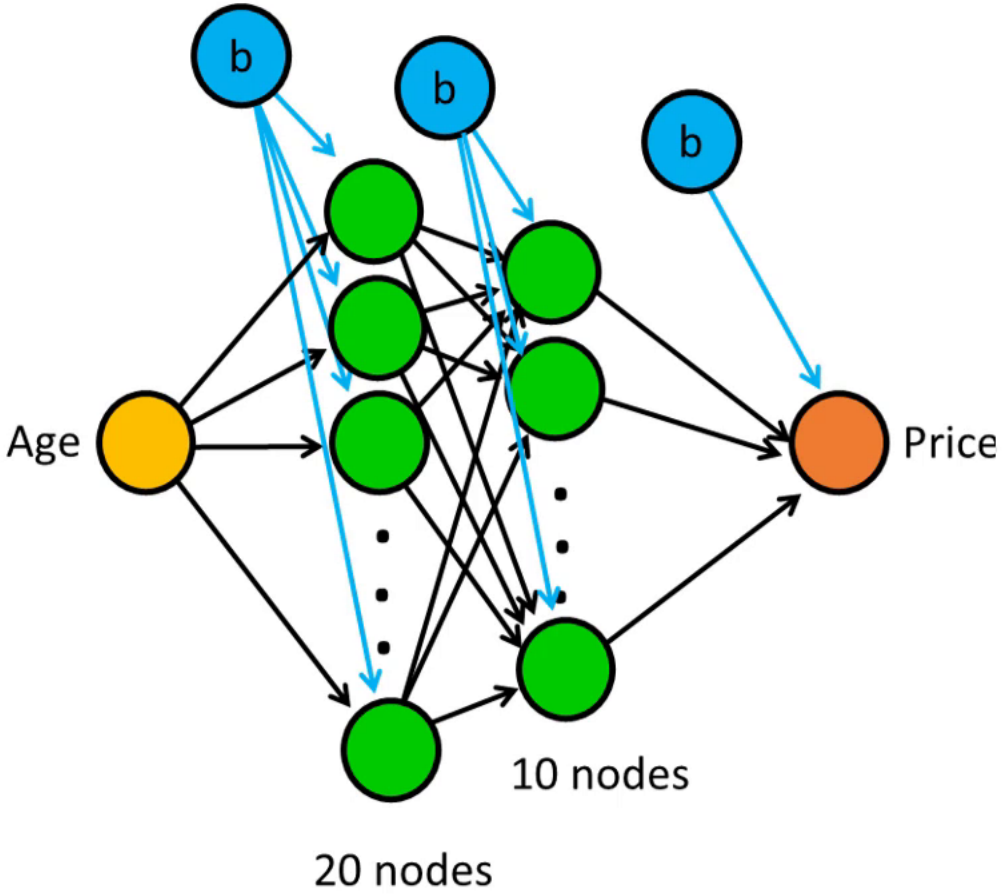


but does not fit well to data sets that the network has not been trained on.

A deep neural network

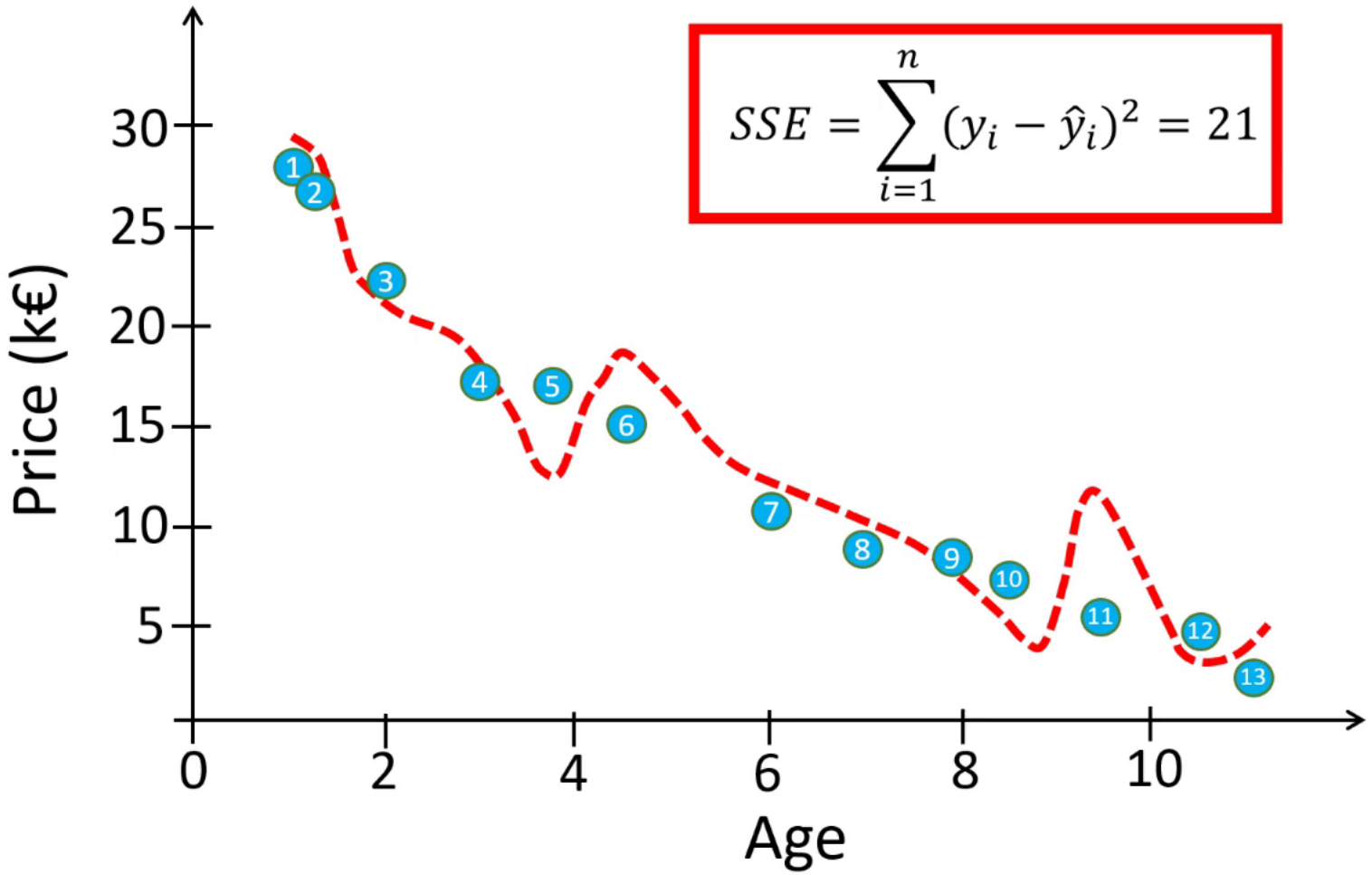


$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \approx 0$$

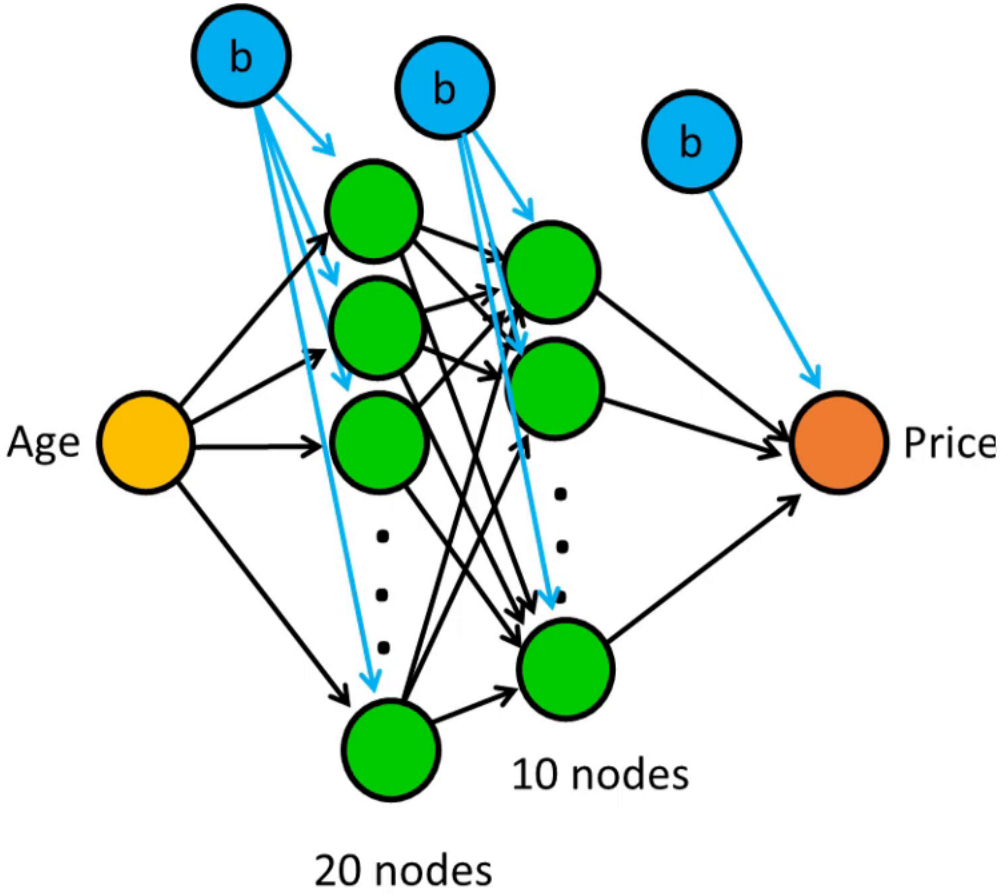


The sum of the squared errors is therefore very small when we compute it based on the training data,

A deep neural network

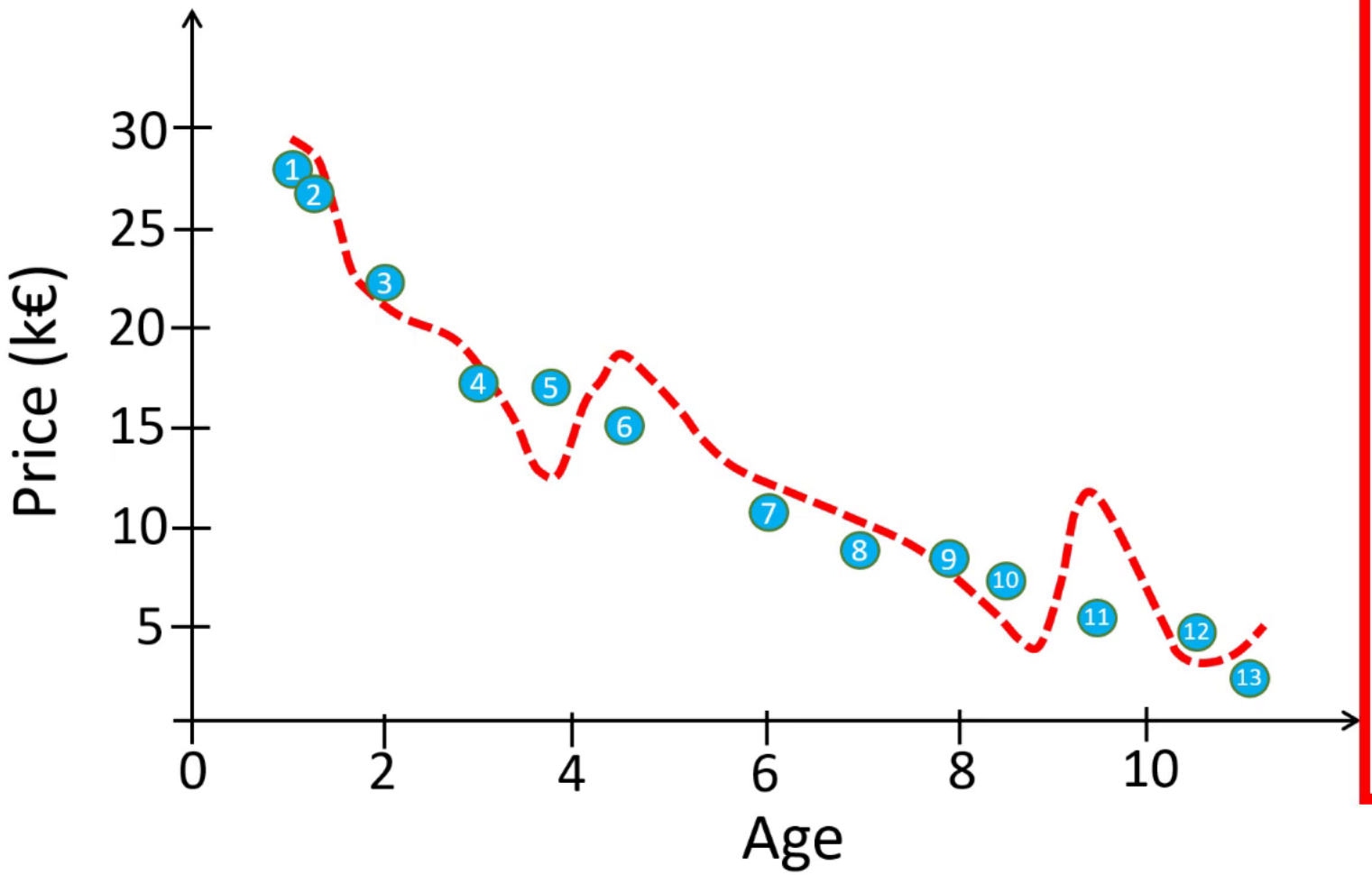


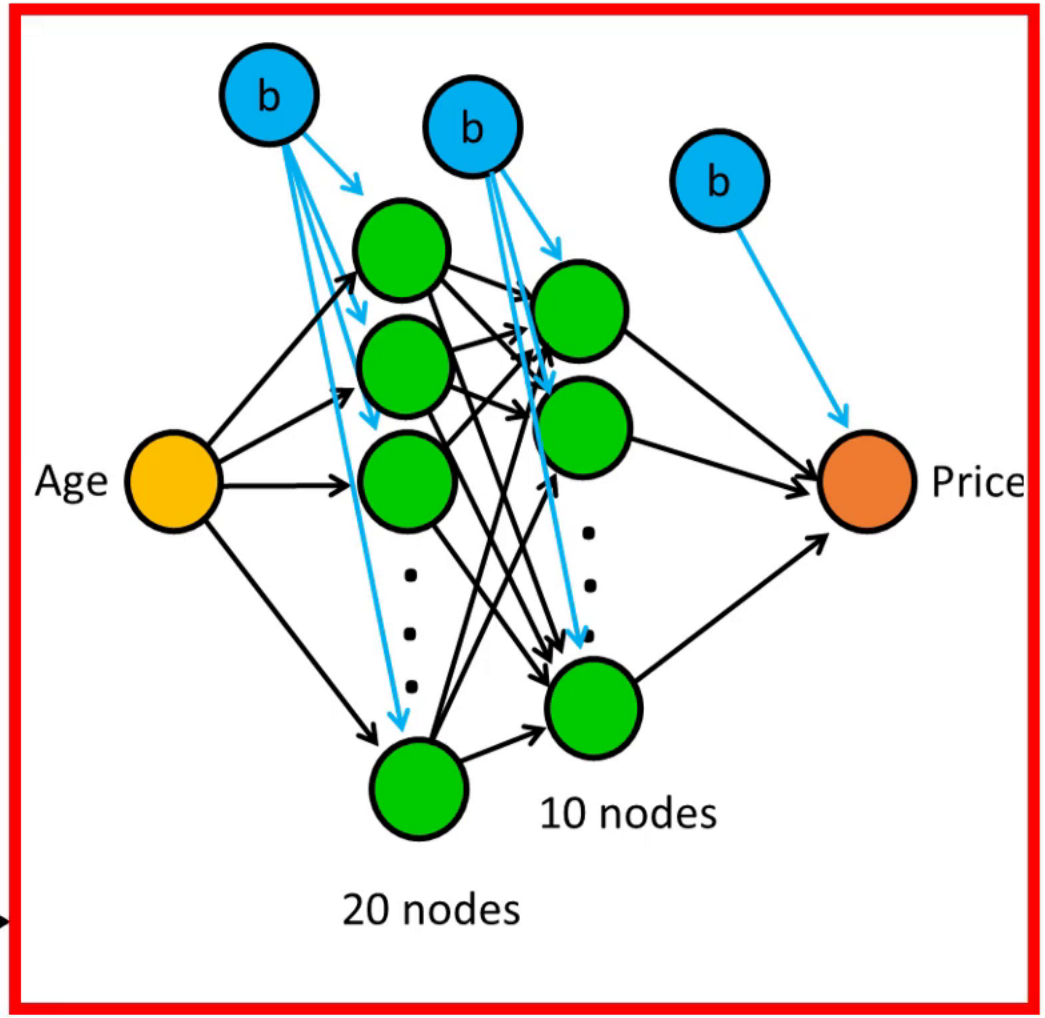
$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 21$$



but much larger when we compute it based on the new data.

A deep neural network



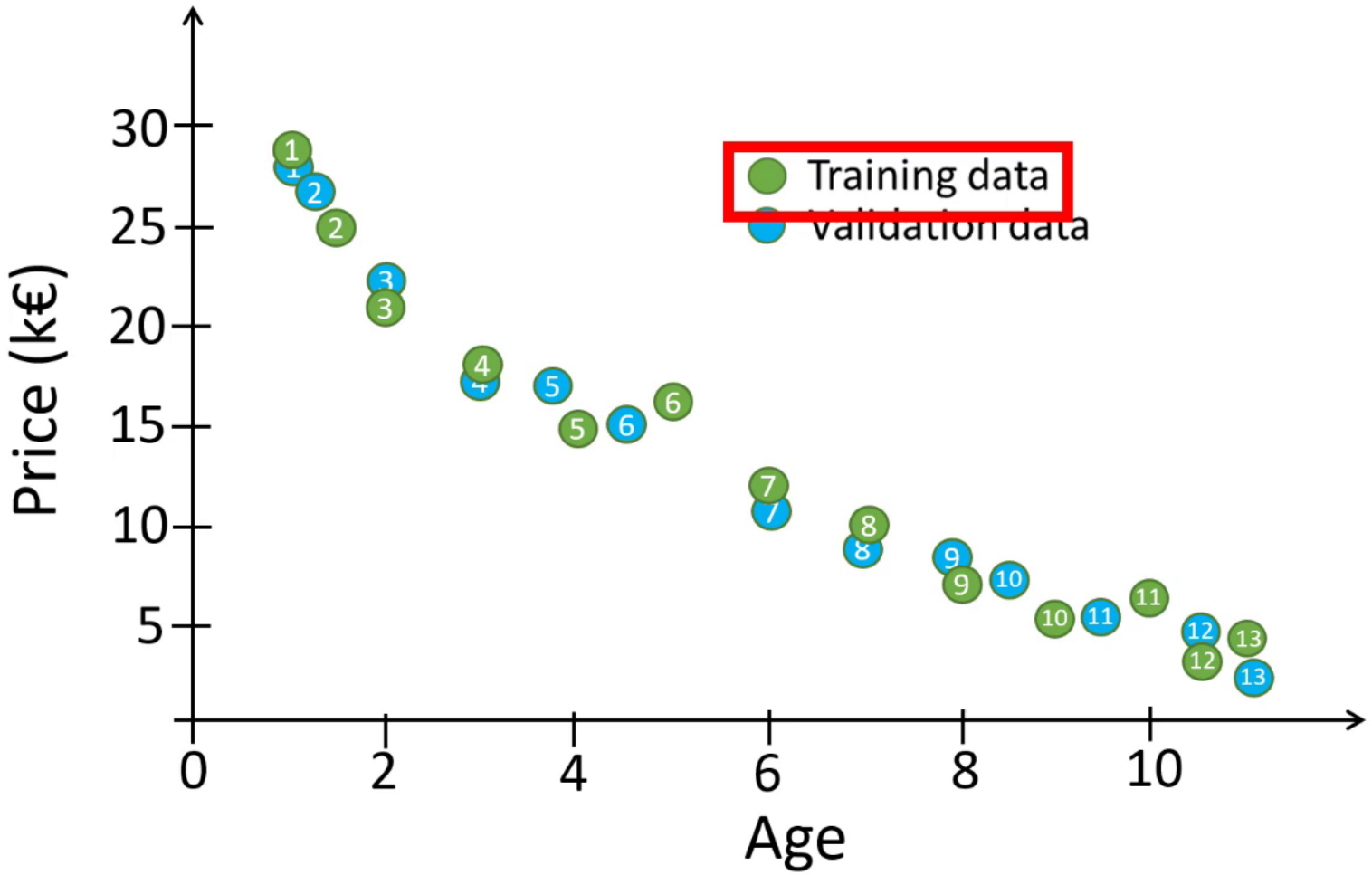


So how do we prevent a deep neural network from overfitting.

- Early stopping
- Dropout

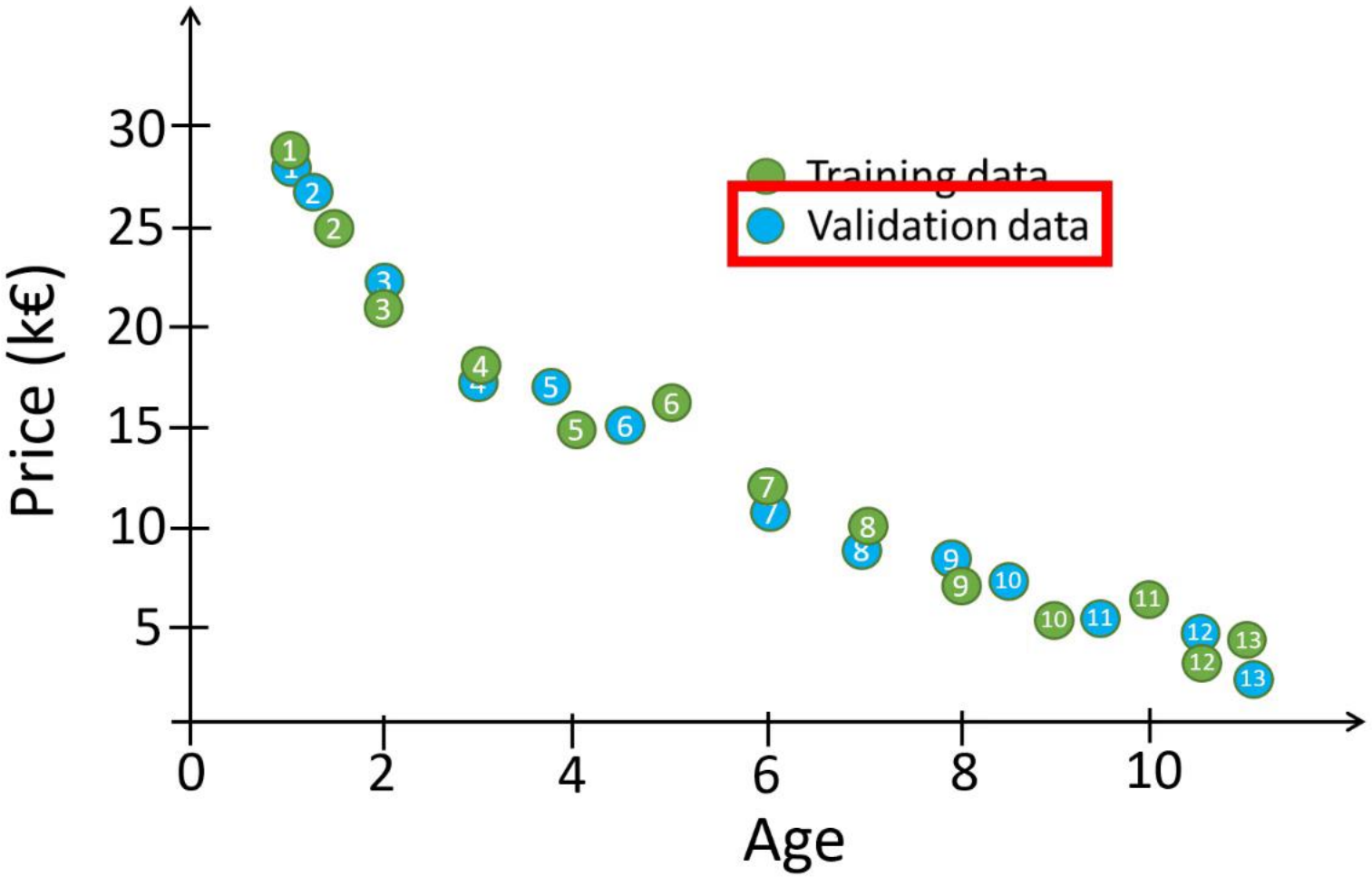
Well, there are several ways we can do this but I will here explain two common methods, early stopping and dropout.

Early stopping



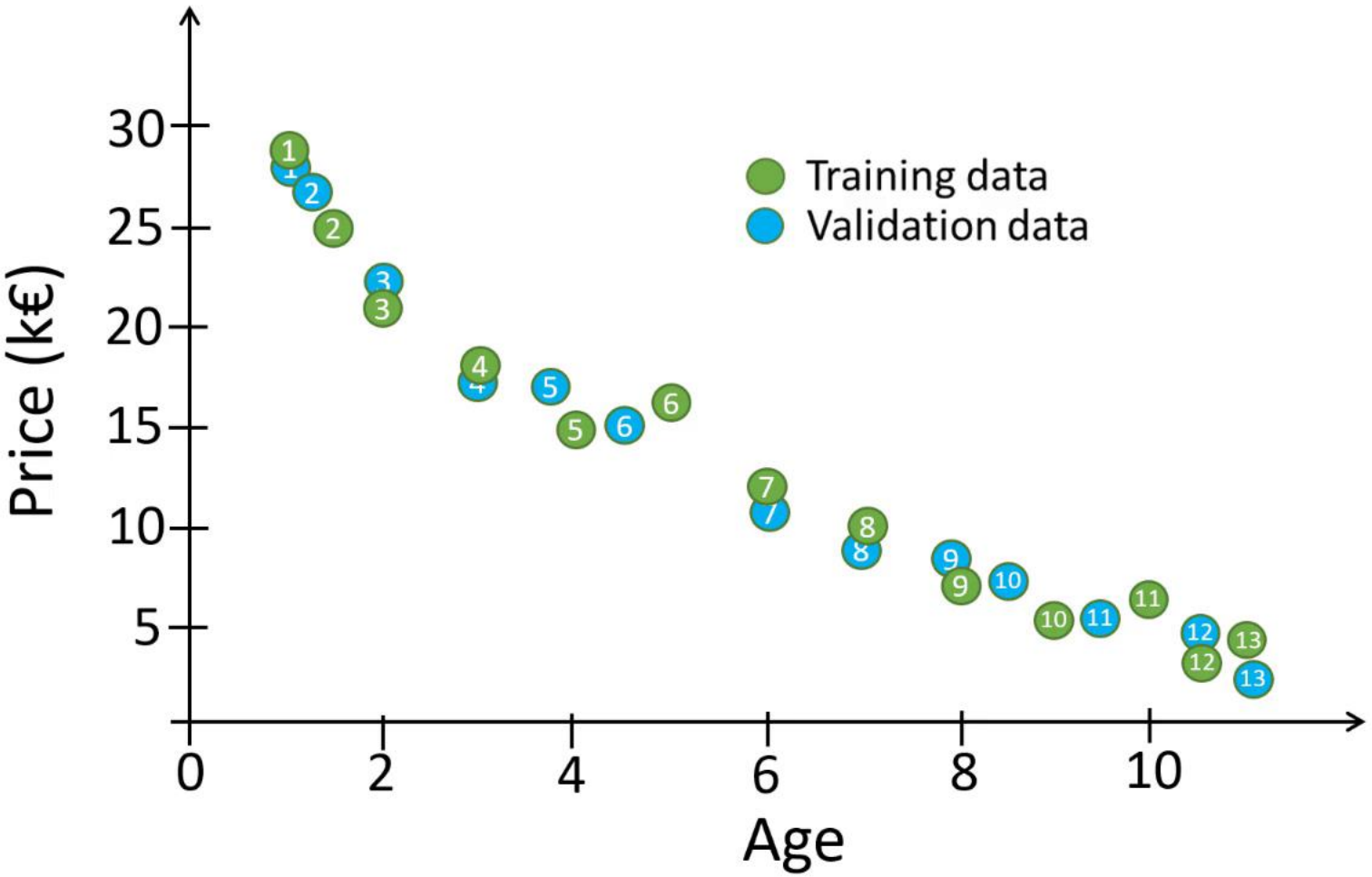
I will now show the method of early stooping, which requires that we have access to training data, which the network will train on,

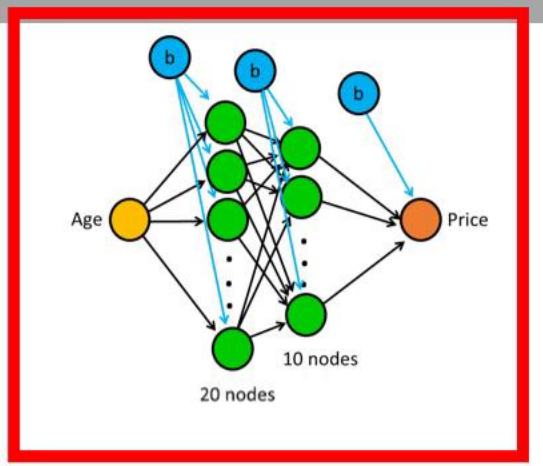
Early stopping



and to some validation data that is not used during the training. The validation data is instead plugged into the network after it has been trained to see how well it can predict new data.

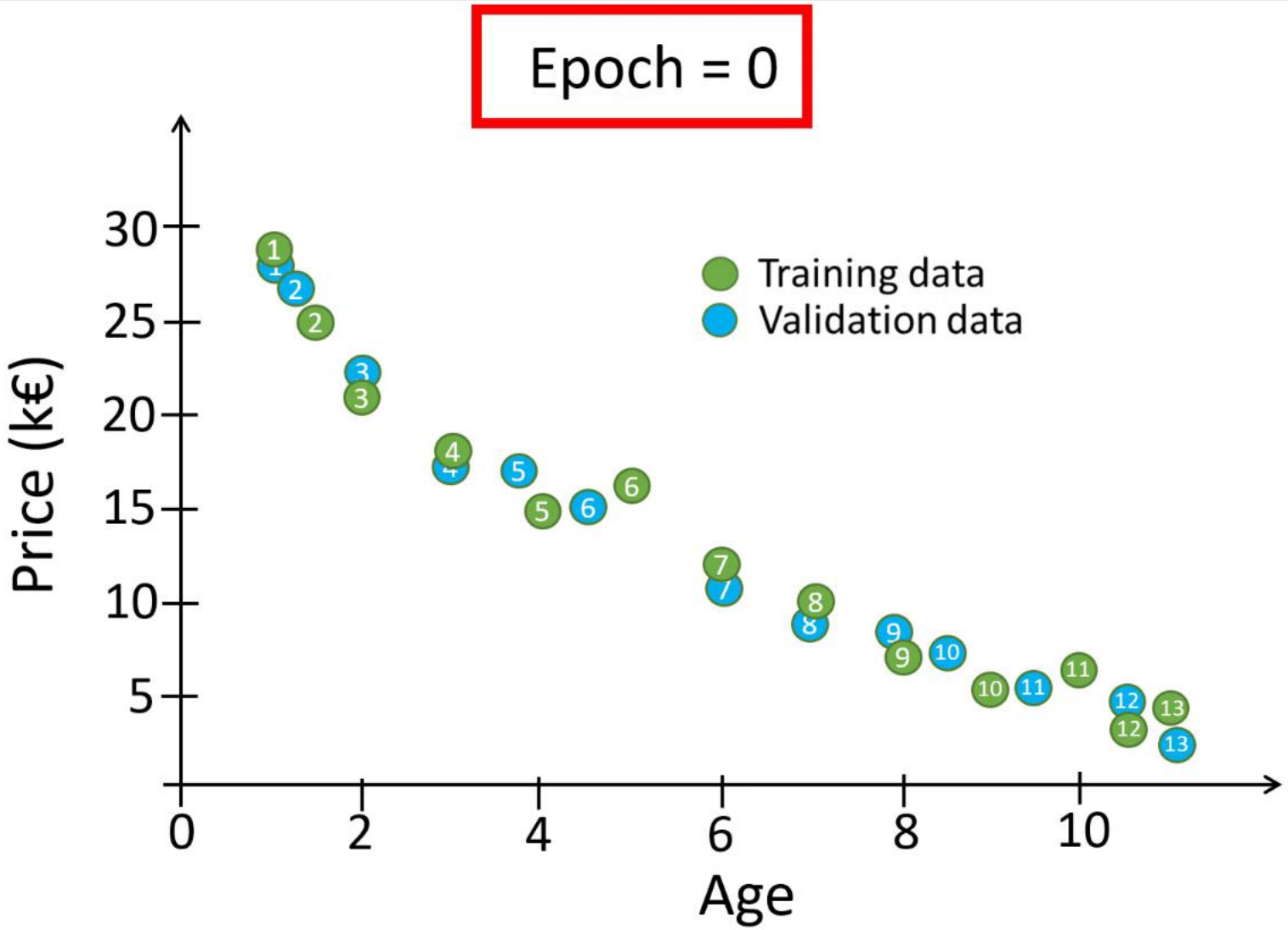
Early stopping

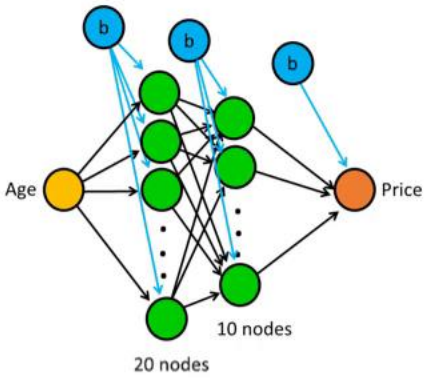




A neural network is usually initialized with random values of the weights,

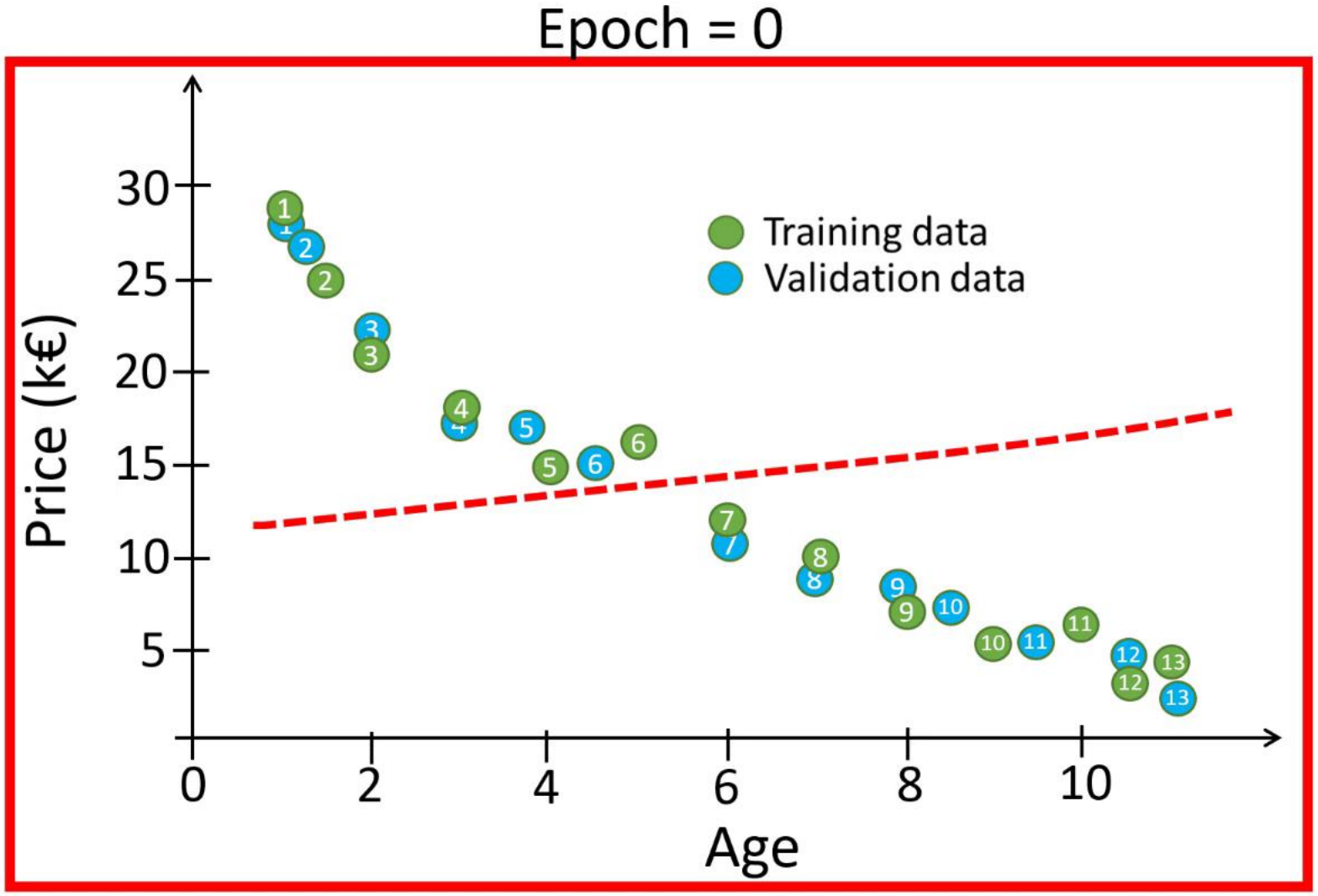
Early stopping

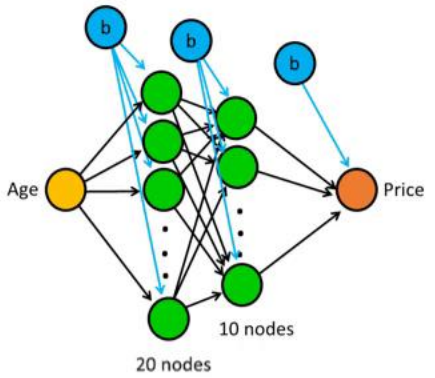




which means that before the training starts,

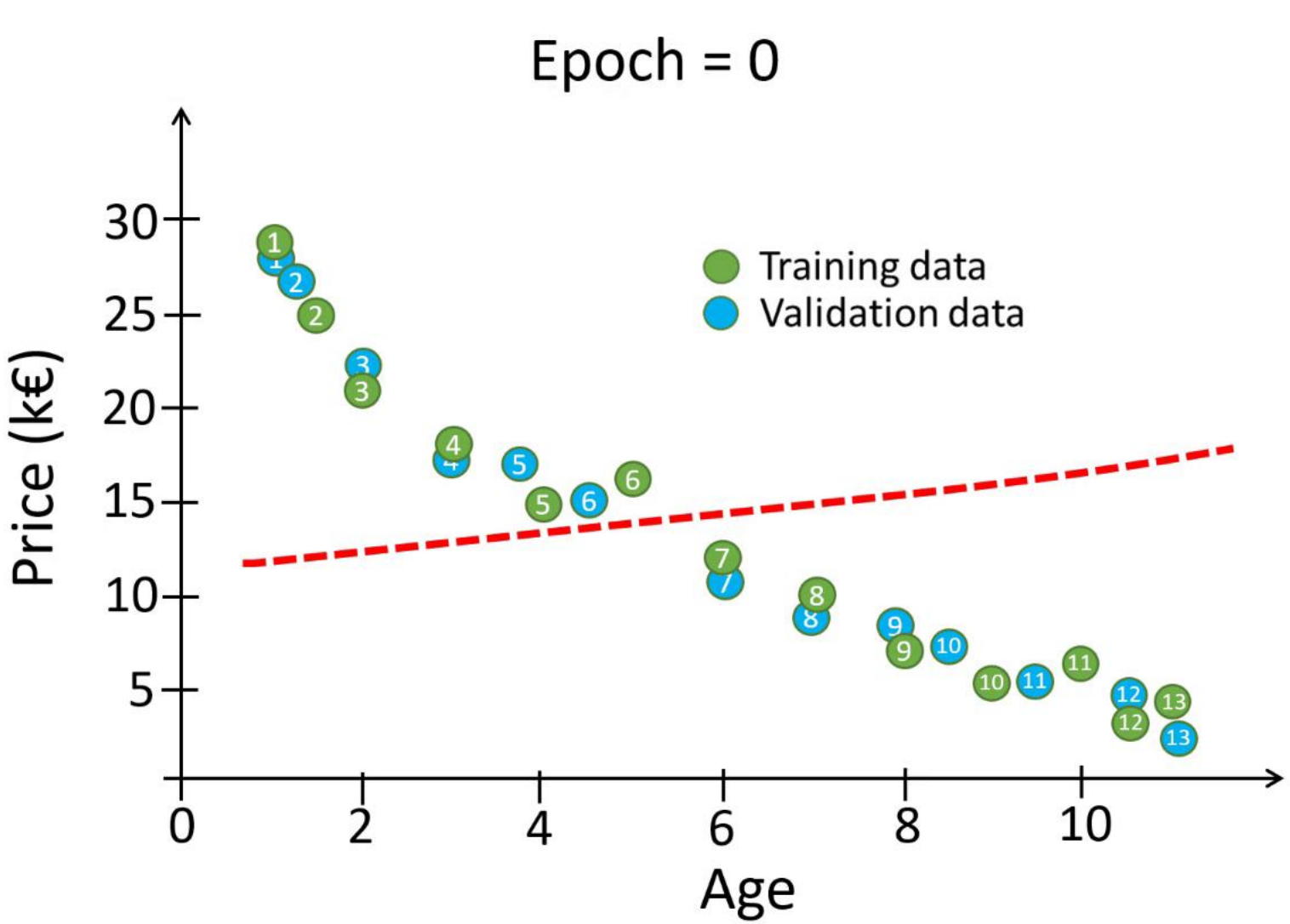
Early stopping

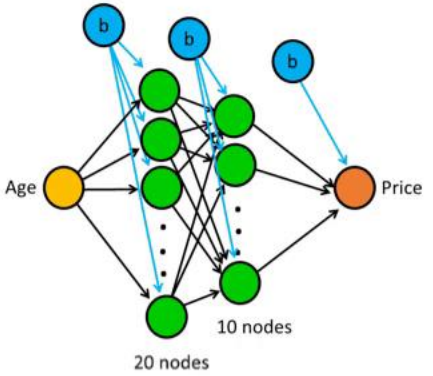


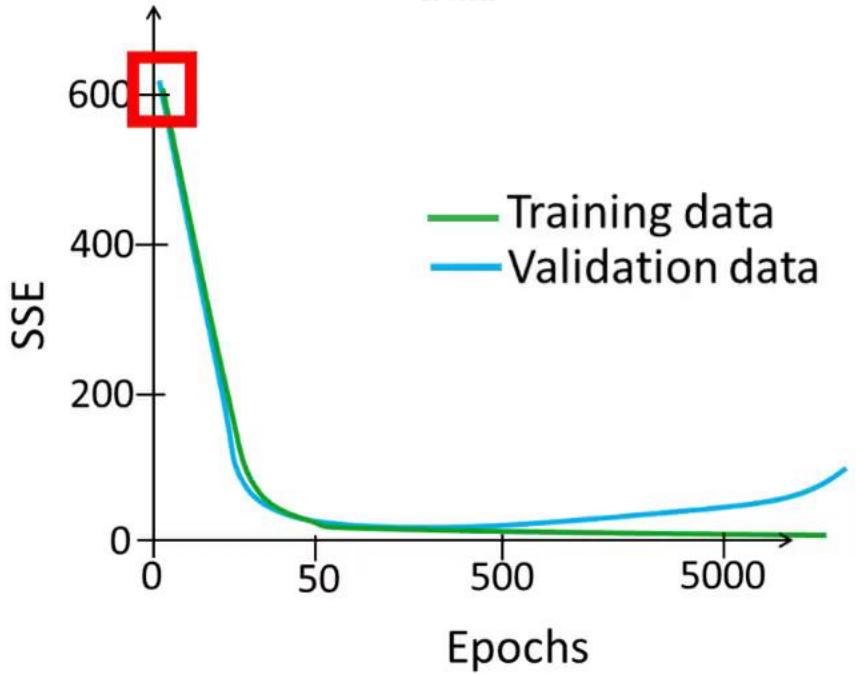


we usually get a bad fit to both the training data and the validation data, because the data points are far away from the curve.

Early stopping

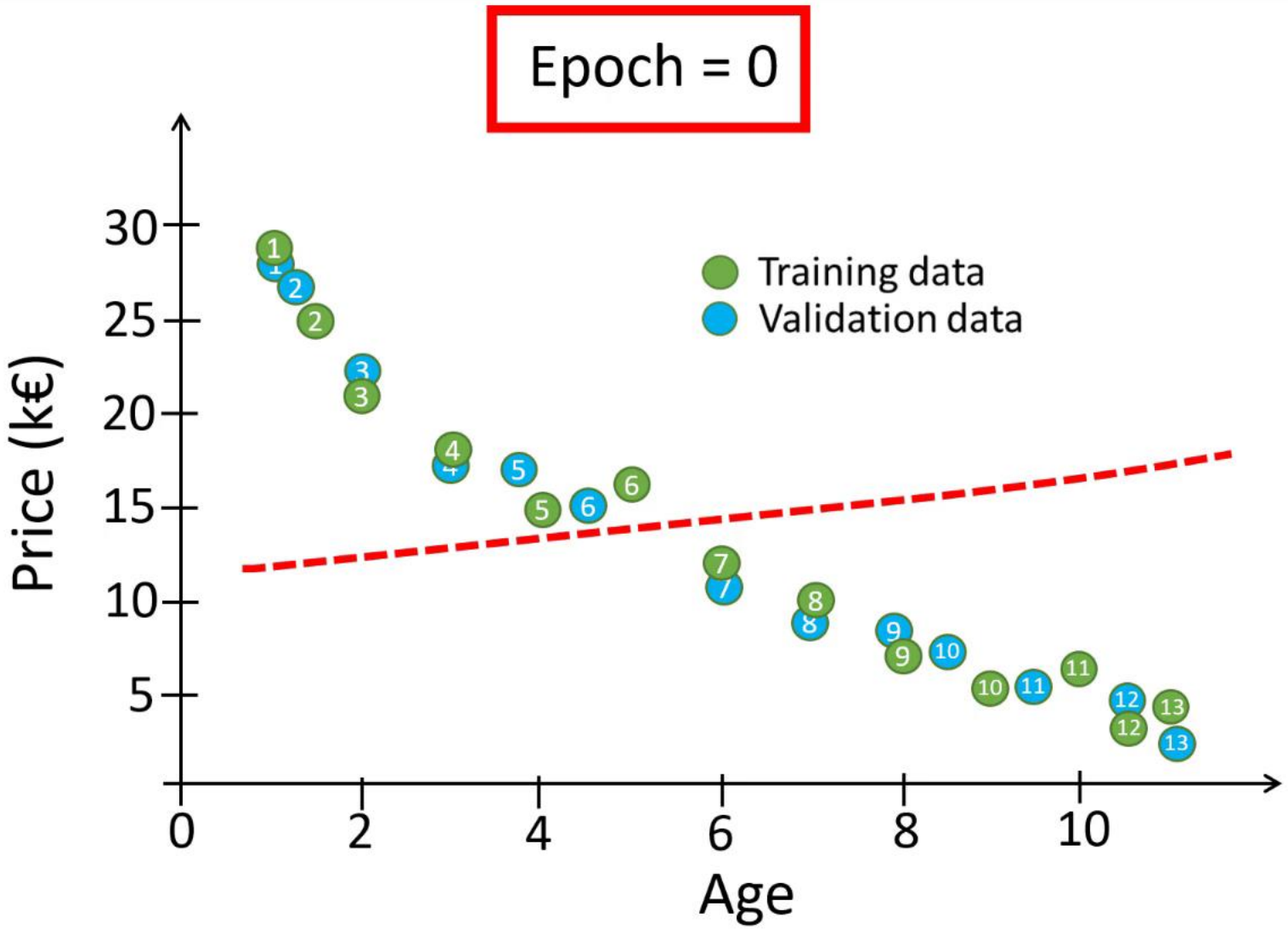


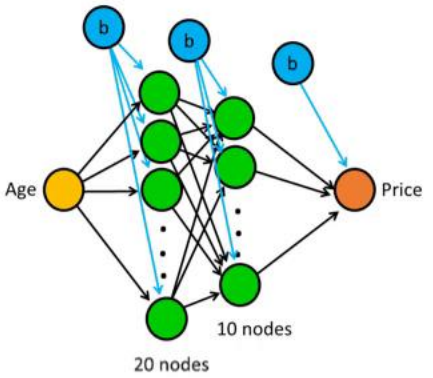


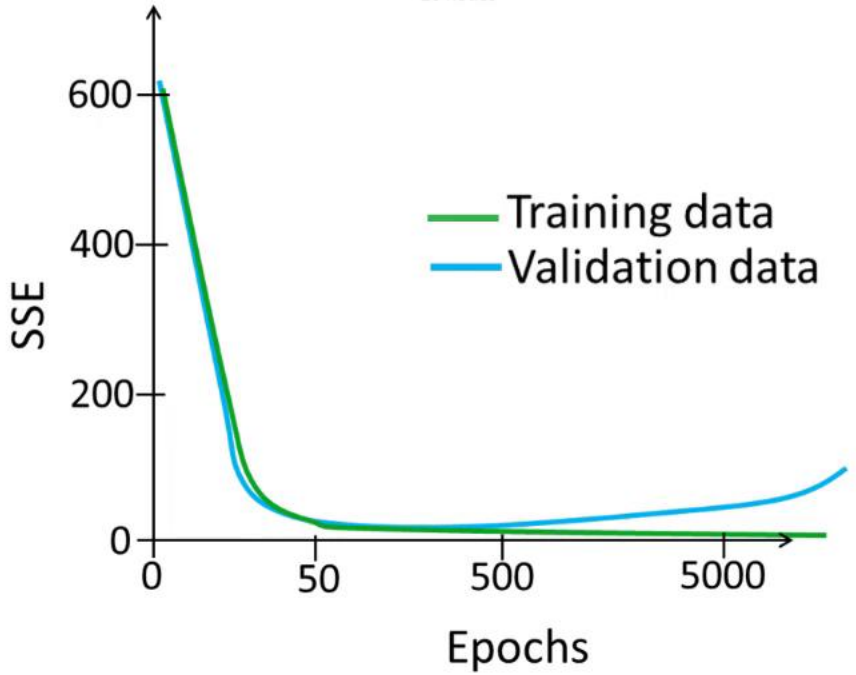


Initially, the sum of the squared errors, SSE, will therefore be very large for both the training data and the validation data.

Early stopping

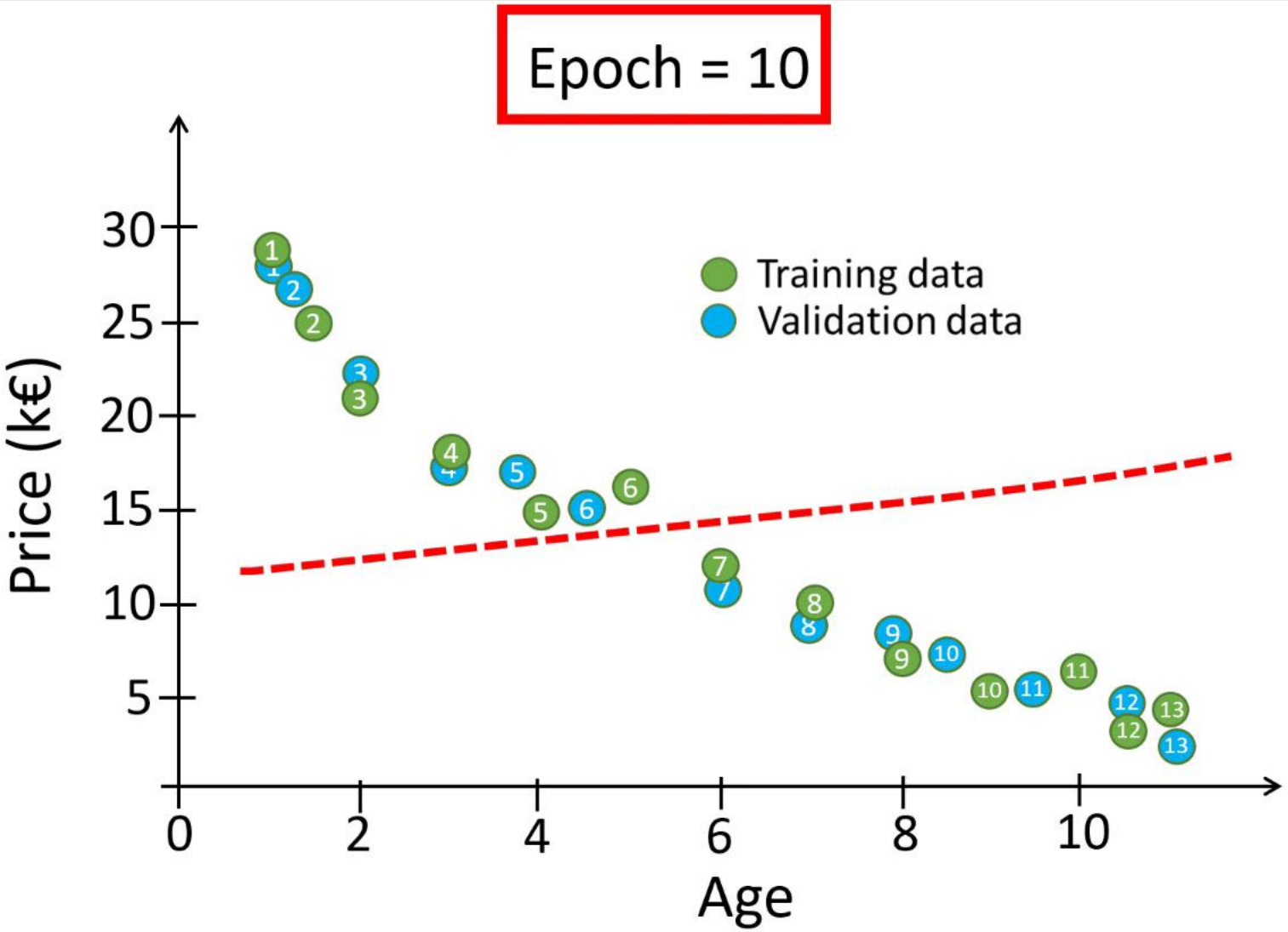


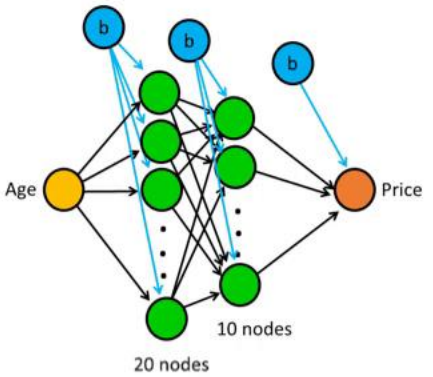


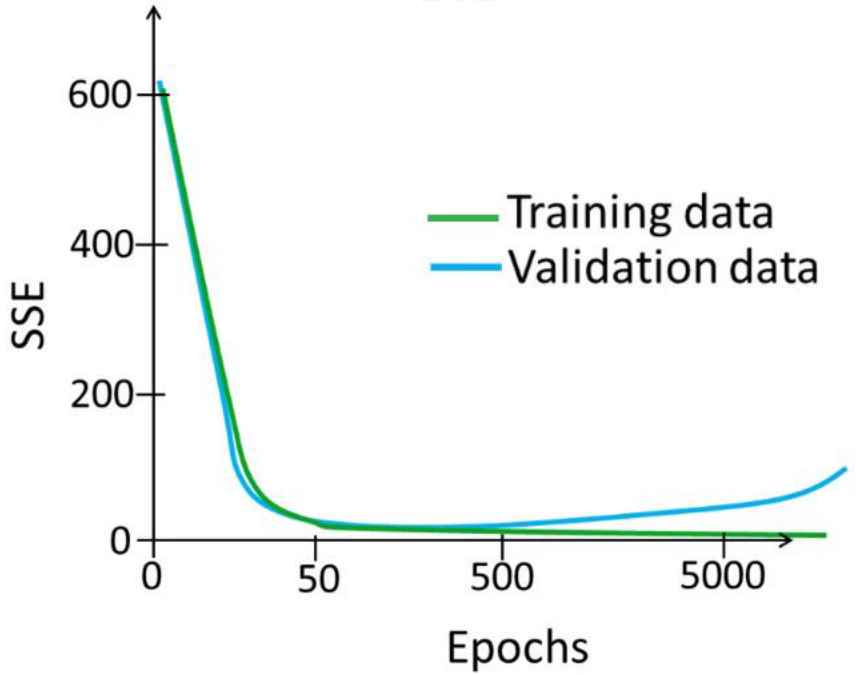


Training the network means, in this case, that its weights will be updated to reduce the error. This is done with an iterative procedure. Once all data points have been used during the iterations, that is called one epoch.

Early stopping

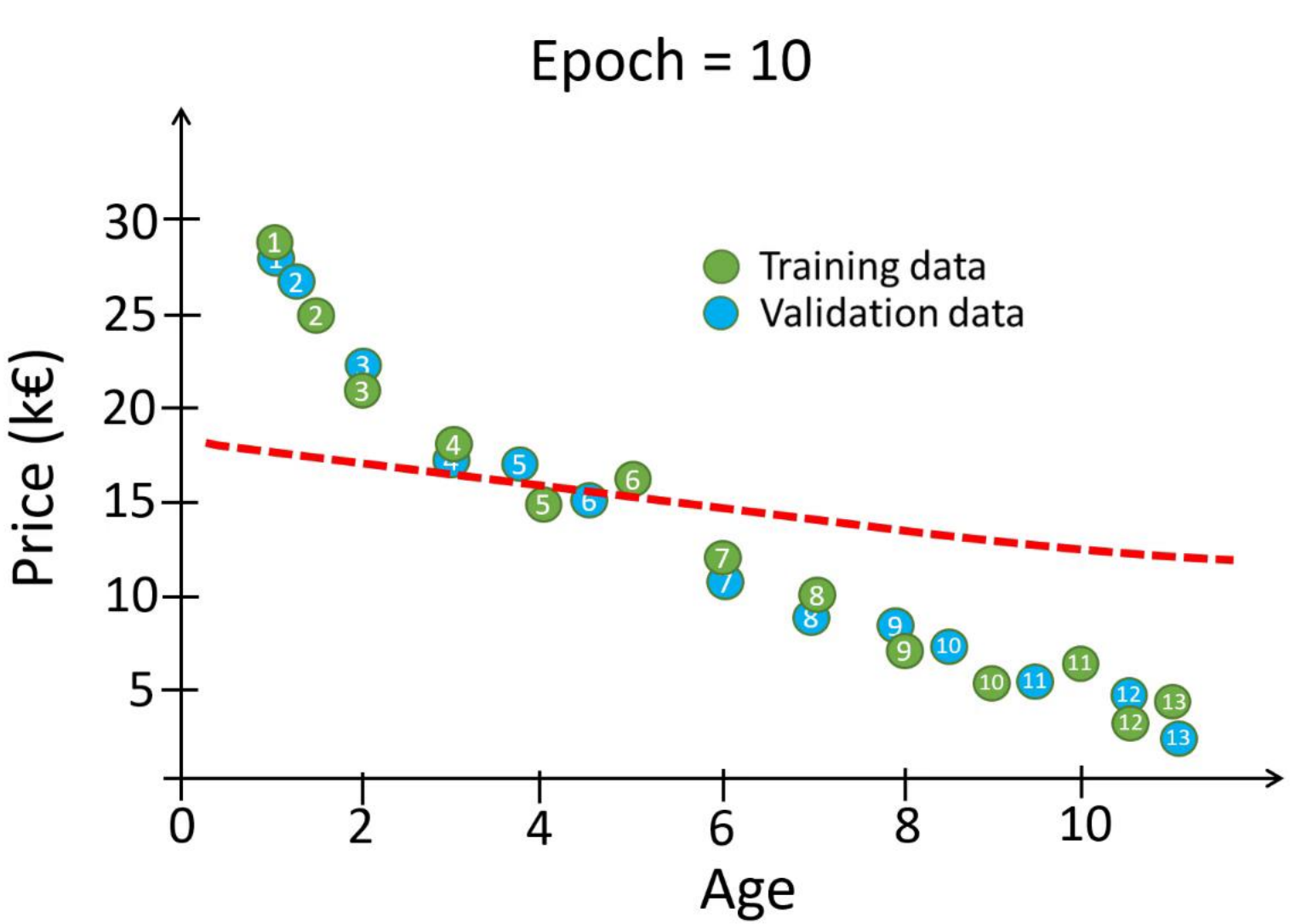


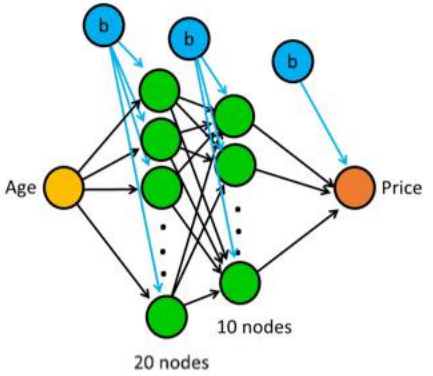


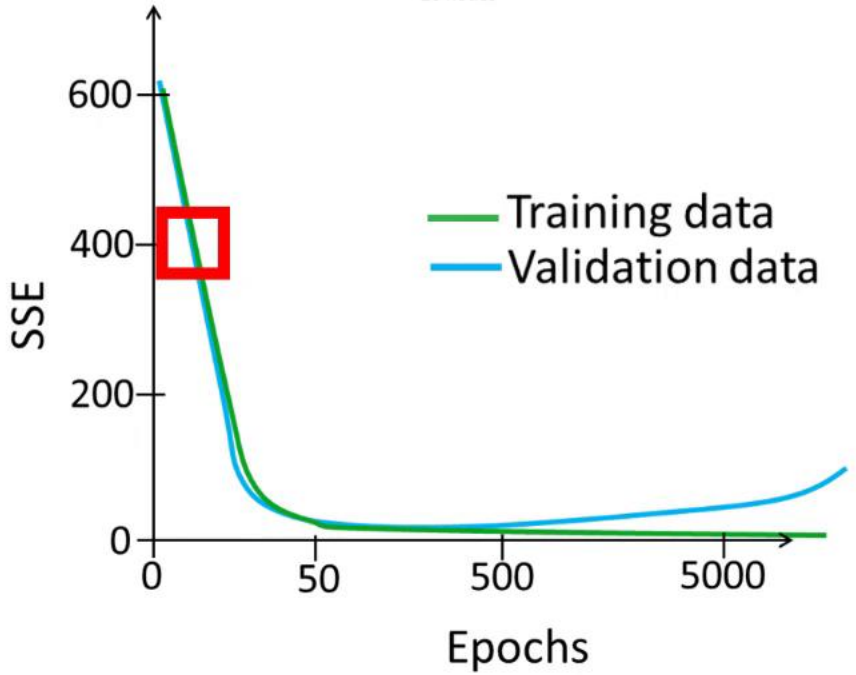


After a few epochs, where the weights have been updated,

Early stopping

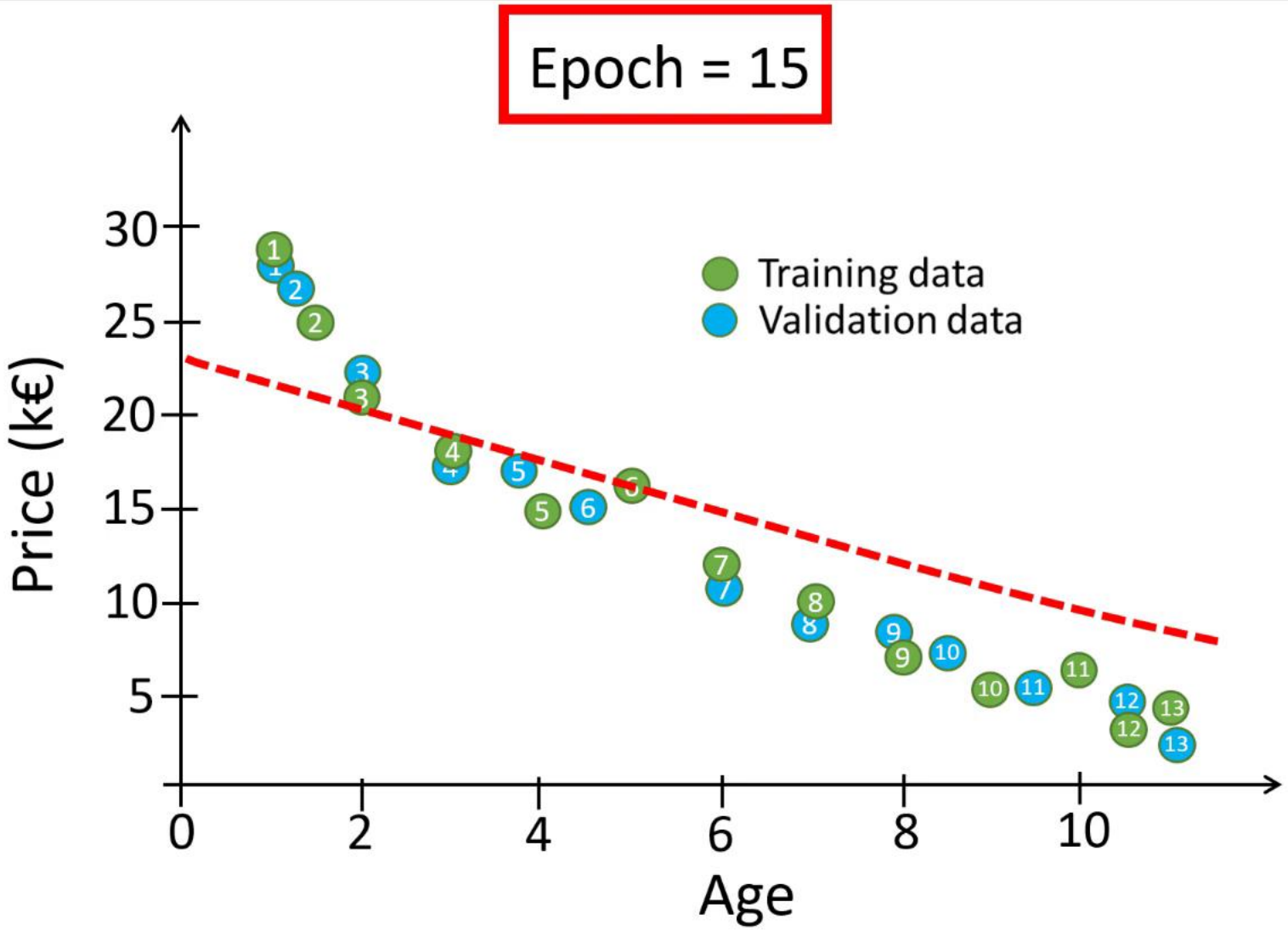


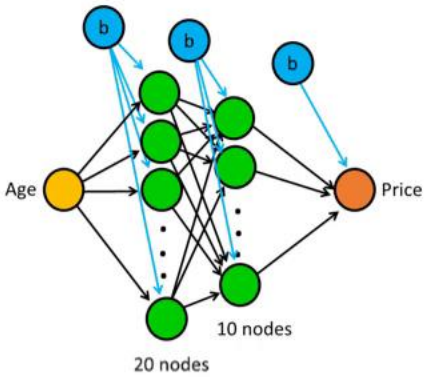


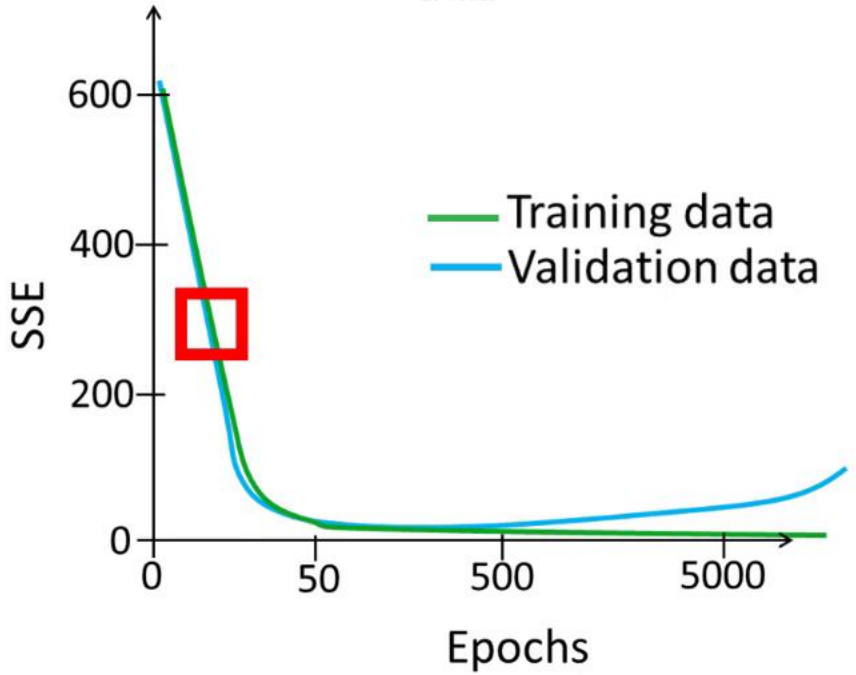


and the SSE value is therefore reduced.

Early stopping

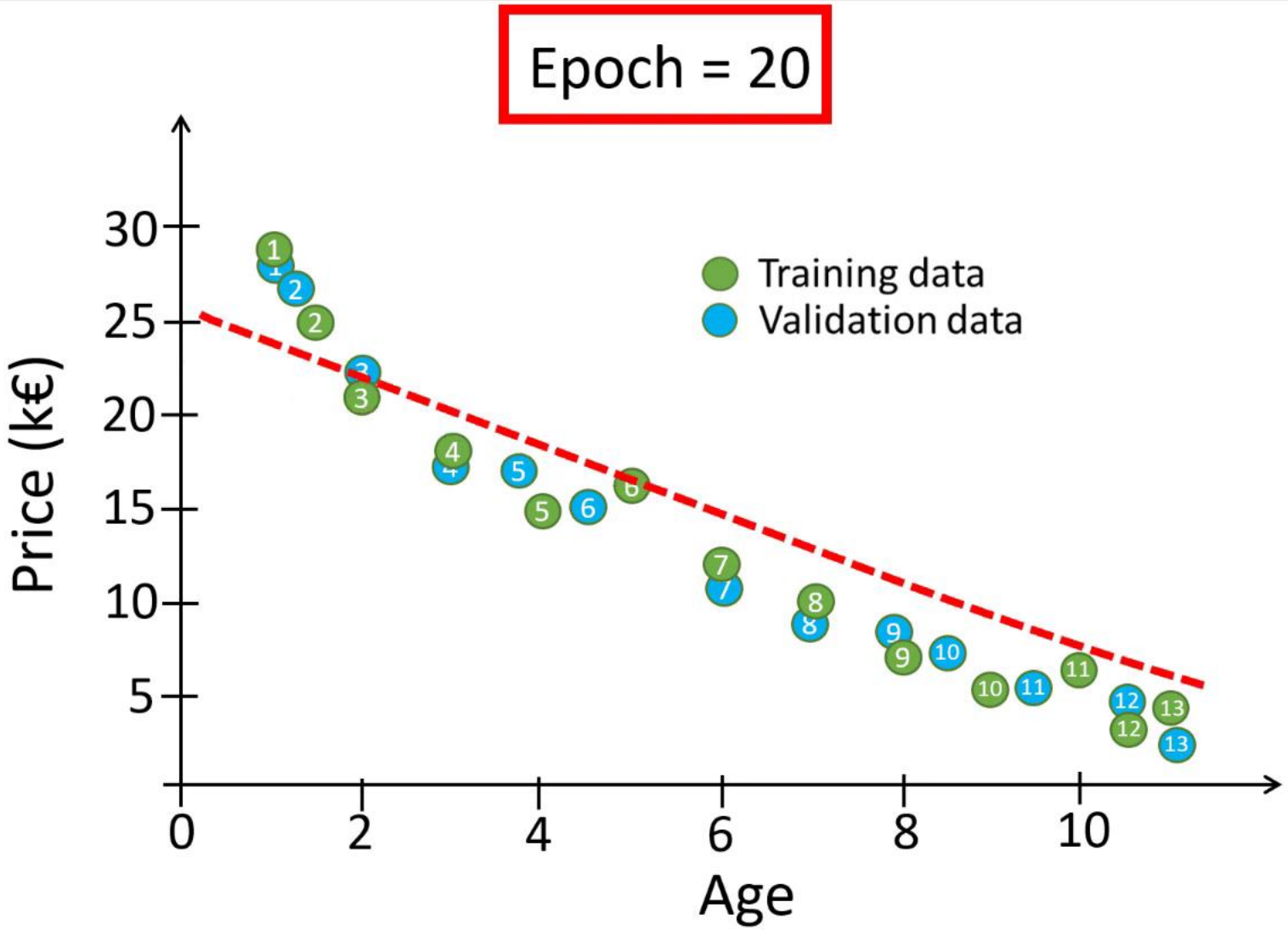


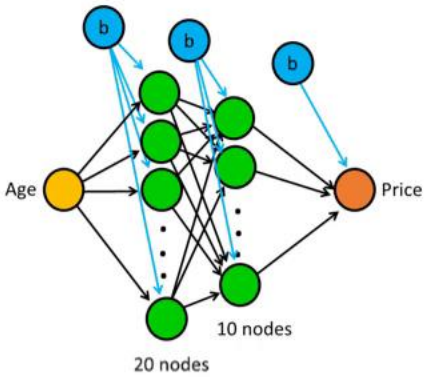


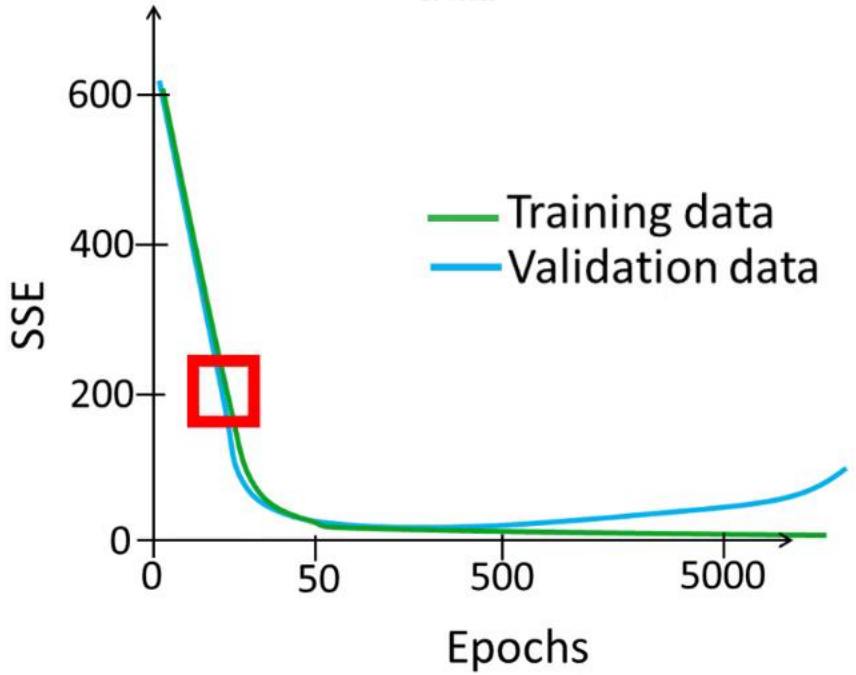


After more training, the curve starts to fit better,

Early stopping

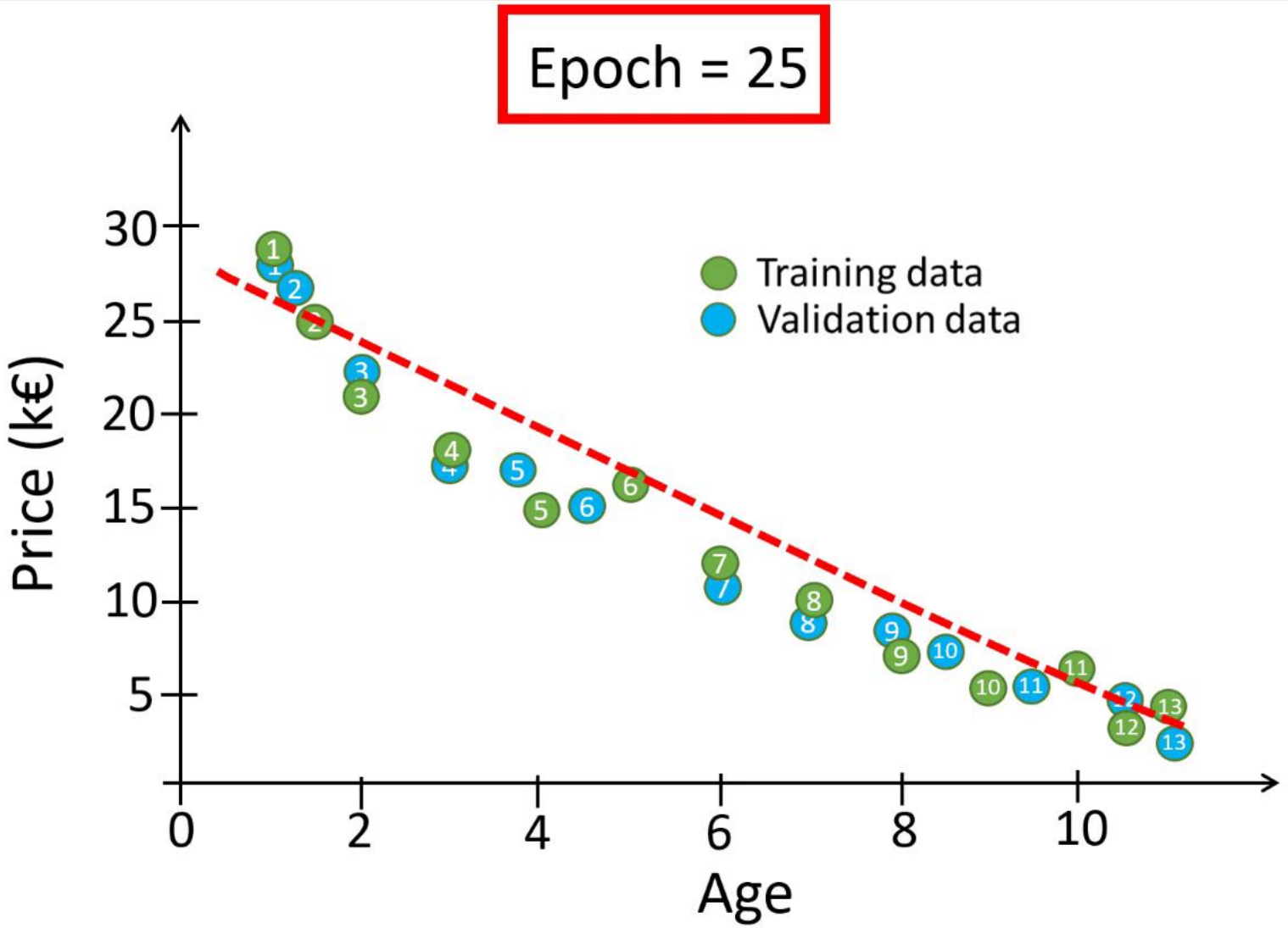


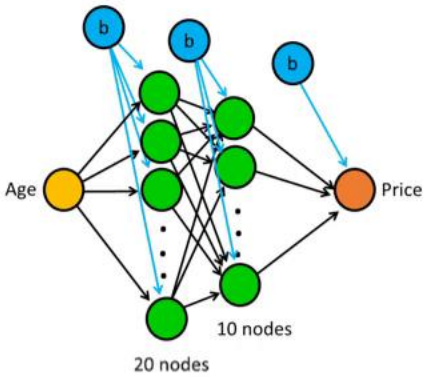


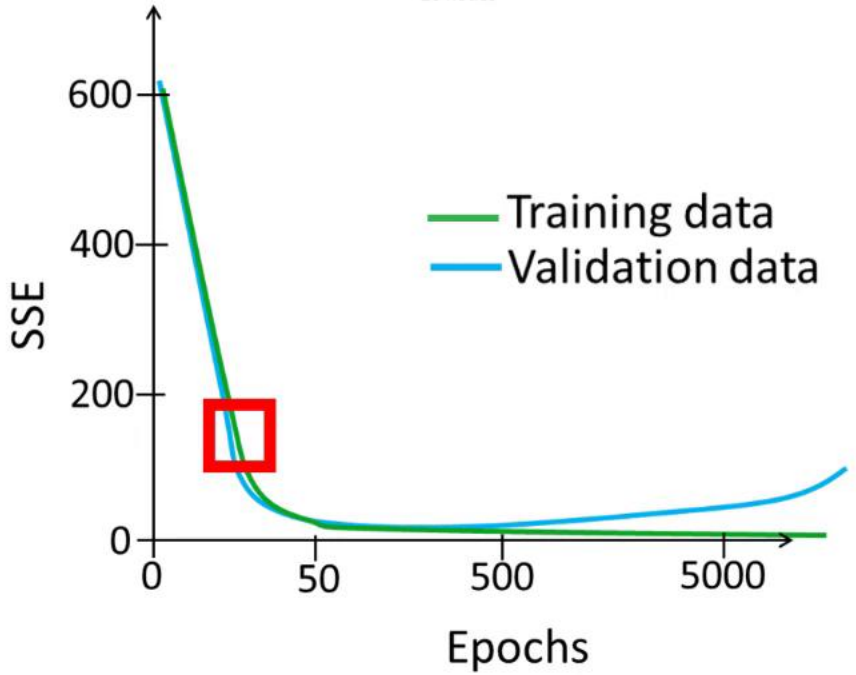


and better,

Early stopping

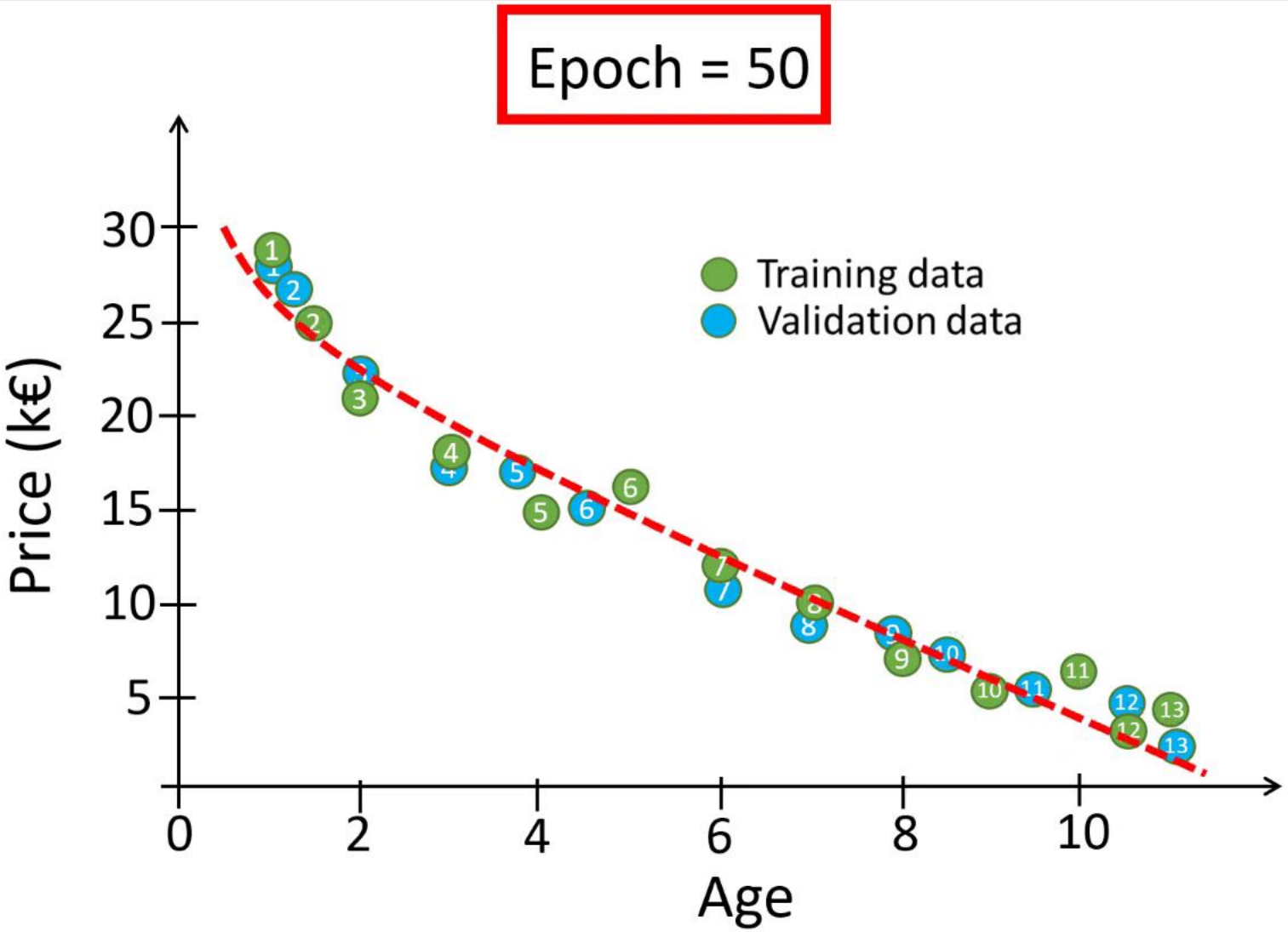


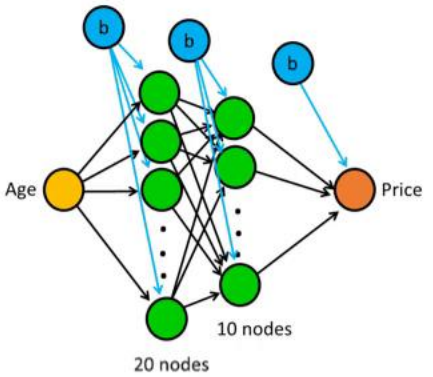


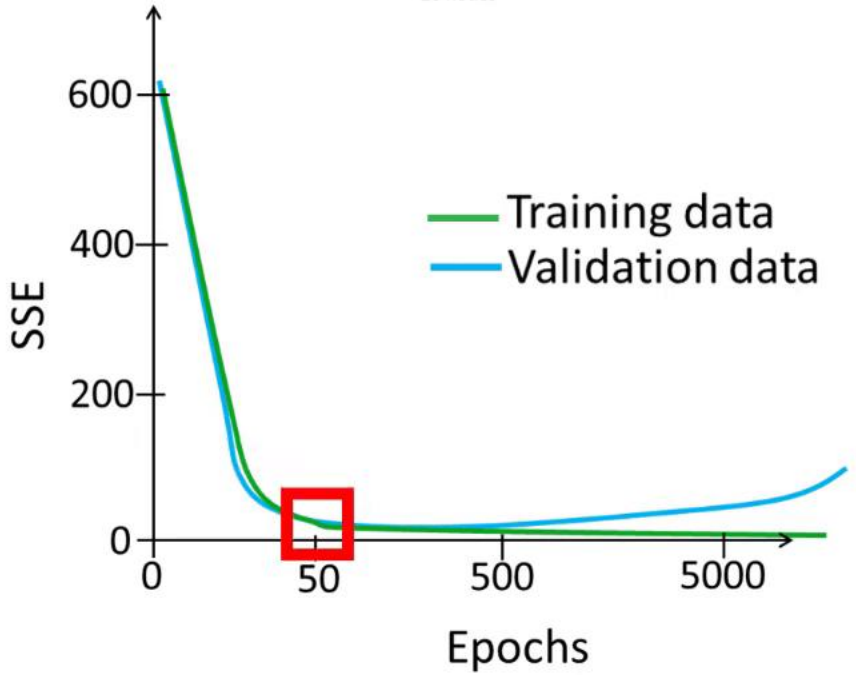


and better.

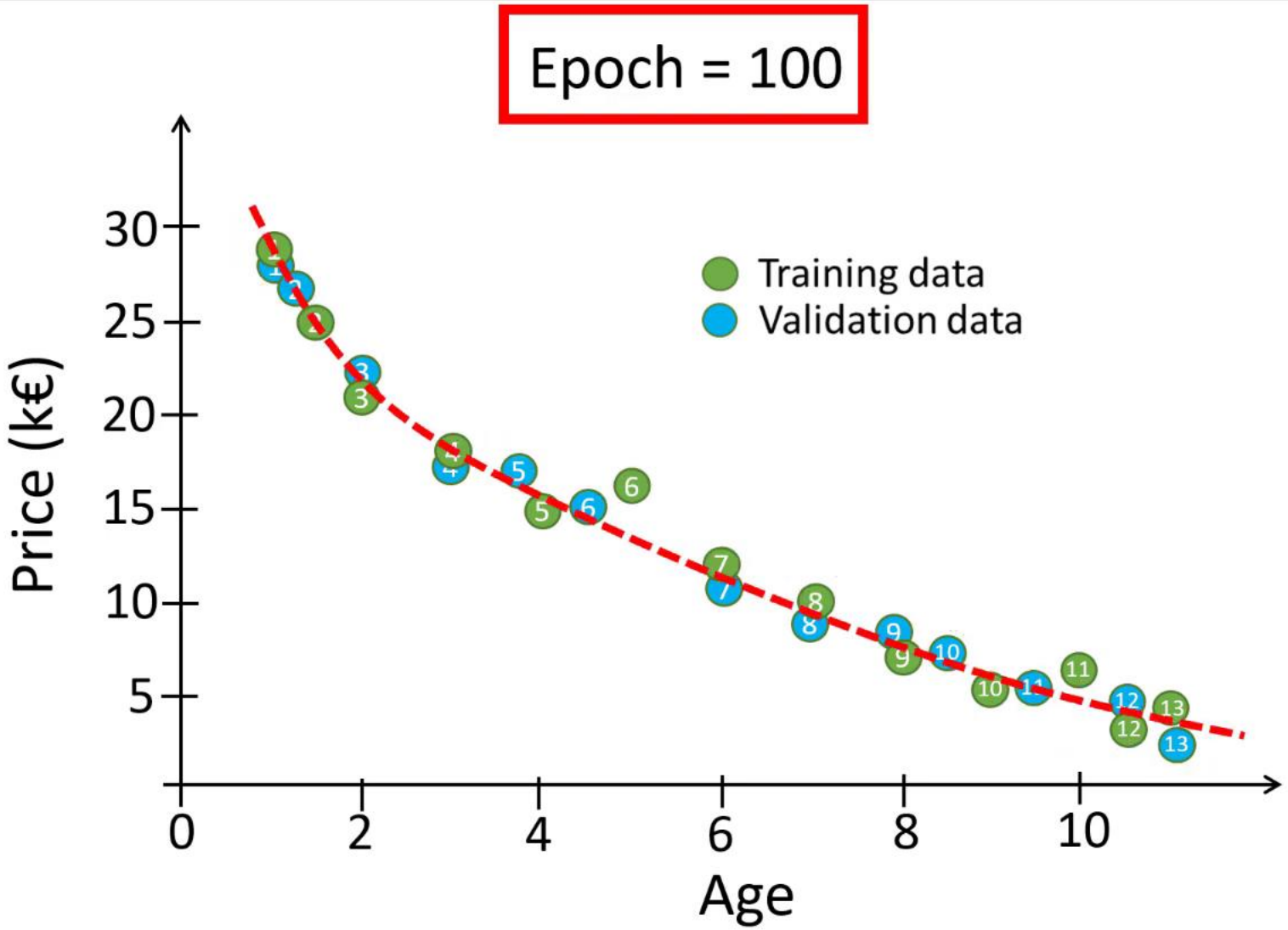
Early stopping

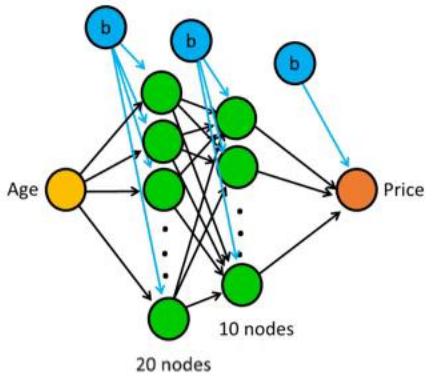


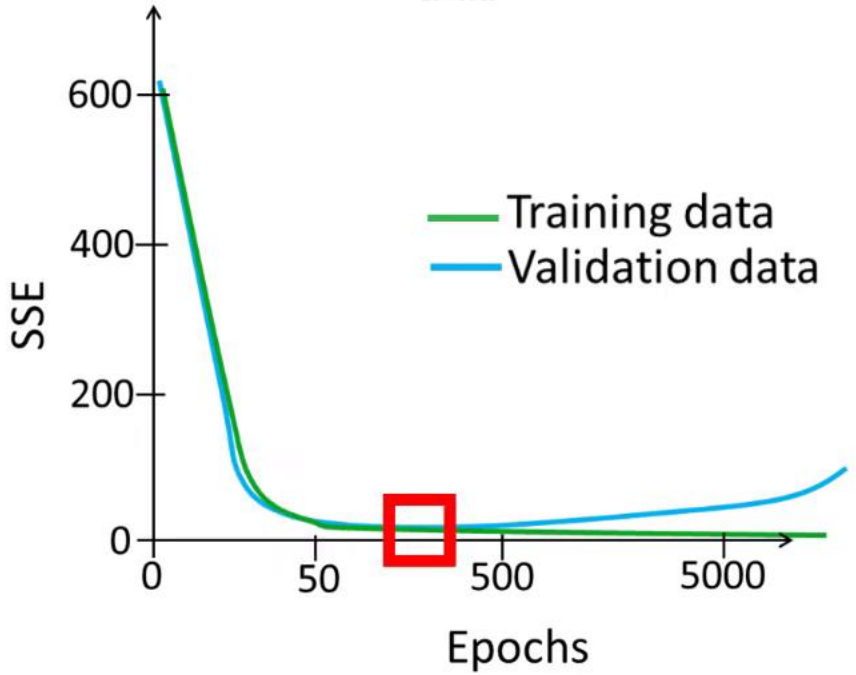




Early stopping



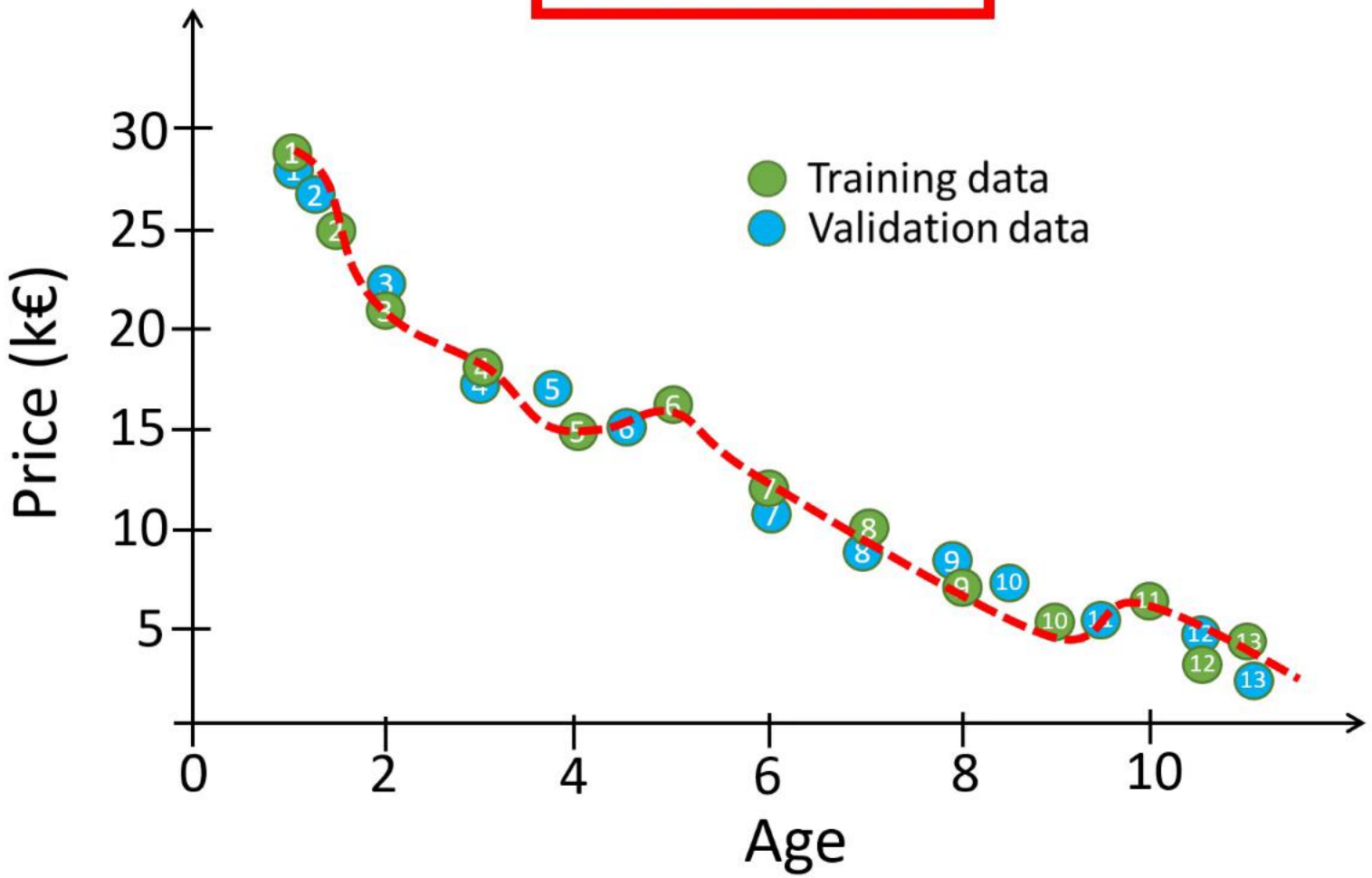


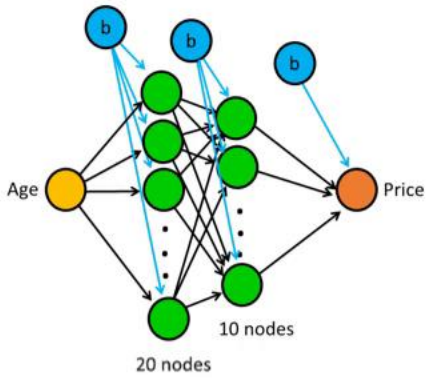


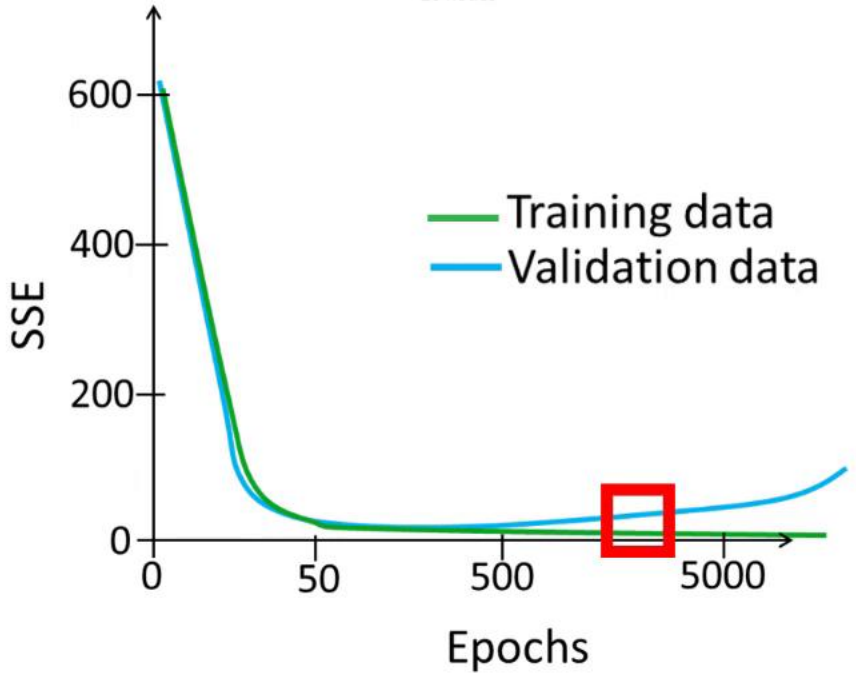
After about 100 epochs, the network starts to fit quite well with both the training data and the validation data.

Early stopping

Epoch = 1000

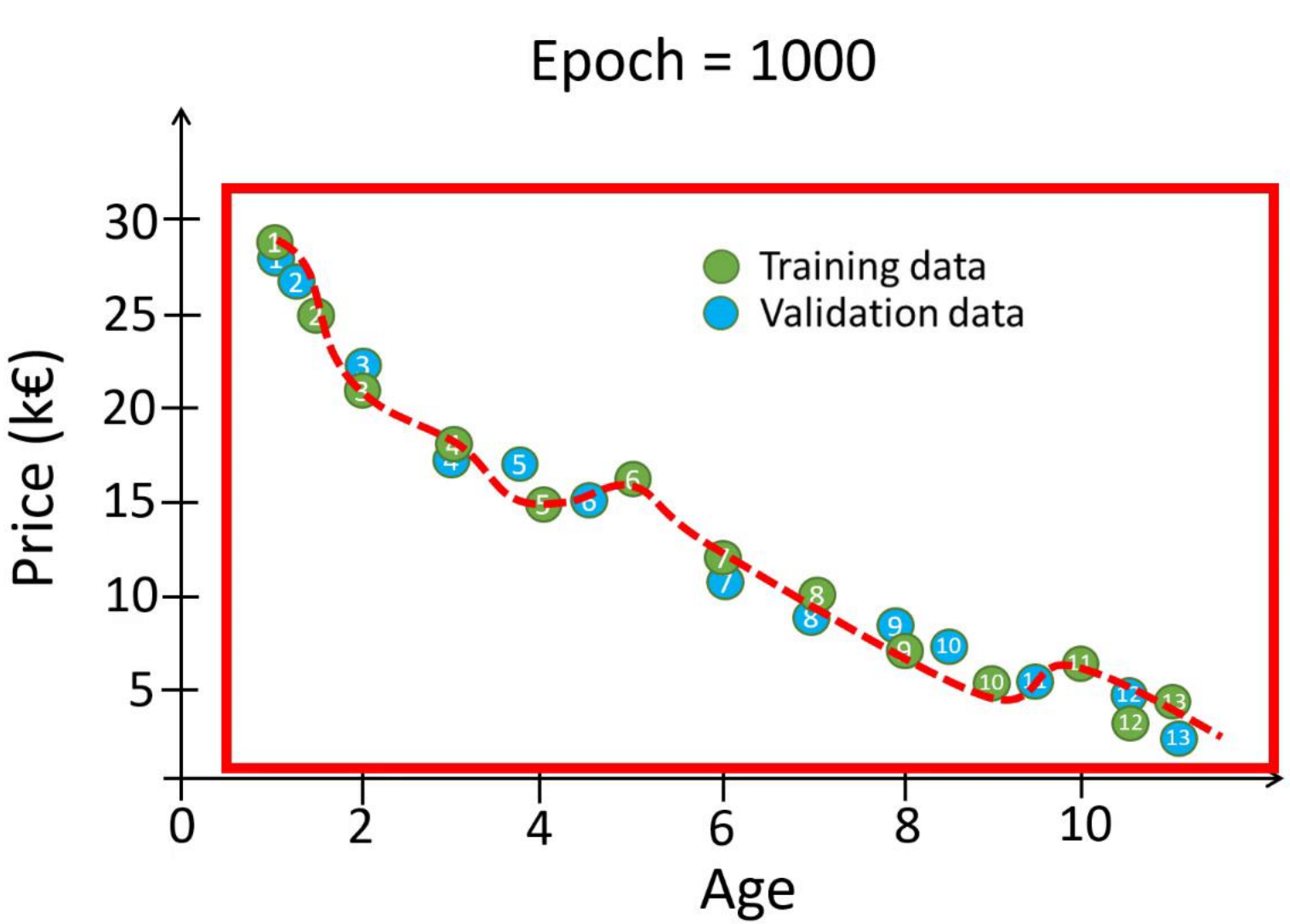


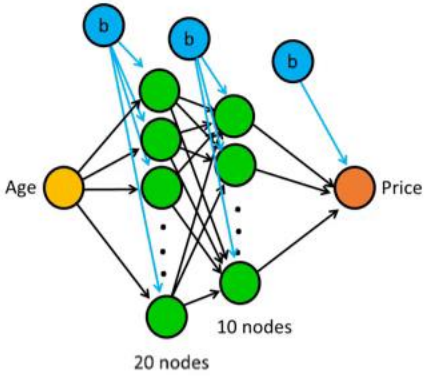


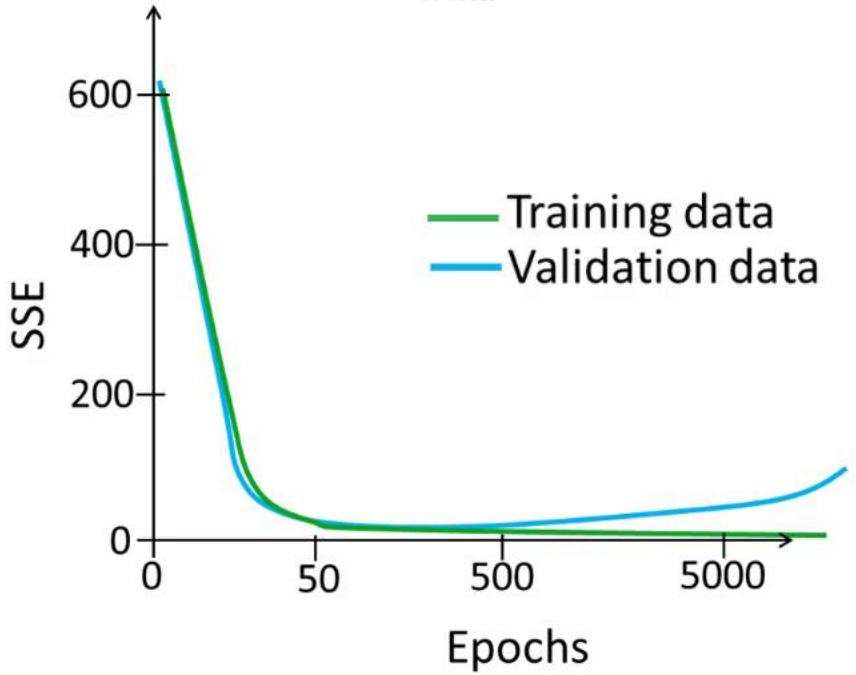


However, if we continue to train, the model starts to overfit because the SSE value starts to go up for the validation data,

Early stopping

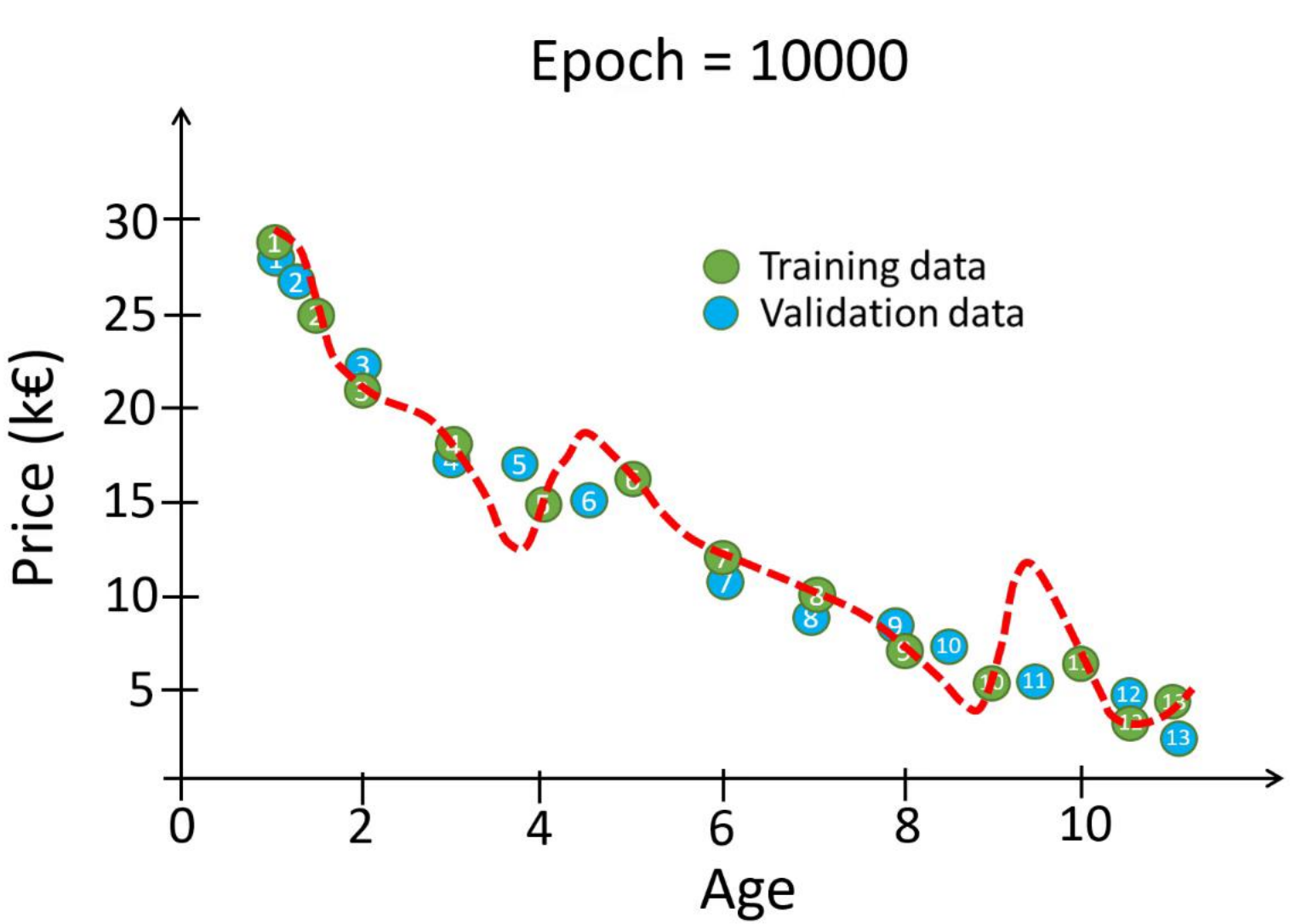


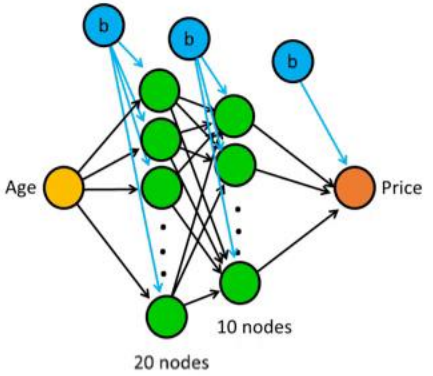


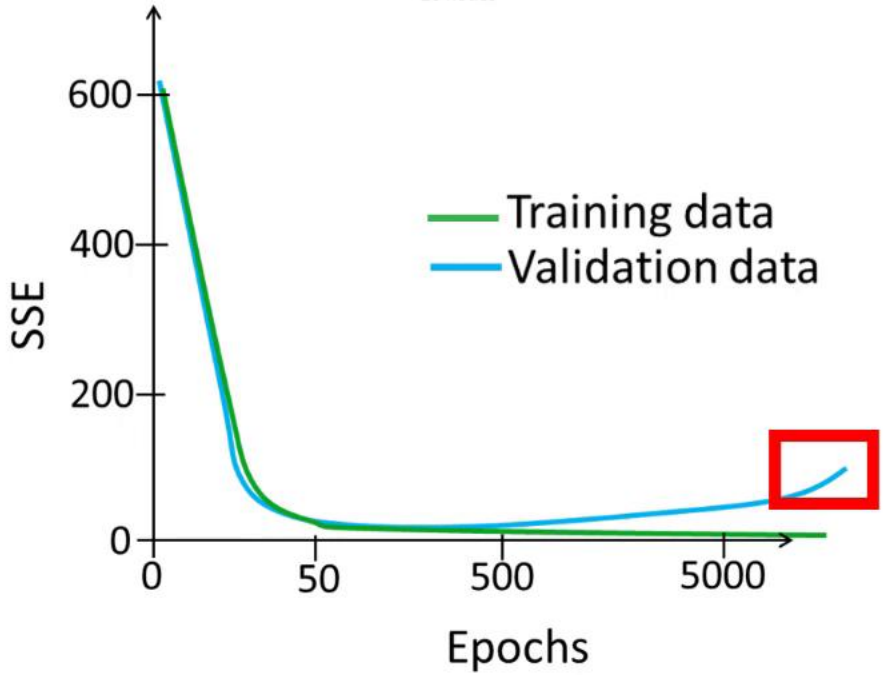


and the curve starts to look a bit strange.

Early stopping

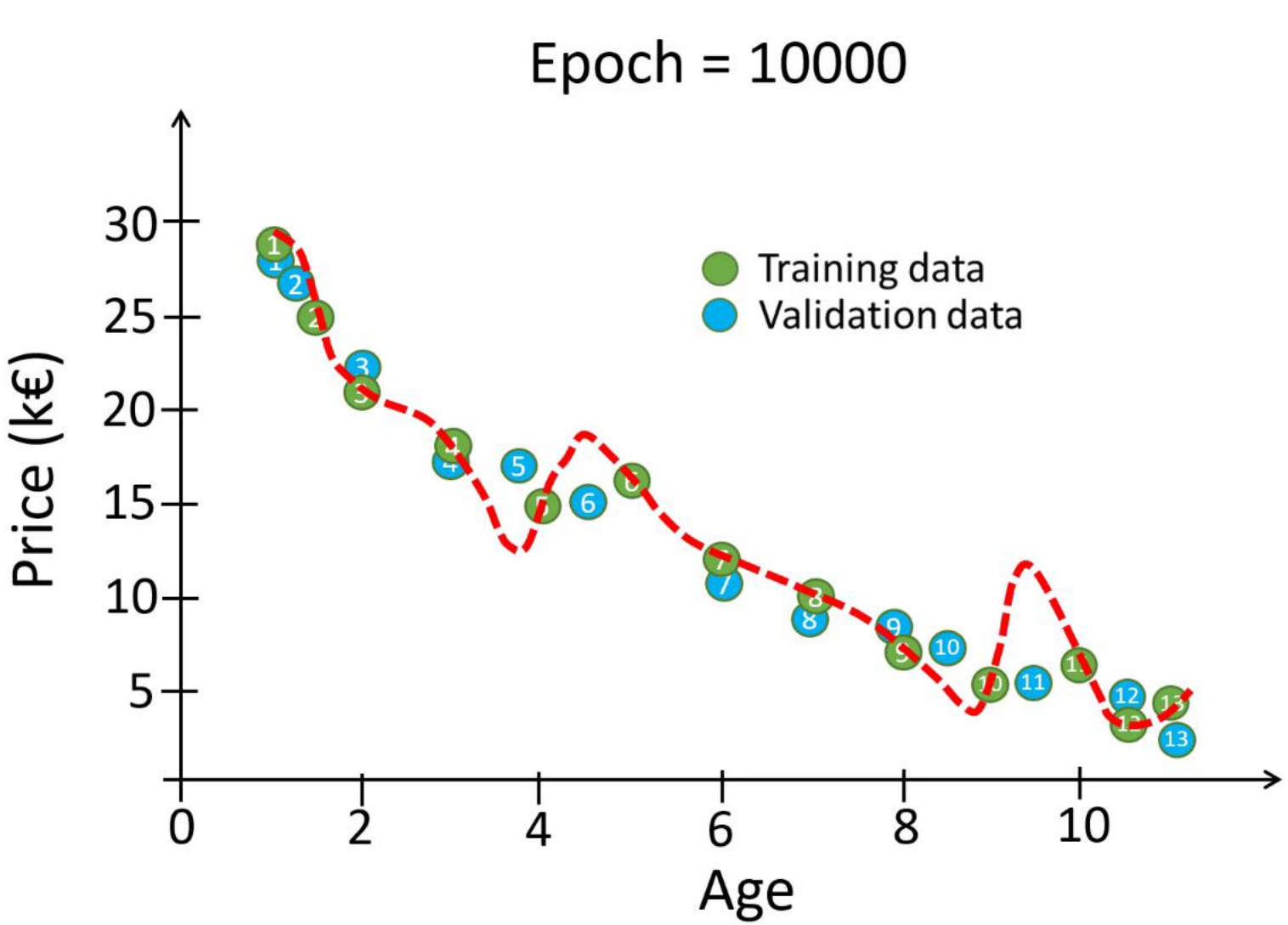


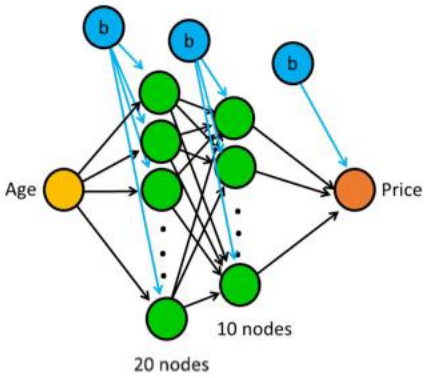


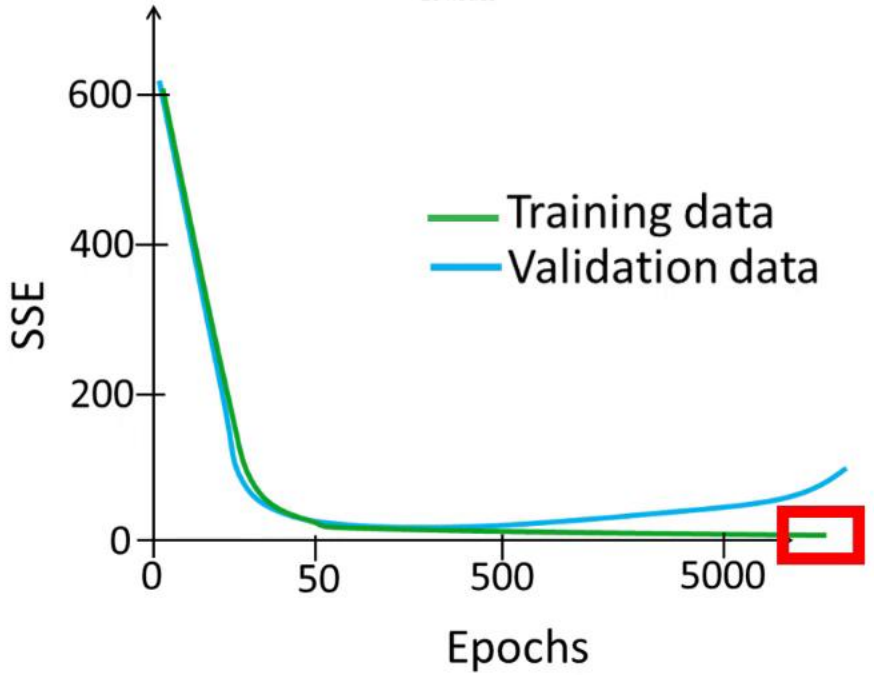


If we continue to train even more, we start to get a bad fit to the validation data,

Early stopping

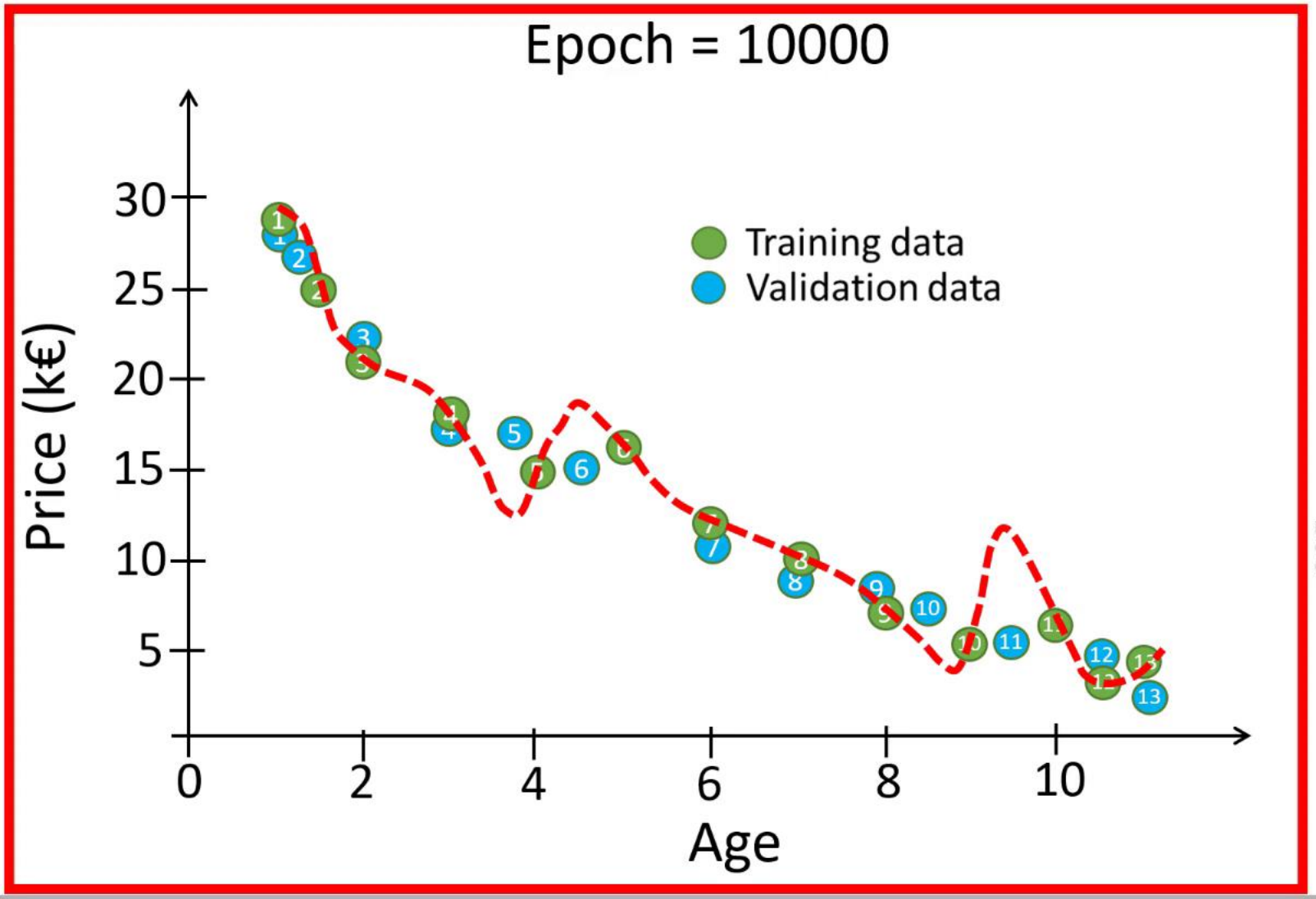


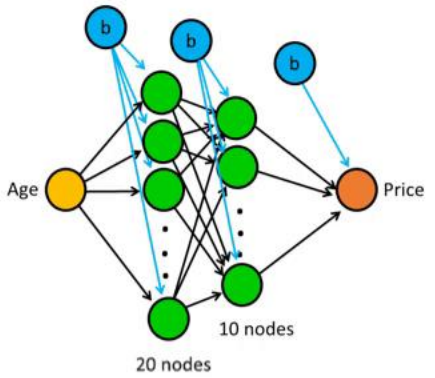


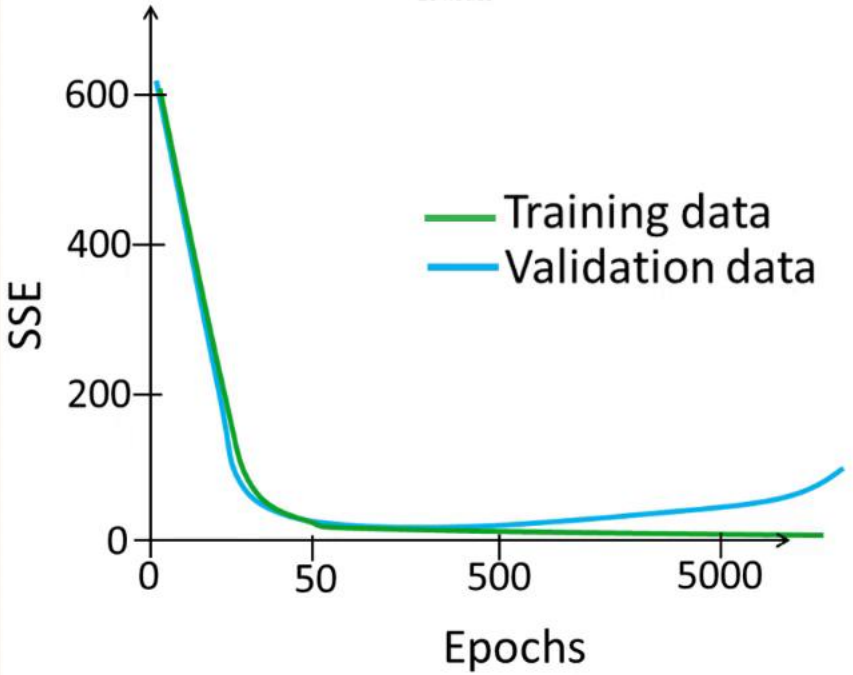


but almost a perfect fit to the training data because the training error is almost zero.

Early stopping

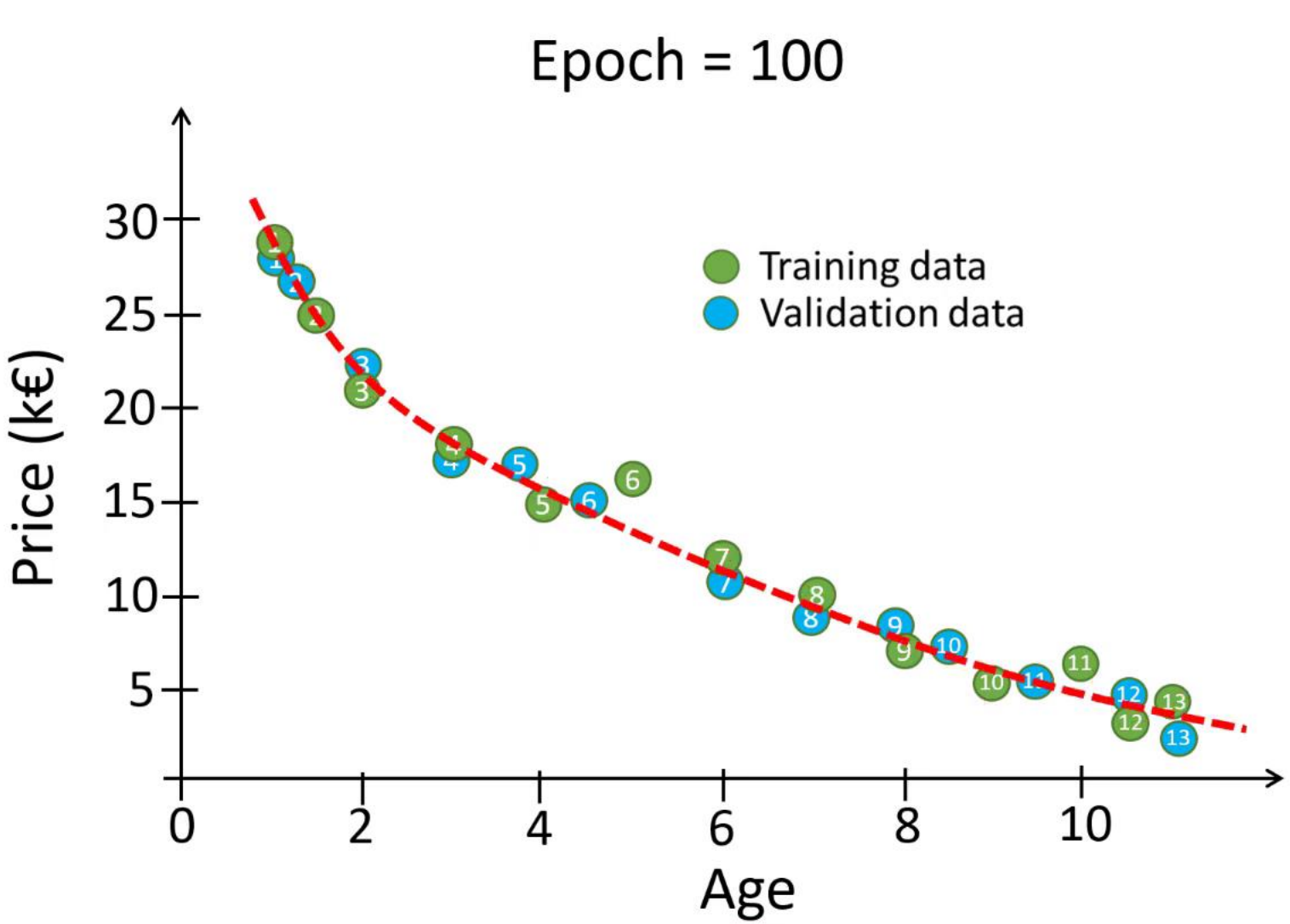


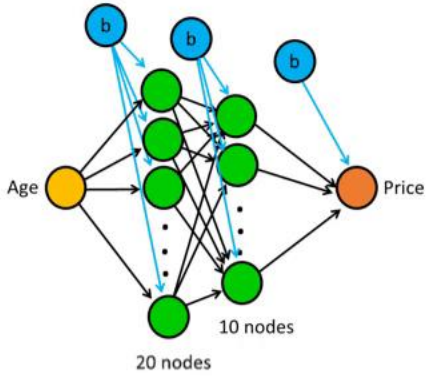


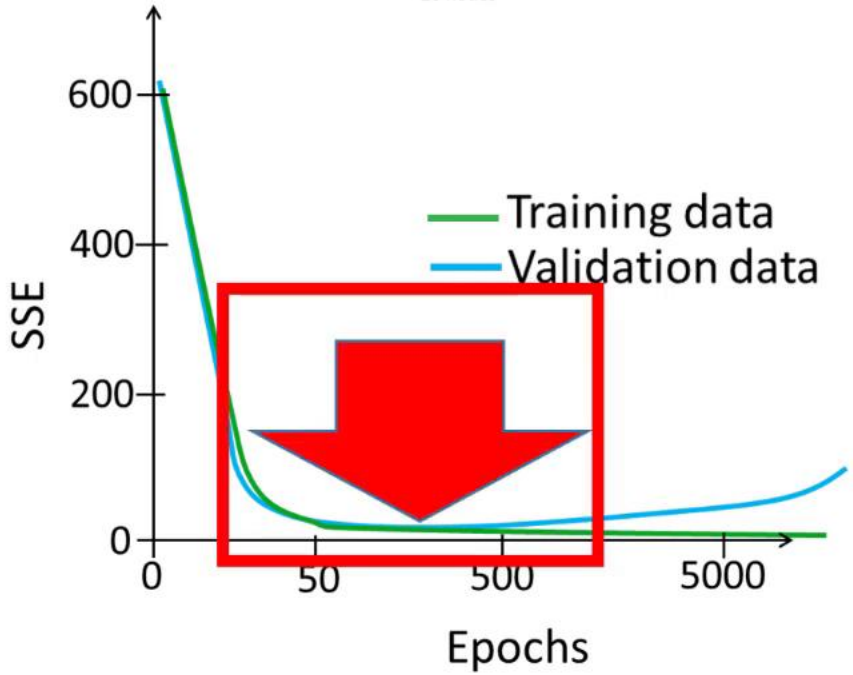


Although the network fits perfectly with the training data, it fails to generalize well on unseen data. This explains the concept of overfitting.

Early stopping



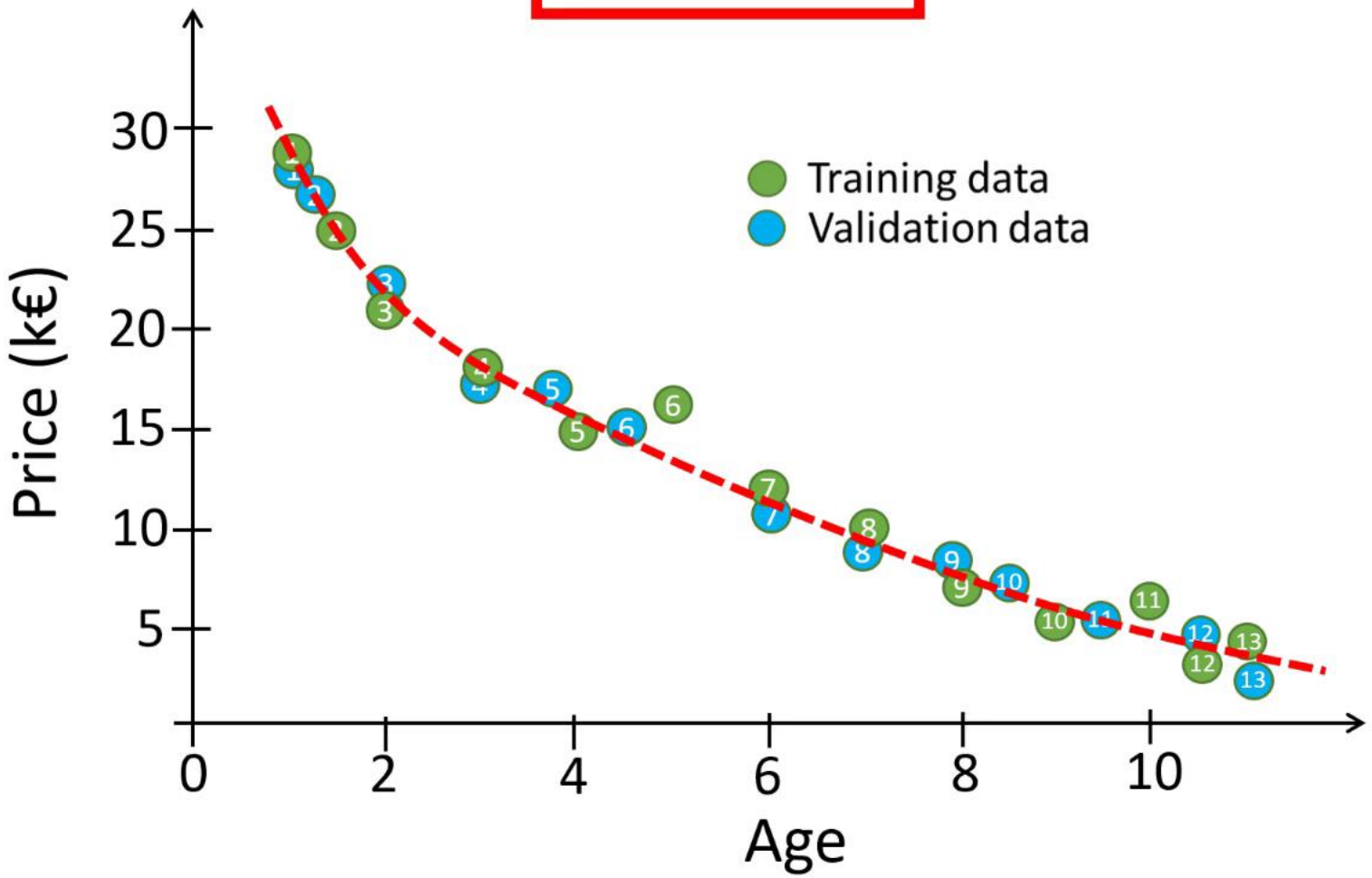


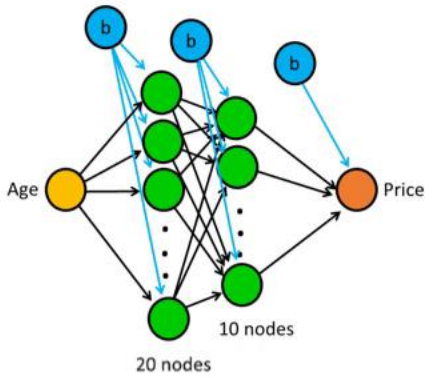


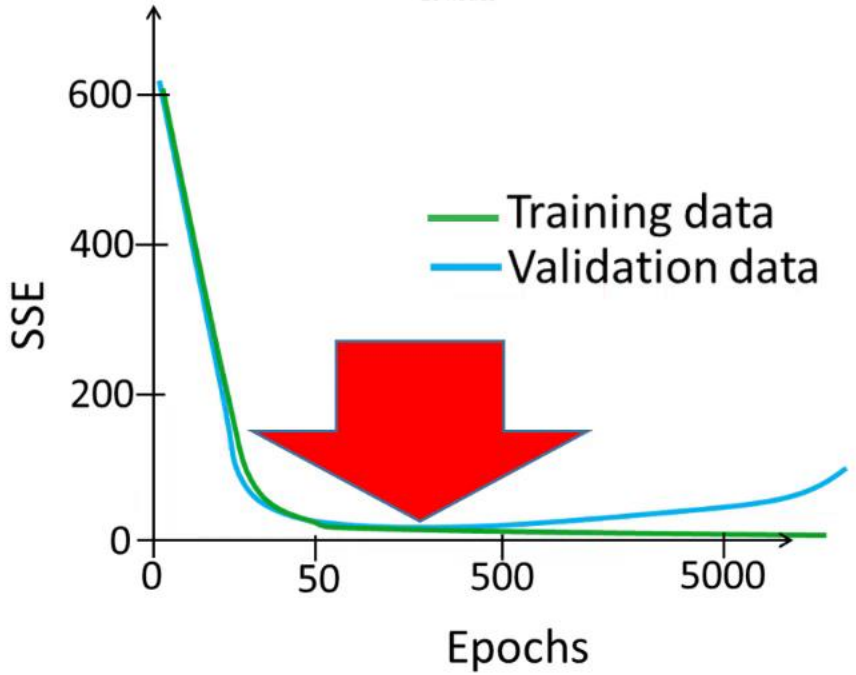
Early stopping means that we stop training the network when the error for the validation data starts to go up.

Early stopping

Epoch = 100

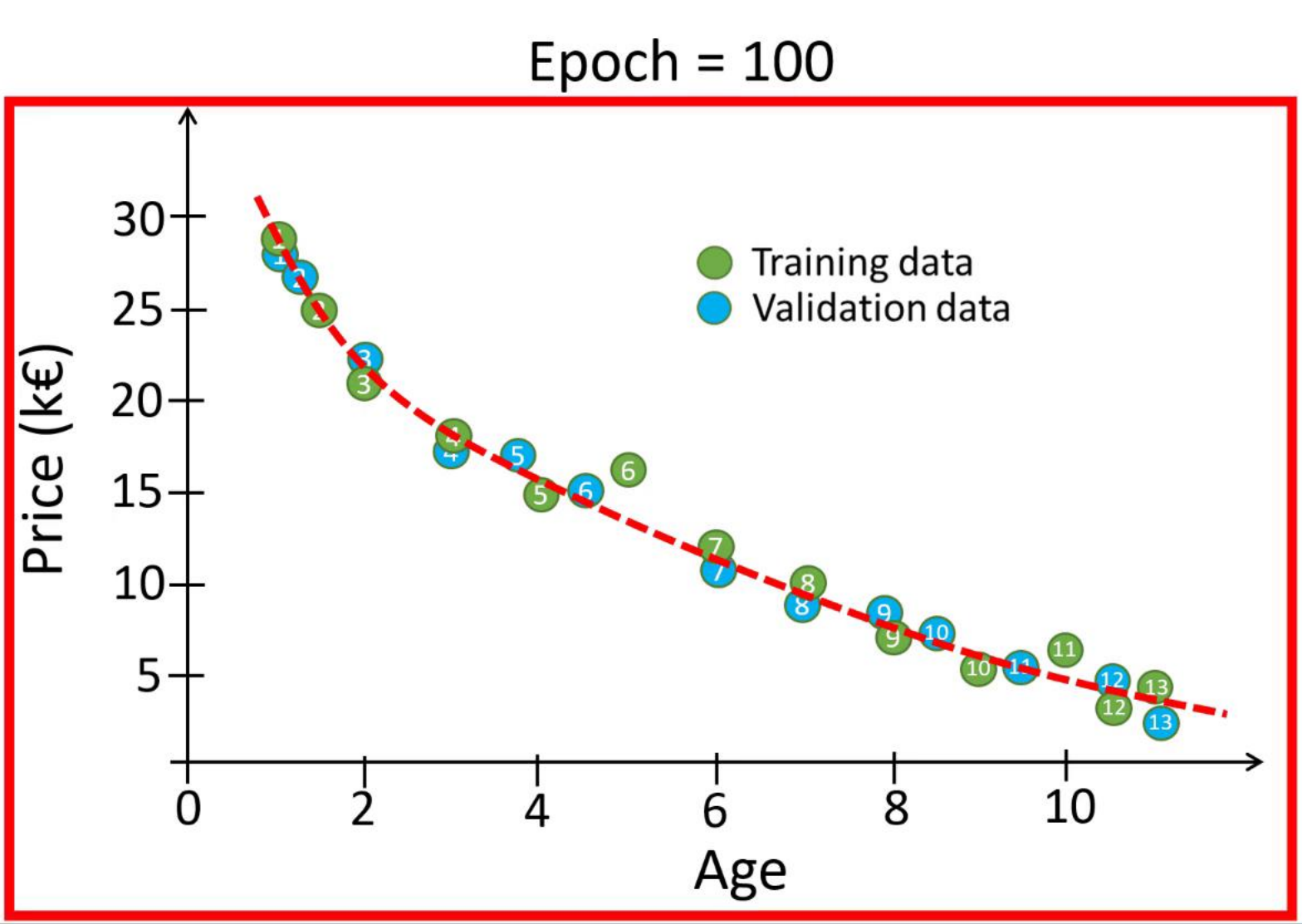


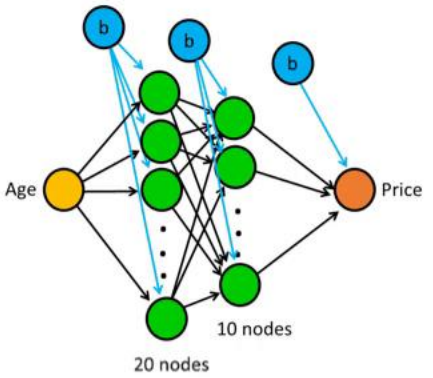


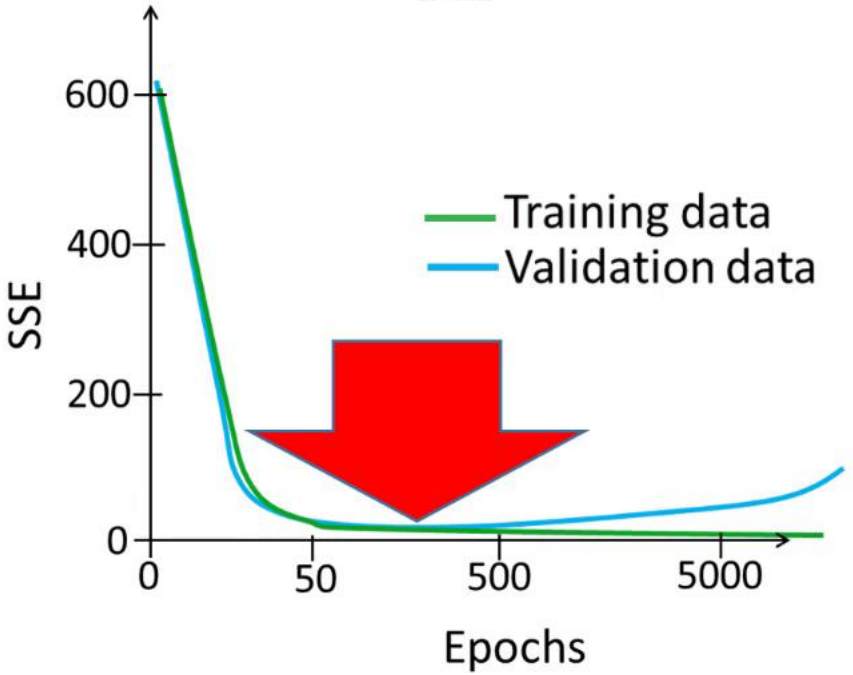


In this example, we would run only 100 epochs because that amount of training results in the lowest validation error.

Early stopping







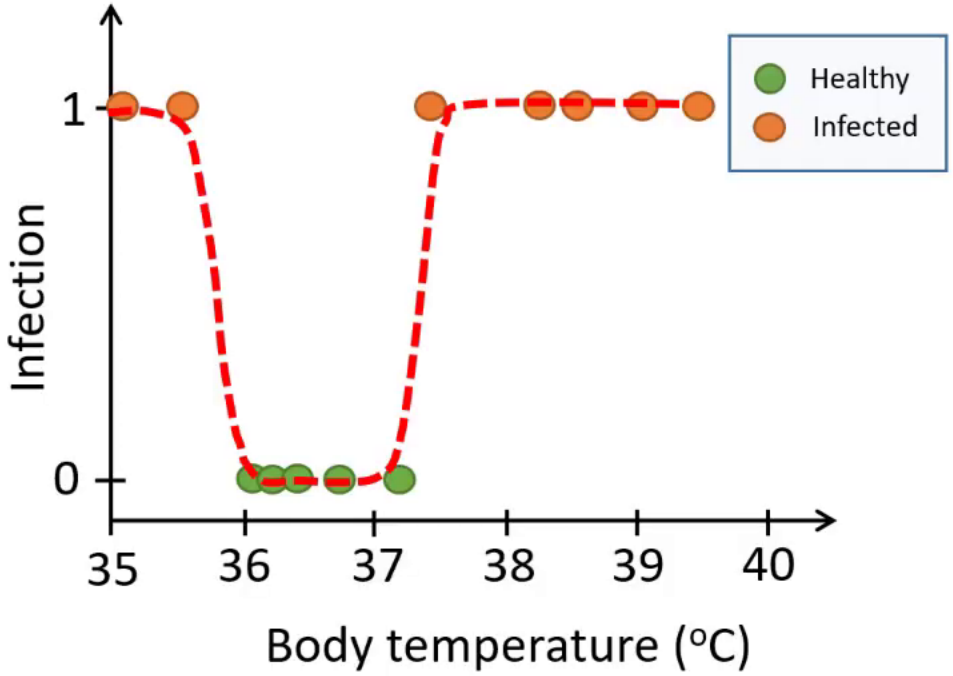
The shape of the curve resembles quite well how we would expect the price should decline for old cars. By using early stopping, we have therefore obtained a network that can generalize well on unseen data.

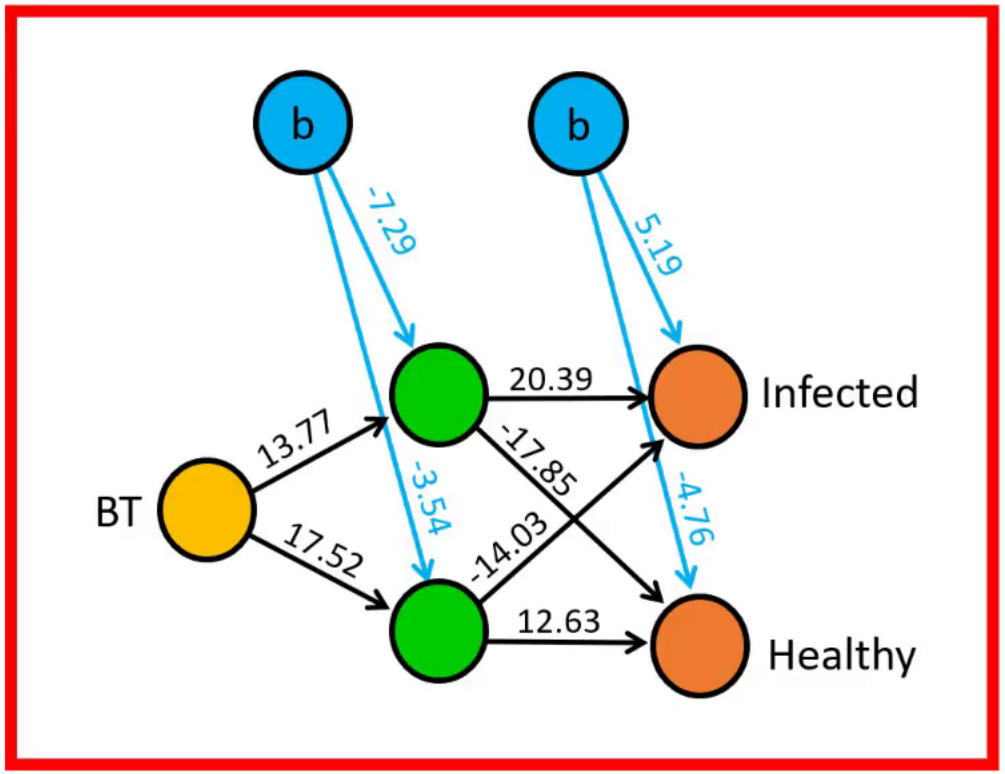
Dropout

- Early stopping
- Dropout

Another common method to prevent overfitting is to use the dropout method.

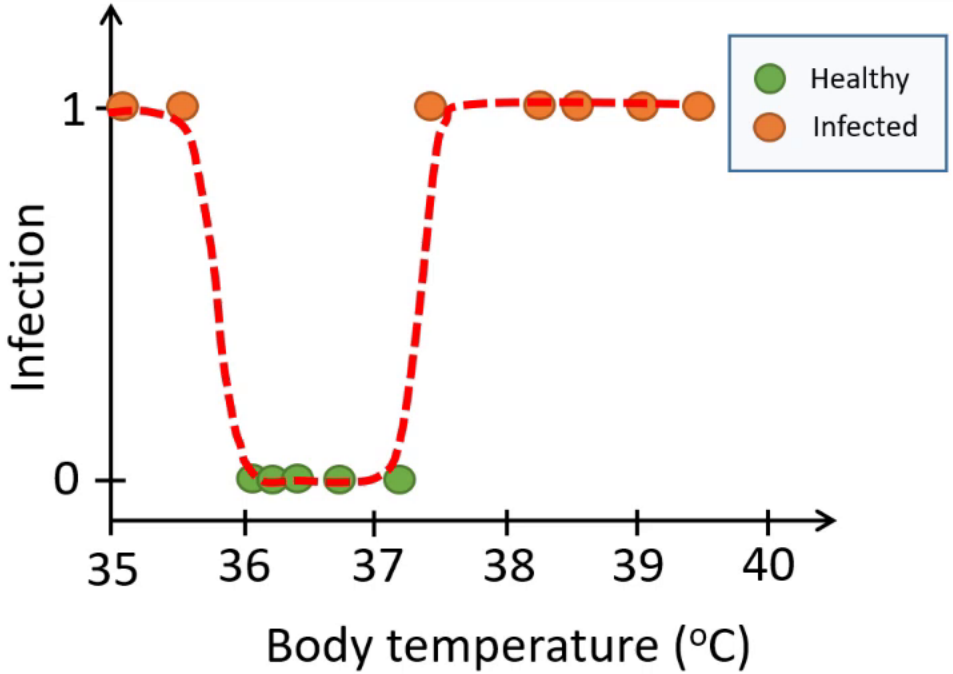
Dropout

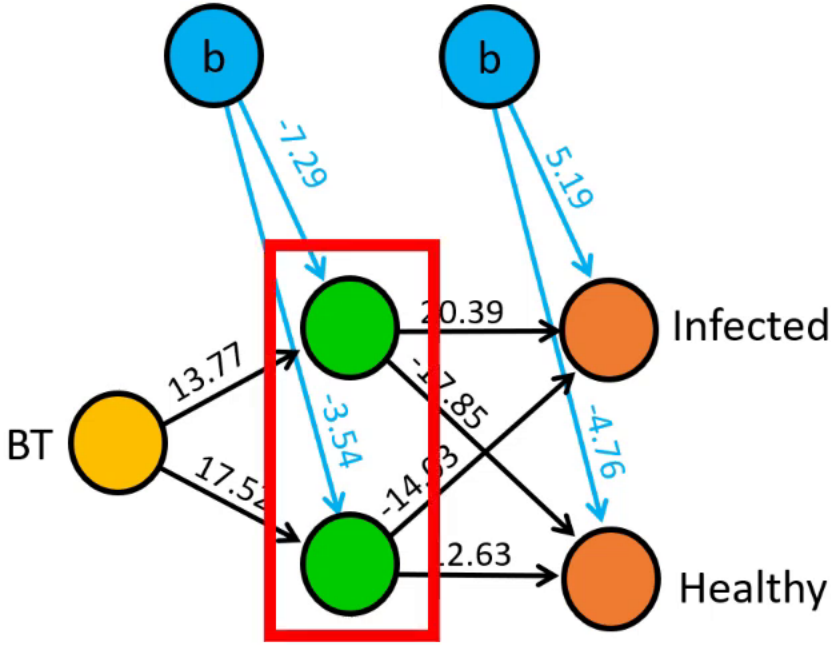




I will use our previous example to explain how the method of dropout works.

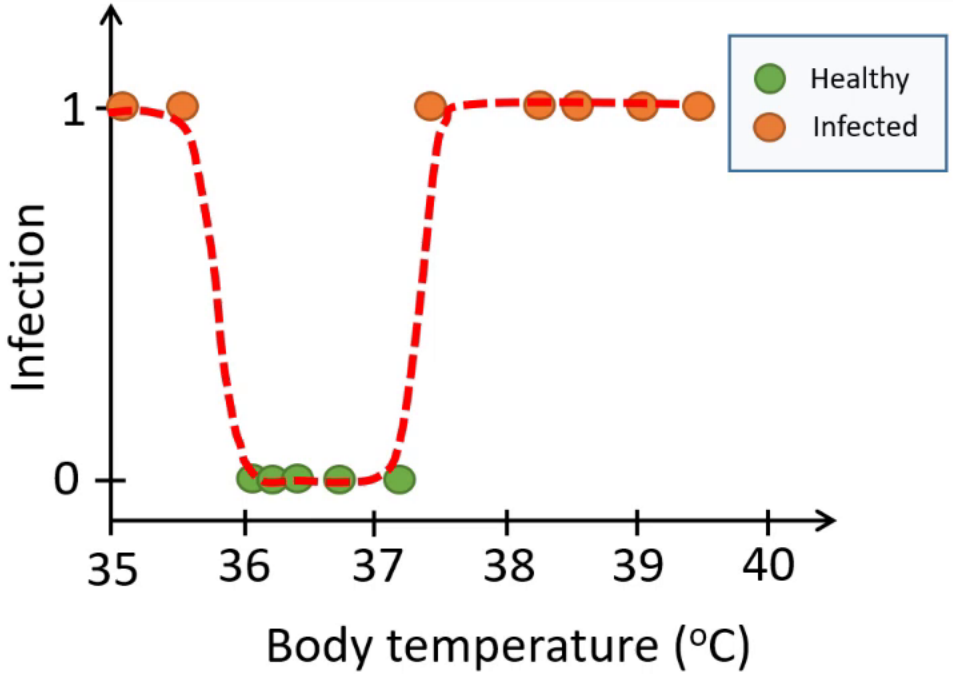
Dropout

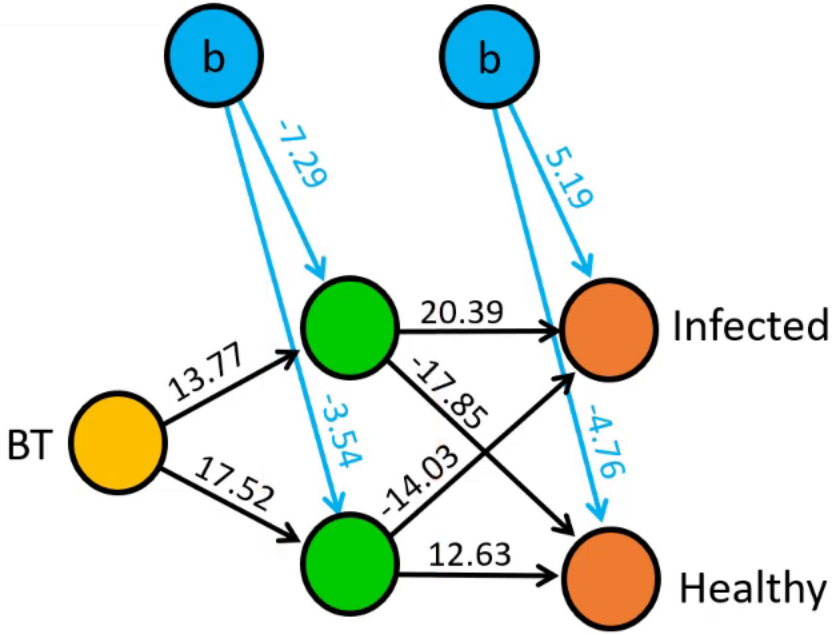




During training, the dropout method randomly deactivates a fraction of nodes, usually the nodes in the hidden layers, during each training iteration. The probability of a node being deactivated is determined by the dropout rate.

Dropout

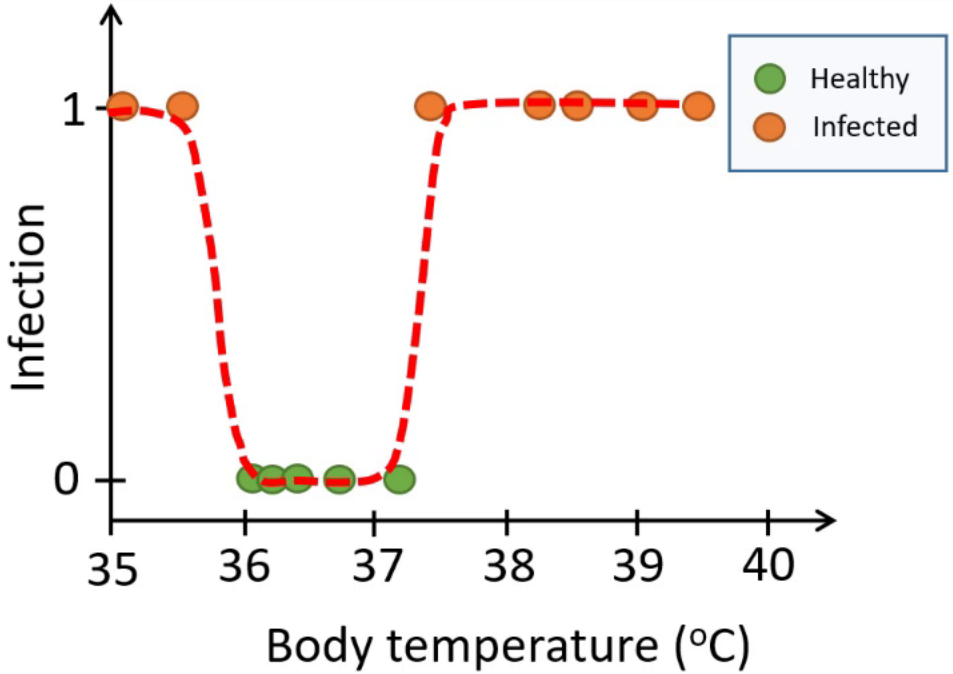


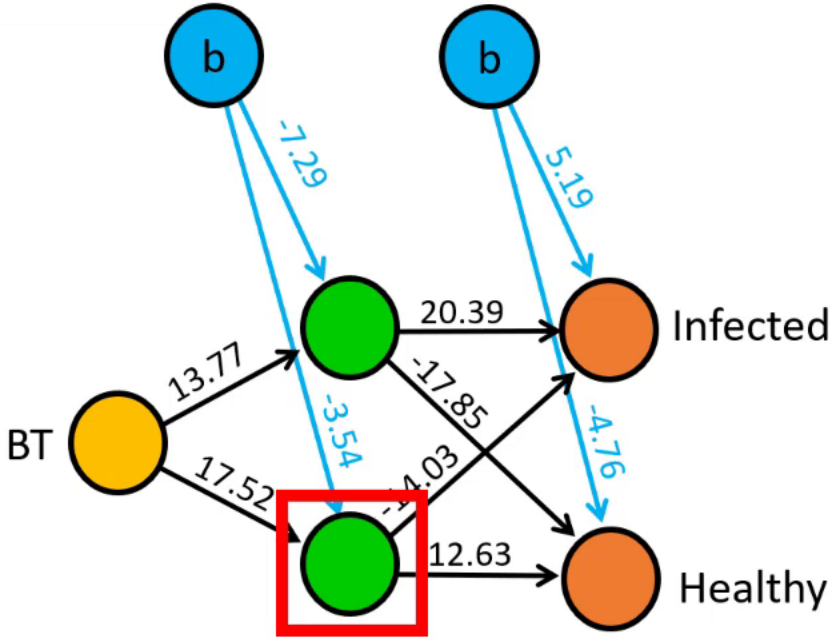


Dropout rate = 0.5

Usually, one uses a dropout rate of 0.5,

Dropout

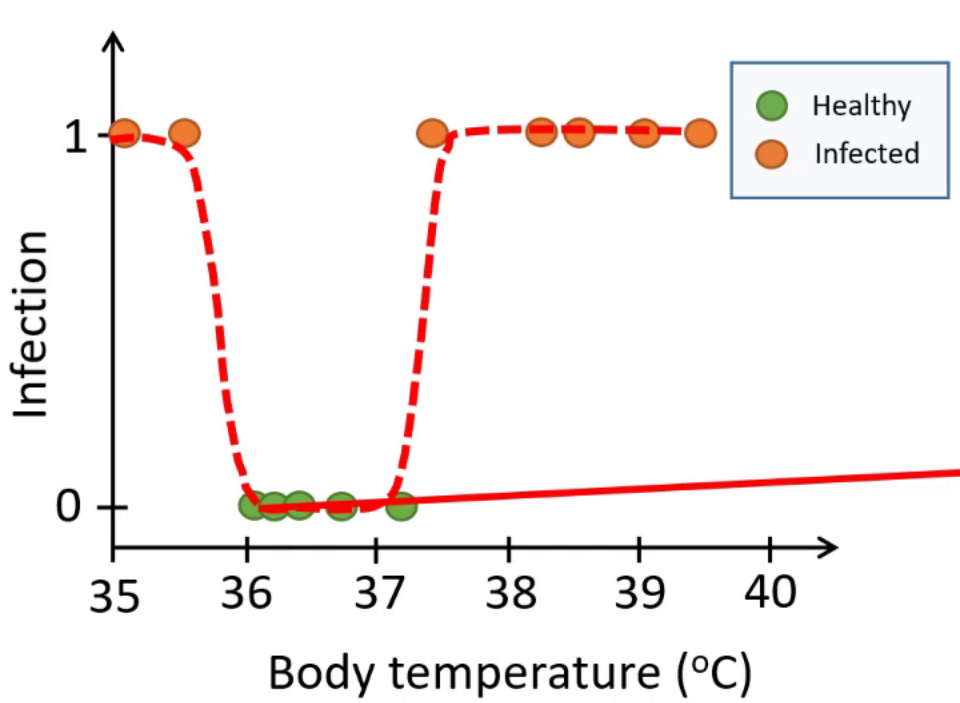


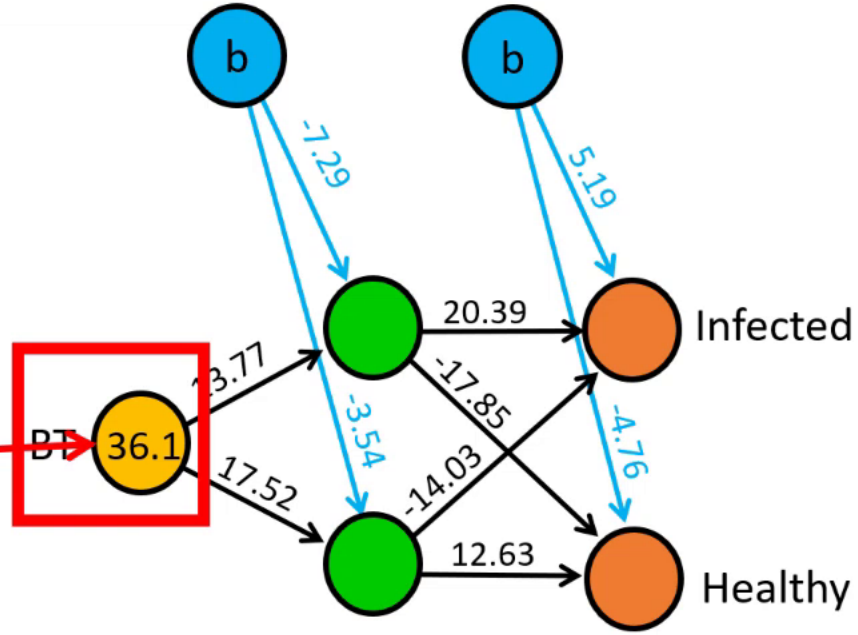


Dropout rate = 0.5

which means that, in every iteration during the training, a hidden node is deactivated by a probability of 50%.

Dropout

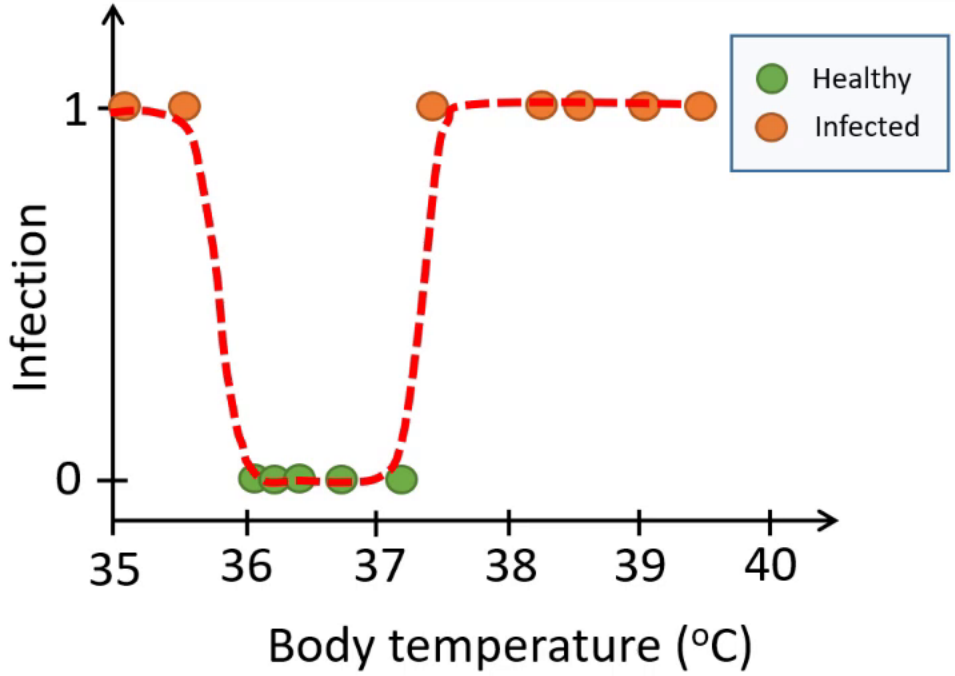


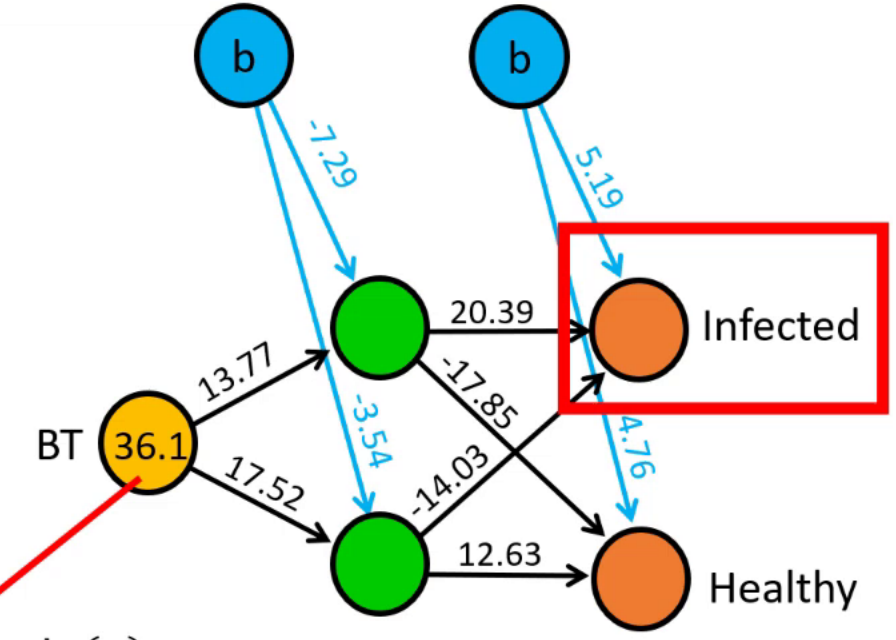


Dropout rate = 0.5

Suppose that we would train this network, where we here use the body temperature of this patient.

Dropout



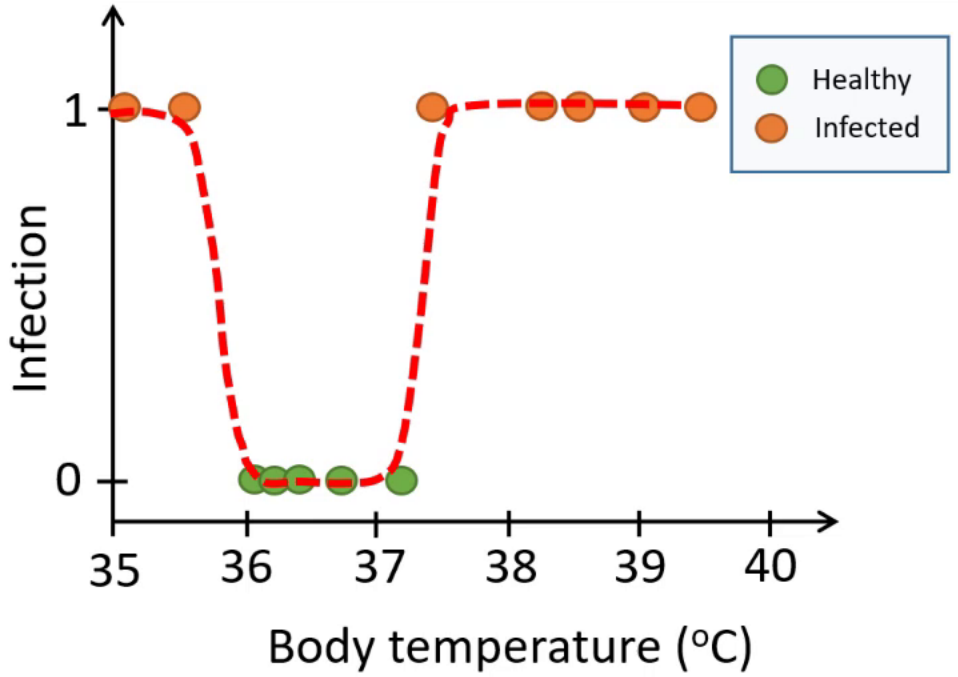


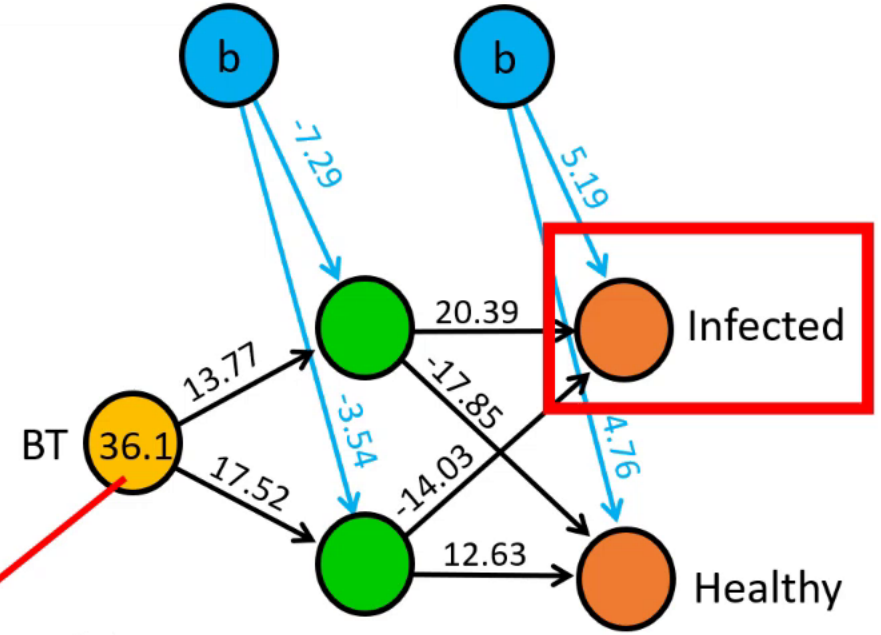
$$x_{norm} = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

Dropout rate = 0.5

we normalize 36.1,

Dropout





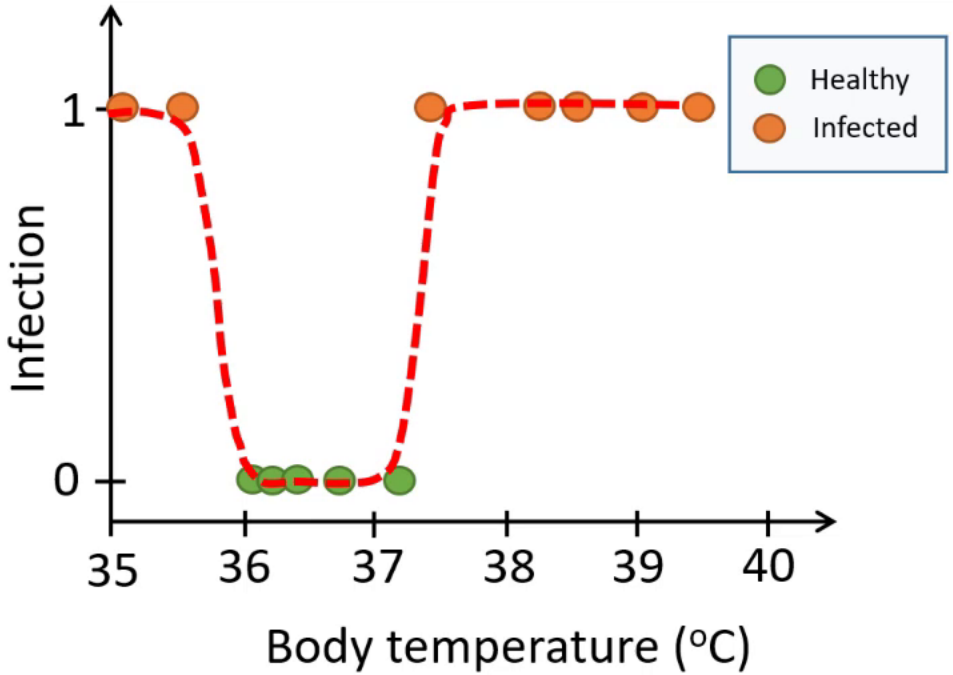
Dropout rate = 0.5

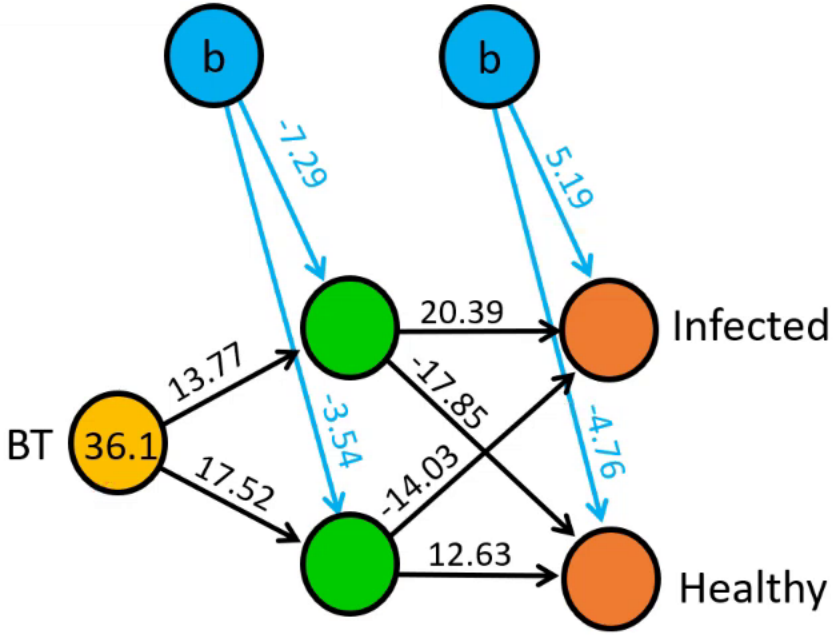
$$x_{norm} = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} + (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}}$$

and plug in the normalized value in the equation,

Dropout



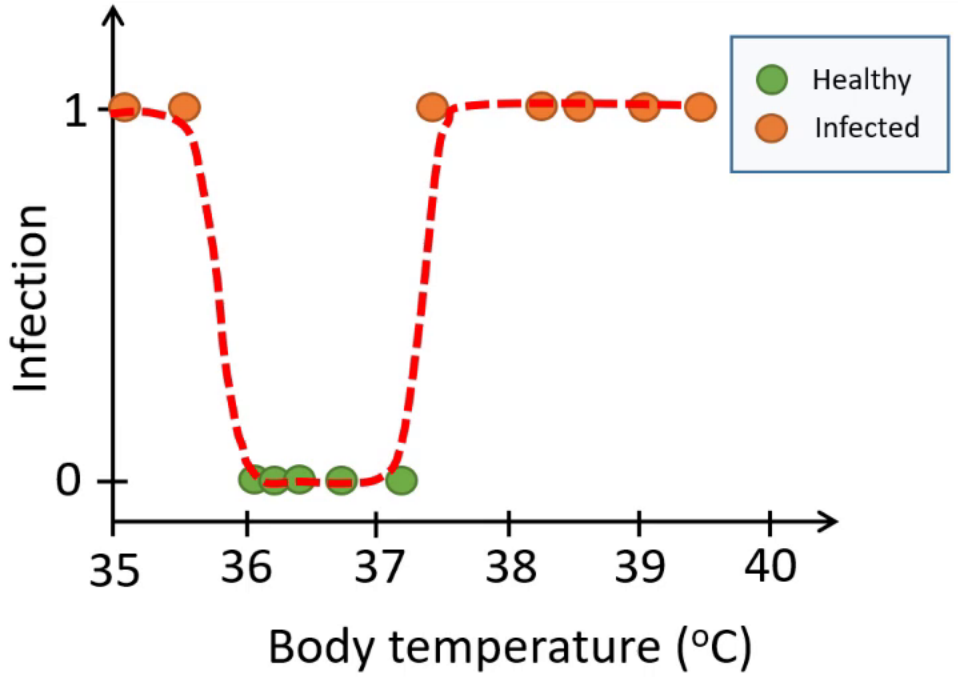


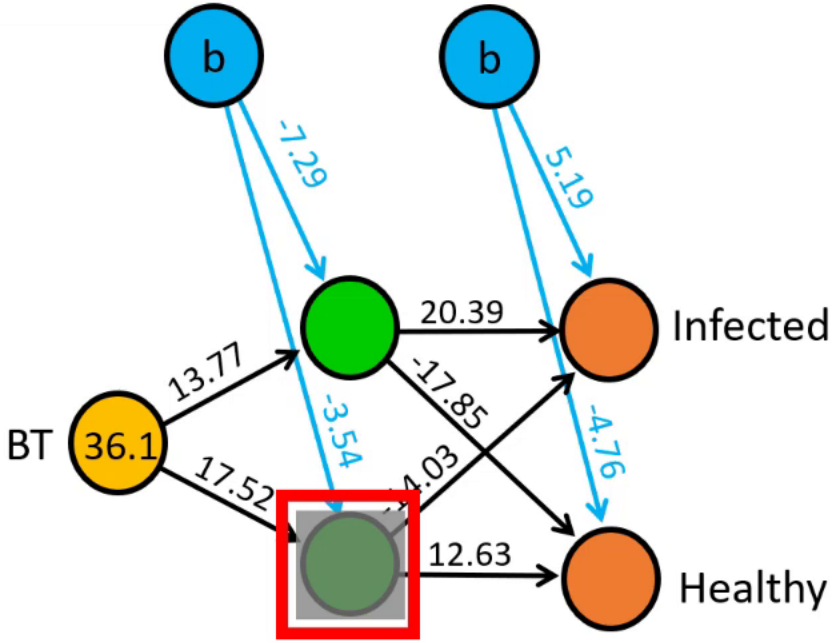
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} + (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}} = 0.019$$

and do the math. Since this value is close to zero, the network correctly predicts that the person is healthy, which means that the network barely needs to change the values of the weights because the predicted value is close to zero.

Dropout



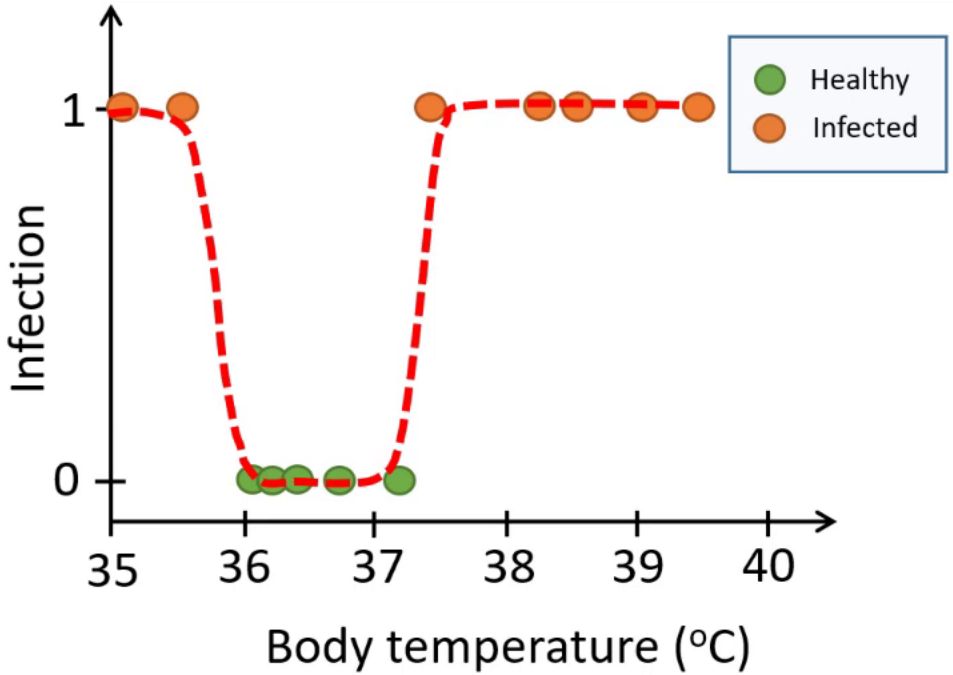


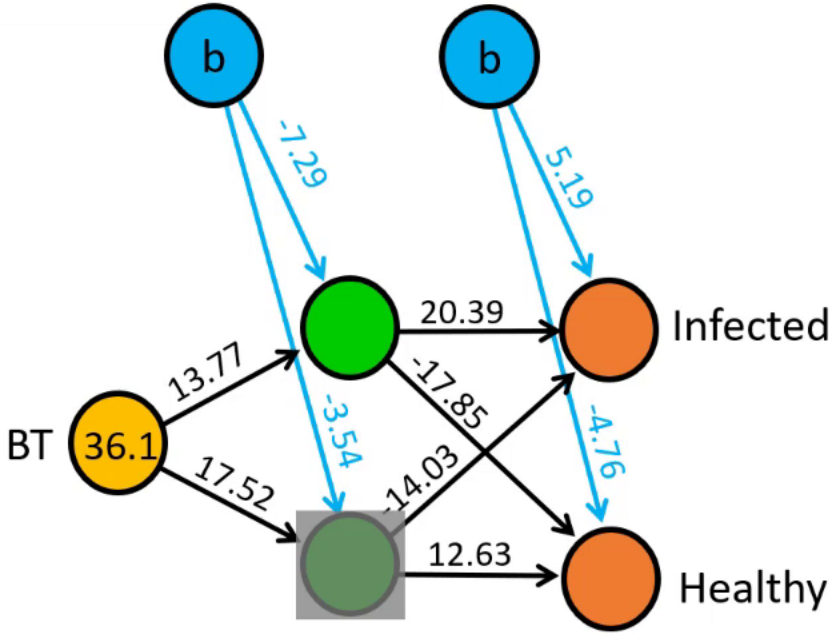
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} \quad + \quad (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}} = 0.019$$

But what would happen if the dropout process deactivates, for example, this hidden node?

Dropout



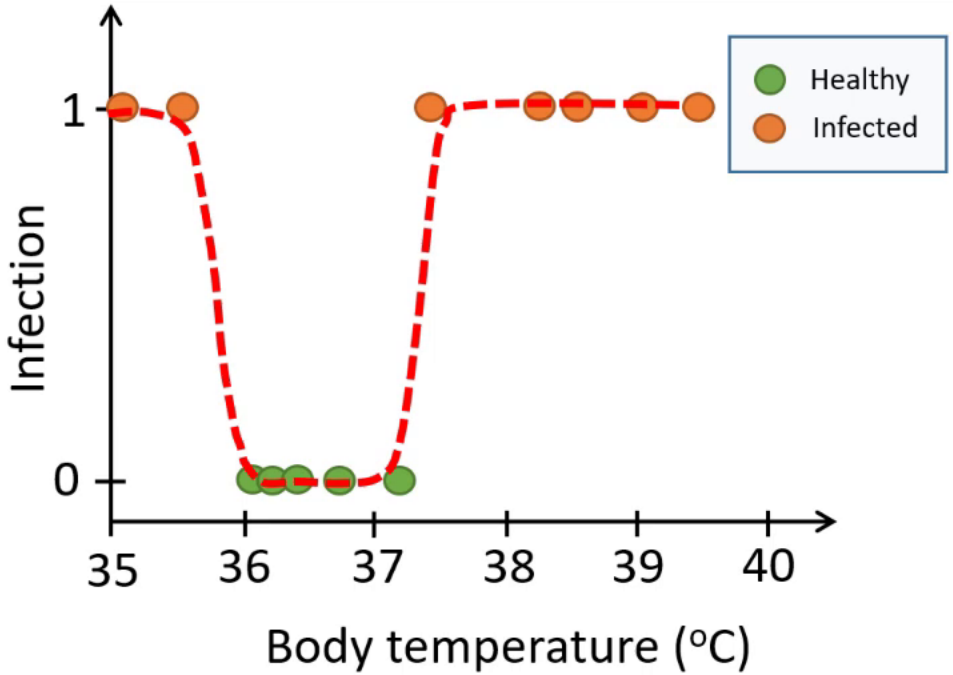


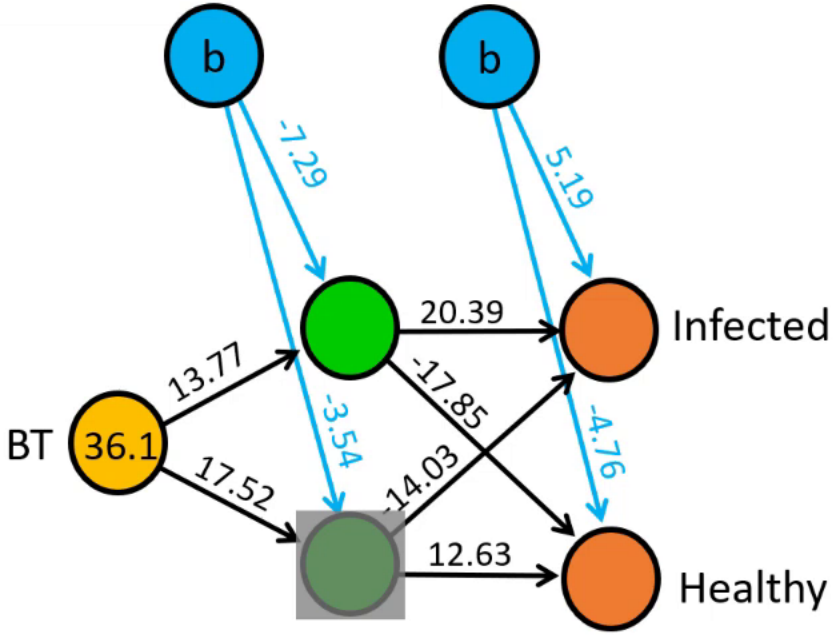
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} + (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}}$$

Well, that would mean that this part of the equation would be set to zero,

Dropout



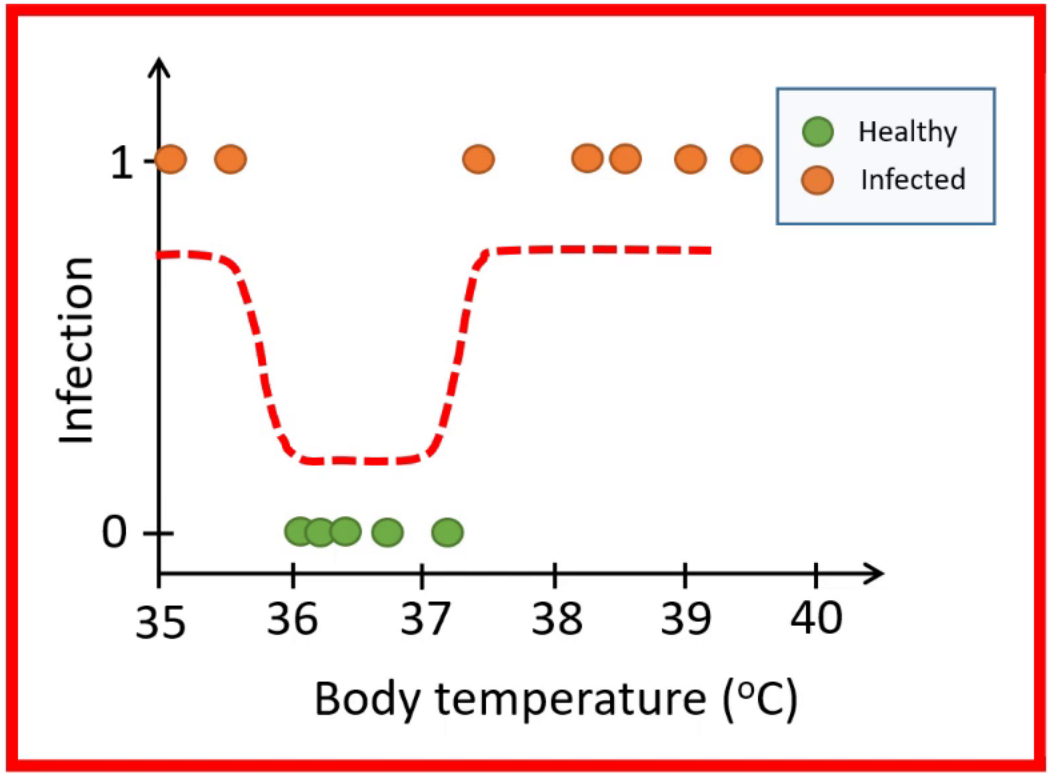


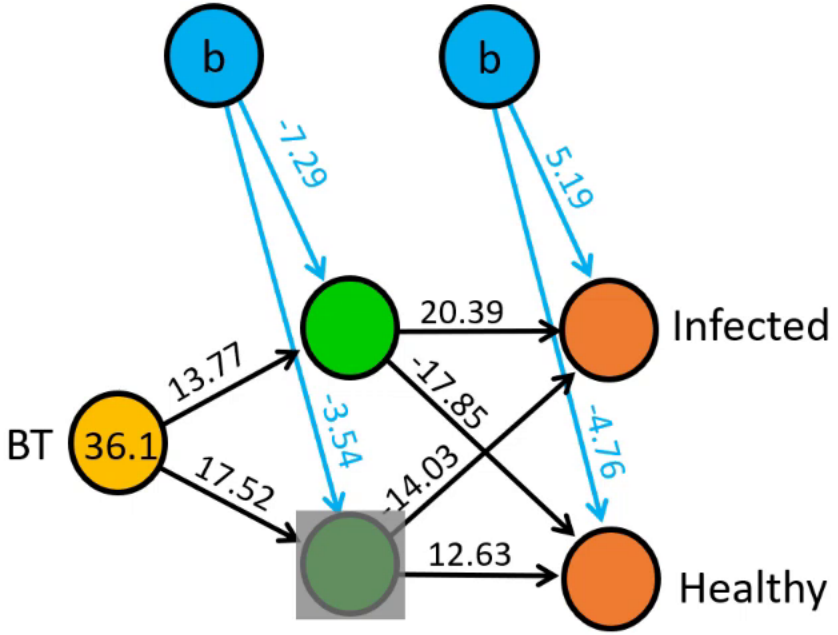
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} + (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}} = 0.996$$

and that the predicted value is now 0.996, which is a bad prediction because the healthy person is now predicted as being infected.

Dropout



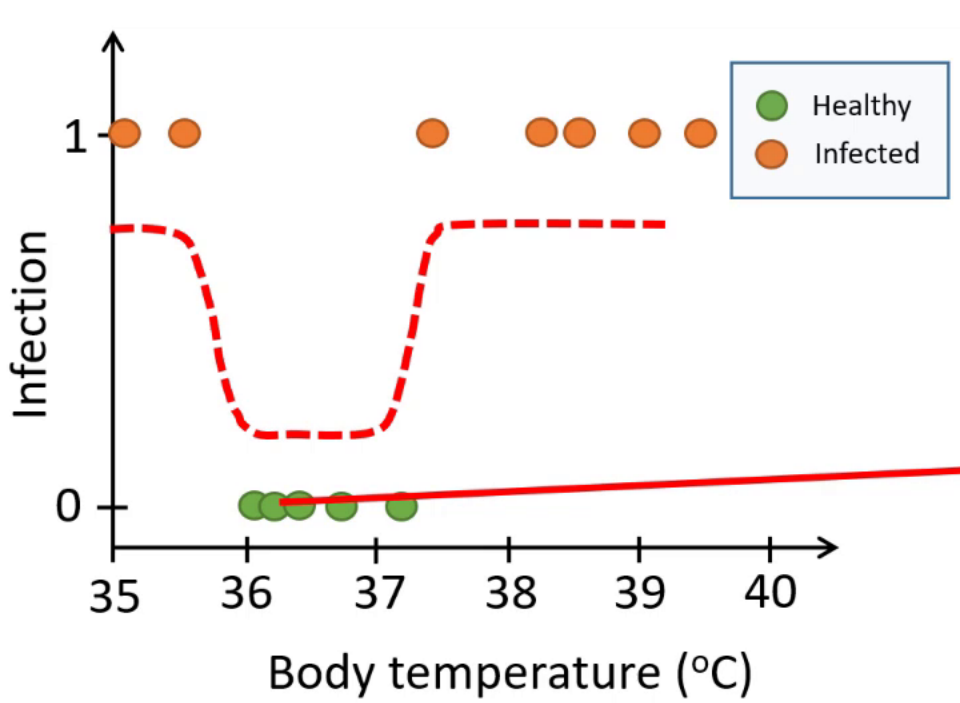


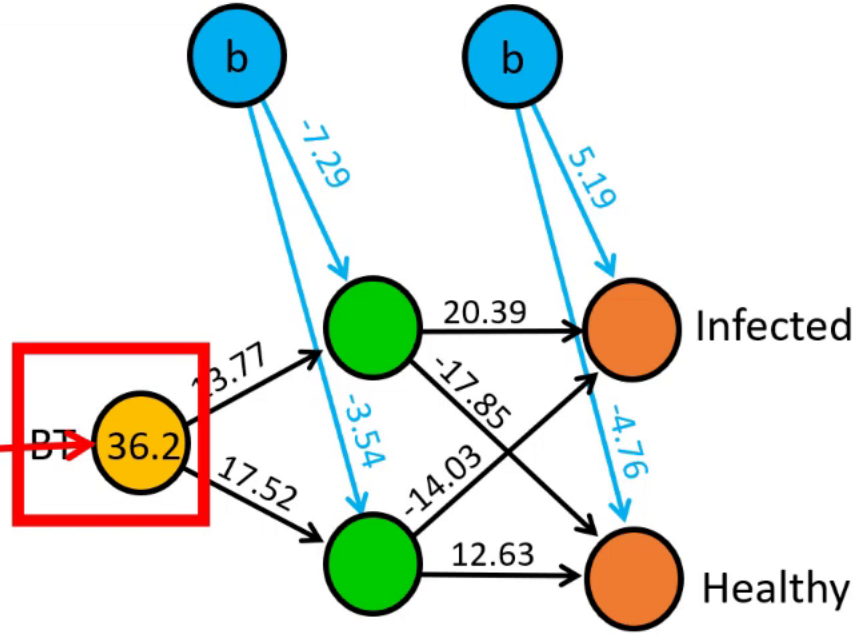
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} + (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}} = 0.996$$

This means that the network will start to change the weights towards a worse fit to the data because it tries to correct the "error".

Dropout



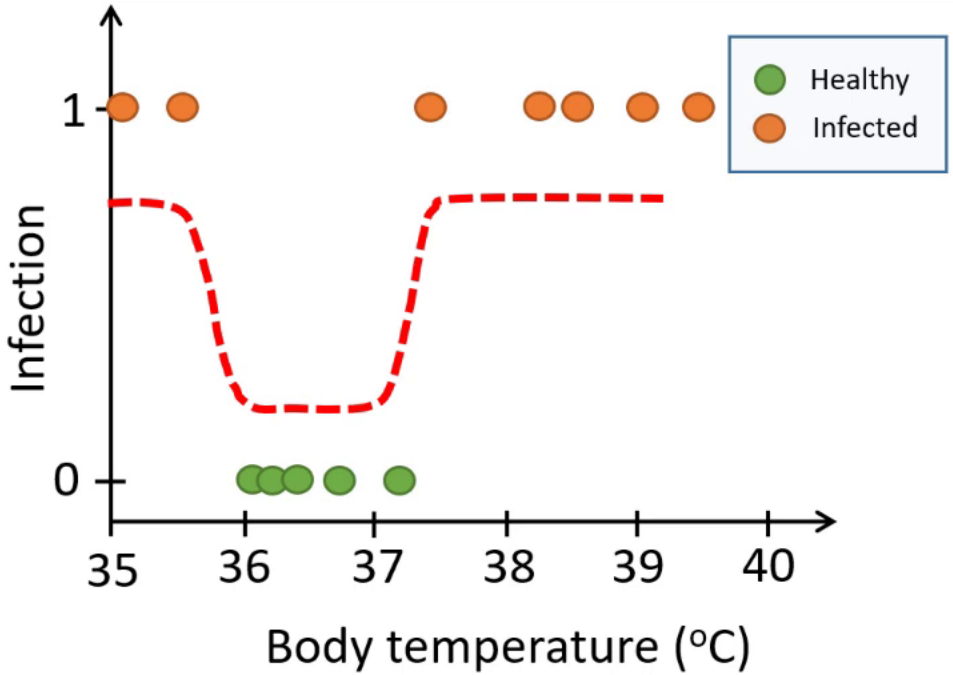


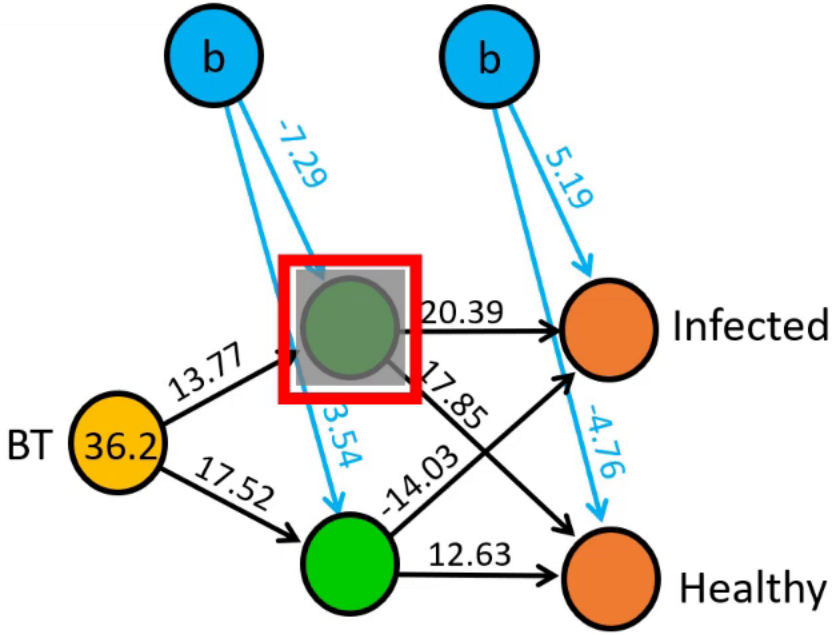
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} + (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}}$$

After the network has updated its weights, it takes the next input value.

Dropout



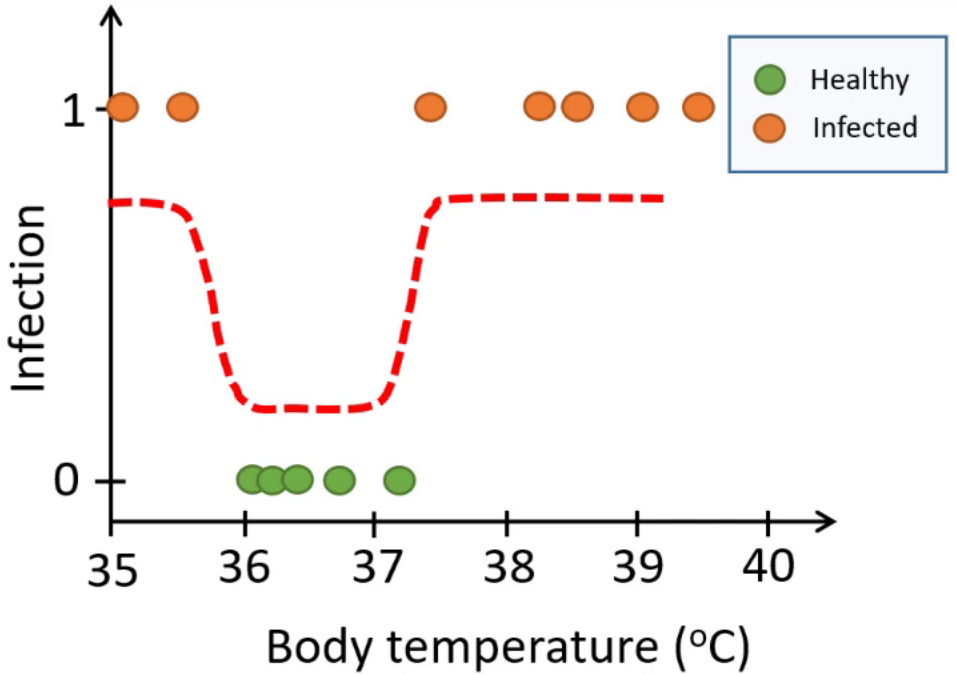


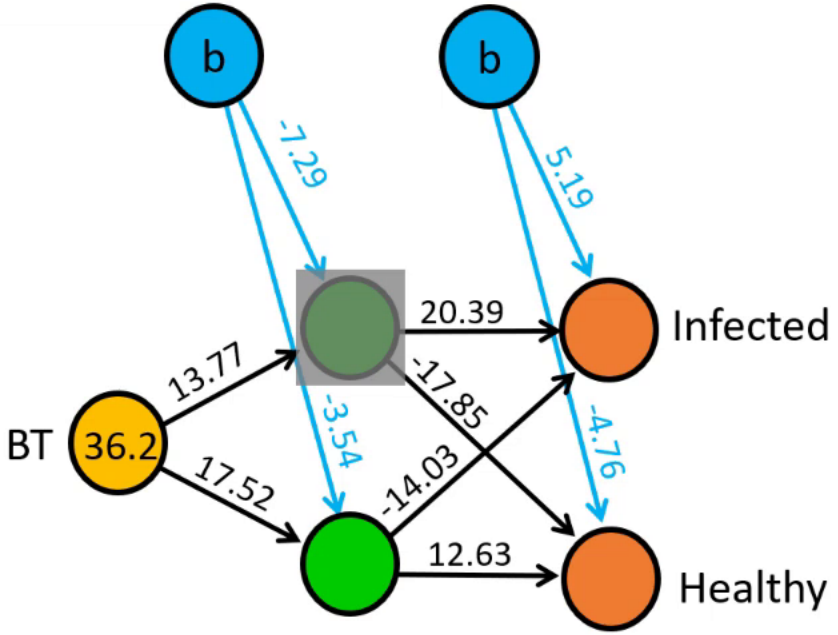
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} \quad + \quad (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}}$$

Now, suppose that this node would be deactivated,

Dropout



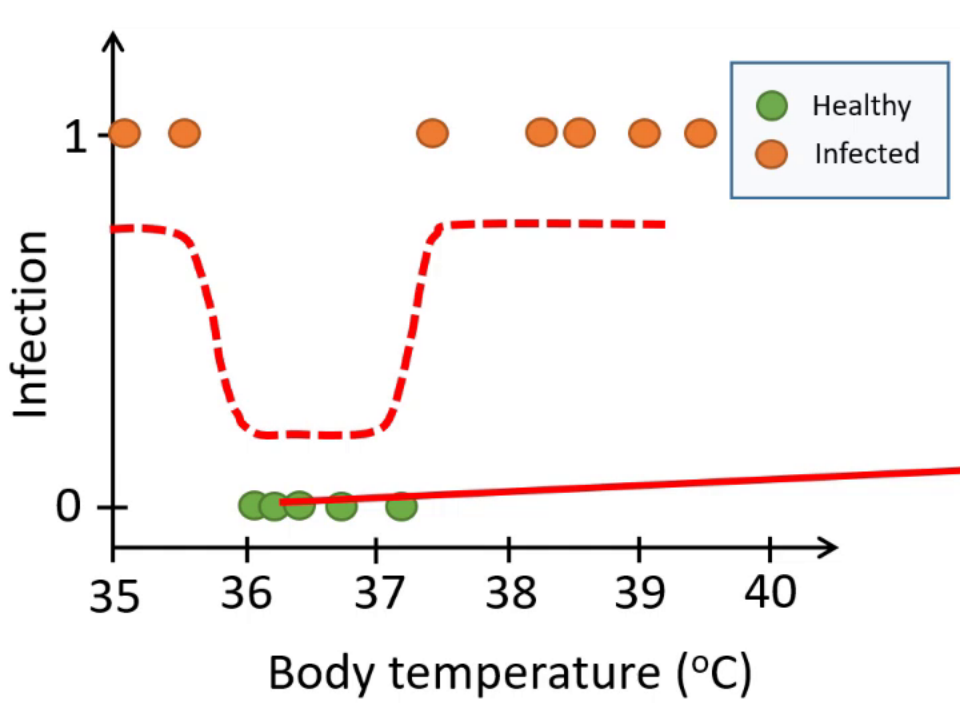


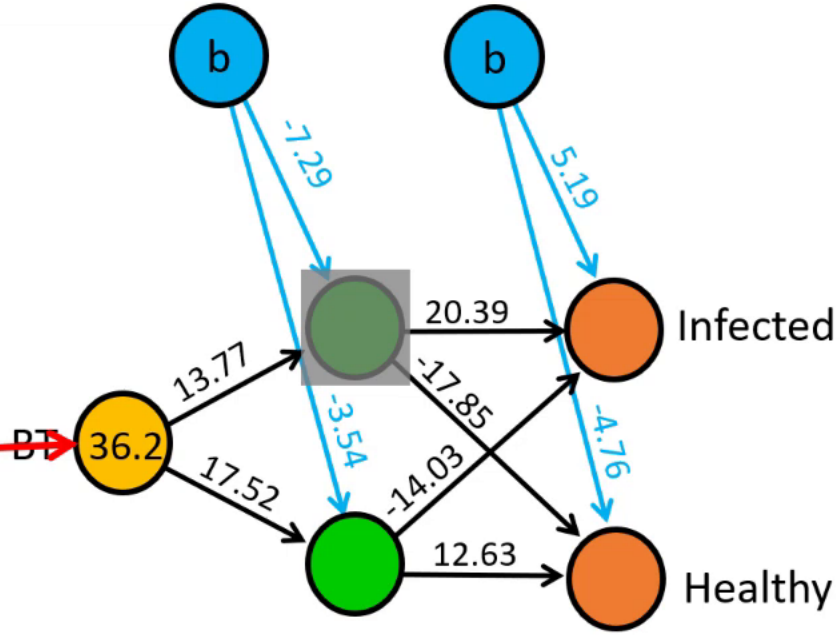
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} + (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}}$$

which means that this part of the equation is set to zero.

Dropout



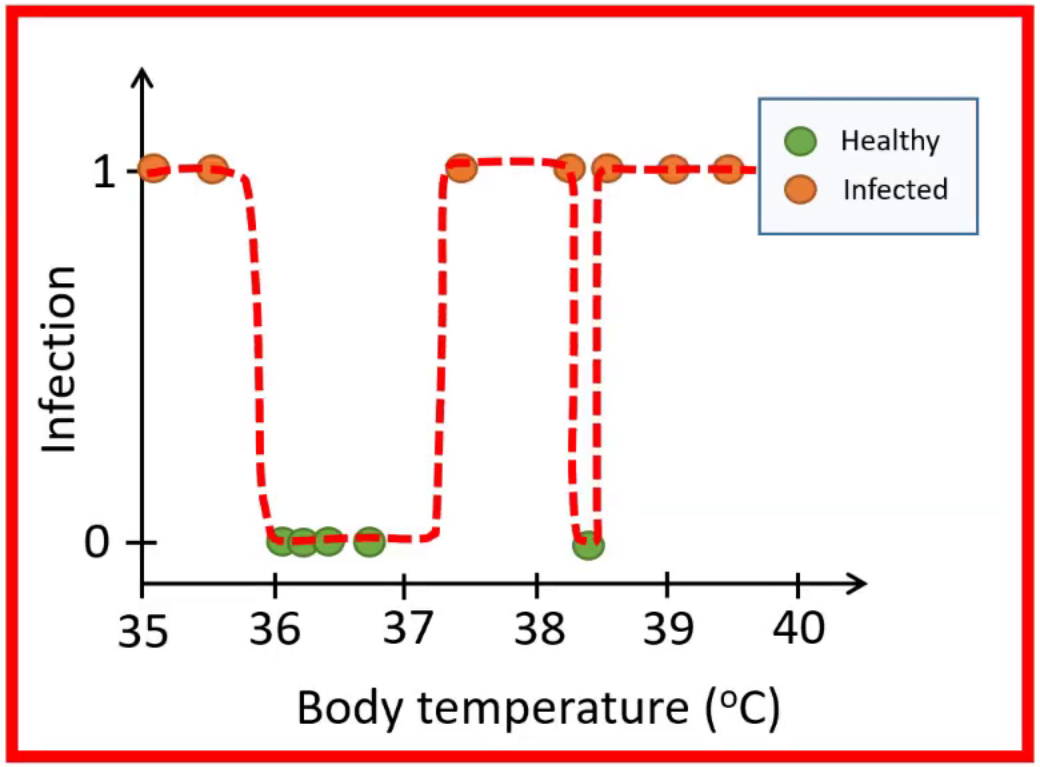


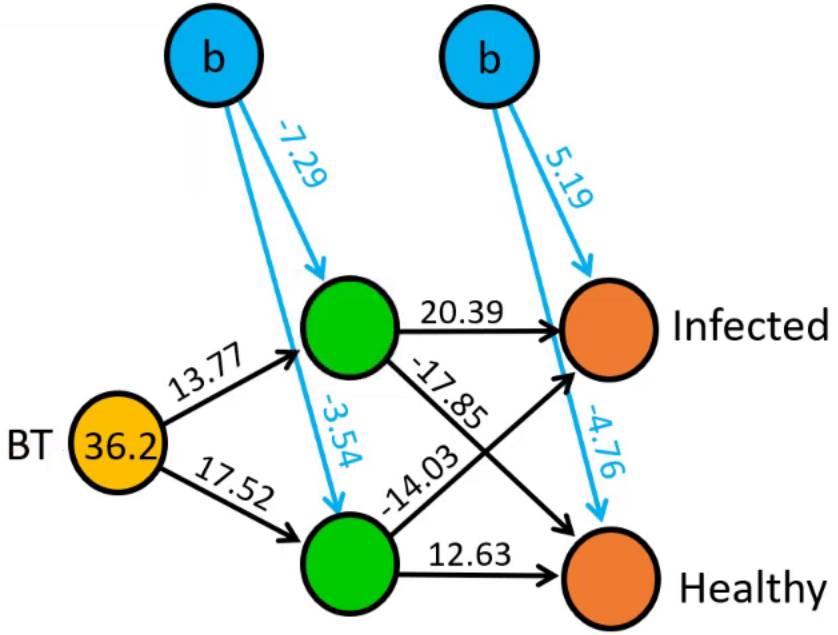
Dropout rate = 0.5

$$f(x_{norm}) = \frac{1}{1 + e^{-\left(5.19 + (20.39)\cdot\frac{1}{1+e^{-(-7.29+13.77\cdot x_{norm})}} + (-14.03)\cdot\frac{1}{1+e^{-(-3.54+17.52\cdot x_{norm})}}\right)}} = 0.0044$$

However, deactivating this node does not affect the predicted value that much. Although we have deactivated one node, the network still correctly predicts this observation.

Dropout

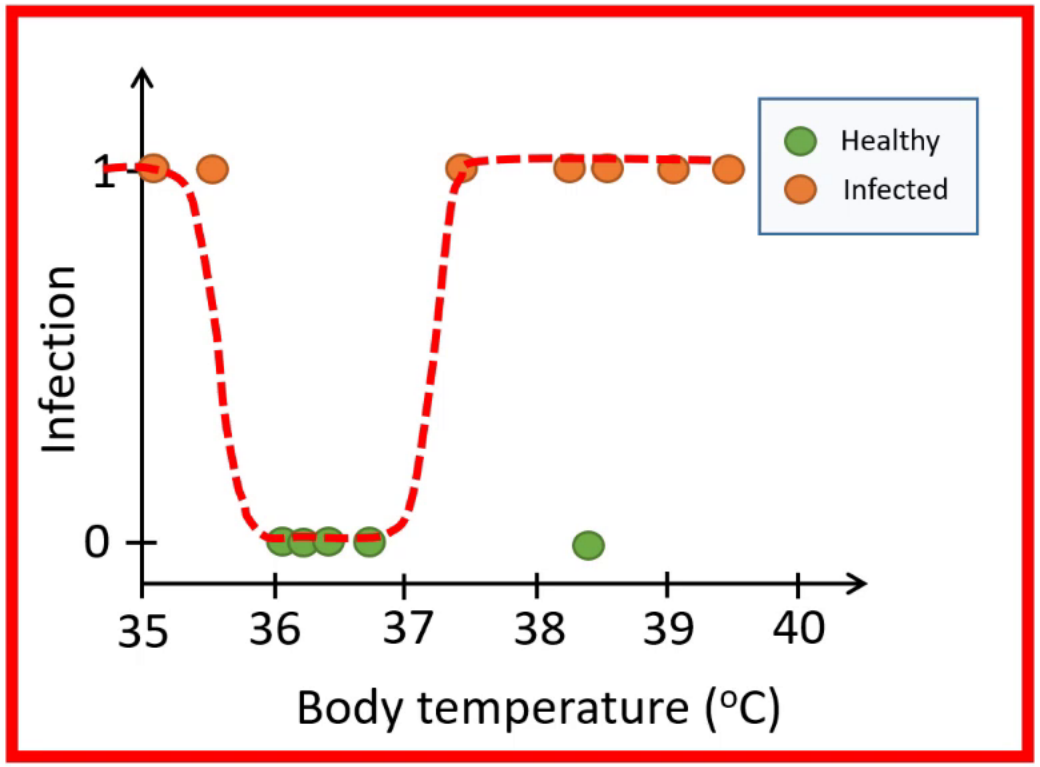


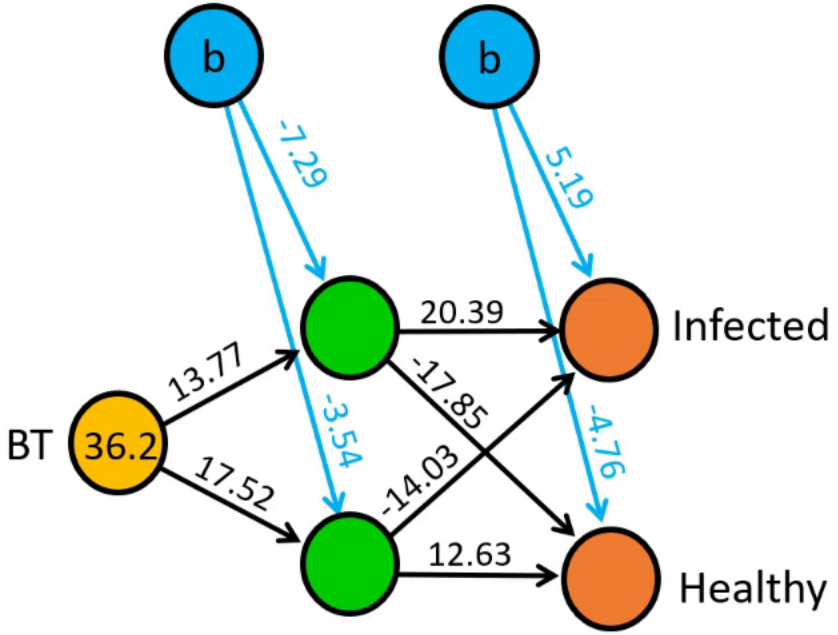


Dropout rate = 0.5

By using the dropout method, we disrupt the training process which prevents the network from learning too well.

Dropout





Dropout rate = 0.5

By using dropout, we can therefore reduce the risk of overfitting.